The infrared triangle on the lattice: the campaign labbook

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Contents

1 Overview: the lattice infrared triangle

1.1 What the continuum triangle is

In four-dimensional quantum field theory three families of statements that look unrelated turn out to be one statement seen from three sides. The first family is the *soft theorems*: when one massless quantum in a scattering process has its frequency taken to zero, the amplitude factorises into a universal kinematic multiplier times the amplitude without that quantum. The second is the *asymptotic symmetries*: transformations that act nontrivially on the data at null or spatial infinity, taken modulo those that die off there, form a genuine symmetry group of the theory rather than a gauge redundancy. The third is the *memory effects*: after a burst of radiation has passed, detectors do not return to their initial configuration but retain a permanent, zero-frequency offset. The triangle is the assertion that the soft theorem is the Ward identity of the asymptotic symmetry, and that the memory is the Fourier-zero-mode content of the same Ward identity. Each edge is a change of variables, not a new physical input.

Two features of that story matter for what follows. First, the triangle is, in the continuum, a family of formal relations: the asymptotic symmetry algebra is defined through boundary conditions at infinity whose function spaces are delicate, and the soft factorisation is an all-orders statement about an S-matrix that is not itself constructed. Second, there are two quite different versions of it. The version usually advertised is the gauge one (photons, gravitons), in which the soft factor has a $1/\omega$ pole because a massless mediator is exchanged. The version relevant here is the *global* one: the soft-pion, or Goldstone, triangle, in which the soft factor has no pole at all but an *Adler zero* — it vanishes linearly as the momentum goes to zero, because the zero-momentum Goldstone mode is exactly a global symmetry rotation of the vacuum and therefore acts trivially.

1.2 Why the lattice version must be the Goldstone one

This campaign asks whether the triangle is a theorem anywhere, and answers the question on a one-dimensional quantum spin chain written in matrix-product-state (MPS) language. There the three corners become finite, checkable statements about tensors: the symmetry acts on-site, the vacua are injective MPS, the excitations are ansatz vectors, and the Ward identity is an exact algebraic rearrangement of a tensor network.

The gauge triangle has no home here. A one-dimensional lattice spin system carries no massless gauge mediator — the natural candidate, the Schwinger model, confines and gives its photon a mass — so there is no mechanism for a $1/\omega$ pole, and any claim of one in this setting is simply wrong. The correct antecedent is the global-symmetry triangle: the soft object is a Goldstone-like magnon, the expected behaviour is an Adler zero, and the leading soft datum is a *slope* rather than a residue. Two further structural facts fix the setting. Mermin–Wagner and Coleman forbid continuous type-A symmetry breaking in a one-dimensional ground state, so the triangle must live either on a type-B Goldstone (the isotropic Heisenberg ferromagnet, exact product vacuum, magnon dispersion $\omega \sim k^2$, exact Bethe scattering data available as an independent oracle) or on a discretely broken symmetry (the easy-axis XXZ ferromagnet, two product vacua, domain walls, gapped magnons, so that "soft" means a low-frequency expansion about the gap). And there are no boosts on a lattice, so the universal factor cannot take a Lorentz-covariant form; its correct dependence on the legs' group velocities is part of the problem rather than an obstacle to it.

The two model classes are used throughout. The ferromagnetic chain supplies exact magnons and an exact two-body scattering matrix; the easy-axis chain supplies exact kinks and a wall-position observable whose displacement is the memory. Neither model alone exhibits every corner, and that tension is structural: it is respected in what follows, not papered over.

1.3 The four headline findings

(i) Memory quantization needs no scattering theory. The naive expectation, inherited from the continuum, is that the memory is quantized because it is the zero-frequency limit of a soft factor, so that one must first have a soft theorem and an S-matrix before one can say anything about the permanent displacement. That expectation is wrong here, and its failure is the campaign's most robust positive result. Fix a finite window $W = [a, b]$ of the chain, a cut $c_0 \in W$, and the windowed wall-position observable

$$\mathfrak{X}_W \;=\; a - 1 + \frac{1}{2s}\sum_{x=a}^{b}\big(S^z_x + s\big), \qquad \hat{Q}_{W,c_0} \;=\; 2s\big(\mathfrak{X}_W - c_0\big), \tag{1}$$

where $\pm s$ are the charge densities of the two vacua. If the on-site charge merely generates a circle — that is, if $e^{2\pi i S^z_x}$ is a scalar, which holds for every spin-S chain — then the spectrum of $\hat{Q}_{W,c_0}$ lies in a single coset of $\mathbb{Z}$, and the increment recorded by measuring this same observable before and after an event is an *integer*. No wave operators, no channel inventory, no completeness, no gap, no Lieb–Robinson velocity: the offsets at the two measurement times are the same number and cancel. This is the finite-window memory-index theorem (Section 8), and it is unconditional. Its infinite-volume companion — that every ordered limit law of the measurement protocol is a probability distribution supported on $\mathbb{Z}$, with $\delta x = -(2s)^{-1}\sum_\nu \nu\, p_\nu$ — is proved as a conditional implication whose hypotheses are relaxation hypotheses on the charge history, never scattering hypotheses; and one of those hypotheses turns out to be a free theorem, provable from separability of the algebra and strong continuity of the dynamics alone (Section 9). Symmetry, not dynamics, does the quantizing.

(ii) Scattering only computes the value. What the quantized ledger does not supply is the number multiplying it. That number is on-shell data, and it is obtained by honest two-body scattering. For the fully polarised bilinear ferromagnet at every site spin $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$ the exact two-magnon ratio in the regular channel is computed directly from the separated, double-occupancy and adjacent equations, with no integrability or completeness assumed, and the physical phase has the exact soft slope

$$\partial_{k_s}\delta_{\text{phys}}\big|_{k_s=0} \;=\; \frac{\operatorname{sgn}(v_h - v_s)}{S}. \tag{2}$$

This is the exact spin-S two-magnon slope theorem (Section 10), [PROVED]. The same value reaches the protocol datum only through an identification step that is itself proved on a displayed separated-preparation subclass (Section 12): the value is transported into the infrared statement by matching, never installed by stipulation. The division of labour is the campaign's spine sentence: *symmetry quantizes the infrared data; the excitation ansatz supplies the kinematics; dynamics only picks the values.*

(iii) The naive universality conjecture is refuted. The brief's Conjecture M — that the wall displacement equals the zero-frequency limit of the soft factor, a lattice Braginsky–Thorne relation — is [REFUTED] as literally stated (Section 7). What survives is weaker and sharper: an exact identity expressing the displacement as the finite-time direct-current weight of the *physical* boundary current, and a conditional charge-bookkeeping law. Independently, the stronger claim that the soft factorisation is universal over unrestricted local or quasi-local sources is also [REFUTED] (Section 13): an explicit four-site source leaves the one-hard amplitude untouched while shifting the linear soft coefficient. Universality survives only relative to a named class of sources with no contact term, and even there the class constant is left open by the factorisation argument itself. These refutations are presented here as results, with the surviving weaker statements attached, because a sharp refutation plus its survivor is worth more than a pile of analogies.

(iv) An SPT rigidity dichotomy. When the symmetry is realised projectively on the virtual space — the one-dimensional symmetry-protected-topological (SPT) situation — the campaign separates two kinds of infrared coefficient (Section 14). Registered *edge* residues are rigid: in a fixed Schmidt/edge register the compensated group residue is the projective representation itself, the Hermitian residue has spectrum congruent to a fixed offset modulo $\mathbb{Z}$, and its irreducible dimension is bounded below by the minimal dimension of a projective irreducible representation of the class. Normalised *bulk* coefficients are not rigid: along a path with a common gap and continuous external data they vary continuously, and become topological only after a separate local-constancy argument that is not supplied. Quantized edge residue, drifting bulk coefficient — that dichotomy, rather than a blanket "the class appears in all soft data", is what the analysis supports.

1.4 How to read this document

Every result in this labbook is stated under a descriptive English name and carries a coloured status tag. The five tags, and exactly what each asserts, are:

[proved] a structured proof exists and survived the capped hostile review round with no open fatal or major objection on the promoted statement.

[proved, conditional] the same, but what is proved is an implication. The antecedent is displayed inside the statement; it is never moved into a footnote and never assumed away.

[sketch] an argument exists and its gaps are named. The statement says precisely what is established and what is missing.

[conjecture] a precise statement with no proof. Whatever evidence exists — a numerical falsifier that was survived, an oracle agreement — is described as evidence and not as proof.

[refuted] false as stated. The mechanism that kills it is given, together with the weaker statement that survives.

Immediately after each definition, theorem, conjecture or refutation a small grey *provenance block* records four things: the campaign identifier used in the repository's claims file, the file and proof step where the statement is established, the checker or numerical experiment that tests it (or a dash, if none does), and the review record. That block is the only place in this document where campaign identifiers carry weight. The prose never argues from an identifier; it argues from named results and cross-references.

Because the campaign has accumulated a large private vocabulary — claim identifiers, numbered definitions, named hypotheses in parentheses — Section 2 is a complete dictionary. Every identifier that appears anywhere in the campaign's claim ledger is listed there with a plain-English gloss and a pointer to the section that owns it. A reader who meets an unfamiliar tag in a provenance block should look it up there and nowhere else; in particular, this document is meant to be readable front to back without ever opening the repository.

The remaining sections follow the logical order rather than the historical one. Sections 3 and 4 fix the setting and the lattice Noether pair; Section 5 is the asymptotic-symmetry corner; Sections 6 to 9 build the memory corner, from the kink model through the quantization theorem to the relaxation hypotheses it rests on; Sections 10 to 13 build the soft corner, from exact two-body data through wave operators to the soft index and the universality class; Section 14 is the rigidity dichotomy; Section 15 records what generalises to two spatial dimensions; Section 16 collects the refutations and obstructions in one place; Section 17 the numerical instruments; and Section 18 states, corner by corner and edge by edge, exactly where the triangle stands.

1.5 The honesty contract

One rule governs this document above all others: **every status written here matches the campaign's claim ledger exactly**. A conditional theorem is labelled conditional and displays its condition. A statement whose proof holds only on a restricted register, a restricted packet class, a projected component, or a single model says so in the statement itself, not in a remark. Nothing is rounded up: a sketch is never described as "essentially proved", a survived falsifier is never described as a verification, and a theorem about an ansatz-conditioned construction is never described as a theorem about the true model. Where a claim was once stated more strongly and has since been narrowed — and there are several such cases, including one projection identity that is exact for a single magnon and false for two — the erratum is written out rather than quietly absorbed.

The reason for the rule is not decorum. The campaign's one non-negotiable constraint is that it must never carry a label stronger than its evidence: a claim resting at sketch or conjecture indefinitely costs nothing, whereas a single mislabelled statement would make the whole ledger worthless. This labbook inherits that constraint, and any commit that changes a status in the ledger is required to change the corresponding statement here in the same breath.

2 The reader's dictionary

This section exists so that no other section needs a glossary. Every identifier the campaign uses — the claim tags that appear in provenance blocks, the numbered definitions, and the named hypotheses that appear in parentheses inside theorem statements — is listed here exactly once, with a one- or two-sentence explanation in plain English and a pointer to the section that states the object in full. Nothing here is a proof, and nothing here carries a status: statuses live with the statements.

The registry is in three parts, each ordered so that an unfamiliar tag can be found by eye. Table 1 lists the claim identifiers alphabetically. Table 2 lists the numbered definitions in numerical order, which is also their logical order. Table 3 lists the named hypotheses, protocols and imported technical terms alphabetically, ignoring the parentheses that usually surround them.

A note on reading the names. Claim identifiers are historical: they were assigned as lanes of work opened, not as a taxonomy, so a prefix is a weak hint and never a promise. A suffix such as `-r1` or `-r2` marks a revision round, and a row whose identifier ends in `-r1` is frequently the withdrawn form of a statement whose repaired form lives elsewhere. A prime, as in `SPT-E'`, marks a rebuilt row that supersedes the unprimed one of the same name.

2.1 Claim identifiers

Table 1: Every claim identifier in the campaign's claim ledger, with a plain-English gloss and the section that owns the statement.

identifier	what it says, in English	owned by
`A-INDEX-PEPS`	Selection rule for string endpoints in a fixed-point tensor network with anyonic string operators: an endpoint can only move the boundary label along an allowed fusion channel. Conditional on displayed typing hypotheses for the tensor network.	Section 15
`A-INDEX-TC-fin`	The exact toric-code version of the same statement: a ribbon operator with one endpoint inside a disk shifts the disk's electric/magnetic label by exactly the ribbon's type, and a closed probe reads off the braiding character.	Section 15
`A1`	The unbroken-symmetry endpoint theorem: half-infinite symmetry strings act on states exactly as single virtual bond insertions, they are not implemented by any strongly convergent operator sequence unless the virtual representation is scalar, and the endpoint data form a projective-linear torsor.	Section 5
`A2`	The broken-symmetry counterpart: every finite string stays in the vacuum sector, but the half-infinite limit exists only weakly and lands in a disjoint kink sector — a sector jump. Vacuum pairs are classified by a double coset.	Section 5
`A2-orbit-r1`	The withdrawn first-round form of the classification, which named the coset space $(G_L \times G_R)/G_{\text{diag}}$ as the orbit of vacuum pairs. Refuted.	Section 16
`AC-EX`	Existence of kink–magnon wave operators built from exact ansatz band data, and the resulting transmitted-channel projection. Conditional on a displayed two-cluster hypothesis that no model has been shown to satisfy.	Section 9
`AC-EX-2M`	The two-magnon counterpart over a single vacuum: the fixed-packet wave operators exist, are isometries, and their product is exactly multiplication by the physical scattering multiplier. Conditional, but the hypotheses are instantiated on the ferromagnet.	Section 11

identifier	what it says, in English	owned by
`AC-EX-2M-D29`	The interface between those wave operators and the fixed-time measurement protocol. Establishes a compactness statement and a creator-choice theorem, and records precisely why neither closes the identification.	Section 11
`ACE-LD-abs`	Abstract spectral diagonality: if a vector splits into channels with distinct charges and each channel's charge escapes in first moment, then the charge's spectral projections asymptotically agree with the channel projections. Distinct charges are necessary.	Section 9
`ACE-LD-eps`	Quantitative index bound: a state whose deviations from the two tail vacua decay exponentially away from a finite core is an approximate eigenvector of the window charge, with an explicit error. Two forced consequences are proved, one of them an incompatibility.	Section 9
`ACE-LD-nec`	The necessity half of the abstract diagonality result: two channels sharing a charge refute the displayed identity, so the distinct-charge hypothesis cannot be dropped.	Section 9
`ACE-LD-obst-prime`	The mean-escape obstruction: if the wall's mean transport grows with the window, the tightness clause of the relaxation hypothesis fails and the ordered displacement is undefined. Its positive form bounds mean transport on any relaxing state.	Section 9
`ACE-LD-sharp`	Sharp on-site charge: a kink state with exponentially decaying tail deviations forces both vacua to have zero on-site charge variance, hence the tail density is an on-site eigenvalue. A spin-1 chain with density 1/2 therefore admits no such state.	Section 9
`AD3-ex`	The named local-decay hypothesis used by the kink–magnon wave-operator theorem: at each fixed window, the charge measured in a channel converges to that channel's projection before the window is enlarged. Assumed, never proved.	Section 9
`AMP`	The soft-leg amputation conjecture: amputating a charge-created soft leg should contribute the per-site order-parameter density to the external flux, fixing the class constant without the jet-identification bridge. Currently an obstruction rather than a theorem.	Section 13
`B3`	Finite-time rigidity of the vacuum-pair label, plus the exact charge ledger $2s\,\delta x + (q_{\mathrm{out}} - q_{\mathrm{in}}) = 0$ for a separated scattering event.	Section 7
`Bc`	The "two 2s" conjecture: the soft phase slope and the memory channel quantum may be one and the same charge datum $\lvert q_{\mathrm{hard}}\rvert/s$. Its falsifier has been survived, not passed into a proof.	Section 7
`D24-VAL`	The conditional matched value of the amputation constant: given a jet-identification bridge and the existence of a class member, the constant is $1/(2S)$ at the tested spins. The implication is a theorem; its antecedents are not.	Section 13
`FUSION-SOFT`	The windowed, momentum-space corollary of the categorical selection rule — the same selection statement read in a Fourier/window topology, not an independent result.	Section 15
`G0`	The Goldstone and lattice-Noether theorem: the zero-momentum Goldstone tensor is pure gauge exactly on unbroken directions; an exact finite-window boundary identity; and the exact lattice continuity equation for any finite-range invariant Hamiltonian.	Section 5
`G0-soft-r1`	The withdrawn overclaim that the kinematic factor $(e^{ik} - 1)$ is a universal soft factor forcing a hard-independent linear coefficient. Hard data can enter at first order.	Section 16
`K1`	The kink bond operator of the easy-axis model is positive, with an explicitly computed kernel.	Section 6
`K2`	Every neighbouring factor of the exact kink family lies in that kernel, so the family is annihilated bond by bond — these are exact zero-energy kinks.	Section 6
`K3`	The boundary field of the kink Hamiltonian telescopes away outside any local observable, so the quasi-local derivation is unchanged.	Section 6

identifier	what it says, in English	owned by
`K4`	Conjectured thermodynamic uniqueness and flatness: exactly one zero-energy kink per regularised-charge sector, with no recoil. Only finite-volume evidence exists.	Section 6
`LD-ID`	The window-charge identity: the regularised window charge is literally the windowed lift of the conserved first-moment wall coordinate, no offset and no factor. Its corollaries kill the escaping-leg loophole class.	Section 9
`LR-D16-EDW`	Energy bounds the *number* of domain walls in the easy-axis model uniformly in time. It says nothing about their length, and so does not supply the tightness clause.	Section 9
`LR-D16-NR`	A repaired two-clause no-runaway condition implies the tightness clause of the relaxation hypothesis, with an explicit tail bound.	Section 9
`LR1-GEN`	Clause one of the relaxation hypothesis is a free theorem: separability of the algebra, strong continuity of the dynamics, and finiteness of the window charge's spectrum already give a common subsequence along which all the averaged states and protocol laws converge.	Section 8
`M`	The brief's literal memory conjecture — displacement equals the direct-current limit of the soft factor. Refuted; the surviving candidates are the physical-current flux identity and a conditional quantization law.	Section 7
`M-ESC-NR`	Composition of two conditional rows: mean escape and the no-runaway condition cannot both hold, so at least one no-runaway bound is infinite on an escaping state.	Section 9
`M-flux`	The displacement equals the finite-time direct-current weight of the *physical* boundary current — an exact identity, and the surviving half of the refuted memory conjecture.	Section 7
`M-IDX-density`	A Lieb–Schultz–Mattis-flavoured by-product: for a circle-covariant injective MPS vacuum pair with opposite tail densities, the density obeys $2\rho \in \mathbb{Z}$. The antisymmetry of the pair is load-bearing.	Section 8
`M-INDEX-2D-fin`	The finite two-dimensional window-integrality theorem: on a disk or annulus the escaped increment of the measurement protocol is an integer, with no assumption that the two measured observables commute.	Section 15
`M-INDEX-2D-spec`	The conditional two-dimensional limit law. Its extra relaxation hypotheses are assumed and no nonzero model instance is exhibited, which caps its status.	Section 15
`M-INDEX-fin`	The finite-window memory-index theorem: the window charge has spectrum in one coset of $\mathbb{Z}$, and the protocol's escaped increment is an integer because the two fixed-window offsets cancel.	Section 8
`M-INDEX-LA-folium`	In the representation generated by the bare kink, the circle symmetry is implemented by a strongly continuous unitary group whose generator is pure point with spectrum in one coset of $\mathbb{Z}$. A statement about implementers, not a limit of window charges.	Section 8
`M-INDEX-LA-strong`	The withdrawn operator form: existence of a self-adjoint regularised total charge in every kink representation. Refuted by two independent mechanisms.	Section 8
`M-INDEX-spec`	The ordered-limit memory law: under the circle-charge and relaxation hypotheses, every subsequential limit of the protocol law is a probability on $\mathbb{Z}$ and $\delta x = -(2s)^{-1} \sum_\nu \nu p_\nu$.	Section 8
`M-INDEX-torus`	The higher-rank version: for a common effective central torus the joint escaped-charge vector lies in the central weight lattice, hence in $\mathbb{Z}^r$ in primitive coordinates.	Section 8
`M-quant`	The conditional memory law for the easy-axis kink model: assuming asymptotic completeness, the displacement operator has spectrum in $\{0, -1/s\}$ with Bernoulli variance.	Section 7

identifier	what it says, in English	owned by
`M-quant-G`	The same law for an arbitrary compact symmetry group and arbitrary bond dimension, proved as a conditional implication with the channel charges displayed.	Section 7
`M-SCOPE-center`	The scope theorem: which symmetry directions can carry a calibrated memory at all. Tail-density jumps descend to the Lie algebra of the effective unbroken centre; semisimple directions and finite groups supply none.	Section 8
`M-tk`	The exact transmission amplitude on the projected component of the kink model, and its soft behaviour $T(k) = 16(\Delta - 1)^2 k^2 + O(k^4)$ — the memory Adler zero.	Section 7
`ML1`	The canonical two-magnon decomposition into a scattering summand and a bound band, with incoming and outgoing maps that are isometries onto the same summand and whose product is multiplication by the physical multiplier.	Section 11
`ML1-D31-kernel`	Generalised kernels obtained by transporting a rigging through the exact band isometry. Two named defects keep it at sketch: a hypothesis mismatch and a rigging that is not preserved.	Section 11
`ML2`	Complete two-magnon resolution by direct Jacobi diagonalisation, with no assumed Bethe completeness: the finite ring's count is exact and the infinite chain is scattering states plus one bound band.	Section 10
`ML3`	Conjectured regularity of the packet-smeared infinite-volume current form factor, which must control momenta of order $1/N$ and exclude physical soft poles.	Section 11
`ML4`	The one-hard-magnon Ward reduction. What is established is an off-shell analytic interpolation at fixed volume; the on-shell, packet-smeared, infinite-volume estimate is open, and the first physical momentum sequence already refutes volume uniformity.	Section 11
`ML4-A`	The abstract matching-plus-regularity cancellation lemma that turns a first-order bound into a second-order one.	Section 11
`ML4-Ward`	The exact finite-sector Ward projection identity, together with the erratum that its convenient scalar form is valid for one magnon only and false for two, and the register-dependent correction.	Section 11
`ML5`	Unrestricted local or quasi-local universality of the soft factorisation is false; an explicit four-site source is the counterexample.	Section 13
`ML5-A`	The repaired criterion: factorisation holds for a source exactly when its connected amplitude vanishes at zero momentum and its contact first jet vanishes. The criterion is agnostic about the value of the class constant.	Section 13
`ML5-B`	On the no-contact class, the connected hard-plus-soft amplitude is linear in the soft momentum and proportional to the one-hard amplitude. This fixes the *shape* of the factorisation and no number.	Section 13
`ML6`	Conjectured control of the order in which volume, packet width and soft momentum limits are taken, and of the bound and off-shell channels.	Section 11
`ML6-HS-SEP`	The proved special case: on the separated packet class, one displayed order of limits removes the entire bound, string and off-selected weight, and every limit equals the packet average of the multiplier.	Section 11
`Mq-AD3`	The projected component of the easy-axis kink model satisfies the asymptotic-completeness hypothesis. Proved for the projection, not for the full chain.	Section 7
`Mq-E`	The explicit unitary equivalence between the projected kink–magnon component and a Fano graph, identifying the incoming, reflected and transmitted legs and their kink bonds.	Section 7
`N1`	Numerics conjecture, not yet run: excitation-ansatz magnon amplitudes should reproduce the Bethe scattering matrix as the momentum goes to zero.	Section 17

identifier	what it says, in English	owned by
`N2`	The recorded easy-axis wavepacket scan agrees with the conditional quantization expectation at the measured precision. Numerical evidence, not a theorem.	Section 7
`NO-CAT-SOFT`	The recorded fence that categorical data fix no Taylor coefficient inside an allowed channel: a gapless input is needed for that.	Section 16
`O1`–`O10`	The Bethe oracle facts: independently recomputed exact statements about the ferromagnet's two-magnon data, used to check the campaign's own derivations and never as premises for them.	Section 10
`OR1`	The oracle cross-check: the campaign's reconstructed soft expansion agrees with the oracle term by term. This proves the two formulas equal, not that either is universal.	Section 10
`OR2`	The oracle fact that the two-magnon amplitude tends to one from either side — a plain limit, weaker than an Adler-zero theorem.	Section 10
`S-A1`	The load-bearing sketch box attached to the unbroken endpoint theorem: the split property, needed for a genuine edge Hilbert space realisation.	Section 5
`S-A2`	The load-bearing sketch box attached to the broken case: uniformity over a continuous vacuum manifold.	Section 5
`S-general`	The n-leg lattice soft theorem as a conjecture: an $(n+1)$-point amplitude equals a soft multiplier times the n-point amplitude plus a controlled remainder, universal only on the repaired no-contact class.	Section 13
`S-IDX-D29-value-HS-SEP`	On the separated-preparation subclass, the protocol's scalar datum has phase slope $\mathrm{sgn}(v_h - v_s)/S$; the number enters only through the two-body slope theorem.	Section 12
`S-IDX-fin-G`	The finite soft-index identity for an arbitrary compact Lie symmetry on any finite graph or periodic lattice, root by root, in both Ward registers, with the normalised index exactly one.	Section 12
`S-IDX-fin-r2`	The same identity in its original setting: a finite ring with $SU(2)$ symmetry, both registers displayed, and the charge-created row ratio exactly one.	Section 12
`S-IDX-G-label`	Sector-label integrality: a charge component spanning a character line of the unbroken centre shifts the central character by a fixed amount, giving an integer label on every cocharacter.	Section 12
`S-IDX-HR-value-r2`	A value instance built from the constructed scattering channel rather than from the protocol: its packet average has the same slope. Explicitly not a protocol instance.	Section 12
`S-IDX-MATCH-HS-SEP`	The matching theorem: on the separated subclass the fixed-time protocol readout converges to the integral of the physical multiplier, with zero remainder. This is the step that transports an on-shell number into the infrared statement.	Section 12
`S-IDX-PROTO-SCALAR-HS-SEP`	The exact scalar factorisation of the protocol datum on the same subclass, with a soft factor $(e^{ik} - 1)$ and a quotient that extends continuously through zero momentum.	Section 12
`S-IDX-spec-r2`	The conditional value statement for the general protocol: adding a class-membership antecedent and a matching hypothesis yields slope $\mathrm{sgn}(v_h - v_s)/S$ and the matched amputation constant. Sketch.	Section 12
`S-IDX-spec-struct-r2`	The structural half of the same: under an uninstantiated decomposition hypothesis, every limit point is linear in the soft momentum and has an Adler zero. The constant is left open.	Section 12
`S2`	A superseded label for the minimal two-magnon core, withdrawn so that one statement does not carry two statuses.	Section 16
`S2-2body`	The spin-$\frac{1}{2}$ two-body soft expansion with cubic remainders, derived from the local current and contact equations and matching the oracle.	Section 10
`S2-2body-S`	The exact regular-channel ratio at every site spin S, fixed without any integrability assumption, giving the soft slope $\mathrm{sgn}(v_h - v_s)/S$.	Section 10

identifier	what it says, in English	owned by
`SHAPE-FLAT`	Isotopy flatness at a fixed point: two string paths with the same resolved endpoints, related by contractible local moves, give the same endpoint morphism. A path around another insertion is outside the hypothesis.	Section 15
`SPT-B'`	Normalised finite, thermodynamic and soft coefficients vary *continuously* along a common-gap path with continuous external data, and become topological only after a separate local-constancy argument.	Section 14
`SPT-B-mult`	For ordered closed symmetry insertions the endpoint action is a representation tensored with its conjugate, so the projective multipliers cancel exactly and the closed register is rephasing-invariant.	Section 14
`SPT-B-r1`	The withdrawn claim that closed adjoint contractions are pointwise blind to the projective class. Refuted by explicit examples.	Section 14
`SPT-D'`	Ordered registered endpoint products realise the cocycle; for a compact semisimple Lie algebra the infinitesimal cocycle is a coboundary and is removed only in a stated phase section.	Section 14
`SPT-E'`	The edge-residue theorem: in a fixed Schmidt/edge register the compensated group residue is the projective representation, the Hermitian residue is centred and phase-gauge invariant with spectrum congruent modulo $\mathbb{Z}$, and its irreducible dimension is bounded below by the class.	Section 14
`SPT-E-AKLT`	The exact registered partial-charge compression for the anisotropic AKLT family, with its explicit family limit.	Section 14
`SPT-M'`	The boundary-window memory theorem in the SPT setting: the protocol's memory change is an integer at every finite length unconditionally, and every ordered limit law is a probability on $\mathbb{Z}$ under the displayed relaxation clauses.	Section 14
`SPT-M'-ch`	The optional channel corollary: given a physical edge split and definite channel charges, the edge charge change is minus the bulk change and fixed-system differences are integral.	Section 14
`SPT-M'-dyn`	The conjectured dynamical instance: for an explicit open AKLT-like boundary Hamiltonian, an edge-changing magnon reflection amplitude is nonzero on an open momentum interval.	Section 14
`SPT-nogo`	The withdrawn all-orders claim that the projective class can appear in no coefficient whatsoever. Refuted: only the infinitesimal cocycle class is trivialised.	Section 14
`SPT-T'`	Twisting relations for a registered endpoint, obtained by eliminating the composite implementer; a physical observable remains conditional on the edge hypotheses.	Section 14
`WI`	The truncated-symmetry (window Ward) identity: applying the symmetry only on an interval leaves exactly two virtual insertions, one at each boundary bond, the bulk having telescoped. Everything in the asymptotic-symmetry corner descends from it.	Section 5

2.2 Numbered definitions

Table 2: The numbered definitions. These are the campaign's single source for every symbol and every convention; a section that uses one restates it in full rather than pointing at it.

number	what it fixes	owned by
`D1`	The setting: the quasi-local algebra of the chain, uniform injective MPS tensors and their transfer matrix, the decorated window vectors (including insertions on the two edge bonds), and the two-sided half-line decorations.	Section 3

number	what it fixes	owned by
`D2`	On-site symmetry, a covariant family of vacua, the intertwining relation, the phase and multiplier cocycles it generates, normal ordering, and the smoothness hypothesis.	Section 3
`D3`	The function-space discipline for symmetry profiles: finitely supported, eventually constant, or of finite total variation. A plane wave is in none of them and is admitted only inside displayed limits.	Section 3
`D4`	Bond implementers on padded windows, the effective asymptotic symmetry group, and the twisted group algebra of asymptotic charges.	Section 3
`D5`	The excitation ansatz, its kink-sector form with different vacua on the two sides, and the null directions of its tangent-space gauge freedom.	Section 3
`D6`	The Heisenberg ferromagnet used as an oracle model: Hamiltonian, vacuum, coordinate and momentum bases, dispersion. Frozen.	Section 10
`D7`	The ordered-coordinate two-magnon Bethe wave, the coefficient ratio that defines the scattering amplitude, the incoming/outgoing convention tied to group velocities, and the rapidity. Frozen.	Section 10
`D8`	The soft limit convention: hard momentum fixed in a half-zone, signed soft momentum sent to zero, and the resulting identification of the ratio as the physical outgoing/incoming amplitude. Frozen as amended.	Section 10
`D9`	Vacuum and kink superselection sectors defined by factorised boundary conditions at the two infinities, the endpoint torsor, and the double-coset label of a vacuum pair.	Section 3
`D10`	The lattice Noether pair: on-site charge density, the cut current defined as a commutator of the Hamiltonian with a half-infinite charge, and the modulated charge and current.	Section 3
`D11`	The Goldstone tensor as the derivative of the symmetry action on the vacuum tensor, its identification with a soft charge insertion, and the count of independent Goldstone directions.	Section 3
`D12`	The precise senses in which the ansatz gauge remainder vanishes — a bound on the remainder, not an identity — together with the corrected profile classes for which the summation-by-parts identity converges.	Section 3
`D13`	The memory observables: the windowed wall position (1), the spatial memory as its change across an event, and the conserved first-moment wall coordinate.	Section 6
`D14`	Asymptotic leg content: the transmitted and reflected magnon numbers as sums of tail deviations, and the fact that the expected transmitted number is a packet *average* of the transmission probability.	Section 6
`D15`	Kink–magnon scattering data: reflection and transmission amplitudes, the transmission probability, and the transmission phase whose derivative is a Wigner–Eisenbud shift and is *not* the wall displacement.	Section 6
`D16`	The easy-axis XXZ kink model with all its conventions: bond term, dispersion, gap, anisotropy, the two exact product vacua, and the symmetry group with its broken reflection.	Section 6
`D17`	The summable kink class: states whose tail deviations from the two vacua are absolutely summable, on which the half-infinite charge and the leg content converge. Not preserved by the soft limit.	Section 6
`D18`	The asymptotic-completeness hypothesis: existence of wave operators, channel projections, a completeness decomposition, and local decay, in a displayed order of limits.	Section 6
`D19`	Soft profiles, the fixed Schmidt/edge register defined before any matrix element is taken, the bulk and edge profile families, and the order in which the limits are taken.	Section 14
`D20`	Operator-valued soft insertions, the scalar form factors built from them, the continuity requirements on all external data along a path, and the Hermitian modulated charge.	Section 14
`D21`	The endpoint module and its shifted charge lattice, the distinction between the edge register and the padded-window matrix module, and the hypothesis that identifies the register with a physical edge space.	Section 14
`D22`	Twists and ordered endpoint products, the phase relation obeyed by conjugation at one endpoint, and the boundary-window edge-memory protocol.	Section 14

number	what it fixes	owned by
D23	The exact comparison tensors: the AKLT tensor, a same-phase anisotropic path, an exactly injective trivial-phase tensor, and the boundary Hamiltonians used for the dynamical test.	Section 14
D24	The universality apparatus: the source classes, the normalisation in which amputated amplitudes and remainders are measured, the contact first jet, the five-clause no-contact class $\mathcal{S}_W(\rho)$, and the explicit refuting source.	Section 13
D25	The name and arguments of the soft multiplier S appearing in the n-leg conjecture. Naming it does not strengthen its status.	Section 13
D26	The circle-integral condition on the on-site charge: the on-site symmetry closes into a scalar at 2π, equivalently the charge spectrum lies in one coset of $\mathbb{Z}$. True of every spin-S chain.	Section 8
D27	Local relaxation of the charge history: one common sequence along which the time-averaged states and protocol laws converge, plus the two substantive clauses — a convergence clause and a tightness clause — that carry the real content.	Section 8
D28	The exact fixed-packet kink–magnon band data: a covariant positive-energy realisation of the kink folium, an exact band map rather than a variational one, velocity separation, and the displayed two-cluster inequality.	Section 9
D29	*Proposed, quarantined.* The windowed, packet-smeared, charge-created ratio datum: a finite-ring protocol that prepares a hard magnon packet, inserts one scale-tied soft charge packet, evolves, and reads off a row ratio in a labelled momentum window.	Section 12
D30	*Proposed, quarantined.* The regularity-only target condition attached to that protocol. It constrains the limit's smoothness and cannot by itself supply any value.	Section 12
D31	The exact fixed-packet two-magnon band data over a single vacuum: an exact magnon band map, almost-local momentum-filtered creators, packet supports with separated velocities, and an inventory of competing channels with spectral margins.	Section 11

2.3 Named hypotheses, protocols, and imported terms

Table 3: Hypotheses that appear by name inside theorem statements, together with the protocols and standard scattering-theory terms the campaign imports.

name	what it says, in English	owned by
`(2D-INT)`	The two-dimensional form of the circle-charge condition: the selected on-site charge has spectrum in one coset of $\mathbb{Z}$ on the planar lattice.	Section 15
`(2M-1P)`	A one-particle property required of a candidate creator family before it can be treated as an asymptotic creator.	Section 11
`(ACE2M-LSZ)`	The creator-choice theorem: at fixed soft scale, an admissible asymptotic creator family with disjoint velocity support may be replaced by another without changing any connected on-shell pairing. It does *not* apply to a fixed-time inserted charge.	Section 11
`(ACE2M-SR)`	The uniformity hypothesis the same interface would need. No instance of it has been verified for the fixed-time protocol family.	Section 11
Adler zero	The statement that a soft amplitude *vanishes* at zero momentum, linearly, rather than having a pole — the signature of a spontaneously broken global symmetry. This, and not a $1/\omega$ pole, is what a lattice soft theorem can assert.	Section 13
Bethe oracle	The exact two-magnon solution of the ferromagnet, recomputed independently and used only to check the campaign's derivations. An oracle is never a premise of a campaign proof.	Section 10
Cesàro states	The time-averaged states over a late window and an early window, whose difference the measurement protocol compares. Averaging, rather than pointwise-in-time limits, is what makes the convergence clause provable.	Section 8

name	what it says, in English	owned by
Cook limits	The standard construction of wave operators as strong limits of the free-to-interacting comparison, controlled by an integrable derivative estimate. Every Cook estimate in this campaign is at fixed packets and none is uniform in the soft scale.	Section 11
`(D28-C)`	The displayed two-sided two-cluster inequality on which the kink–magnon wave-operator theorem rests. It is the load-bearing hypothesis and is unverified on any model.	Section 9
`(D29-den)`	A pair of assumed bounds — a lower bound on the selected denominator and an upper bound on the numerator — under which the protocol data are compact. Both are assumptions, not derivations.	Section 12
`(D29-HS-SEP)`	The separated-preparation subclass of the protocol: the two packets are prepared far apart and the settling time is long enough that a displayed Schwartz-seminorm estimate closes. This is the class on which the matching theorem is proved.	Section 12
`(D29-order)`	The mandated order in which the protocol's limits are taken: volume, then time, then window and width, then the soft scale.	Section 12
`(E-LR2)`, `(E-LR3)`	The two substantive relaxation clauses in the SPT boundary-window setting: a convergence clause and a tightness clause, assumed and not derived.	Section 14
Fano graph	The exactly solvable one-particle graph — two half-line legs joined through a resonant site — to which the projected kink–magnon component is unitarily equivalent, and from which the transmission amplitude is read off.	Section 7
Haag–Ruelle	The standard construction of asymptotic particle states from almost-local, energy-momentum-filtered creators acting on the vacuum. The campaign's fixed-time inserted charge is deliberately *not* of this form, and that mismatch is an open interface.	Section 11
`H-ACE`	Shorthand for the exact fixed-packet kink–magnon band data listed under D28.	Section 9
`H-ACE2M`	Shorthand for the exact fixed-packet two-magnon band data listed under D31. Unlike its kink counterpart it is instantiated on the ferromagnet.	Section 11
`(H-AD)`	Shorthand for the four-clause asymptotic-completeness hypothesis of D18: wave operators and completeness, channel projections, local decay, and the order in which windows are enlarged.	Section 6
`H-AD-edge`	The corresponding hypothesis at a boundary: asymptotic channel structure for an open chain with an edge. Assumed by the optional SPT channel corollary.	Section 14
`H-AD-G`	The same four-clause hypothesis for a selected kink packet with the channel charges fixed to one incoming leg of charge -1, one reflected leg of charge -1, and one transmitted leg of charge $+1$.	Section 7
`H-dress`	The hypothesis that dressed endpoint overlaps are nonzero with endpoint separation taken first — what is needed to turn a registered endpoint statement into a physical one.	Section 14
`H-MQG(1)--(5)`	The standing register of the general memory law: (1) a covariant family of injective MPS vacua and a fixed kink sector; (2) a common unbroken circle direction with tail densities $\pm s$, $s > 0$; (3) a translation-invariant, finite-range, invariant Hamiltonian with stationary vacua; (4) a fixed wave packet in the summable kink class, no plane waves; (5) the asymptotic hypothesis `H-AD-G` with its displayed channel charges.	Section 7
`H-soft-p`	The extra regularity assumed when a derivative of a soft coefficient, rather than the coefficient itself, is asserted to have a limit.	Section 14
`H-split`	The hypothesis that the half-chain low-energy subspace splits normally, so that the fixed edge register may be read as a physical edge Hilbert space. Every "physical edge" statement is conditional on it.	Section 14

name	what it says, in English	owned by
`(INT)`	The circle-integral condition of D26 — the single arithmetic input behind every integrality statement in the memory corner.	Section 8
`(IT)`	The intertwining relation: applying the on-site symmetry to an injective MPS tensor is the same as conjugating it by a projective representation on the virtual space, up to a phase. The fundamental theorem of MPS supplies it; the whole Ward apparatus rests on it.	Section 4
`(K-Q)`	The hypothesis that a state is an approximate eigenvector of the limiting relative charge, with a displayed error.	Section 9
`(K-TAIL)`	The hypothesis that a state's on-site deviations from the two tail vacua have exponentially decaying two-point data away from a finite core. It is satisfiable — the exact zero-energy kinks satisfy it — and it is incompatible with an escaping charged leg.	Section 9
λ–D chain	The running non-integrable model of the numerics: a spin-1 chain with single-ion anisotropy D and exchange anisotropy, whose phase diagram contains Néel, Haldane and large-D phases. Chosen precisely because integrability-free hypotheses should be tested on a non-integrable model.	Section 17
$\mathfrak{a}_{\mathrm{leg}}(\rho)$	The soft-leg amputation constant: the one number the no-contact factorisation leaves undetermined. It is defined only relative to a fixed normalisation convention, and rescaling that convention rescales it.	Section 13
`(LR)`	The local-relaxation hypothesis of D27, in three clauses. Its first clause is a theorem; the content is entirely in the other two.	Section 8
(LR1$_{2D}$)--(LR3$_{2D}$)	The planar analogues of those three clauses. The first is again free; the other two are assumed, with no nonzero model instance exhibited.	Section 15
LSZ reduction	The standard passage from correlation functions to on-shell amplitudes by amputating external legs. The campaign needs an *exhaustive* normed version of it, and that requirement is one of the live obligations on the soft edge.	Section 11
M1 and M2	The two model classes of the brief: M1 is the isotropic spin-$\frac{1}{2}$ Heisenberg ferromagnet (type-B Goldstone, exact product vacuum, exact magnons, no kinks); M2 is the easy-axis XXZ ferromagnet (two product vacua, kinks, gapped magnons). Neither alone exhibits every corner.	Section 1
`(M-ESC)`	Mean escape: the hypothesis that the wall's averaged transport grows proportionally to the window size. No model or state realising it has been exhibited anywhere in the campaign.	Section 9
`(MATCH-S)`	The matching hypothesis identifying the fixed-time protocol readout with the on-shell multiplier to first order. It contains no number; the number arrives afterwards, from two-body scattering. Proved on the separated subclass.	Section 12
`(NR)`	The two-clause no-runaway condition: a uniform second-moment bound and a uniform conditional bound on the window's charge content. It is sufficient for the tightness clause and is itself unproved.	Section 9
`(PROTO-LSZ)`	The one uninstantiated hypothesis on which the general soft limit law still rests: that the limit datum decomposes exhaustively into a descendant term of the expected shape plus orthogonal, direct-contact and boundary terms that are of second order.	Section 12
`(PT1)--(PT4)`	The displayed typing hypotheses for the planar tensor-network selection theorem; the fourth is needed only for the instrument-normalisation clause.	Section 15
`(S)`	The smoothness hypothesis: when the symmetry group is a Lie group, the vacuum tensor and the virtual representation admit continuously differentiable local sections through the identity. Stated wherever used.	Section 3

name	what it says, in English	owned by
$\mathcal{S}_W(\rho)$	The Ward-covariant no-contact class of sources: the five-clause class on which the soft factorisation survives. Membership is open at every density — no microscopic member has been exhibited — so every statement about the class is currently a constraint on future work.	Section 13
`(SEP)`	The displayed separation estimate defining the separated-preparation subclass: preparation distance, settling time and packet seminorms are tied together so that a stationary-phase remainder vanishes.	Section 12
`(T)`	Transitivity: the hypothesis that the symmetry group acts transitively on the vacuum labels. Without it, the vacuum-pair classification is made orbit by orbit.	Section 3
`tns-...`	Tracker identifiers for open lanes of work, quoted in some statements to say where an acknowledged gap is being pursued. They name tasks, never results.	Section 18
TPM protocol	The two-projective-measurement protocol: measure the windowed charge once before the event and once after, using the *same* window, cut and background both times, and record the difference. Because the two measurements use the same observable, their spectral offsets cancel; no assumption that the two Heisenberg observables commute is needed. This protocol is the operational definition of "memory" throughout.	Section 8
Wigner–Eisenbud shift	The spatial shift of a transmitted wavepacket given by the derivative of the transmission phase. It is smooth and non-quantised and must not be confused with the wall displacement.	Section 6
Z_ρ	The per-site order-parameter density 2ρ appearing in the leg-conversion computation. The exact conversion factor supplies only $Z_\rho^{-1/2}$, which is why the amputation conjecture needs a second, non-leg mechanism.	Section 13

3 The setting: uniform matrix-product states, windows, and excitations

A uniform matrix-product state is a translation-invariant state of an infinite spin chain built from a single small tensor. One fixes a physical dimension d and a *bond dimension* χ, and one array $A = (A^s)_{s=1}^d$ of $\chi \times \chi$ matrices. Reading a physical configuration $s_1 s_2 \cdots s_w$ of w consecutive sites as the ordered matrix word $A^{s_1} A^{s_2} \cdots A^{s_w}$, and closing that word against two fixed matrices, produces the amplitudes of a state on any window; the state on the whole chain is the compatible family of all such windows. All correlations of that state are governed by a single completely positive map on $\chi \times \chi$ matrices, the *transfer matrix* $E(Y) = \sum_s A^s Y (A^s)^\dagger$. When E has a unique, simple eigenvalue of largest modulus — the *injective* case — the resulting state is pure, translation invariant, and exponentially clustering, with correlation length set by the gap between the leading eigenvalue 1 and the rest of the spectrum. Injectivity is the whole reason such states are tractable: it makes the state unique, it makes the symmetry data of Section 4 unique, and it makes long matrix words span the full matrix algebra, which is what lets a bond insertion be recovered from the window vector it produces.

The campaign is about what a symmetry does when it is applied to a *finite region* rather than the whole chain, so nothing may be said only about the infinite chain. This is why the basic object below is not the state but the *window vector*: a finite interval $\Lambda = [a, b]$, the ordered product of the tensors sitting on it, arbitrary invertible matrices inserted on any of its bonds, and two boundary vectors closing the word. Everything the campaign proves — the truncated-symmetry identity, the bond implementers, the soft limits — is first an exact identity between window vectors, and only afterwards a statement about states. Two technical points earn their keep immediately. First, the $|\Lambda| + 1$ bonds available for insertion must include the two *edge* bonds $(a-1|a)$ and $(b|b+1)$; without them the truncated-symmetry identity is simply false on the window $\Lambda = R$ that it is about. Second, the decay of E^m towards its fixed point must be quoted honestly: E may carry Jordan blocks at the subleading modulus, so the correct statement carries a rate $\tilde{\lambda}$ strictly above the transfer gap λ_E, never λ_E itself. Both points are recorded in the definitions below because both were live errors at some stage of the campaign.

3.1 Vacua: uniform injective matrix-product states

The first definition fixes the arena — the algebra of observables on the chain — and the class of states the whole campaign treats as "vacua". The content of the injectivity clause is that the transfer matrix has a spectral gap, so that the state clusters exponentially and long matrix words are unconstrained.

Definition 3.1 (Uniform injective matrix-product vacuum (D1(a)–(d)))**.** *(a) The chain.* Fix $d \in \mathbb{N}$ and let the lattice be $\mathbb{Z}$. The *quasi-local algebra* is

$$\mathfrak{A} := \bigotimes_{x \in \mathbb{Z}} M_d(\mathbb{C}), \tag{3}$$

the C*-inductive limit of the window algebras $\mathfrak{A}_\Lambda := \bigotimes_{x \in \Lambda} M_d(\mathbb{C})$ over finite $\Lambda \subset \mathbb{Z}$; the norm-dense *local subalgebra* is $\mathfrak{A}_{\mathrm{loc}} := \bigcup_\Lambda \mathfrak{A}_\Lambda$. Translation by x sites is the automorphism $\tau_x \in \mathrm{Aut}(\mathfrak{A})$. A *state* is a normalised positive linear functional on $\mathfrak{A}$; states are compared in the weak-$*$ topology, i.e. pointwise on $\mathfrak{A}_{\mathrm{loc}}$. For a state ϱ we write $(\mathcal{H}_\varrho, \pi_\varrho, \Omega_\varrho)$ for its GNS triple.

(b) Tensor and transfer matrix. A *uniform matrix-product tensor* of bond dimension χ is

$A = (A^s)_{s=1}^d$ with $A^s \in M_\chi(\mathbb{C})$. Its *transfer matrix* is the completely positive map $E : M_\chi \to M_\chi$,

$$E(Y) \;:=\; \sum_s A^s\, Y\, (A^s)^\dagger, \tag{4}$$

equivalently the $\chi^2 \times \chi^2$ matrix $\sum_s A^s \otimes \overline{A^s}$.

(c) Injectivity, canonical form, gap. A is *injective* (equivalently *normal*) if E has spectral radius 1, the eigenvalue 1 is the unique eigenvalue of modulus 1, it is non-degenerate, and the left and right eigenmatrices l, r — i.e. $E^*(l) = l$ and $E(r) = r$ — are positive definite. We fix the *canonical form*

$$l = \mathbb{1}, \qquad r > 0, \qquad \operatorname{tr} r = 1, \tag{5}$$

after which the residual matrix-product gauge freedom is $A^s \mapsto Y^{-1} A^s Y$ with Y unitary commuting with r, together with an overall phase. The *transfer gap* is

$$\lambda_E \;:=\; \max\big\{\, |\mu| \;:\; \mu \in \operatorname{spec}(E),\ \mu \neq 1 \,\big\} \;<\; 1, \qquad \xi_c := -1/\log \lambda_E, \tag{6}$$

with ξ_c the correlation length. Because E may have Jordan blocks at modulus λ_E, the honest decay statement is: *for every $\tilde\lambda \in (\lambda_E, 1)$ there is $C_{\tilde\lambda} < \infty$ with*

$$\|E^m - P\| \;\le\; C_{\tilde\lambda}\, \tilde\lambda^{\,m} \qquad \text{for all } m \ge 0, \tag{7}$$

where $P(Y) = \operatorname{tr}(Y)\, r$ is the spectral projection onto the fixed point. A bare $O(\lambda_E^m)$ is *false* in general: it needs the polynomial prefactor m^p with p equal to the Jordan block size minus one. Every rate quoted anywhere in this labbook is in the $\tilde\lambda$ form. Injectivity is equivalent to the existence of an n_0 with

$$\operatorname{span}\big\{\, A^{s_1} \cdots A^{s_n} \;:\; s_1 \ldots s_n \,\big\} \;=\; M_\chi(\mathbb{C}) \qquad \text{for all } n \ge n_0. \tag{8}$$

(d) The state. For injective A in canonical form, ω_A is the unique state on $\mathfrak{A}$ with, for $O \in \mathfrak{A}_{[1,w]}$,

$$\omega_A(O) \;=\; \sum_{s,s'} \langle s'|O|s\rangle \; \operatorname{tr}\Big[\, (A^{s'_w})^\dagger \cdots (A^{s'_1})^\dagger \cdot \mathbb{1} \cdot A^{s_1} \cdots A^{s_w} \cdot r \,\Big], \tag{9}$$

extended by translation invariance. The state ω_A is pure, translation invariant, and exponentially clustering with rate λ_E (in the sense of (7)). We write $(\mathcal{H}_A, \pi_A, \Omega_A)$ for its GNS triple and suppress π_A when harmless.

campaign id:	D1(a)–(d)
proved in:	`definitions.md` D1, clauses (a)–(d); FROZEN at the 2026-08-26 freeze
tested by:	—
reviewed in:	`theory/verdicts/corner-a-r3.md` (PASS)

3.2 Windows, decorations, and window vectors

A decoration is a window with things put on it: possibly different tensors at some sites, possibly invertible matrices squeezed onto some bonds, and two boundary vectors at the ends. This is the working object of the campaign because every symmetry statement below turns out to be an identity between a plain window vector and a decorated one. The edge-bond clause is not cosmetic: it is what makes "the symmetry applied to R acts only on the two boundary bonds of R" a statement about the window R itself rather than about some larger window.

Definition 3.2 (Window vectors, decorations, and decorated states (D1(e)))**.** For a finite window $\Lambda = [a, b]$, a *decoration* is a tuple $T = (T_a, \ldots, T_b)$ of tensors $T_x = (T_x^s)$, together with *bond insertions* $M_{x|x+1} \in GL(\chi)$ on any subset of the $|\Lambda| + 1$ bonds

$$(a-1|a), \quad (a|a+1), \quad \ldots, \quad (b|b+1) \tag{10}$$

— **including the two edge bonds** — and boundary vectors $b_l, b_r \in \mathbb{C}^\chi$. The *window vector* is

$$|\psi_\Lambda(T; b_l, b_r)\rangle \;:= \sum_{s_a \ldots s_b} \langle b_l| M_{a-1|a}\, T_a^{s_a}\, M_{a|a+1} \cdots T_b^{s_b}\, M_{b|b+1} |b_r\rangle \; |s_a \ldots s_b\rangle, \tag{11}$$

absent factors being $\nVdash$. An edge insertion is equivalently a redefinition of the boundary vector,

$$b_l \mapsto M_{a-1|a}^\dagger\, b_l, \qquad b_r \mapsto M_{b|b+1}\, b_r, \tag{12}$$

and is included so that identities whose insertions land on $\partial\Lambda$ are statements about the same window rather than about a larger one.

There is a constant $C_\partial = C_\partial(A, b_l, b_r) < \infty$ with

$$\big\| |\psi_\Lambda(T; b_l, b_r)\rangle \big\| \;\le\; C_\partial \prod_M \|M\| \tag{13}$$

for every Λ and every decoration differing from the uniform A on boundedly many sites — uniformly in $|\Lambda|$, because the contraction is of the form $(b_l \otimes \bar{b}_l | E^{|\Lambda|} | b_r \otimes \bar{b}_r)$ and E^m is power-bounded by Definition 3.1(c).

A *decorated state* $\omega_A[T]$ is the state on $\mathfrak{A}$ obtained from a decoration that differs from the uniform A only on finitely many sites and bonds, by the prescription of Definition 3.1(d) with the modified string and with l, r as the two infinite environments; it is well defined by the $\tilde{\lambda}$-estimate (7). We write $\omega_A^{M@b}$ for the state decorated by a single bond insertion M on the bond b.

Remark 3.3 (Why the edge bonds are in the definition)*.* The clause admitting insertions on $(a-1|a)$ and $(b|b+1)$ was added in revision r2 of the Corner A loop. The earlier window vector carried interior bonds only, which made the earlier statement of the truncated-symmetry identity false for the window $\Lambda = R$ — the two surviving virtual operators then lie outside the expression altogether. The measured discrepancy for the $\chi = 2$ Pauli tensor was 1.658; admitting edge insertions restores exactness to 0.0.

campaign id:	D1(e)
proved in:	`definitions.md` D1(e); repair recorded at `theory/corner-a.md` step (1.3), sub-step (2.7)
tested by:	`theory/checks/corner_a_check.py`, checks C2, C2b
reviewed in:	`theory/verdicts/corner-a-r2.md` objection 1; `theory/verdicts/corner-a-r3.md`

A kink state has a different vacuum on each half-line, so it is not a decoration in the above sense at all: it differs from any uniform tensor on a whole half-line. It is defined instead by its finite-window restrictions, and those restrictions are matrix-environment contractions rather than window vectors — a distinction that cost the campaign one round of corrections.

Definition 3.4 (Two-sided (half-line) decorations (D1(e′)))**.** A *two-sided decoration* carries an injective canonical-form tensor A_α on $(-\infty, m]$ and an injective canonical-form tensor A_β on $[m + 1, \infty)$ (possibly $\alpha = \beta$), plus finitely many further site and bond modifications. For a finite window $W \ni m$ let $P_W(s)$ be the ordered product of the window tensors, with the modifications inserted, for the physical index string s, and define the *unnormalised completely positive contraction*

$$\tilde{\varrho}_W := \sum_{s, s'} \mathrm{tr}\big[\, l_\alpha\, P_W(s)\, r_\beta\, P_W(s')^\dagger \,\big] \; |s\rangle\langle s'|, \tag{14}$$

$$\omega_{\alpha|\beta}^{(m)}[T](O) := \frac{\mathrm{tr}[\tilde{\varrho}_W O]}{\mathrm{tr}[\tilde{\varrho}_W]} \qquad \text{for } O \in \mathfrak{A}_W. \tag{15}$$

Lemma 3.5 (The two-sided prescription defines a state)**.** *$\tilde\varrho_W \geq 0$, $\operatorname{tr}\tilde\varrho_W > 0$, and the functionals are consistent under $W \subseteq W'$; hence they define a unique state on $\mathfrak{A}$.*

Idea of proof. Positivity. For $c \in \mathbb{C}^{d^{|W|}}$ put $T := \sum_s c_s P_W(s)$. Then

$$\langle c|\tilde\varrho_W|c\rangle \;=\; \operatorname{tr}\big[\, l_\alpha T r_\beta T^\dagger \,\big] \;=\; \operatorname{tr}\Big[\, \big(l_\alpha^{1/2} T r_\beta^{1/2}\big)\big(l_\alpha^{1/2} T r_\beta^{1/2}\big)^\dagger \,\Big] \;\geq\; 0, \tag{16}$$

using $l_\alpha, r_\beta > 0$ from Definition 3.1(c). *Nondegeneracy.* $\operatorname{tr}\tilde\varrho_W = \operatorname{tr}[\, l_\alpha\, \mathfrak{E}_W(r_\beta)\,] > 0$ with $\mathfrak{E}_W$ the window completely positive map, since $l_\alpha, r_\beta > 0$ and $\mathfrak{E}_W$ is faithful on positive definite inputs. *Consistency.* Enlarging W by one site on the left tail applies $\mathfrak{E}^*_{A_\alpha}$ to l_α, which fixes it; on the right tail it applies $\mathfrak{E}_{A_\beta}$ to r_β, which fixes it. □

Remark 3.6 (A corrected argument)*.* The r2 text justified positivity by saying that each entry is $\langle v|\cdot|v\rangle$ for a finite-window vector v. That is wrong: with full-rank matrix environments the restriction is a positive completely positive contraction, and becomes a vector expression only after purifying the environments — not a rank-one boundary-vector window vector. The argument above is the corrected one. Independently, the weak-$*$ limit constructed in the Corner A endpoint theorem supplies positivity, so Lemma 3.5 is a convenience rather than a load-bearing hypothesis.

campaign id:	D1(e′)
proved in:	`definitions.md` D1(e′), with the well-definedness lemma stated and proved there
tested by:	`theory/checks/corner_a_check.py`, check C11 (Hermiticity 0, minimal eigenvalue $-1.4 \cdot 10^{-16}$, positive trace)
reviewed in:	`theory/verdicts/corner-a-r2.md` objection 8 (conceded); `theory/verdicts/corner-a-r3.md`

3.3 Excitations: the ansatz and its gauge freedom

One-particle states are built by replacing the vacuum tensor at one site by a different tensor B and superposing over the position of that replacement with a momentum phase. If the tensors to the left and to the right of B are the same, the result is a local excitation over a single vacuum; if they are two different vacua, it is a kink. The parametrisation is redundant, and the redundancy — the *null directions* — is the technical heart of the soft theorem: at zero momentum the Goldstone tensor of Definition 3.15 becomes a null direction, which is exactly the statement that the zero-momentum Goldstone mode is a global symmetry rotation and therefore carries no independent excitation.

Definition 3.7 (Excitation ansatz and kink sectors (D5(a)))**.** For injective canonical-form tensors A_α, A_β, a tensor $B = (B^s)$ with $B^s \in M_\chi$, and $k \in (-\pi, \pi]$, the *excitation-ansatz vector* is defined on windows by

$$|\Phi_k(B; A_\alpha, A_\beta)\rangle_\Lambda \;:=\; \sum_{n\in\Lambda} e^{ikn}\, |\psi_\Lambda\big(\ldots A_\alpha A_\alpha\, B\, A_\beta A_\beta \ldots;\, b_l, b_r\big)\rangle, \tag{17}$$

with B at site n, A_α at all sites $< n$ and A_β at all sites $> n$. For $\alpha = \beta$ we write $|\Phi_k(B)\rangle$ and call it the *trivial* (local) sector; for $\alpha \neq \beta$ it is the *kink* (topological) sector. On the infinite chain these are δ-normalised generalised vectors,

$$\langle \Phi_k(B)|\Phi_{k'}(B')\rangle \;=\; 2\pi\,\delta(k-k')\cdot B^\dagger N_k B', \tag{18}$$

with N_k the Gram form of the ansatz at momentum k.

campaign id:	D5(a)
proved in:	`definitions.md` D5(a); convention matched against the local TEX ground truth `refs/arxiv-1103.2286` (`dispersionrelation_final.tex`) and `refs/arxiv-1810.07006` (`p5_excitations.tex`, Eq. (kink))
tested by:	—
reviewed in:	`theory/verdicts/corner-a-r3.md`

Definition 3.8 (Null directions of the ansatz (D5(b)))**.** For $X \in M_\chi$ set

$$\mathcal{N}_k^{\alpha\beta}(X) \;:=\; e^{ik} A_\alpha X - X A_\beta, \qquad \text{i.e.} \qquad \mathcal{N}_k(X)^s = e^{ik} A_\alpha^s X - X A_\beta^s. \tag{19}$$

This is the gauge freedom $B \simeq B + e^{ik}AX - XA$ of the ansatz. The rank of $X \mapsto \mathcal{N}_k(X)$ is χ^2 for $k \neq 0$ and $\chi^2 - 1$ for $k = 0$ with $\alpha = \beta$: the rank *drops* by one as $k \to 0$.

Remark 3.9 (The vanishing of a null direction is a limit statement, not a finite-window identity)*.* The identity $|\Phi_k(\mathcal{N}_k(X))\rangle = 0$ is **false as a finite-window identity**. On $\Lambda = [a, b]$ the reindexing that would prove it leaves exactly two boundary terms,

$$e^{ik(b+1)}|\psi_\Lambda;\, X@(b|b+1)\rangle \;-\; e^{ika}|\psi_\Lambda;\, X@(a-1|a)\rangle, \tag{20}$$

which do not vanish for generic b_l, b_r: the measured norm is 1.77 for the $\chi = 2$ Pauli tensor at $k = 0.37$ (an earlier measurement with different boundary vectors gave 0.591). The vanishing statement holds only in the limits specified in Definition 3.16. Shards of the first revision asserted the exact identity and were wrong; the exact identity, with its boundary terms, is the summation-by-parts lemma of the Goldstone shard.

campaign id:	D5(b)
proved in:	`definitions.md` D5(b); exact identity with boundary terms at `theory/corner-a-goldstone.md` step (1.5) (Lemma SBP); rank statement ibid.; ground truth `refs/arxiv-1103.2286` and `refs/arxiv-1810.07006` Eq. (gaugeexcitations)
tested by:	`theory/checks/corner_a_check.py`
reviewed in:	`theory/verdicts/corner-a-r2.md` (warning conceded); `theory/verdicts/corner-a-r3.md`

3.4 Superselection: vacuum and kink sectors

The next definition says what it means for a state to "look like vacuum α far to the left and vacuum β far to the right". The factorised form of the boundary condition — with a spectator observable D inserted — is what makes the label stable under normal perturbations, and hence a genuine superselection label rather than a property of one particular state. It refers forward to the covariant vacuum family of Definition 4.2.

Definition 3.10 (Vacuum and kink superselection sectors (D9(a),(b)))**.** Let $\{\omega_\alpha\}_{\alpha \in \Omega_{\mathrm{vac}}}$ be a G-covariant vacuum family (Definition 4.2). For $\alpha, \beta \in \Omega_{\mathrm{vac}}$ the *kink sector* $\mathcal{K}_{\alpha\beta}$ is the set of states ϱ on $\mathfrak{A}$ obeying the *factorised* boundary conditions at infinity,

$$\lim_{n \to -\infty} \varrho\big(D \cdot \tau_n(O)\big) = \varrho(D)\, \omega_\alpha(O), \qquad \lim_{n \to +\infty} \varrho\big(D \cdot \tau_n(O)\big) = \varrho(D)\, \omega_\beta(O), \tag{21}$$

for all $D, O \in \mathfrak{A}_{\mathrm{loc}}$. Taking $D = \mathbb{1}$ gives the plain conditions $\varrho(\tau_n(O)) \to \omega_{\alpha/\beta}(O)$. The sector $\mathcal{K}_{\alpha\alpha}$ is the *vacuum sector of* α; it contains ω_α, every state $\omega_\alpha \circ \mathrm{Ad}(W^\dagger)$ with $W \in \mathfrak{A}$ unitary, and every decorated state $\omega_{A_\alpha}[T]$ of Definition 3.2. A state in $\mathcal{K}_{\alpha\beta}$ with $\alpha \neq \beta$ is a *kink state* (equivalently a domain-wall or topological state).

Disjointness. States in $\mathcal{K}_{\alpha\beta}$ and $\mathcal{K}_{\alpha'\beta'}$ with $(\alpha, \beta) \neq (\alpha', \beta')$ are disjoint, i.e. mutually non-normal. Hence "superselection sector" is used here in the precise sense: the label (α, β) is a classical observable at infinity, unchangeable by any element of $\mathfrak{A}$.

campaign id: D9(a),(b)
proved in: `definitions.md` D9(a),(b); stability of the factorised label and disjointness proved at `theory/corner-a-kinks.md` step (1.8), with rate λ_E for every decorated matrix-product state at sub-step (1.8)(iii)
tested by: —
reviewed in: `theory/verdicts/corner-a-r3.md`

The next object is the set of states obtained from one vacuum by inserting a single invertible matrix on one bond. It is defined here as a bare set with a bare action; *that* the action descends, and what its orbits are, are theorems of the Corner A shard, restated below only for the reader's convenience.

Definition 3.11 (Endpoint space and endpoint action (D9(c), definition only))**.** Fix a bond b and a vacuum label α. The *endpoint space* is

$$\mathcal{E}_b^\alpha \;:=\; \left\{\, \omega_{A_\alpha}^{M@b} \;:\; M \in GL(\chi) \,\right\}, \tag{22}$$

and the *endpoint action* of G on representatives is

$$g \star M \;:=\; V_\alpha(g)\, M. \tag{23}$$

Nothing beyond this is definitional.

Remark 3.12 (What the endpoint theorem adds)*.* The following are proved in the Corner A endpoint theorem and are recorded here only because they are what makes Definition 3.11 interesting. (i) $\omega^{M@b} = \omega^{M'@b}$ if and only if $M' \in \mathbb{C}^\times M$; hence $[M] \mapsto \omega^{M@b}$ is a canonical bijection $PGL(\chi) \to \mathcal{E}_b^\alpha$, and $\mathcal{E}_b^\alpha$ is a $PGL(\chi)$-torsor. (ii) The action $\star$ descends to $\mathcal{E}_b^\alpha$ and acts there as left translation by $\rho_\alpha(g) = [V_\alpha(g)] \in PGL(\chi)$, a genuine G-action with kernel N_α; the orbit of ω_α is $\rho_\alpha(G) \cong \mathcal{A}_{\text{eff}} = G/N_\alpha$, on which $\mathcal{A}_{\text{eff}}$ acts simply transitively. (iii) The lift of ρ_α to the padded window space furnished by Definition 4.14 is a *projective* action with multiplier ω_α; it exists always, including when $[\omega_\alpha] \neq 0$. What the class $[\omega_\alpha]$ obstructs is *removing* the multiplier, i.e. lifting ρ_α to an honest homomorphism $G \to U(\chi)$. The earlier phrasing, that $[\omega_\alpha]$ obstructs "lifting the state action to the window action", is wrong and has been retracted.

Two sanity checks. For $\chi = 1$, or whenever V_α is a scalar for every g, one has $N_\alpha = G$ and a one-point orbit. For the AKLT chain with $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ and $V_\alpha \in \{\mathbb{1}, X, Z, XZ\}$ one has $N_\alpha = \{e\}$ and a four-point orbit.

When it is said that the asymptotic symmetry "relabels charge superselection sectors", that is asserted about this torsor and about the module structure of $\mathbb{C}^\chi$ over the charge algebra of Definition 4.19 — never about $\mathcal{K}_{\alpha\beta}$, which is trivial in the unbroken case.

campaign id: D9(c),(c′)
proved in: `definitions.md` D9(c) (definition) and D9(c′) (corollaries); proved at `theory/corner-a.md` step (1.4), sub-steps (d2) and (f)
tested by: `theory/checks/corner_a_check.py`
reviewed in: `theory/verdicts/corner-a-r2.md` objections 2 and 7; `theory/verdicts/corner-a-r3.md`

Finally, when the symmetry is broken there is a question of which pairs of vacua are genuinely different once the global symmetry is quotiented out. The naive answer — the difference $g_L g_R^{-1}$ — is wrong for nonabelian groups; the correct invariant is a double coset.

Definition 3.13 (Vacuum-pair invariant (D9(d)))**.** $G_L \times G_R$ acts on $\Omega_{\text{vac}} \times \Omega_{\text{vac}}$ componentwise. Carry explicitly the *transitivity hypothesis* (T): *G acts transitively on* Ω_{vac}. The covariance requirement of Definition 4.2 does *not* imply (T) — the vacuum family may split into several G-orbits — so without (T) everything below holds per orbit. Under (T), $\Omega_{\text{vac}} \cong G/H_\alpha$ and

$$\Omega_{\text{vac}} \times \Omega_{\text{vac}} \;\cong\; (G/H_\alpha) \times (G/H_\alpha), \tag{24}$$

transitively, with the stabiliser of (α,β) equal to $H_\alpha\times H_\beta$. This is **not** $(G\times G)/G_{\mathrm{diag}}$: for $G=SU(2)$ with $H_\alpha=U(1)$ it is $S^2\times S^2$, of dimension 4, whereas $(G\times G)/G_{\mathrm{diag}}\cong SU(2)$ has dimension 3.

The complete invariant of a vacuum pair *modulo the global (diagonal) symmetry* is the *double coset*

$$\mathfrak{d}(\alpha_L,\alpha_R) \;:=\; H_\alpha\, g_L^{-1} g_R\, H_\alpha \;\in\; H_\alpha\backslash G/H_\alpha, \tag{25}$$

where $\alpha_L=g_L\cdot\alpha$ and $\alpha_R=g_R\cdot\alpha$. It is well defined — changing $g_L\mapsto g_L h_1$ and $g_R\mapsto g_R h_2$ conjugates within the double coset — and diagonal-invariant, since $g_i\mapsto h g_i$ cancels. For $G=SU(2)$ with $H_\alpha=U(1)$ it is the relative polar angle, $\cos\theta=\hat n_L\cdot\hat n_R\in[-1,1]$.

Remark 3.14 (The corrected classification). An earlier version asserted that the orbit of the coset space $(G_L\times G_R)/G_{\mathrm{diag}}$ is the set of vacuum pairs, and that $[g_L g_R^{-1}]$ is the diagonal-invariant label. Both are false for nonabelian G: the componentwise stabiliser is $H_\alpha\times H_\beta$, not G_{diag}, and $g_L g_R^{-1}\mapsto h\, g_L g_R^{-1} h^{-1}$ under the diagonal action. Both the double-coset invariant and the failure of $g_L g_R^{-1}$ to be diagonal-invariant were verified numerically.

campaign id:	D9(d)
proved in:	`definitions.md` D9(d)
tested by:	`theory/checks/corner_a_check.py`
reviewed in:	`theory/verdicts/corner-a-r2.md` objection 5; `theory/verdicts/corner-a-r3.md`

3.5 Goldstone tensors and broken directions

Differentiating the symmetry action on the vacuum tensor produces a tensor in the excitation ansatz. This is the object whose zero-momentum limit is pure gauge, and whose small-momentum behaviour is the Adler zero the campaign is chasing.

Definition 3.15 (Goldstone tensor, broken and unbroken directions (D11))**.** Assume the smoothness hypothesis (S) of Definition 4.8. Fix $\alpha\in\Omega_{\mathrm{vac}}$, write $\mathfrak{h}_\alpha:=\mathrm{Lie}(H_\alpha)$, and choose an $\mathrm{Ad}(H_\alpha)$-invariant complement $\mathfrak{m}_\alpha$ with $\mathfrak{g}=\mathfrak{h}_\alpha\oplus\mathfrak{m}_\alpha$; the elements of $\mathfrak{m}_\alpha$ are the *broken directions.*

(a) Goldstone tensor. For $\xi\in\mathfrak{g}$,

$$B_{\mathrm{G}}(\xi) \;:=\; \frac{d}{d\varepsilon}\Big[\mathcal{U}\big(\exp(\varepsilon\xi)\big)A_\alpha\Big]\Big|_{\varepsilon=0}, \qquad \text{equivalently} \qquad B_{\mathrm{G}}(\xi)^s=\sum_{s'} q(\xi)_{ss'}\,A_\alpha^{s'}. \tag{26}$$

(b) Soft insertion. In the sense of Definition 3.7 and Definition 4.22,

$$|\Phi_k\big(B_{\mathrm{G}}(\xi)\big)\rangle \;=\; Q_k(\xi)\triangleright\omega_\alpha\, : \tag{27}$$

replacing A_α by $B_{\mathrm{G}}(\xi)$ at site n is exactly the action of the charge density $q_n(\xi)$ there.

(c) Goldstone count. The number of independent Goldstone tensors at α is

$$\dim_{\mathbb{C}}\mathrm{span}_{\mathbb{C}}\big\{\,B_{\mathrm{G}}(\xi)\;:\;\xi\in\mathfrak{m}_\alpha\,\big\}\quad\text{modulo }\mathcal{N}_0(\cdot), \tag{28}$$

which may be strictly smaller than $\dim_{\mathbb{R}}\mathfrak{m}_\alpha$. The deficiency is the *type-B* phenomenon; the isotropic ferromagnet realises $\dim_{\mathbb{R}}\mathfrak{m}_\alpha=2$ broken directions but only one Goldstone mode.

campaign id:	D11
proved in:	`definitions.md` D11; the count and the ferromagnetic example at `theory/corner-a-goldstone.md` step (1.7)
tested by:	`theory/checks/corner_a_check.py`
reviewed in:	`theory/verdicts/corner-a-r3.md`

3.6 In which limits the ansatz gauge remainder vanishes

The exact statement about null directions is the summation-by-parts identity, whose remainder consists of exactly two boundary window vectors. Whether that remainder can be dropped depends on the profile class, and different parts of the campaign need different classes. The definition below states only what is proved — a bound on the remainder — and then names the three regimes in which that bound is used. Its type discipline is binding: quoting a fixed-k identity with a decay hypothesis, or the reverse, is a defect.

Definition 3.16 (The three regimes in which the gauge remainder is negligible (D12))**.** Let $\Lambda_L := [-L, L]$ and

$$|\Phi_f^\Lambda(B)\rangle \;:=\; \sum_{n\in\Lambda} f(n)\,|\psi_\Lambda(\ldots B@n\ldots)\rangle. \tag{29}$$

The summation-by-parts lemma shows that the gauge remainder $\mathfrak{B}_\Lambda[f, X]$ is a sum of exactly two boundary window vectors, and gives the **upper bound**

$$\big\|\mathfrak{B}_\Lambda[f,X]\big\| \;\le\; 2\,C_\partial\,\|X\|\,\max\big(|f(a)|,|f(b)|\big), \tag{30}$$

with C_∂ the window-norm constant (13). Everything below is a consequence of this bound alone. Three regimes are used, and every statement citing this definition must name which.

(a) Vanishing remainder for decaying profiles. If $f \in c_0(\mathbb{Z})$ then $\|\mathfrak{B}_\Lambda[f, X]\| \to 0$ as $\Lambda_L \nearrow \mathbb{Z}$. **This is a statement about the remainder only.** It does *not* assert that either side of the summation-by-parts identity converges to a vector: for that one needs a summability class, as in (a′), *and* the split property, which remains a sketch. Measured decay with the centred profile $(1 + |n - c|)^{-3}$: $\|\mathfrak{B}_\Lambda\|/\|\mathrm{bulk}\| = 3.4 \cdot 10^{-1}$, $5.7 \cdot 10^{-2}$, $8.5 \cdot 10^{-3}$, $1.2 \cdot 10^{-3}$ for $L = 4, 8, 16, 32$.

(a′) Norm-convergent wave packets. If in addition $f \in \ell^1(\mathbb{Z})$ **and** $\Delta f \in \ell^1(\mathbb{Z})$ — that is, f is summable of bounded variation — then both sides of the summation-by-parts identity are absolutely convergent sums of window vectors of norm at most $C_\partial\|X\|$, uniformly in Λ; the identity then holds between the two limits at fixed Λ and remains remainderless as $\Lambda \nearrow \mathbb{Z}$. A smooth compactly supported momentum packet $f = \hat\varphi$ with $\varphi \in C_c^\infty((-\pi,\pi])$ has rapidly decreasing f and so lies in $\ell^1 \cap BV$. The class $\mathfrak{F}_{\ell^1}$ of Definition 4.9 is exactly the bounded-variation condition. **This class, not c_0, is the one to cite for wave-packet statements.**

(b) δ-normalised plane wave. For $f(n) = e^{ikn}$ one has $|f| \equiv 1$, so (30) gives $\|\mathfrak{B}_\Lambda\| = O(1)$ *uniformly in* Λ — and that is all that is needed:

$$\big\|\,|\Lambda|^{-1/2}\,\mathfrak{B}_\Lambda\,\big\| \;=\; O\big(|\Lambda|^{-1/2}\big) \;\longrightarrow\; 0. \tag{31}$$

No claim is made, or needed, about how the bulk term grows. Equivalently, per site,

$$\lim_{\Lambda\nearrow\mathbb{Z}} |\Lambda|^{-1}\langle\Phi_k^\Lambda(B')|O\,|\mathfrak{B}_\Lambda\rangle \;=\; 0 \tag{32}$$

for every $O \in \mathfrak{A}_{\mathrm{loc}}$ and every fixed B'. Measured: $\|\mathfrak{B}_\Lambda\| \in [1.31,\ 2.03]$ for $L = 4, \ldots, 32$.

Type discipline. A fixed-k identity may be quoted only with (b); a c_0 or ℓ^1 identity only with (a) or (a′), and then in the real-space summation-by-parts form or as a Fourier superposition of the fixed-k identities — never as a fixed-k equation with a c_0 hypothesis. Statements quoted "exactly" without naming a regime are defects.

Remark 3.17 (What this definition deliberately no longer says)*.* The r2 version over-quantified in three ways, all conceded. It asserted that the identity "holds exactly" for every $f \in c_0$, although $c_0 \not\subset \ell^1$ and neither side need converge in norm — take $f(n) = (1+|n|)^{-1/4}$. It asserted a universal

$\Theta(|\Lambda|^{1/2})$ bulk growth, which is *false*: for $\chi = 1$ one has $A_\alpha X \propto A_\alpha$, so the plane-wave bulk sum is a bounded geometric sum. And it attached a fixed-k formula to a c_0 hypothesis, even though a plane wave is not in c_0. What survives, and all that is used, is the remainder bound (30).

campaign id:	D12
proved in:	`definitions.md` D12 (rewritten in r3); remainder bound and summation-by-parts lemma at `theory/corner-a-goldstone.md` step (1.5); the sketch split property at `theory/corner-a.md` step (1.4), sub-step (2.9)
tested by:	`theory/checks/corner_a_check.py`, checks C4 (decay), C5 (plane-wave boundedness), C10 (the withdrawn bulk-growth assertion)
reviewed in:	`theory/verdicts/corner-a-r2.md` objection 4; `theory/verdicts/corner-a-r3.md`

4 Symmetry on a window: intertwiners, bond implementers, and lattice Noether

An on-site symmetry acts on a matrix-product state in a way that is completely determined at the level of the single tensor. If the symmetry leaves the state invariant, then applying the physical unitary $u(g)$ to the physical index of the tensor cannot change the tensor by more than the residual gauge freedom of the parametrisation: a phase and a conjugation by some invertible matrix acting on the virtual (bond) indices. That statement is the *fundamental theorem of matrix product states*, and the resulting relation — the *intertwining relation* of Definition 4.3 — is the engine that drives the entire campaign. Because the conjugating matrices $V_\alpha(g)$ are determined only up to a phase, they compose only up to a phase: they form a projective representation, whose cohomology class $[\omega_\alpha] \in H^2(H_\alpha, U(1))$ is the one-dimensional symmetry-protected-topological index. On the lattice this class is the exact analogue of an anomaly of the asymptotic symmetry, and it is where the campaign's genuinely new prediction lives.

The reason the intertwining relation matters so much is what happens when the symmetry is applied only to a finite interval $R = [a, b]$ instead of the whole chain. Substituting the relation at each of the $|R|$ sites of R produces an alternating string $V_\alpha(g)^{-1} A\, V_\alpha(g) V_\alpha(g)^{-1} A\, V_\alpha(g) \cdots$, in which every interior pair $V_\alpha(g)V_\alpha(g)^{-1}$ collapses to the identity. Only the two outermost factors survive, and they sit on the two *boundary bonds* of R. Concretely, with $T_R^{(g)}$ denoting the decoration that carries A_α on $\Lambda \setminus R$, $A_{g\cdot\alpha}$ on R, the bond insertion $V_\alpha(g)^{-1}$ on $\partial_- R = (a-1|a)$ and $V_\alpha(g)$ on $\partial_+ R = (b|b+1)$,

$$U_R(g)\,|\psi_\Lambda(A_\alpha; b_l, b_r)\rangle \;=\; e^{\,i|R|\theta_\alpha(g)}\,|\psi_\Lambda\big(T_R^{(g)}; b_l, b_r\big)\rangle, \tag{33}$$

exactly, as window vectors — and, as states, with no phase at all, since the extensive factor $e^{i|R|\theta_\alpha(g)}$ appears in bra and ket and cancels. This is the lattice Ward identity. It says that a symmetry applied to a region acts *only at the region's boundary*, and acts there by virtual-space operators that are invisible to any local physical measurement in the bulk. Every selection rule, every current-conservation statement, and the whole soft-theorem mechanism of the campaign is this telescoping plus transfer-matrix calculus. Three points about (33) deserve emphasis. The identity is exact on a finite window, provided either that the window is enlarged to contain $[a-1, b+1]$, or that the window vector admits insertions on its edge bonds — which is precisely why Definition 3.2 was written to admit them. The telescoping uses only the intertwining relation; injectivity is needed to *obtain* that relation and to make the pair $(\theta_\alpha, V_\alpha)$ unique, not for the algebra. And the orientation matters: the surviving operator on the *left* boundary bond is $V_\alpha(g)^{-1}$ and on the *right* it is $V_\alpha(g)$, fixed by the convention of Definition 4.3. The original campaign brief states the opposite orientation; that document is historical and is not edited, so the correction is recorded here. A $\mathbb{Z}_2$ example is blind to the flip, because there $V = Z = V^{-1}$; a $U(1)$ symmetry with $V(t) = e^{itZ/2}$ separates them cleanly, the correct orientation giving an error of $5.6 \cdot 10^{-17}$ and the flipped one 0.267.

The rest of this section fixes, in order: the symmetry data and the intertwining relation; the classes of *profiles* $f : \mathbb{Z} \to G$ for which a modulated symmetry operation makes sense (a discipline that is not bureaucratic — half-infinite strings are not elements of the algebra at all); the bond implementers and the group they generate; and the lattice Noether pair of charge density and current.

4.1 The symmetry data and the intertwining relation

Definition 4.1 (On-site symmetry and its action on a tensor (D2, preamble))**.** Let G be a compact group (finite, or a compact Lie group) with a unitary representation $u : G \to U(\mathbb{C}^d)$ on one site, and let $U_\Lambda(g) := \bigotimes_{x\in\Lambda} u_x(g)$. Write $\mathcal{U}(g)$ for the induced linear map on tensors,

$$\big(\mathcal{U}(g)A\big)^s := \sum_{s'} u(g)_{ss'} A^{s'}, \qquad u(g)_{ss'} := \langle s|u(g)|s'\rangle, \tag{34}$$

so that $\mathcal{U}(g)\mathcal{U}(h) = \mathcal{U}(gh)$.

Definition 4.2 (Covariant vacuum family (D2(a)))**.** A *G-covariant vacuum family* consists of a finite or compact set Ω_{vac} of labels, injective canonical-form tensors $\{A_\alpha\}_{\alpha\in\Omega_{\mathrm{vac}}}$ of common bond dimension χ generating pairwise distinct states $\omega_\alpha := \omega_{A_\alpha}$, together with a G-action $(g,\alpha)\mapsto g\cdot\alpha$ on Ω_{vac} such that $U_\Lambda(g)$ maps ω_α to $\omega_{g\cdot\alpha}$ for every finite Λ, in the sense

$$\omega_\alpha \circ \mathrm{Ad}\big(U_\Lambda(g)^\dagger\big) \;=\; \omega_{g\cdot\alpha} \qquad \text{on } \mathfrak{A}_\Lambda, \text{ for all } \Lambda. \tag{35}$$

The *stabiliser* $H_\alpha := \{g\in G : g\cdot\alpha = \alpha\}$ is the *unbroken subgroup*. The family is *unbroken at* α if $H_\alpha = G$, in which case $|\Omega_{\mathrm{vac}}|$ may be 1.

campaign id:	D2(a)
proved in:	`definitions.md` D2(a); FROZEN at the 2026-08-26 freeze
tested by:	—
reviewed in:	`theory/verdicts/corner-a-r3.md` (PASS)

The next relation is the one everything rests on. It is not an assumption of this campaign: it is the fundamental theorem of matrix product states, and it is quoted here against the local TEX sources rather than from memory.

Definition 4.3 (The intertwining relation (D2(b)))**.** For every $\alpha\in\Omega_{\mathrm{vac}}$ and every $g\in G$ there exist a phase $\theta_\alpha(g)\in\mathbb{R}/2\pi\mathbb{Z}$ and a unitary $V_\alpha(g)\in U(\chi)$ with

$$\sum_{s'} u(g)_{ss'}\, A_\alpha^{s'} \;=\; e^{\,i\theta_\alpha(g)}\, V_\alpha(g)^{-1}\, A^s_{g\cdot\alpha}\, V_\alpha(g) \qquad \text{for all } s. \tag{36}$$

The pair $(\theta_\alpha(g), V_\alpha(g))$ is unique up to $V_\alpha(g)\mapsto e^{i\varphi}V_\alpha(g)$.

Remark 4.4 (Ground truth for (64)). Existence: `refs/arxiv-0802.0447` (`StringOrder-v10.tex`), Lemma 1, states that the spectral radius of the mixed transfer map E_u is at most 1, with equality if and only if there are a unitary V and a phase θ with $V^\dagger\tilde{A}_j = e^{i(\theta-\theta_j)}\tilde{A}_jV^\dagger$, together with the accompanying condition $(u\otimes\nVdash)B = (\nVdash\otimes V)BV^\dagger$; and `refs/arxiv-2011.12127` (`TN-Review-main.tex`), which for $U(g)^{\otimes N}|\psi_N\rangle \simeq |\psi_N\rangle$ states $\sum_j U_{ij}(g)A^j = e^{i\phi(g)}X^\dagger(g)A^iX(g)$. Uniqueness: `refs/arxiv-2011.12127`, the equation labelled `XAX=B`, states that

$$X^{-1}A^iX = e^{i\varkappa}\,Y^{-1}A^iY \quad\Longrightarrow\quad \varkappa = 0 \;\text{ and }\; X = e^{i\varphi}Y \text{ for some } \varphi. \tag{37}$$

campaign id:	D2(b)
proved in:	`definitions.md` D2(b); consequences derived at `theory/corner-a.md` step (1.2)
tested by:	`theory/checks/corner_a_check.py`, checks C0, C1, C1b, C1c (orientation)
reviewed in:	`theory/verdicts/corner-a-r3.md`

Composing the intertwining relation with itself forces the phase and the virtual matrices to compose in a definite way. The virtual matrices compose only projectively, and the class of that projective multiplier is the symmetry-protected-topological index — the lattice anomaly.

Definition 4.5 (Cocycles and the symmetry-protected index (D2(c)))**.** Composition of (64) gives, for all $g,h\in G$,

$$\theta_\alpha(hg) = \theta_\alpha(g) + \theta_{g\cdot\alpha}(h), \qquad V_{g\cdot\alpha}(h)\,V_\alpha(g) = e^{\,i\omega_\alpha(h,g)}\,V_\alpha(hg). \tag{38}$$

Restricted to $g, h \in H_\alpha$ these say that $\theta_\alpha|_{H_\alpha} : H_\alpha \to U(1)$ is a homomorphism and that $\omega_\alpha : H_\alpha \times H_\alpha \to U(1)$ is a 2-cocycle. Its class

$$[\omega_\alpha] \;\in\; H^2\big(H_\alpha, U(1)\big) \tag{39}$$

is the *symmetry-protected-topological index* of A_α.

Remark 4.6 (A notation hazard, stated once). The symbol ω carrying *two group* arguments is this cocycle; the symbol ω carrying a *momentum* argument, $\omega(k)$, is a magnon dispersion. The two never occur with the same argument type, and this is the mechanical disambiguation rule used throughout the repository.

Definition 4.7 (Normal ordering (D2(d))). For $g \in H_\alpha$ put $\check{u}_\alpha(g) := e^{-i\theta_\alpha(g)}u(g)$. Because $\theta_\alpha|_{H_\alpha}$ is a homomorphism, $\check{u}_\alpha$ is again a unitary representation of H_α, and it satisfies (64) with phase 1. All charges below are those of $\check{u}_\alpha$ unless stated otherwise; physically this is the charge measured from its vacuum value.

Definition 4.8 (Smoothness hypothesis (S) (D2(e))). When G is a Lie group we assume, and state whenever it is used, that $\varepsilon \mapsto A_{\exp(\varepsilon\xi)\cdot\alpha}$ and $\varepsilon \mapsto V_\alpha(\exp(\varepsilon\xi))$ admit C^1 local sections with $V_\alpha(e) = \nVdash$. We then write

$$X_\alpha(\xi) := \frac{d}{d\varepsilon}V_\alpha\big(\exp(\varepsilon\xi)\big)\Big|_{\varepsilon=0} \quad \text{(anti-Hermitian)}, \qquad \theta'_\alpha(\xi) := \frac{d}{d\varepsilon}\theta_\alpha\big(\exp(\varepsilon\xi)\big)\Big|_{\varepsilon=0}. \tag{40}$$

campaign id:	D2(c),(d),(e)
proved in:	`definitions.md` D2(c)–(e); the cocycle relations (38) derived at `theory/corner-a.md` step (1.2)
tested by:	`theory/checks/corner_a_check.py`
reviewed in:	`theory/verdicts/corner-a-r3.md`

4.2 Which profiles are allowed

A "modulated symmetry" means applying $u(f(x))$ at site x for some profile $f : \mathbb{Z} \to G$. Whether the resulting object is an operator, a map on states, or nothing at all depends entirely on how f behaves at infinity. The following classes are not bookkeeping: a half-infinite symmetry string is *not* an element of the quasi-local algebra, and a plane wave is in none of the classes and is admissible only inside a Fourier transform. Every soft statement in the campaign must say which class it uses.

Definition 4.9 (Admissible profile classes (D3(a))). Let

$$\mathfrak{F}_c := \big\{\, f : \mathbb{Z} \to G \;:\; f(x) = e \text{ for all but finitely many } x \,\big\}, \tag{41}$$

$$\mathfrak{F}_{\mathrm{ec}} := \big\{\, f : \mathbb{Z} \to G \;:\; f \text{ is constant on some } (-\infty, -n] \text{ and on some } [n, \infty) \,\big\}, \tag{42}$$

and, for a Lie algebra element $\xi \in \mathfrak{g}$,

$$\mathfrak{F}_c(\xi) := \big\{\, f : \mathbb{Z} \to \mathbb{R} \;:\; f \text{ finitely supported} \,\big\}, \tag{43}$$

$$\mathfrak{F}_{\mathrm{ec}}(\xi) := \big\{\, f : \mathbb{Z} \to \mathbb{R} \;:\; f \text{ constant near } \pm\infty \,\big\}, \tag{44}$$

$$\mathfrak{F}_{\ell^1}(\xi) := \Big\{\, f : \mathbb{Z} \to \mathbb{R} \;:\; \sum_x |f(x+1) - f(x)| < \infty \,\Big\} \quad \text{(finite total variation)}. \tag{45}$$

One has $\mathfrak{F}_c \subset \mathfrak{F}_{\mathrm{ec}}$ and $\mathfrak{F}_{\mathrm{ec}}(\xi) \subset \mathfrak{F}_{\ell^1}(\xi)$. A profile is *admissible for an operator statement* if and only if it lies in $\mathfrak{F}_c$, in which case the operator lies in $\mathfrak{A}_{\mathrm{loc}}$; it is *admissible for a state statement* if and only if it lies in $\mathfrak{F}_{\mathrm{ec}}$, in which case the operator is a weak-$*$ limit of elements of $\mathfrak{A}_{\mathrm{loc}}$. The plane wave $f(x) = e^{ikx}$ lies in none of these for $k \neq 0$; it is admitted only inside a Fourier transform, i.e. as the distributional kernel of a wave packet $\sum_k \varphi(k) f_k$ with $\varphi \in C_c^\infty((-\pi, \pi])$.

Definition 4.10 (Truncated symmetry operations (D3(b)))**.** For a finite interval $R = [a, b] \subset \mathbb{Z}$ put $U_R(g) := \prod_{x \in R} u_x(g) \in \mathfrak{A}_{\mathrm{loc}}$. Its *boundary bonds* are $\partial_- R := (a-1|a)$ and $\partial_+ R := (b|b+1)$, and $|R| = b - a + 1$. More generally, for $f \in \mathfrak{F}_c$, $U[f] := \prod_x u_x(f(x)) \in \mathfrak{A}_{\mathrm{loc}}$.

Definition 4.11 (Half-infinite strings act on states only (D3(c)))**.** For $f \in \mathfrak{F}_{\mathrm{ec}}$ the product $U[f]$ is *not* an element of $\mathfrak{A}$. It is used only through the induced map on states,

$$\varrho \longmapsto \mathrm{w}^*\text{-}\lim_{n \to \infty} \varrho \circ \mathrm{Ad}\big(U[f_n]^\dagger\big), \tag{46}$$

where $f_n \in \mathfrak{F}_c$ agrees with f on $[-n, n]$ and equals e outside. Existence of this limit is a theorem, not a definition; when it exists the limit state is written $f \triangleright \varrho$. The canonical case is $f = 1_{[x,\infty)} \cdot g$, written $U_{[x,\infty)}(g)$.

campaign id:	D3
proved in:	`definitions.md` D3(a)–(c); existence of the half-infinite limit proved at `theory/corner-a.md` steps (1.2)(c) and (1.4)
tested by:	—
reviewed in:	`theory/verdicts/corner-a-r3.md`

4.3 Bond implementers and the effective asymptotic symmetry group

The telescoping identity (33) leaves virtual operators on bonds. To speak of them as *operators* one needs the map "$M \mapsto$ the window vector with M inserted on bond b_0" to be injective, so that left multiplication on M transports to a well-defined map on window vectors. That injectivity is genuinely conditional: it needs enough uniform sites on each side of the bond, and without them the construction is ill defined — not merely unproved. The definition therefore carries its padding hypothesis explicitly.

Definition 4.12 (Padded windows and the insertion map (D4(a1),(a2)))**.** A window $\Lambda = [a, b]$ is *padded about the bond* $b_0 = (m|m+1)$ if it contains at least n_0 sites on each side of b_0, with n_0 the word-spanning length of (8), and if the boundary vectors satisfy $b_l \neq 0$ and $b_r \neq 0$. For such Λ let $\mathcal{W}_{\Lambda, b_0}$ be the linear span of the window vectors carrying one insertion on b_0, and let

$$\iota_{\Lambda, b_0} : M_\chi(\mathbb{C}) \longrightarrow \mathcal{W}_{\Lambda, b_0}, \qquad M \longmapsto |\psi_\Lambda(A; b_l, b_r; M@b_0)\rangle. \tag{47}$$

On a padded window, ι_{Λ, b_0} is **injective**.

Idea of proof of injectivity. The coefficient of $|s\rangle$ is $b_l^\dagger P(s)\, M\, Q(s)\, b_r$, with P and Q the matrix words to the left and to the right of b_0. By (8) each of P, Q ranges over a spanning set of $M_\chi(\mathbb{C})$, and the coefficient is linear in each; since $b_l \neq 0$ the set $\{b_l^\dagger P : P \in M_\chi\}$ is all of $(\mathbb{C}^\chi)^*$, and since $b_r \neq 0$ the set $\{Q b_r\}$ is all of $\mathbb{C}^\chi$. So $\iota(M) = \iota(M')$ forces $v^\dagger(M - M')w = 0$ for all v, w, i.e. $M = M'$. □

Remark 4.13 (Padding is necessary, not decorative). Without padding the kernel of ι need not be invariant under left multiplication, so the rule "$\iota(M) \mapsto \iota(NM)$" is ill defined. An explicit injective, canonical-gaugeable, $\mathbb{Z}_2$-symmetric counterexample: take

$$A^0 = \mathrm{diag}(1, 2), \quad A^1 = X, \quad u(g) = \mathrm{diag}(1, -1), \quad V = Z, \quad b_l = (\sqrt{2},\, 1), \quad b_r = (1,\, 0), \tag{48}$$

one site on each side of the bond, and

$$N = \begin{pmatrix} -\sqrt{2} & 0 \\ 1 & 0 \end{pmatrix}. \tag{49}$$

Then $\iota(N) = 0$ while $\|\iota(ZN)\|_\infty = 4$. The transfer spectrum is $\{4.303, 3, 1, 0.697\}$ — a unique top eigenvalue, with $\lambda_E = 0.697$ after rescaling — and the length-two words have rank 4, so this genuinely is a tensor of the class of Definition 3.1. Padding both sides to $n_0 = 2$ restores $\operatorname{ran}\iota$ of full dimension $\chi^2 = 4$.

Definition 4.14 (Bond implementers (D4(a4)))**.** On a padded window define, for $M \in GL(\chi)$,

$$\mathcal{V}_{b_0}(M) \;:=\; \iota_{\Lambda,b_0} \circ L_M \circ \iota_{\Lambda,b_0}^{-1} \qquad \text{on } \operatorname{ran}\iota_{\Lambda,b_0} = \mathcal{W}_{\Lambda,b_0}, \tag{50}$$

where L_M is left multiplication on $M_\chi(\mathbb{C})$. By Definition 4.12 this is well defined and linear, and $\mathcal{V}_{b_0}(M)\mathcal{V}_{b_0}(M') = \mathcal{V}_{b_0}(MM')$, with scalars acting by scalars. Put

$$\mathcal{V}_{b_0}(g) \;:=\; \mathcal{V}_{b_0}\big(V_\alpha(g)\big) \qquad (\textbf{not } V_\alpha(g)^{-1}). \tag{51}$$

With the convention (38), this gives

$$\mathcal{V}_{b_0}(h)\,\mathcal{V}_{b_0}(g) \;=\; e^{\,i\omega_\alpha(h,g)}\,\mathcal{V}_{b_0}(hg), \tag{52}$$

a linear representation of the twisted group algebra $\mathfrak{a}_\alpha$ of Definition 4.19 on $\mathcal{W}_{\Lambda,b_0}$, in which the multiplier acts nontrivially. On *states* the multiplier is invisible, since $\omega^{cM@b} = \omega^{M@b}$ for a scalar c; the state-level action is Definition 4.16.

Remark 4.15 (Two corrections carried in this definition). The r1 version used the implementer $V_\alpha(g)^{-1}$ with left multiplication, which reverses the composition law, and asserted a twisted-algebra action on states, where phases are invisible. The r2 version declared the implementer on an unpadded insertion span, where it is not well defined — see Remark 4.13. Both are superseded by the form above. This definition applies in the unbroken case $H_\alpha = G$; the broken case is handled through Definition 3.10 and the Corner A endpoint theorem.

campaign id:	D4(a)
proved in:	`definitions.md` D4(a1)–(a4); consumed at `theory/corner-a.md` step (1.4)
tested by:	`theory/checks/corner_a_check.py`, checks C8 (kernel not invariant without padding) and C8b (padding to $n_0 = 2$ restores injectivity)
reviewed in:	`theory/verdicts/corner-a-r2.md` objections 1, 3, 4, 7, 8; `theory/verdicts/corner-a-r3.md`

Passing from window vectors to states loses exactly the phases, and with them the multiplier. What survives is an honest group action, and the group that acts is not G but G modulo those elements whose virtual matrix is a scalar.

Definition 4.16 (The effective asymptotic symmetry group (D4(b)))**.** Set

$$N_\alpha \;:=\; \big\{\, g \in G \;:\; V_\alpha(g) \in \mathbb{C}^\times \mathbb{1} \,\big\}, \tag{53}$$

a normal subgroup of G — the preimage of the centre under a projective representation. The induced map on states is the group homomorphism

$$\rho_\alpha : G \longrightarrow PGL(\chi), \qquad \rho_\alpha(g) := [V_\alpha(g)], \tag{54}$$

which is *not* projective, the cocycle having been quotiented away, and which has $\ker\rho_\alpha = N_\alpha$. The *effective asymptotic symmetry group* is

$$\mathcal{A}_{\text{eff}} \;:=\; G/N_\alpha \;\cong\; \rho_\alpha(G) \;\subseteq\; PGL(\chi). \tag{55}$$

This is a genuine group.

Remark 4.17 (The superseded object)*.* The object $\mathcal{A} = (G_L \times G_R)/G_{\text{diag}}$ of the original brief is *not* the vacuum orbit unless $N_\alpha = \{e\}$, and it is a group only for abelian G. It is retained only as a name for the coset space and is never used as a classifying object; the classifying objects are $\mathcal{A}_{\text{eff}}$ in the unbroken case and the double coset (70) in the broken case.

Definition 4.18 (The two asymptotic copies (D4(c)))**.** G_L and G_R act through half-infinite strings on $(-\infty, b]$ and $[b+1, \infty)$ respectively. By the telescoping identity (33), the left string leaves $V_\alpha(g_L)$ on the bond b and the right string leaves $V_\alpha(g_R)^{-1}$, so the composite residue is $V_\alpha(g_L)V_\alpha(g_R)^{-1}$, which is trivial on states if and only if $g_L g_R^{-1} \in N_\alpha$. Hence the true stabiliser is

$$S_\alpha := \big\{(g_L, g_R) \ : \ g_L g_R^{-1} \in N_\alpha\big\} \ \supseteq \ G_{\text{diag}}, \tag{56}$$

with $S_\alpha = G_{\text{diag}}$ if and only if $N_\alpha = \{e\}$.

Definition 4.19 (Asymptotic charge algebra (D4(d)))**.** $\mathfrak{a}_\alpha := \mathbb{C}_{\omega_\alpha}[H_\alpha]$ is the twisted group algebra with basis $\{v_g\}$ and product $v_h v_g = e^{i\omega_\alpha(h,g)} v_{hg}$. This definition supplies only the name; the content — that $g \mapsto V_\alpha(g)$ is a $*$-representation of it, and that ρ_α is the induced state-level action — is a theorem of the Corner A endpoint shard. For a Lie group, under the smoothness hypothesis of Definition 4.8, the *asymptotic charges* are $\mathfrak{q}_b(\xi) := X_\alpha(\xi)$, acting by left multiplication on the bond insertion, and

$$\big[\,\mathfrak{q}_b(\xi),\, \mathfrak{q}_b(\zeta)\,\big] \ = \ \mathfrak{q}_b\big([\xi, \zeta]\big) \ + \ c_\alpha(\xi, \zeta)\mathbb{1}. \tag{57}$$

Remark 4.20 (What the central term is, and what it is not)*.* The term c_α in (57) is only the *local infinitesimal image* of the class $[\omega_\alpha]$, and it generally loses it. For $\mathfrak{h}_\alpha$ compact semisimple, Whitehead's second lemma says that its Lie-algebra cohomology class in $H^2(\mathfrak{h}_\alpha, \mathbb{R})$ is trivial; the displayed central cocycle is then a coboundary and can be gauged away by choosing a phase section, although it need not vanish in every local section. The global class $[\omega_\alpha]$ may nevertheless be nontrivial torsion: for the AKLT chain, $H^2(SO(3), U(1)) = \mathbb{Z}_2$, while a suitable phase convention gives the ordinary $su(2)$ bracket for the spin-one-half edge generators. **The lattice symmetry-protected anomaly is a group-cohomological multiplier, not a Lie-algebra central charge.** This is a genuine disanalogy with the continuum picture of a centrally extended charge algebra, and it must be stated as such rather than glossed over.

campaign id:	D4(b),(c),(d)
proved in:	`definitions.md` D4(b)–(d); the representation content proved at `theory/corner-a.md` step (1.4), sub-steps (d) and (f)
tested by:	`theory/checks/corner_a_check.py`
reviewed in:	`theory/verdicts/corner-a-r2.md` objections 3, 4; `theory/verdicts/corpus-r2.md` note N5 (the central-term caveat); `theory/verdicts/corner-a-r3.md`

4.4 The lattice Noether pair

Differentiating the on-site symmetry gives a charge density; commuting the Hamiltonian with the half-line charge gives a current supported near the cut. The pair satisfies an exact lattice continuity equation — no continuum limit, no approximation. The last clause is the one that connects back to the telescoping mechanism: acting on the vacuum, the physical charge density is the lattice divergence of a purely virtual, bond-supported quantity. That quantity is the lattice analogue of a gauge potential, and the charge density is its field strength.

Definition 4.21 (Charge density and cut current (D10(a),(b)))**.** Write $\mathfrak{g} = \mathrm{Lie}(G)$, and let $H = \sum_x h_{x,x+1}$ be a translation-invariant nearest-neighbour Hamiltonian; the definition extends verbatim to finite range R_h. For $\xi \in \mathfrak{g}$ let

$$q(\xi) := \frac{d}{d\varepsilon} u\big(\exp(\varepsilon\xi)\big)\Big|_{\varepsilon=0} \quad \text{(anti-Hermitian on } \mathbb{C}^d\text{)}, \tag{58}$$

with $q_x(\xi)$ its copy at site x. Assume H is G-invariant on-site: $[\, h_{x,x+1},\, q_x(\xi) + q_{x+1}(\xi)\,] = 0$.

(a) Cut current. For finite range R_h, each h_x supported in $[x, x + R_h - 1]$, and G-invariance $[\, h_x, \sum_{y \in \operatorname{supp} h_x} q_y(\xi)\,] = 0$, define the current across the bond $(m|m+1)$ as

$$j_{m|m+1}(\xi) \;:=\; -\Big[\, H, \sum_{y \le m} q_y(\xi) \Big] \;=\; -\sum_x \Big[\, h_x, \sum_{y \le m} q_y(\xi) \Big]. \tag{59}$$

The sum over x is finite — at most R_h terms, those whose support straddles the cut — because $[\, h_x, \sum_{y \le m} q_y\,] = 0$ whenever $\operatorname{supp} h_x$ lies wholly on one side. Hence $j_{m|m+1}(\xi) \in \mathfrak{A}_{\mathrm{loc}}$. In the nearest-neighbour case $R_h = 2$ this reduces to

$$j_{x|x+1}(\xi) = -\big[\, h_{x,x+1}, q_x(\xi)\,\big] = \big[\, h_{x,x+1}, q_{x+1}(\xi)\,\big]. \tag{60}$$

(b) Continuity equation. Exactly, on the lattice,

$$\big[\, H, q_x(\xi)\,\big] \;=\; j_{x-1|x}(\xi) \;-\; j_{x|x+1}(\xi). \tag{61}$$

Definition 4.22 (Modulated charge and current (D10(c)))**.** For $f \in \mathfrak{F}_c(\xi)$,

$$Q[f;\xi] := \sum_x f(x)\, q_x(\xi) \in \mathfrak{A}_{\mathrm{loc}}, \qquad J[f;\xi] := \sum_x f(x)\, j_{x|x+1}(\xi). \tag{62}$$

For $f(x) = e^{ikx}$ we write $Q_k(\xi)$ and $J_k(\xi)$, understood in the wave-packet sense of Definition 4.9.

Definition 4.23 (Bond current potential (D10(d)))**.** In the unbroken case, with the intertwining relation (64) and normal ordering (Definition 4.7), let $\mathcal{J}_b(\xi)$ denote the operation "insert $X_\alpha(\xi)$ on bond b", in the sense of Definition 4.14. The Goldstone shard's theorem G0(d) states that

$$q_x(\xi) \triangleright \omega_\alpha \;=\; \big(\mathcal{J}_{x|x+1}(\xi) - \mathcal{J}_{x-1|x}(\xi)\big) \triangleright \omega_\alpha : \tag{63}$$

the physical charge density acting on the vacuum is the lattice divergence of a purely virtual, bond-supported quantity. In this dictionary $\mathcal{J}$ is the lattice analogue of the gauge potential and q is the field strength.

Remark 4.24 (A definition that was repaired). The r1 text defined the current by the nearest-neighbour formula while the hypothesis it served quantified over finite range. The cut-current form (59) is the repaired definition, and it supersedes the nearest-neighbour one, which remains correct in its original restricted setting.

campaign id:	D10
proved in:	`definitions.md` D10(a)–(d); the continuity equation (95) and the bond-potential identity (63) proved at `theory/corner-a-goldstone.md` step (1.6)
tested by:	`theory/checks/corner_a_check.py`
reviewed in:	`theory/verdicts/corner-a-r2.md` objection 19; `theory/verdicts/corner-a-r3.md`

5 Corner A: asymptotic symmetry as boundary-bond insertions

5.1 What Corner A is about

A global on-site symmetry, applied to a *finite* region of the chain, does nothing at all in the interior of that region and everything at its two edges. This is the whole of Corner A in one sentence, and on a matrix-product state it is a theorem rather than a slogan: the symmetry is absorbed into the virtual (bond) indices, the absorptions cancel pairwise along the region, and what survives is a single virtual matrix sitting on each of the region's two boundary bonds. Everything else in this section — the endpoint torsor, the projective charge algebra, the superselected kink sectors, the lattice Noether identity — is a consequence of that cancellation, read in one register or another.

Notation recalled. We work on the infinite spin chain with quasi-local algebra $\mathfrak{A}$ and its local subalgebras $\mathfrak{A}_W$, W finite, as fixed in Section 3. A vacuum is described by an injective canonical-form tensor $A_\alpha = (A^s_\alpha)$, $A^s_\alpha \in M_\chi(\mathbb{C})$, whose transfer operator E_α has a unique modulus-one eigenvalue, second-largest modulus $\lambda_E < 1$, and injectivity length n_0 (products of n_0 tensors span $M_\chi(\mathbb{C})$). Window vectors $|\psi_\Lambda(A; b_l, b_r)\rangle$ carry boundary vectors $b_l, b_r \in \mathbb{C}^\chi$ and may be *decorated*: a matrix M inserted on a bond $(m \,|\, m+1)$, or a different tensor placed at some sites; $\omega_\alpha^{M@b}$ denotes the state of the vacuum decorated by M on the bond b, and C_∂ the uniform norm constant for one such insertion. The on-site symmetry is a compact group G acting by $u(g)$, with $U_R(g) = \prod_{x\in R} u_x(g)$ on a finite region R and $(\mathcal{U}(g)A)^s = \sum_{s'} u(g)_{ss'} A^{s'}$ on tensors. The covariant vacuum family $\{A_\alpha\}_{\alpha\in\Omega_{\mathrm{vac}}}$ obeys the intertwining relation

$$\mathcal{U}(g)A_\alpha \;=\; e^{i\theta_\alpha(g)}\, V_\alpha(g)^{-1}\, A_{g\cdot\alpha}\, V_\alpha(g), \tag{64}$$

with $V_\alpha(g)$ unitary and $\theta_\alpha(g)$ real, the pair unique up to $V \mapsto e^{i\varphi}V$. We write H_α for the stabiliser of α, $\mathfrak{h}_\alpha$ for its Lie algebra and $\mathfrak{m}_\alpha$ for a complementary (broken) subspace. Following the campaign's overload convention, ω_α with no argument is the vacuum *state*, while $\omega_\alpha(h, g)$ with two group arguments is the projective *two-cocycle* of Equation (65) below; the argument types never coincide. The definitions themselves live in Section 3 and Section 4 and are not restated here.

Large versus small, on a lattice. The background viewpoint we keep (a symmetry counts as physical, or "large", exactly when it acts on the state space with a well-defined charge, not because of a fall-off slogan) has a sharp lattice translation. A truncated symmetry operation is genuinely non-trivial at infinity precisely when the virtual matrix it deposits on the surviving bond is *not* a multiple of the identity: by Theorem 5.4(b) the half-infinite string is then implemented on states but by no strongly convergent sequence of operators, and by Theorem 5.4(c) the state it produces really is a new one. If instead $V_\alpha(g)$ is scalar, the two insertions cancel exactly, the string sequence is constant, and the operation is invisible — "small". The discriminator is thus a finite-dimensional linear-algebra condition on $V_\alpha(g)$, not a decay rate. What plays the role of the choice of function spaces at infinity is the class of profiles f allowed in a modulated charge: summable-of-bounded-variation packets and δ-normalised plane waves behave differently, and Theorem 5.11(c) is stated separately in each.

The charge-algebra viewpoint. Read as a charge algebra, the content of Corner A is that the bond insertions carry a *projective* representation of G: on padded windows the operators $\mathcal{V}_b(g)$ compose with the multiplier $e^{i\omega_\alpha(h,g)}$, whose class $[\omega_\alpha] \in \mathrm{H}^2(H_\alpha, \mathrm{U}(1))$ is the lattice analogue of a central extension of the charge algebra — and is the one-dimensional symmetry-protected-topological index of Section 14. Two disciplines must be kept. First, the extension is group-cohomological, not Lie-algebraic: for $\mathfrak{h}_\alpha$ compact semisimple Whitehead's second lemma forces the infinitesimal cocycle to be a coboundary, so a nonzero torsion class $[\omega_\alpha]$ can coexist with an entirely ordinary Lie bracket for the edge generators (the AKLT edge is the standard example). Second, the algebra is so far shown to act on finite-window vectors and on states, *not* on the GNS Hilbert space of the vacuum; the missing step is the split property, recorded as a load-bearing sketch in Scope 5.16.

5.2 The telescoping identity

The following lemma is the algebraic engine. It is what makes the phase in Equation (64) removable, and it produces the cocycle whose class is the SPT index.

Lemma 5.1 (Composition of intertwiners, the cocycle, and normal ordering)**.** *Assume the setting above. For all $g, h \in G$:*

(i) $\theta_\alpha(hg) = \theta_\alpha(g) + \theta_{g\cdot\alpha}(h)$ *and*

$$V_{g\cdot\alpha}(h)\, V_\alpha(g) \;=\; e^{i\omega_\alpha(h,g)}\, V_\alpha(hg); \tag{65}$$

(ii) restricted to H_α, θ_α is a homomorphism $H_\alpha \to \mathrm{U}(1)$ and ω_α is a $\mathrm{U}(1)$-valued two-cocycle whose class $[\omega_\alpha] \in \mathrm{H}^2(H_\alpha, \mathrm{U}(1))$ is independent of the phase convention for V_α, of the matrix-product gauge $A^s \mapsto Y^{-1}A^sY$, and of blocking sites;

(iii) in the unbroken case $H_\alpha = G$, the normally ordered *operators $\check{u}_\alpha(g) := e^{-i\theta_\alpha(g)}u(g)$ form a unitary representation of H_α satisfying Equation* (64) *with phase* 1*;*

(iv) in the unbroken case, $V_\alpha(g)\, r\, V_\alpha(g)^\dagger = r$ for the fixed point r of the transfer operator, and $E\big(V_\alpha(g)YV_\alpha(g)^\dagger\big) = V_\alpha(g)E(Y)V_\alpha(g)^\dagger$.

campaign id:	— (supporting lemma of WI and A1; no separate claim row)
proved in:	`theory/corner-a.md`, step (1.2)
tested by:	`theory/checks/corner_a_check.py` C0
reviewed in:	`theory/verdicts/corner-a-r3.md`

Idea of proof. Apply Equation (64) twice and compare with its single application at hg: the on-site action composes as a representation, so the two right-hand sides agree, and the uniqueness of the pair $(\theta_\alpha, V_\alpha)$ up to a phase forces (i). Associativity of the triple product $V(g_1)V(g_2)V(g_3)$ is the cocycle identity; rephasing $V_\alpha \mapsto e^{i\varphi}V_\alpha$ changes ω_α by a coboundary, a gauge change conjugates V_α and leaves ω_α alone, and blocking n sites reproduces Equation (64) with the same V_α and phase $n\theta_\alpha(g)$, which gives (ii). Part (iv) is the covariance of the transfer operator, from unitarity of $u(g)$ together with uniqueness of the fixed point. ☐

Theorem 5.2 (Truncated symmetry telescopes to the boundary (WI))**.** [PROVED] *Let $R = [a, b]$ be a finite interval, $g \in G$, and $b_l, b_r \in \mathbb{C}^\chi$ arbitrary. Assume* either *(W1) a finite window $\Lambda \supseteq R$ together with edge-bond insertions,* or *(W2) a window $\Lambda \supseteq [a-1, b+1]$, in which case both insertions land on interior bonds. Then, exactly, as window vectors,*

$$U_R(g)\,\big|\psi_\Lambda(A_\alpha; b_l, b_r)\big\rangle \;=\; e^{\,i|R|\theta_\alpha(g)}\,\big|\psi_\Lambda(T_R^{(g)}; b_l, b_r)\big\rangle, \tag{66}$$

where the decoration $T_R^{(g)}$ carries A_α on $\Lambda \setminus R$, $A_{g\cdot\alpha}$ on R, the bond insertion $V_\alpha(g)^{-1}$ on $\partial_- R = (a-1\,|\,a)$ and $V_\alpha(g)$ on $\partial_+ R = (b\,|\,b+1)$. As states, *with* ***no phase****, for every $O \in \mathfrak{A}$,*

$$\omega_\alpha\big(U_R(g)^\dagger\, O\, U_R(g)\big) \;=\; \omega_\alpha\big[T_R^{(g)}\big](O). \tag{67}$$

In particular, in the unbroken case the interior of R is unchanged and the entire effect of a symmetry applied to a region is two virtual operators on the region's two boundary bonds.

Scope 5.3. The window hypothesis is not cosmetic. For $\Lambda = R$ with interior insertions only, the two boundary matrices fall outside the expression and Equation (66) is *false*: the measured discrepancy for a $\chi = 2$ Pauli tensor is 1.658, against 0.0 once edge-bond insertions are admitted. The extensive phase $e^{i|R|\theta_\alpha(g)}$ belongs to the *vector* identity only; a phase times a state is the same state, so Equation (67) carries none, and asserting a limit of the *operator* $U_{[x,y]}(g)$ when $\theta_\alpha(g) \neq 0$ is meaningless. The orientation — $V_\alpha(g)^{-1}$ on the left bond, $V_\alpha(g)$ on the right — is fixed by the convention in Equation (64); it is the opposite of the assignment in the campaign brief, which is a historical document and is not edited. A $\mathbb{Z}_2$ test tensor with $V = Z = V^{-1}$ cannot see the difference and must not be used to confirm it; a U(1) test with $V(t) = e^{itZ/2}$ gives error $5.6 \cdot 10^{-17}$ for the correct orientation and 0.267 for the flipped one. Finally, only Equation (64) is used: injectivity is needed to *obtain* Equation (64) and to make $(\theta_\alpha, V_\alpha)$ unique, not for the algebra.

campaign id:	WI
proved in:	`theory/corner-a.md`, step (1.3)
tested by:	`theory/checks/corner_a_check.py` C1, C1b, C2, C2b
reviewed in:	`theory/verdicts/corner-a-r3.md`

Idea of proof. Expand $U_R(g)|\psi_\Lambda\rangle$ in the physical basis: the effect is to replace $A_\alpha^{s_x}$ by $(\mathcal{U}(g)A_\alpha)^{s_x}$ at each $x \in R$, and by Equation (64) each such replacement is $e^{i\theta_\alpha(g)}V_\alpha(g)^{-1}A_{g\cdot\alpha}^{s_x}V_\alpha(g)$. In the ordered product over $x = a, \dots, b$ every interior pair $V_\alpha(g)V_\alpha(g)^{-1}$ collapses to $\mathbf{1}$ — this is the telescoping — leaving exactly one factor at each end and $|R| = b - a + 1$ copies of the phase. Passing to states, the phase cancels between bra and ket and the decoration differs from A_α on finitely many sites and bonds. ☐

5.3 The unbroken case: endpoints, charge algebra, and the obstruction

Throughout this subsection $H_\alpha = G$ and we work normally ordered, so $\theta_\alpha \equiv 0$. Recall from Section 4 the objects $N_\alpha := \{g \in G : V_\alpha(g) \in \mathbb{C}^\times \mathbf{1}\}$ (a normal subgroup), the induced map $\rho_\alpha : G \to \mathrm{PGL}(\chi)$, $\rho_\alpha(g) = [V_\alpha(g)]$, the effective group $\mathcal{A}_{\text{eff}} := G/N_\alpha \cong \rho_\alpha(G)$, the endpoint space $E_b^\alpha = \{\omega_\alpha^{M@b} : M \in \mathrm{GL}(\chi)\}$ at a bond b, the twisted group algebra $\mathfrak{a}_\alpha = \mathbb{C}_{\omega_\alpha}[G]$, and the space $\mathcal{W}_{\Lambda,b}$ spanned by window vectors carrying one insertion on b, with $\iota_{\Lambda,b} : M_\chi(\mathbb{C}) \to \mathcal{W}_{\Lambda,b}$ the map that places the insertion. A window is *padded about b* if it contains at least n_0 sites on each side of b and $b_l, b_r \neq 0$.

Theorem 5.4 (Endpoint torsor, projective charge algebra, and the cohomological obstruction (A1))**.** [PROVED] *In the unbroken case, normally ordered, with every clause mentioning $\mathcal{W}_{\Lambda,b}$ or $\mathcal{V}_b$ additionally assuming a window padded about b:*

(a) (charges live on one bond) *For finite R the right insertion is invisible to observables to its left and the left insertion invisible to those to its right. Hence the half-infinite operations exist* as maps on states,

$$1_{[x,\infty)}g \triangleright \omega_\alpha \;=\; \omega_\alpha^{V_\alpha(g)^{-1}@(x-1\,|\,x)},$$

and the convergence is exact *— eventually constant on each $\mathfrak{A}_W$ — not merely asymptotic.*

(b) (non-implementability, as an equivalence) *The sequence* $\big(U_{[x,y]}(g)\Omega_A\big)_y$ *fails to be Cauchy in* $\mathcal{H}_A$ ***if and only if*** $V_\alpha(g) \notin \mathbb{C}^\times \mathbf{1}$. *For scalar* $V_\alpha(g)$ *the strings stabilise exactly. Thus the half-infinite symmetry is implemented on states, but by no strongly convergent sequence of operators.*

(c) (the endpoint bijection) $\omega_\alpha^{M@b} = \omega_\alpha^{M'@b}$ ***iff*** $M' \in \mathbb{C}^\times M$; *hence* $[M] \mapsto \omega_\alpha^{M@b}$ *is a canonical bijection and* E_b^α *is a* $\mathrm{PGL}(\chi)$*-torsor.*

(d) (the charge algebra, in the two registers it actually has)

(d1) *On a window padded about b, the map* $\iota_{\Lambda,b}$ *is injective, so* $\mathcal{V}_b(M) := \iota_{\Lambda,b} \circ L_M \circ \iota_{\Lambda,b}^{-1}$ *is a well-defined linear operator, and with* $\mathcal{V}_b(g) := \mathcal{V}_b(V_\alpha(g))$

$$\mathcal{V}_b(h)\,\mathcal{V}_b(g) \;=\; e^{i\omega_\alpha(h,g)}\,\mathcal{V}_b(hg): \tag{68}$$

a linear *representation of* $\mathfrak{a}_\alpha = \mathbb{C}_{\omega_\alpha}[G]$ *in which the multiplier acts nontrivially.* Padding is necessary, not cosmetic*: without it the kernel of* $\iota_{\Lambda,b}$ *need not be invariant under left multiplication and* $\mathcal{V}_b$ *is not an operator at all.*

(d2) *The induced action on* states *kills the multiplier and is the genuine homomorphism* $\rho_\alpha : G \to \mathrm{PGL}(\chi)$ *with* $\ker\rho_\alpha = N_\alpha$*; it is bond-independent and padding-independent.*

(d3) *The window action (d1)* is *a lift of* ρ_α*, but a projective one, and such a lift exists for every class* $[\omega_\alpha]$*. What* $[\omega_\alpha]$ *obstructs is* ***removing the multiplier****: it is exactly the obstruction to lifting* $\rho_\alpha : G \to \mathrm{PGL}(\chi)$ *to an honest homomorphism* $G \to \mathrm{U}(\chi)$*, being the pullback under* ρ_α *of the extension class of* $1 \to \mathrm{U}(1) \to \mathrm{U}(\chi) \to \mathrm{PU}(\chi) \to 1$*. It does* not *obstruct the projective window action, which always exists.*

(e) (stabiliser and orbit) *The stabiliser of* ω_α *in* $G_L \times G_R$ *is*

$$S_\alpha = \{(g_L, g_R) : g_L g_R^{-1} \in N_\alpha\} \;\supseteq\; G_{\mathrm{diag}},$$

with equality ***iff*** $N_\alpha = \{e\}$*, and the orbit is in bijection with* $\mathcal{A}_{\mathrm{eff}} = G/N_\alpha$*, a genuine group. The coset space* $\mathcal{A} = (G_L \times G_R)/G_{\mathrm{diag}}$ *is* not *the orbit unless* $N_\alpha = \{e\}$*, and is a group only for abelian G.*

(f) (what acts on what) $\mathcal{A}_{\mathrm{eff}}$ *acts simply transitively on the orbit* $\rho_\alpha(G) \subseteq E_b^\alpha$*, and does not move the sector label* (α,α)*, which is trivial in the unbroken case.*

(g) (invariance of the index) $[\omega_\alpha]$ *is constant along any continuous path of G-symmetric injective canonical-form tensors, provided* $\mathrm{H}^2(G, \mathrm{U}(1))$ *is discrete.*

Scope 5.5. The claims register carries this theorem with the scope qualifier "padded-window and state registers as stated", and the clauses displayed in its headline row are (a)–(e); (f) and (g) are proved in the same shard and are recorded here because Section 14 uses (g). What is *not* proved is the physical, GNS-level realisation: that $\omega_\alpha^{M@b}$ is a normal state of $\pi_\alpha(\mathfrak{A})''$, equivalently that $\mathfrak{a}_\alpha$ acts on the vacuum Hilbert space rather than on finite-window spaces. That is the split-property sketch of Scope 5.16, and it is load-bearing: the background desideratum of a charge algebra with a well-defined action on the physical state space is *not* met by this theorem. Reading (d) as an action on an edge Hilbert space is therefore an over-read. Completeness of $[\omega_\alpha]$ as a classifier of one-dimensional SPT phases is cited from the literature, not proved here.

campaign id:	A1
proved in:	`theory/corner-a.md`, step (1.4)
tested by:	`theory/checks/corner_a_check.py` C6, C8, C8b
reviewed in:	`theory/verdicts/corner-a-r3.md`

Idea of proof. Clause (a) is a transfer-operator computation on top of Theorem 5.2: in canonical form the left environment is $\mathbf{1}$ and the right environment is the fixed point r, and a unitary insertion M replaces them by $M^\dagger \mathbf{1} M = \mathbf{1}$ and $MrM^\dagger$; since $V_\alpha(g) r V_\alpha(g)^\dagger = r$ by Lemma 5.1(iv), the far insertion simply disappears once the region's edge has passed the support of the observable, which is why the limit is exact rather than asymptotic.

Clause (c) is the multilinear step. Writing the reduced density matrix of a window containing the bond, its matrix elements are $\mathrm{tr}[P_1 M P_2 r Q_2^\dagger M^\dagger Q_1^\dagger]$ with the four words P_1, P_2, Q_1, Q_2 varying *independently*; by injectivity each ranges over a spanning set of $M_\chi(\mathbb{C})$, and both sides of the equality with the undecorated state are multilinear, so the equality extends to all of $M_\chi(\mathbb{C})^{\times 4}$. Specialising the words to matrix units turns this into $MZM^\dagger = \gamma Z$ for every Z, whence $M \in \mathbb{C}^\times \mathbf{1}$. Note the normalisation: the correct target is γZ with $\gamma > 0$, not Z.

Clause (b) uses the overlaps $\langle \Psi_y, \Psi_{y'} \rangle = \rho_{y'-y}$ coming from translation invariance. Being Cauchy forces $\rho_n = 1$ for *every* $n \geq 1$, not merely $\rho_n \to 1$; that in turn makes the one-sided string act trivially on every local algebra, and (c) then forces $V_\alpha(g)$ scalar. Conversely a scalar $V_\alpha(g) = c\mathbf{1}$ gives boundary insertions $c^{-1}\mathbf{1}$ and $c\mathbf{1}$ whose product is the identity, so the sequence is constant.

Clause (d1) rests on injectivity of $\iota_{\Lambda,b}$ on padded windows: the coefficient $b_l^\dagger PMQb_r$ is linear in each of the two words, which span $M_\chi(\mathbb{C})$, and $b_l, b_r \neq 0$, so $v^\dagger(M - M')w = 0$ for all v, w. Without padding this fails concretely: for the injective tensor $A^0 = \mathrm{diag}(1,2)$, $A^1 = X$ with $V = Z$, one site each side, and a suitable N, one has $\iota(N) = 0$ while $\|\iota(ZN)\|_\infty = 4$, so $\ker \iota$ is not invariant and the rule "$\iota(M) \mapsto \iota(NM)$" defines nothing. Given injectivity, Equation (68) is Equation (65) transported through ι. Clause (d2) is (c) again — states see M only modulo scalars — and (d3) is the observation that a lift with $\omega_\alpha \equiv 0$ exists iff the class vanishes.

Clause (e) composes the two half-infinite residues: the left string leaves $V_\alpha(g_L)$ on the bond and the right string leaves $V_\alpha(g_R)^{-1}$, so the composite residue is $V_\alpha(g_L)V_\alpha(g_R)^{-1}$, trivial on states exactly when $g_L g_R^{-1} \in N_\alpha$; the map $(g_L, g_R) \mapsto g_L g_R^{-1} N_\alpha$ is then a bijection $(G \times G)/S_\alpha \to G/N_\alpha$. Clause (g) is continuity of an isolated spectral projection along the path, giving a continuous map into a discrete set. □

Remark 5.6 (Two instances that pin the orbit down). A $\chi = 1$ product vacuum with a $\mathbb{Z}_2$ symmetry has $V_\alpha \equiv 1$, hence $N_\alpha = G$, $S_\alpha = G \times G$, and a *one-point* orbit. The AKLT vacuum with $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ has $V_\alpha \in \{\mathbf{1}, X, Z, XZ\}$, hence $N_\alpha = \{e\}$ and a four-point orbit of pairwise distinct states. Any reading of the orbit as "a copy of the symmetry group" fails on the first example.

5.4 The broken case: sector jumps and the classification of vacuum pairs

Now let $|\Omega_{\mathrm{vac}}| \geq 2$, fix α and $g \in G$ with $\beta := g \cdot \alpha \neq \alpha$. Recall the kink sectors $\mathcal{K}_{\alpha\beta}$ of Section 4: states with factorised vacuum asymptotics ω_α at $-\infty$ and ω_β at $+\infty$. Sectors with distinct labels are disjoint, and the label of a decorated matrix-product state is read off from its tensors at $\pm\infty$ at exponential rate.

Theorem 5.7 (A broken truncated symmetry creates a kink only in the limit (A2)). [PROVED] *For each fixed $g \notin H_\alpha$:*

- *(a)* (finite regions never leave the sector) *For every finite $R = [a,b]$, $\omega_\alpha \circ \mathrm{Ad}(U_R(g)^\dagger) \in \mathcal{K}_{\alpha\alpha}$; explicitly it is the state decorated with A_α outside R, A_β inside R, $V_\alpha(g)^{-1}$ on $\partial_- R$ and $V_\alpha(g)$ on $\partial_+ R$ — a kink at one edge and an antikink at the other, of total topological charge zero.*

(b) (the half-infinite limit exists, weak-$*$ only, with a rate) *For $O \in \mathfrak{A}_W$, $w := \max W$, $y > w$, and for **every** $\tilde\lambda \in (\lambda_E, 1)$ there is $C_{\tilde\lambda} < \infty$ with*

$$\Big|\, \omega_\alpha\big(U_{[x,y]}(g)^\dagger O\, U_{[x,y]}(g)\big) - \varrho_x^{(g)}(O) \,\Big| \;\le\; C_{\tilde\lambda}\, \|O\|\, \tilde\lambda^{\,y-w}, \tag{69}$$

where $\varrho_x^{(g)}$ is the two-sided decorated state with A_α on $(-\infty, x-1]$, $V_\alpha(g)^{-1}$ on the bond $(x-1\,|\,x)$, and A_β on $[x,\infty)$. Hence $1_{[x,\infty)} g \triangleright \omega_\alpha = \varrho_x^{(g)}$ in the weak-$*$ topology.

(c) (the limit is a kink) $\varrho_x^{(g)} \in \mathcal{K}_{\alpha\beta}$.

(d) (the sector jump) *Every finite-R state is normal with respect to ω_α, while the limit is disjoint from it. A weak-$*$ limit of a path of vector states thus leaves the folium of the vacuum; the jump is invisible at every finite $|R|$ and is created entirely by the surviving end.*

(e) (vacuum pairs, under hypothesis (T)) *$G_L \times G_R$ acts componentwise on $\Omega_{\mathrm{vac}} \times \Omega_{\mathrm{vac}}$. Assume **(T)**: G acts transitively on Ω_{vac} (this does not follow from covariance — the family may split into several G-orbits — and without it every statement here holds per G-orbit). Then*

$$\Omega_{\mathrm{vac}} \times \Omega_{\mathrm{vac}} \;\cong\; (G/H_\alpha) \times (G/H_\alpha),$$

transitively, with stabiliser of (α', β') equal to $H_{\alpha'} \times H_{\beta'}$; and the complete invariant of a vacuum pair modulo the global diagonal symmetry is the double coset

$$\mathfrak{d}(\alpha_L, \alpha_R) \;=\; H_\alpha\, g_L^{-1} g_R\, H_\alpha \;\in\; H_\alpha \backslash G / H_\alpha, \tag{70}$$

which for $G = SU(2)$, $H_\alpha = \mathrm{U}(1)$ is the relative angle $\cos\vartheta = \hat n_L \cdot \hat n_R$.

(f) (every such sector is reached) *Every kink sector $\mathcal{K}_{\alpha\beta}$ with $\beta \in G\cdot\alpha$ arises this way, by choosing g with $g \cdot \alpha = \beta$.*

Scope 5.8. The convergence in Equation (69) is weak-$*$: the constant depends on the observable's window through w, and the convergence cannot be improved to norm convergence in $\mathfrak{A}^*$, because the approximants are normal with respect to the vacuum and the limit is not. The operators $U_{[x,y]}(g)$ have no strong limit on the vacuum Hilbert space. The rate is $\tilde\lambda \in (\lambda_E, 1)$ and *not* λ_E itself, which would be false if the transfer operator has a Jordan block at the subleading modulus. Hypothesis (T) is carried explicitly and is not implied by covariance of the vacuum family. Finally, the slogan "a kink is the contact term of a broken truncated symmetry" is a mnemonic, not a result: no contact term is defined anywhere in this campaign, no distributional Ward identity is written, and no matrix element is shown to be supported at coincident insertions. It may not appear as the content of a proved claim.

campaign id:	A2
proved in:	`theory/corner-a-kinks.md`, steps (1.8)–(1.9)
tested by:	`theory/checks/corner_a_check.py` C7, C11
reviewed in:	`theory/verdicts/corner-a-r3.md`

Idea of proof. Clause (a) is Theorem 5.2 read as a state: the decoration equals A_α on both far tails, so both asymptotic labels are α, and conjugation by a local unitary preserves normality. Clause (b) is a transfer-matrix gap estimate. The finite-string state and the limit state carry identical decorations to the left of the observable and differ only in the environment to its right, which is $E_\beta^{y-w}\big(V_\alpha(g) r_\alpha V_\alpha(g)^\dagger\big)$ for the first and r_β for the second. Since $\mathrm{tr}(V_\alpha(g) r_\alpha V_\alpha(g)^\dagger) = 1$, the difference is controlled by the convergence $\|E_\beta^m(Y) - \mathrm{tr}(Y) r_\beta\| \le C_{\tilde\lambda} \tilde\lambda^m \|Y\|$, valid for every

$\tilde{\lambda} > \lambda_E$. One further point makes this uniform along the orbit: by Equation (64) and unitarity of $u(g)$ one has $E_\beta = \mathrm{Ad}_{V_\alpha(g)} \circ E_\alpha \circ \mathrm{Ad}_{V_\alpha(g)^{-1}}$, a similarity, so the two transfer operators have the same spectrum and the same gap. Clauses (c), (d) then follow by reading the asymptotic labels and invoking disjointness of distinct sectors. For (e), orbit–stabiliser in each factor gives the product structure; and while $g_L^{-1} g_R$ is invariant under the diagonal action and changes to $h_1^{-1}(g_L^{-1} g_R)h_2$ under the residual stabilisers, which is precisely the double-coset statement, the quantity $g_L g_R^{-1}$ transforms by conjugation and is not diagonal-invariant at all. □

Remark 5.9 (Which model is which). For the easy-axis chain the broken group is the discrete spin flip, $\Omega_{\mathrm{vac}} = \{\uparrow, \downarrow\}$, and Theorem 5.7 says that the half-infinite flip applied to the polarised state produces a sharp domain wall disjoint from the ferromagnetic vacuum: a genuinely gapped topological sector, and the setting in which the superselection language is sharp (Section 6). For the isotropic ferromagnet $G = SU(2)$ is broken to U(1) with $\Omega_{\mathrm{vac}} \cong S^2$ a continuum, $\chi = 1$ and $\lambda_E = 0$, so the estimate Equation (69) is exact at every $y > w$; but the kinks interpolate between infinitesimally different vacua and cost arbitrarily little energy. The phrase "the broken symmetry" is ambiguous in this campaign and is avoided: every statement names G, H_α, or the broken complement $\mathfrak{m}_\alpha$.

5.5 Goldstone tensors, the lattice Noether identity, and the kinematic factor

The infinitesimal shadow of Theorem 5.2 is a summation by parts. The next lemma is stated with its two boundary terms because they are $\Theta(1)$ and dropping them is the error the round-one draft made.

Lemma 5.10 (Summation by parts, with the exact gauge remainder)**.** *Let $\Lambda = [a, b]$ be finite, $X \in M_\chi(\mathbb{C})$, and $f : \mathbb{Z} \to \mathbb{C}$ any profile. Write $|\psi; X@m\rangle$ for the window vector with X inserted on the bond $(m \,|\, m+1)$, and $(\Delta f)(m) := f(m+1) - f(m)$. Then, exactly,*

$$\sum_{n\in\Lambda} f(n)\,\big|\psi\big((A_\alpha X - XA_\alpha)@n\big)\big\rangle = -\sum_{m=a}^{b-1} (\Delta f)(m)\,|\psi; X@m\rangle + f(b)\,|\psi; X@b\rangle - f(a)\,|\psi; X@(a-1)\rangle, \tag{71}$$

the last two terms being the gauge remainder $\mathfrak{B}_\Lambda[f, X]$. In particular, for a plane wave $f(n) = e^{ikn}$ and the null map $\mathcal{N}_k(X)^s := e^{ik} A^s_\alpha X - XA^s_\alpha$,

$$\big|\Phi^\Lambda_k(\mathcal{N}_k(X))\big\rangle = e^{ik(b+1)}|\psi; X@b\rangle - e^{ika}|\psi; X@(a-1)\rangle \;\neq\; 0. \tag{72}$$

Moreover $\|\mathfrak{B}_\Lambda[f, X]\| \le 2C_\partial \|X\| \max(|f(a)|, |f(b)|)$ uniformly in Λ; if $f \in c_0(\mathbb{Z})$ the remainder vanishes in norm as $\Lambda \nearrow \mathbb{Z}$ (a statement about the remainder only), and if moreover $f \in \ell^1 \cap BV$ both sides are absolutely convergent and the identity is remainderless in the limit; for a plane wave the remainder is $O(1)$, hence $O(|\Lambda|^{-1/2})$ after δ-normalisation. Finally $\operatorname{rank} \mathcal{N}_k = \chi^2$ *for $k \neq 0$ and* $\operatorname{rank} \mathcal{N}_0 = \chi^2 - 1$*: the rank* drops *by one as $k \to 0$.*

campaign id:	— (supporting lemma of G0; no separate claim row)
proved in:	`theory/corner-a-goldstone.md`, step (1.5)
tested by:	`theory/checks/corner_a_check.py` C3, C3b, C4, C5, C10
reviewed in:	`theory/verdicts/corner-a-r3.md`

Idea of proof. Placing $A^s_\alpha X$ at site n is the same as placing X on the bond immediately to its right, and placing XA^s_α at site n is placing X on the bond to its left; subtracting the two finite sums after a shift of index leaves the difference coefficient $-(\Delta f)(m)$ on the interior bonds and one uncancelled term at each end. The uniform bound is one insertion's worth of norm, uniformly in the window because the transfer powers are power-bounded. Numerically the identity holds to $3\cdot 10^{-17}$, the plane-wave remainder has norm 1.771 for a $\chi = 2$ Pauli tensor at $k = 0.37$, and the ratio of remainder to bulk for the centred profile $(1+|n-c|)^{-3}$ falls as $3.4\cdot 10^{-1}$, $5.7\cdot 10^{-2}$, $8.5\cdot 10^{-3}$, $1.2\cdot 10^{-3}$ for window half-widths $4, 8, 16, 32$. No lower bound on the growth of the bulk term is claimed or needed: for $\chi = 1$ the plane-wave bulk sum is a bounded geometric sum. The rank statement is injectivity of $\mathcal{N}_k$ for $k \neq 0$ (iterating the relation forces $e^{ik} = 1$) together with $\ker \mathcal{N}_0 = \mathbb{C}\mathbf{1}$; the factor $1 - e^{ik}$ of Theorem 5.11(c) is algebraic and is *not* caused by this rank discontinuity. $\square$

Theorem 5.11 (Goldstone tensors, the exact finite-window identity, and lattice Noether (G0))**.** [PROVED] *Let $\xi \in \mathfrak{g}$ and let $B_G(\xi)$ be the Goldstone tensor of Section 4, with $X_\alpha(\xi)$ the associated virtual generator.*

(a) (exact form on unbroken directions) *For $\xi \in \mathfrak{h}_\alpha$, $B_G(\xi) = A_\alpha X_\alpha(\xi) - X_\alpha(\xi) A_\alpha + i\theta'_\alpha(\xi) A_\alpha$, and after normal ordering the last term is absent, so $B_G(\xi) = \mathcal{N}_0(X_\alpha(\xi))$.*

(b) (the dichotomy) *$B_G(\xi) \in \operatorname{ran} \mathcal{N}_0 + \mathbb{C}A_\alpha$ **iff** $\xi \in \mathfrak{h}_\alpha$. For a broken direction it is* not *the case that the intertwining relation fails — Equation (64) holds for every g, with target tensor $A_{g\cdot\alpha}$. What fails is the* same-vacuum return $g \cdot \alpha = \alpha$, *so $B_G(\xi)$ is a tangent to the vacuum manifold rather than to the gauge orbit.*

(c) (the k-dependence, with the limit named) *For $\xi \in \mathfrak{h}_\alpha$, normally ordered, and any k, $B_G(\xi) = \mathcal{N}_k(X) + (1 - e^{ik}) A_\alpha X$ with $X := X_\alpha(\xi)$; hence on a finite window $\Lambda = [a, b]$ the **exact** identity is*

$$\big|\Phi_k^\Lambda(B_G(\xi))\big\rangle = (1 - e^{ik}) \sum_{m=a}^{b-1} e^{ikm} |\psi; X@m\rangle + e^{ikb} |\psi; X@b\rangle - e^{ika} |\psi; X@(a-1)\rangle. \tag{73}$$

*The clean form $(1 - e^{ik})\big|\Phi_k(A_\alpha X_\alpha(\xi))\big\rangle$ holds **only** in the δ-normalised sense; for decaying profiles the correct statement is the real-space identity Equation (71) with $f \in \ell^1 \cap BV$, not a fixed-k equation.*

(d) (lattice Noether: the bond potential) *Exactly, on the vacuum, with no limit and no boundary term, for $\xi \in \mathfrak{h}_\alpha$ normally ordered,*

$$q_x(\xi) \triangleright \omega_\alpha \;=\; \big(\mathcal{J}_{x|x+1}(\xi) - \mathcal{J}_{x-1|x}(\xi)\big) \triangleright \omega_\alpha : \tag{74}$$

the physical *charge density acting on the vacuum is the lattice divergence of the* virtual, *bond-supported quantity $\mathcal{J}$.*

(e) (continuity equation, finite range) *For a translation-invariant G-invariant Hamiltonian of finite range, with the cut current $j_{m|m+1}(\xi) := -[H, \sum_{y \le m} q_y(\xi)] \in \mathfrak{A}_{\mathrm{loc}}$, one has $[H, q_x(\xi)] = j_{x-1|x}(\xi) - j_{x|x+1}(\xi)$ and, for every admissible profile f of compact support and **any** $\xi \in \mathfrak{g}$,*

$$\big[H,\ Q[f;\xi]\big] \;=\; \sum_x (\Delta f)(x)\, j_{x|x+1}(\xi), \qquad \textit{whence} \qquad [H, Q_k(\xi)] = (e^{ik} - 1)\, J_k(\xi) \tag{75}$$

in the wave-packet sense.

Scope 5.12. Clauses (a), (b), (d) are asserted for $\xi \in \mathfrak{h}_\alpha$ and normally ordered; clause (e) holds for any $\xi \in \mathfrak{g}$ and any finite-range Hamiltonian. The prefactor $(e^{ik} - 1)$ in Equation (75) is *kinematic*: it comes from the profile alone, it is convention-dependent (a kernel e^{-ikx} gives $e^{-ik} - 1$, and reversing the current's orientation flips the sign), and its independence of the Hamiltonian is the tautology that a discrete difference was factored out *after* an H-dependent current had been defined. It is not a soft factor; see Refuted claim 5.15. Clauses (a)–(d) are statements about the matrix-product vacuum and the linear ansatz space; only (e) is an operator identity in the local algebra, true of the model itself whatever its ground state and needing no matrix-product structure — which is why (e), and nothing else in Corner A, is the candidate seed for the soft-theorem corner (Section 12).

campaign id:	G0
proved in:	`theory/corner-a-goldstone.md`, steps (1.5)–(1.7)
tested by:	`theory/checks/corner_a_check.py` C3, C3b, C4, C5, C9, C10; oracle O1 rederived at step (1.7)(2.3)
reviewed in:	`theory/verdicts/corner-a-r3.md`

Idea of proof. Clause (a) is Equation (64) differentiated at the identity along an unbroken direction, where the target tensor returns to A_α; the product rule produces the commutator plus the phase derivative, which normal ordering removes. For the forward direction of (b), the state map is constant on gauge orbits and on rays, so its differential annihilates $\operatorname{ran}\mathcal{N}_0+\mathbb{C}A_\alpha$; the resulting curve of vacua has vanishing derivative everywhere by the group law, hence is constant, and since distinct labels give distinct states the direction must stabilise α. Clause (c) is the algebraic decomposition $\mathcal{N}_k(X) = (A_\alpha X - XA_\alpha) + (e^{ik}-1)A_\alpha X$ fed into Equation (71) and Equation (72), with care over the two summation ranges: the bulk sum for $A_\alpha X$ runs to b, the null-map remainder contributes at b as well, and combining the two right-edge coefficients through $(1-e^{ik})e^{ikb}+e^{ik(b+1)} = e^{ikb}$ gives Equation (73). An earlier draft truncated the bulk sum at $b-1$ while keeping the uncombined right coefficient, thereby losing the term $(1-e^{ik})e^{ikb}|\psi; X@b\rangle$; the omission was measured as 0.45058621, exactly that term's norm, while both corrected displays agree with the left-hand side to $6\cdot 10^{-17}$. Clause (d) is a single-site statement — the charge density replaces the tensor at x by $B_G(\xi)$, i.e. by $X_\alpha(\xi)$ on the right bond minus $X_\alpha(\xi)$ on the left bond — so no reindexing and no boundary term arise; summing over a region telescopes it back to the two insertions of Theorem 5.2. Clause (e) is the observation that the cut current is local, because G-invariance kills every term of the Hamiltonian lying entirely inside the half-line and locality kills every term lying entirely outside, leaving only the finitely many straddling terms; one Abel summation then gives Equation (75). □

Proposition 5.13 (The isotropic ferromagnet, and the one oracle Corner A rederives)**.** [PROVED]
For the isotropic ferromagnet with $G = SU(2)$ in the spin-$\frac{1}{2}$ representation and the polarised product vacuum ($\chi = 1$, $\lambda_E = 0$): (i) $H_\alpha = \mathrm{U}(1)$, the rotations about z; (ii) $[\omega_\alpha] = 0$ and the unbroken $\mathrm{U}(1)$ produces no Goldstone tensor; (iii) the two broken real generators produce one *complex Goldstone tensor, $B_G(\xi_y) = i\,B_G(\xi_x)$ — the type-B count, derived rather than assumed; (iv) $|\Phi_k(B_G(\xi_x))\rangle \propto |k\rangle$ is the exact one-magnon state and Equation (75) reproduces the dispersion $\omega(k) = J(1-\cos k)$, which is the content of oracle fact O1.*

campaign id:	— (supporting proposition of G0; no separate claim row)
proved in:	`theory/corner-a-goldstone.md`, step (1.7)
tested by:	`theory/checks/corner_a_check.py` C0; oracle O1
reviewed in:	`theory/verdicts/corner-a-r3.md`

Idea of proof. The z-rotation acts on the polarised site by a phase, so Equation (64) holds with $V_\alpha = 1$ and $\theta'_\alpha = \frac{1}{2} \neq 0$ — the extensive phase of Theorem 5.2, which is exactly the vacuum magnetisation density that normal ordering subtracts. Rotations off the axis send the tensor to an inequivalent one, so the stabiliser is the circle; its second cohomology vanishes and $V_\alpha \equiv 1$, giving (ii). For (iii), the raising direction annihilates the vacuum site-wise and the lowering direction creates the magnon, so the real two-dimensional broken space maps onto a one-dimensional complex space of Goldstone tensors. For (iv), the ferromagnetic current itself is a difference of broken charge densities, supplying a *second* factor of $(1 - e^{ik})$; the two factors together produce the quadratic dispersion. This is the only oracle fact that Corner A rederives, and only in the one-magnon sector. □

5.6 Two claims that were withdrawn

Both of the following were asserted in earlier rounds of this corner, are false as stated, and are recorded here because each killed a mechanism the campaign had been relying on.

Refuted claim 5.14 (The naive asymptotic symmetry group is not the classifier (A2-orbit-r1))**.** [REFUTED] *Withdrawn statement.* The coset space $\mathcal{A} = (G_L \times G_R)/G_{\text{diag}}$ is the lattice asymptotic symmetry group; its orbit is the set of vacuum pairs, and it is labelled by the class $[g_L g_R^{-1}]$.

Why it fails. Three independent reasons. (1) $\mathcal{A}$ is a group only for abelian G, since G_{diag} is normal in $G \times G$ exactly then. (2) The componentwise stabiliser of a vacuum pair is $H_\alpha \times H_\beta$, not G_{diag}, so the pair space is $(G/H_\alpha) \times (G/H_\alpha)$ and has the wrong dimension to be $\mathcal{A}$: for $G = SU(2)$, $H_\alpha = \mathrm{U}(1)$ that is $S^2 \times S^2$, of dimension 4, against $\dim SU(2) = 3$. (3) The proposed label is not diagonal-invariant: under $g_i \mapsto hg_i$ the quantity $g_L g_R^{-1}$ transforms by conjugation, $g_L g_R^{-1} \mapsto hg_L g_R^{-1} h^{-1}$. The numerical gate makes the contrast explicit: the relative angle is invariant to $1.4 \cdot 10^{-16}$, while $\|g_L g_R^{-1} - h\, g_L g_R^{-1} h^{-1}\| = 1.364$. In the unbroken case the same error appears in a different guise: the orbit is $\mathcal{A}_{\text{eff}} = G/N_\alpha$, and a $\chi = 1$ symmetric product vacuum has $N_\alpha = G$ and a *one-point* orbit where the withdrawn claim predicted an orbit of size $|G|$.

What survives. Theorem 5.4(e) in the unbroken case: the orbit is $\mathcal{A}_{\text{eff}} = G/N_\alpha$, a genuine group, with stabiliser $S_\alpha \supseteq G_{\text{diag}}$. And Theorem 5.7(e) in the broken case: the pair space is $(G/H_\alpha) \times (G/H_\alpha)$ and the complete diagonal invariant is the double coset Equation (70). Consequently the phrase "asymptotic symmetry group" may not be used in this campaign without saying which of $\mathcal{A}_{\text{eff}}$, $(G/H_\alpha) \times (G/H_\alpha)$, or $H_\alpha \backslash G/H_\alpha$ is meant.

campaign id:	A2-orbit-r1
proved in:	disproved in `theory/corner-a-kinks.md`, step (1.9)(2.5)(3.3)
tested by:	`theory/checks/corner_a_check.py` C7
reviewed in:	`theory/verdicts/corner-a-r3.md`

Refuted claim 5.15 (The naive universal soft factor (G0-soft-r1))**.** [REFUTED] *Withdrawn statement.* The factor $(e^{ik} - 1)$ appearing in Equation (75) is a *universal soft factor*: it yields an Adler zero and forces the hard-independent linear coefficient of the oracle's two-body phase.

Why it fails. In a matrix element the continuity equation says only

$$\langle \text{out}|[H, Q_k]|\text{in}\rangle = (e^{ik} - 1)\,\langle \text{out}|J_k|\text{in}\rangle. \tag{76}$$

All dependence on the Hamiltonian, on the direction ξ, on the vacuum and on the hard legs sits in the current matrix element on the right. If $\langle \text{out}|J_k|\text{in}\rangle = C_{\text{hard}} + O(k)$, then hard data enters the amplitude at order k, not k^2, and there is no universal linear coefficient; if that matrix element has a $1/k$ infrared singularity, there is no zero at all. The factor is moreover convention-dependent, and its independence of the Hamiltonian is a tautology, as recorded in the scope note to Theorem 5.11.

What survives, and what it costs. Theorem 5.11(e) itself, as a lattice continuity equation, exactly and for any finite-range Hamiltonian; and Proposition 5.13, in which the ferromagnetic current happens to supply a second lattice difference and the oracle's dispersion is genuinely rederived. What is lost is the brief's original mechanism for the soft theorem. Recovering it is the burden of the soft corner (Section 12, Section 13), which must supply four things Corner A does not: regularity of $\langle \mathrm{out}|J_k|\mathrm{in}\rangle$ at $k = 0$ on the relevant scattering states; a reduction turning the operator identity Equation (76) into a statement about amplitudes; a Ward identity relating the zero-momentum current to the external legs' charges; and control of the remainder in a stated norm. Until those exist, agreement with the oracle's linear coefficient is a target, not a derivation.

campaign id:	G0-soft-r1
proved in:	withdrawn in `theory/corner-a-goldstone.md`, step (1.6)(2.7)
tested by:	—
reviewed in:	`theory/verdicts/corner-a-r3.md`

5.7 Scope of the corner as a whole

Scope 5.16 (Four amendments, and two sketches that may not be used silently). Four corrections must be absorbed by everything downstream of this section.

(i) *The naive asymptotic symmetry group is not the classifying object.* Unbroken: the orbit is $\mathcal{A}_{\text{eff}} = G/N_\alpha$, a group. Broken, under hypothesis (T) — otherwise per G-orbit — vacuum pairs form $(G/H_\alpha) \times (G/H_\alpha)$ and the diagonal invariant is the double coset in $H_\alpha \backslash G/H_\alpha$. The coset space $\mathcal{A}$ is a group only for abelian G and is never the orbit unless $N_\alpha = \{e\}$ (Refuted claim 5.14).

(ii) *The edge from this corner to the soft corner is not supplied here.* Theorem 5.11(e) is a lattice continuity equation; it gives no Adler zero, no universality, and does not rederive the two-body oracle facts (Refuted claim 5.15).

(iii) *"The zero-momentum Goldstone mode is pure gauge" holds only along unbroken directions.* Along a broken direction the intertwining relation Equation (64) still holds, with target tensor $A_{g\cdot\alpha}$; what fails is the same-vacuum return (Theorem 5.11(b)).

(iv) *The ansatz gauge identity is not a finite-window identity.* It carries two boundary terms of size $\Theta(1)$ (Equation (72)) and holds only in the named limits: norm convergence for summable profiles of bounded variation, and δ-normalisation for plane waves.

Two sketches in this corner are load-bearing and may not be used silently. The first is the split property: it is not proved that a bond-decorated state $\omega_\alpha^{M@b}$ is a normal state of the vacuum's von Neumann algebra, equivalently that the charge algebra $\mathfrak{a}_\alpha$ acts on the GNS Hilbert space rather than on finite-window spaces. The expected route is an infinite-volume Schmidt decomposition across the bond, with the Schmidt index identified with the matrix-product virtual index; at finite windows this is elementary, in the thermodynamic limit it is open. Its absence is exactly why the background desideratum of a charge algebra acting on the physical state space is not met by Theorem 5.4; every downstream use of "the asymptotic charge algebra acting on the vacuum Hilbert space" must cite this gap. It is *not* needed for Theorem 5.2, for clauses (a)–(c), (e)–(g) of Theorem 5.4, for Theorem 5.7, or for Theorem 5.11. The second is uniformity over a continuous vacuum manifold: for the isotropic ferromagnet, Theorem 5.7 produces an uncountable family of mutually disjoint sectors with no uniform separation — the separating observable degrades as $g \to e$, so the sectors are algebraically superselected but not energetically separated, the kink creation energy tending to zero. Missing are a version of Theorem 5.7 uniform in g, with the separation quantified by the distance between the two vacua, and a selection criterion that cuts the continuum down to a physically meaningful family — exactly the question of which function spaces are admissible at infinity. Consequently the ferromagnet's kinks may not be used as superselection sectors in the soft-theorem conjecture, and the memory construction of Section 7 cannot be run on that model through this route. Nothing in Theorem 5.7(a)–(f) is affected: those are proved for each fixed $g \notin H_\alpha$.

campaign id:	WI, A1, A2, G0 (scope common to all four)
proved in:	`theory/corner-a-pitfalls.md` (campaign pitfall list); sketches at `theory/corner-a.md` step (1.4)(2.9) and `theory/corner-a-kinks.md` step (1.10)(2.3)
tested by:	`theory/checks/corner_a_check.py` C0–C11 (all pass)
reviewed in:	`theory/verdicts/corner-a-r3.md` (loop converged; four minor wording notes only)

6 Memory observables and the exactly solvable kink model

This section fixes the observable whose change *is* the memory effect, the bookkeeping that says what has left the observation window, the scattering data of a magnon hitting a domain wall, and the one microscopic model in which all of this can be computed by hand: the easy-axis XXZ chain with its exact zero-energy kink family. It closes with the hypothesis package on scattering theory (called *H-AD*) on which almost every quantitative memory statement of Section 7 is conditional.

Throughout, $s > 0$ is the calibration constant of the wall observable: the two asymptotic vacua carry S^z-densities $+s$ on the left and $-s$ on the right, so that moving the wall by one lattice site moves $2s$ units of charge across a fixed cut. In every concrete statement below the model is the spin-1/2 chain and $s = 1/2$.

Remark 6.1 (The calibration constant is a density, not an on-site dimension). It is tempting to read s through the familiar gloss $d = 2s + 1$ relating a site spin to its on-site dimension. That gloss is the fully polarised special case only. What the definitions below actually use, and what the general theorems of Section 8 assume under the name H-MQG(2), is that s equals the common tail density ρ of the two vacua. No result quoted in this labbook assumes any relation between that density and the on-site dimension.

6.1 The memory observables

Definition 6.2 (Windowed wall position, and its spectral and dynamical dresses)**.** Let $W = [a, b] \subset \mathbb{Z}$ be a finite window in a chain whose S^z-vacuum densities are $s_\alpha = +s$ at $-\infty$ and $s_\beta = -s$ at $+\infty$.

(a) The windowed wall-position observable (the frozen definition). Put

$$\mathfrak{X}_W \;:=\; a - 1 + \frac{1}{2s} \sum_{x=a}^{b} \big(S^z_x + s\big) \;\in\; \mathfrak{A}_{\mathrm{loc}}, \tag{77}$$

normalised so that $\mathfrak{X}_W = m$ on a sharp wall sitting at the bond $(m \,|\, m+1)$ with no other content inside W. The *spatial memory* of an event running from t_i to t_f is

$$\delta x \;:=\; \varrho_{t_f}(\mathfrak{X}_W) - \varrho_{t_i}(\mathfrak{X}_W). \tag{78}$$

Because $\mathfrak{X}_W \in \mathfrak{A}_{\mathrm{loc}}$ unconditionally, δx exists on all of $\mathfrak{A}^*$ and at finite volume, with no asymptotic hypothesis and no order-of-limits clause. For δx to *mean* a wall displacement one needs two further geometric conditions: (i) the wave packet lies outside W at $t \in \{t_i, t_f\}$ up to ε, and (ii) the kink core is padded away from both edges of W. If d_W denotes the minimum core-to-edge distance, then for every $\tilde\lambda \in (\lambda_E, 1)$ the transfer-operator estimate of the MPS setting (Section 3) gives the tail bound $C_{\tilde\lambda} \tilde\lambda^{d_W}$. A bare $O(e^{-(b-a)/\xi_c}) = O(\lambda_E^{b-a})$ is *not* valid, because the transfer operator may have a Jordan block at modulus λ_E. Clause (a) is the definition the campaign freezes; (b) and (c) below are corollary characterisations, not competing definitions.

(b) The spectral dress. With $m_x(t) := \varrho_t(S^z_x)$ and the DC weight

$$D(x) \;:=\; \lim_{\omega \to 0} \int dt\, e^{i\omega t}\, \dot m_x(t) \;=\; m_x(+\infty) - m_x(-\infty), \tag{79}$$

set $\delta x^{\mathrm{spec}} := \frac{1}{2s} \sum_{x \in W} D(x)$. This is identically equal to (a) — the same observable in a Fourier dress — and it requires $\dot m_x \in L^1(dt)$ for each $x \in W$. The limit $\omega \to 0$ must be taken at fixed

x and fixed W, *after* the thermodynamic limit; interchanging $\omega \to 0$ with $|W| \to \infty$ destroys the identity.

(c) The dynamical dress, and its trap. With the first-moment wall coordinate

$$X_1(t) \;:=\; \sum_{x>0} \frac{\varrho_t(S^z_x)+s}{2s} \;-\; \sum_{x\le 0} \frac{s-\varrho_t(S^z_x)}{2s}, \tag{80}$$

convergent on the class of Definition 6.7, and with $V_\pm := \lim_{t\to\pm\infty} \dot X_1(t)$, set

$$\delta x^{\mathrm{dyn}} \;:=\; \lim_{t\to+\infty}\big[X_1(t)-V_+t\big] \;-\; \lim_{t\to-\infty}\big[X_1(t)-V_-t\big]. \tag{81}$$

This requires the thermodynamic limit *before* $t \to \pm\infty$, and is meaningless on a ring. **The trap:** X_1 is the regularised total magnetisation, which is exactly conserved, so $\delta x^{\mathrm{dyn}} \equiv 0$ unless the asymptotic leg charges of Definition 6.4 are subtracted first — that is, unless the scattering hypothesis Definition 6.8 is invoked. Clause (a) performs that subtraction geometrically, by windowing, and needs no such hypothesis.

campaign id:	D13 (historical label Bd1–Bd3)
proved in:	`definitions.md` D13; derivation and discussion in `theory/corner-b-draft.md` §2–§3
tested by:	`theory/checks/mquant_check.py` (operator and finite-time flux residues)
reviewed in:	`theory/verdicts/corpus-r2.md`

Remark 6.3 (Why (a) is the selected observable). All three dresses answer the same physical question, but only (a) is a bounded local observable that exists at finite volume and finite time with no asymptotic input. Dress (b) is convenient for reporting a measured magnetisation history; dress (c) is what one would write down naively as "the asymptotic trajectory of the magnetisation centroid", and it returns identically zero. Explicitly, on the class of Definition 6.7,

$$X_1(t) \;=\; \underbrace{x_{\mathrm{wall}}(t)}_{\text{(a) in the } L\to\infty \text{ limit}} \;+\; \frac{1}{2s}\sum_{\text{legs}} q_{\mathrm{leg}}(t), \qquad q_{\mathrm{leg}} = -1 \text{ (left)},\; +1 \text{ (right)}, \tag{82}$$

so (c) equals (a) only after the asymptotic leg charges have been removed.

Definition 6.4 (Asymptotic leg content: transmitted and reflected magnon number)**.** For a state ϱ in the ℓ^1 kink class $\mathcal{K}^{(1)}_{\alpha\beta}$ of Definition 6.7 and a window $W=[a,b]$, put

$$N_R \;:=\; \frac{1}{2s}\sum_{x<a}\big(s-\varrho(S^z_x)\big), \qquad N_T \;:=\; \frac{1}{2s}\sum_{x>b}\big(\varrho(S^z_x)+s\big), \tag{83}$$

both convergent on $\mathcal{K}^{(1)}$. For a one-magnon initial state $N_R+N_T+N_W=1$, where N_W is the weight remaining inside W. For a packet with momentum amplitude φ,

$$\langle N_T\rangle \;=\; \int_{-\pi}^{\pi} \frac{dk}{2\pi}\,|\varphi(k)|^2\,T(k), \tag{84}$$

a *packet average* of the transmission probability of Definition 6.5, and **not** $T(\langle k\rangle)$. The difference is $\frac{1}{2}T''(\langle k\rangle)\sigma_k^2$ and is largest exactly in the soft region, where T bends most.

campaign id:	D14 (historical label Bd4)
proved in:	`definitions.md` D14; `theory/corner-b-draft.md` §3(d)
tested by:	—
reviewed in:	`theory/verdicts/corpus-r2.md`

Definition 6.5 (Kink–magnon scattering data)**.** Work in the kink sector $\mathcal{K}_{\uparrow\downarrow}$ with one magnon present. At magnon momentum $k \in (0, \pi)$ the stationary scattering solutions define $r(k)$ and $t(k)$ through the asymptotics: an incoming $\downarrow$-magnon of momentum k in the $\uparrow$ region goes to

$$r(k) \cdot (\downarrow\text{-magnon, momentum } -k,\ \uparrow \text{ region}) \ +\ t(k) \cdot (\uparrow\text{-magnon, momentum } k,\ \downarrow \text{ region}).$$

Put $T(k) := |t(k)|^2$ and $R(k) := |r(k)|^2$, so that $T + R = 1$, and let $\delta_t(k) := \arg t(k)$ be the *kink–magnon transmission phase*, fixed continuous with $\delta_t \to 0$ at the point where $t \to 1$. The derivative $d\delta_t/dk$ is the spatial (Wigner–Eisenbud) shift of the *transmitted magnon.* It is a smooth, non-quantised function of k and Δ, and it is **not** the wall displacement (78). Keeping these two apart is the whole content of the refutation opening Section 7.

campaign id:	D15 (historical label Bd5)
proved in:	`definitions.md` D15; `theory/corner-b-draft.md` §5.3
tested by:	—
reviewed in:	`theory/verdicts/corpus-r2.md`

6.2 The easy-axis XXZ kink model

Definition 6.6 (The easy-axis XXZ kink model and its exact conventions)**.** Fix $J > 0$ and $\Delta > 1$, and let

$$h^{XXZ}_{x,x+1} := -J\Big[S^x_x S^x_{x+1} + S^y_x S^y_{x+1} + \Delta\big(S^z_x S^z_{x+1} - \tfrac{1}{4}\big)\Big], \tag{85}$$

$$\omega(k) = J(\Delta - \cos k), \qquad v(k) = J \sin k, \qquad \omega_{\text{gap}} = J(\Delta - 1). \tag{86}$$

Here Δ is the same anisotropy as the ratio $J_z/J_\perp$ of the symbol table; the two conventions agree. At $\Delta = 1$ this reduces to the isotropic ferromagnet used in Section 10. Every "soft" statement in this model is a low-frequency expansion *about the gap*, not about zero energy.

The symmetry is $G = U(1) \rtimes \mathbb{Z}_2$, with $U(1)$ generated by S^z (unbroken) and $\mathbb{Z}_2$ the π-rotation about S^x (broken). The vacua are $\Omega_{\text{vac}} = \{\uparrow, \downarrow\}$, both exact injective product states with bond dimension $\chi = 1$.

Kink normalisation. Put

$$h^{\text{kink}}_{x,x+1} := h^{XXZ}_{x,x+1} + \frac{J}{2}\sqrt{\Delta^2 - 1}\,\big(S^z_x - S^z_{x+1}\big), \qquad H_{\text{kink}} := \sum_x h^{\text{kink}}_{x,x+1}. \tag{87}$$

The added field is a telescoping boundary term: H_{kink} and H_{XXZ} generate the **same** derivation on the quasi-local algebra $\mathfrak{A}$ (Lemma 6.13), so every dynamical statement made with H_{kink} is a statement about the pure XXZ chain, and H_{kink} serves only to normalise the kink to zero energy.

Kink coordinates. Put

$$q \ :=\ \Delta - \sqrt{\Delta^2 - 1}\ \in\ (0, 1), \qquad \text{equivalently} \qquad \Delta = \frac{q + q^{-1}}{2}. \tag{88}$$

The exact zero-energy product family is

$$|K(z)\rangle \ =\ \bigotimes_{n \in \mathbb{Z}} \big(|\uparrow\rangle_n + z\, q^n |\downarrow\rangle_n\big), \qquad z \in \mathbb{C} \setminus \{0\}, \tag{89}$$

and writing $z = q^{-x_0} e^{i\phi}$ exhibits the conjugate pair (x_0, ϕ) with $x_0 \in \mathbb{R}$ the kink centre and ϕ the residual $U(1)$ phase. The crossover momentum of the model is $k_* := 1/(4(\Delta - 1))$.

campaign id:	D16 (historical label Bd6, with §1.1–§1.2 of the Corner B draft)
proved in:	`definitions.md` D16; `theory/corner-b-draft.md` §1.1–§1.2
tested by:	`theory/checks/mq_e_check.py`; `theory/checks/lr_d16_check.py`
reviewed in:	`theory/verdicts/corpus-r2.md`

Definition 6.7 (The ℓ^1 kink class)**.** $\mathcal{K}^{(1)}_{\alpha\beta} \subset \mathcal{K}_{\alpha\beta}$ is the set of states ϱ in the kink sector (Section 5) satisfying

$$\sum_{x<0} \big|\varrho(S^z_x) - s_\alpha\big| + \sum_{x>0} \big|\varrho(S^z_x) - s_\beta\big| < \infty, \tag{90}$$

together with the additional first-moment condition $\sum_x |x| \, \big|\varrho(S^z_x) - s_{\alpha/\beta}\big| < \infty$ whenever the first moment X_1 of Definition 6.2(c) is used. The bare kink sector requires only weak-$*$ convergence to the two vacua, which is too weak for the half-infinite charge to exist; $\mathcal{K}^{(1)}$ is the function-space refinement on which Definition 6.2(c) and Definition 6.4 converge.

Two properties are load-bearing. The class is preserved by the dynamics on finite time intervals, by Lieb–Robinson quasi-locality. It is **not** preserved by the limit $k \to 0$: a plane-wave magnon is not ℓ^1. **Every soft statement about memory must therefore fix the packet first and take $k \to 0$ afterwards; the two limits do not commute.**

campaign id:	D17 (historical label Bd7)
proved in:	`definitions.md` D17; `theory/corner-b-draft.md` §1.3
tested by:	—
reviewed in:	`theory/verdicts/corpus-r2.md`

6.3 The scattering hypothesis

Definition 6.8 (Hypothesis (H-AD): wave operators, channels, and local decay)**.** Fix a conserved regularised-charge sector of a finite-range $U(1)$-invariant dynamics, free left and right one-particle channel spaces $\mathcal{H}_L, \mathcal{H}_R$ with $\mathcal{H}_{\rm as} := \mathcal{H}_L \oplus \mathcal{H}_R$, a channel Hamiltonian $H_{\rm as}$, an identification map J, and channel projections P_L, P_T. The sector satisfies **(H-AD)** for a selected one-kink/one-magnon scattering vector if and only if all four clauses hold.

(AD1) Wave operators and completeness. The strong limits

$$W_\pm \;:=\; \operatorname*{s-lim}_{t\to\pm\infty} e^{itH}\, J\, e^{-itH_{\rm as}} \tag{91}$$

exist as isometries with common range $\mathcal{H}_{\rm sc}$, and the physical sector decomposes orthogonally as $\mathcal{H} = \mathcal{H}_b \oplus \mathcal{H}_{\rm sc}$, where $\mathcal{H}_b$ contains spatially localised bound states. There is no further propagating channel.

(AD2) The selected scattering vector. The vector is $\Psi = W_-(\varphi, 0)$ with $\varphi \in \mathcal{H}_L$; it has no $\mathcal{H}_b$ component, and $W_+^*\Psi = (\varphi_R, \varphi_T)$. On $\mathcal{H}_{\rm sc}$ define

$$N_T \;:=\; W_+ P_T W_+^*, \qquad \langle N_T\rangle \;:=\; \|\varphi_T\|^2. \tag{92}$$

For an on-shell diagonal scattering matrix, $\langle N_T\rangle = \int (dk/2\pi)\, |\varphi(k)|^2 T(k)$; it is a packet average, not in general T evaluated at the mean momentum.

(AD3) Channel charges and local decay. The incoming left leg has charge $q_{\rm in}$ relative to the left vacuum, and each outgoing leg has a stated charge $q_{\rm out}$ relative to its own vacuum. For every fixed window containing the kink, the free leg charge and the non-bound dressing leave the window as $t \to \pm\infty$; the remaining local state is a kink charge eigenstate, and the increasing-window limit of Definition 6.2(a) exists on it.

(AD4) Order of limits. The infinite-volume dynamics and the wave operators are formed before $t \to \pm\infty$; the fixed-window scattering limits are formed before the window is increased to $\mathbb{Z}$.

(H-AD) is narrower than general many-body asymptotic completeness and remains an explicit hypothesis unless a separate theorem supplies AD1–AD4. It is a coherent Hilbert-space statement: reflected and transmitted superpositions are retained, and are not replaced in norm by classical mixtures.

campaign id:	D18
proved in:	`definitions.md` D18; used as the standing hypothesis in `theory/memory-quantization.md` §2 and `theory/memory-quantization-general.md` H-MQG(5)
tested by:	`theory/checks/mq_e_check.py` (discharges AD1–AD4 for the projected component only)
reviewed in:	`theory/verdicts/corpus-r2.md`; `theory/verdicts/blitz-mq-e-r1.md`

Remark 6.9 (This hypothesis is load-bearing everywhere downstream)*.* It is worth saying plainly: Definition 6.8 is the single hypothesis that carries the quantitative half of the memory corner. The conditional quantization theorem for the spin-1/2 chain, its compact-group generalisation, the reduction of the dynamical dress Definition 6.2(c) to the windowed observable Definition 6.2(a), and the channel reading of the charge ledger all assume it. It is currently *proved* only for the projected three-wall component of the spin-1/2 model (Section 7), and remains open for the unprojected chain. Nothing in the flux identity (96) or in the finite-time label rigidity of Theorem 7.4 uses it.

Remark 6.10 (The superseded first formulation)*.* An earlier version of this hypothesis required the in and out states to be norm-close to a convex mixture of product configurations. That condition is withdrawn, and for a good reason. For $|\psi_{\text{out}}\rangle = r|R\rangle + t|T\rangle$ with $rt \neq 0$ and orthogonal separated channels, the pure density matrix retains the cross terms $r\bar{t}|R\rangle\langle T| + \bar{r}t|T\rangle\langle R|$, whose norm distance from the diagonal mixture does not vanish. Clause (AD3) uses local decay without discarding this coherence, and the variance statement in Section 7 depends on that choice.

6.4 Four exact facts about the kink model

The next three lemmas are elementary and completely explicit; together they say that the model of Definition 6.6 is frustration-free, that its kink family (89) is an exact zero-energy manifold, and that the boundary field used to arrange this is dynamically invisible. The fourth statement — that this manifold exhausts the zero-energy states and is flat — is what one would need for a genuinely inertia-free wall, and it is only a conjecture.

Lemma 6.11 (The kink bond is positive, with an explicit kernel)**.** [PROVED] *For the model of Definition 6.6, $h^{\text{kink}}_{x,x+1} \succeq 0$ with spectrum $\{0, 0, 0, J\Delta\}$, and*

$$\ker h^{\text{kink}}_{x,x+1} = \operatorname{span}\{|\uparrow\uparrow\rangle,\ |\downarrow\downarrow\rangle,\ |\uparrow\downarrow\rangle + q^{-1}|\downarrow\uparrow\rangle\}, \tag{93}$$

with q as in (88)*.*

Idea of proof. The bond term is block diagonal in the S^z basis. The states $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ have zero diagonal entry by the $-\frac{1}{4}$ subtraction in (85) and are not connected to anything. On the remaining domain-wall block spanned by $(|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle)$ the matrix is, in units of J,

$$\begin{pmatrix} \Delta/2 + C & -1/2 \\ -1/2 & \Delta/2 - C \end{pmatrix}, \qquad C = \tfrac{1}{2}\sqrt{\Delta^2 - 1}, \tag{94}$$

whose determinant is $\left(\Delta^2/4 - C^2\right) - 1/4 = 0$ and whose trace is $\Delta > 0$. A rank-one positive semidefinite 2×2 matrix has eigenvalues 0 and $J\Delta$, and the null vector is the one displayed in (93).
□

campaign id:	K1
proved in:	`theory/corner-b-draft.md` §1.2, steps (1.1)–(1.3), named computation K1-2x2
tested by:	recorded finite-matrix computation; scratch cross-check $\max\lVert h^{\mathrm{kink}}v\rVert \le 1.2\times10^{-11}$ for $\Delta \in \{1.5, 2, 4, 8\}$
reviewed in:	`theory/verdicts/corpus-r2.md`

Lemma 6.12 (The product family is annihilated bond by bond)**.** [PROVED] *Every vector* $|K(z)\rangle$ *of* (89)*,* $z \in \mathbb{C}\setminus\{0\}$*, lies in the kernel of every bond term* $h^{\mathrm{kink}}_{x,x+1}$*. In particular the whole family has exactly zero energy for* H_{kink}*, and the model is frustration-free.*

Idea of proof. The two-site factor of $|K(z)\rangle$ at the bond $(n, n+1)$ is $(|\uparrow\rangle + a|\downarrow\rangle)\otimes(|\uparrow\rangle + b|\downarrow\rangle)$ with $a = zq^n$ and $b = zq^{n+1} = qa$. Its domain-wall component is $b|\uparrow\downarrow\rangle + a|\downarrow\uparrow\rangle \propto |\uparrow\downarrow\rangle + q^{-1}|\downarrow\uparrow\rangle$, which is precisely the null vector (93) of Lemma 6.11; the two aligned components lie in the kernel as well. Since a product vector is annihilated by a bond term whenever its two-site factor is, every bond term annihilates $|K(z)\rangle$. □

campaign id:	K2
proved in:	`theory/corner-b-draft.md` §1.2, steps (1.1)–(1.2)
tested by:	recorded local residual computation
reviewed in:	`theory/verdicts/corpus-r2.md`

Lemma 6.13 (The boundary field telescopes and is dynamically invisible)**.** [PROVED] H_{kink} *and* H_{XXZ} *generate the same derivation on the quasi-local algebra* $\mathfrak{A}$*. The added field of* (87) *shifts the energy of each superselection sector by the constant* $\frac{J}{2}\sqrt{\Delta^2-1}\,(s_\alpha - s_\beta)$ *and does nothing else.*

Idea of proof. For a local observable O supported in a finite set Λ, the field contributes $\sum_x \big[S^z_x - S^z_{x+1}, O\big]$, which telescopes to $\big[S^z_{-M} - S^z_{M+1}, O\big] = 0$ for any M with $\Lambda \subset (-M, M)$. The same computation runs unchanged in infinite volume, because only finitely many commutators are nonzero for a fixed local O; the sum is not a conditionally convergent rearrangement. Note the contrast with finite volume: on a finite open chain of N sites the two Hamiltonians differ by the genuine boundary operator $\frac{J}{2}\sqrt{\Delta^2-1}\,(S^z_1 - S^z_N)$, so a finite-chain certificate must use the H_{XXZ} propagator (see Section 9). □

campaign id:	K3
proved in:	`theory/corner-b-draft.md` §1.2, steps (1.1)–(1.2), using the finite-support register of the Noether setting (Section 4)
tested by:	direct finite-support telescoping; `theory/checks/lr_d16_check.py` C4(c) finite-chain calibration
reviewed in:	`theory/verdicts/corpus-r2.md`

Conjecture 6.14 (Thermodynamic uniqueness and flatness of the kink band)**.** [CONJECTURE] *Let* $\mathcal{S}_M$ *be the sector of fixed regularised magnetisation* M *inside* $\mathcal{K}_{\uparrow\downarrow}$*. Then* $\mathcal{S}_M$ *contains exactly one zero-energy state of* H_{kink}*, and these states are degenerate across all* M*. Consequently the kink has no dispersion, no effective mass and no recoil velocity: its translation modulus is rigidly locked to the conserved* $U(1)$ *charge.*

Evidence, and what is missing. Frustration-freeness (Lemma 6.11) gives $E \ge 0$, and Lemma 6.12 exhibits states of energy exactly 0, so existence is settled. Uniqueness per sector has been verified only in finite volume: exact diagonalisation at $N = 12$, $\Delta = 3$, for every $m = 0, \dots, 12$, gives a ground energy $E_0 = 0$ to 10^{-13}, unique in each sector, and a first excited energy $E_1 = 2.03407417$ in every sector — the magnon gap $J(\Delta - 1) = 2$ plus a finite-size boundary shift. A proof of uniqueness in the thermodynamic limit is an open interface requirement, and no statement about absence of recoil, about a unique kink torsor, or about all Δ may cite this conjecture as proved.

campaign id:	K4
proved in:	`theory/corner-b-draft.md` §1.2 (statement only; no proof)
tested by:	finite $N = 12$ exact-diagonalisation evidence only
reviewed in:	`theory/verdicts/corpus-r2.md`

Remark 6.15 (Why the lattice trades momentum for charge). On the lattice, translation is not a symmetry *within* a magnetisation sector: translating a kink changes the regularised S^z. Momentum conservation for the wall is therefore traded for charge conservation, and this is what makes the displacement quantized rather than continuous once Definition 6.8 supplies channels. The continuum easy-plane counterpart behaves oppositely — there the wall is inertial, its modulus is locked to no charge, and the displacement is a continuous function of the magnon intensity. The dichotomy "charge-locked gives quantized memory, charge-free gives continuous memory" is the sharpest physical statement of this section, and it is taken up again in Section 7.

7 Memory: what is refuted, what is exact, and what quantizes

The memory corner began with an attractive guess: that the wall displacement should equal the zero-frequency limit of the soft factor, a lattice analogue of the Braginsky–Thorne relation. That guess is false, and saying exactly why it is false is the most useful thing this section does. What survives is sharper and, in a sense, better: an exact flux identity that holds at finite time and finite volume with no hypotheses at all, a charge ledger that constrains the displacement channel by channel, and — conditional on the scattering hypothesis Definition 6.8 — a genuinely quantized displacement whose quantum is a charge, not a phase. Soft data re-enter at exactly one place: the value of the transmission probability $T(k)$.

7.1 The refuted conjecture

Refuted claim 7.1 (The soft-factor reading of the memory)**.** [REFUTED] Withdrawn statement. *The wall displacement δx equals the zero-frequency limit of the soft factor* S *summed over the event — the lattice Braginsky–Thorne relation as written in the campaign brief.*

Why it fails. *Two independent mechanisms kill it in the model of Definition 6.6.*

1. *Once the channels of Definition 6.8 exist, conditional charge bookkeeping fixes the* same *per-channel displacement for every k, every Δ and every packet shape (Theorem 7.6), whereas any soft-factor expression varies with all three.*

2. *δx is insensitive to the transmission phase $\delta_t(k)$ entirely. A purely transmitting wall with $\delta_t \equiv 0$ still displaces by exactly $-1/s$. The soft factor is a phase; the memory quantum is a charge, and the two are not the same kind of object.*

What survives. *Two statements, and no more. The flux identity Theorem 7.2, which says that the memory is the DC weight of the* physical boundary current*; and the conditional quantization theorem Theorem 7.6, into which soft data enter only through the transmission probability $T(k)$. No reading in terms of virtual or bond data is claimed.*

campaign id:	M
proved in:	disproved in `theory/memory-quantization.md` §1 and `theory/corner-b-draft.md` §§2, 6, 10
tested by:	`theory/checks/mquant_check.py` tests the flux identity and the empirical scan only
reviewed in:	`theory/verdicts/corpus-r2.md`

7.2 The exact flux identity

Theorem 7.2 (The memory is the DC weight of the physical boundary current)**.** [PROVED] *Let the chain be finite and open, let $\varrho(t)$ be any state, let $W = [a, b]$ be any interior window, and assume only the local continuity equation*

$$\frac{d}{dt} S^z_x = j_{x-1|x} - j_{x|x+1}. \tag{95}$$

Then for every finite $t_i < t_f$,

$$\varrho_{t_f}(\mathfrak{X}_W) - \varrho_{t_i}(\mathfrak{X}_W) = \frac{1}{2s}\int_{t_i}^{t_f} dt\,\Big[\varrho_t\big(j_{a-1|a}\big) - \varrho_t\big(j_{b|b+1}\big)\Big] = \frac{1}{2s}\Big[\tilde{\jmath}_{a-1|a}(0) - \tilde{\jmath}_{b|b+1}(0)\Big], \tag{96}$$

where $\tilde{\jmath}(0)$ denotes the finite-time Fourier transform of the corresponding bond current evaluated at zero frequency. No kink, spectral, large-time or asymptotic-completeness hypothesis is used.

Idea of proof. The windowed wall observable (77) is a bounded local observable at every finite volume, so its Heisenberg derivative may be computed term by term. Inserting (95) into the finite sum, the interior currents cancel in pairs and only the two boundary bonds survive: $\dot{\mathfrak{X}}_W = \frac{1}{2s}\big(j_{a-1|a} - j_{b|b+1}\big)$. Integrating from t_i to t_f and reading the result as a Fourier transform at zero frequency gives (96). The whole content is that $\mathfrak{X}_W$ can change *only* through its two edges. □

Remark 7.3 (The current here is the physical one). Nothing in (96) identifies the boundary current with the virtual bond potential $\mathcal{J}_b$ of the Noether analysis (Section 4). That identity writes the *vacuum charge density* as a difference of virtual insertions, in a different register. The earlier claim that (96) is a difference of virtual or bond data is withdrawn; extending the virtual-potential reading to a kink-sector dynamical theorem remains future work.

campaign id:	M-flux
proved in:	`theory/memory-quantization.md` §1, Lemma Mq-flux, steps (1.1)–(1.4); `theory/corner-b-draft.md` §2 Eq. (2.2)
tested by:	`theory/checks/mquant_check.py`, named computation Mq-check F1: zero operator residue in (95) and 3.37×10^{-16} finite-time residue in (96) on frozen-boundary chains at $N = 7, 8$
reviewed in:	`theory/verdicts/corpus-r2.md`

7.3 Finite-time label rigidity and the charge ledger

Theorem 7.4 (The vacuum-pair label is rigid at finite time, and the charge ledger holds)**.** [PROVED] *Assume a translation-invariant finite-range dynamics whose two asymptotic vacua are stationary, and a state initially in the kink sector $\mathcal{K}_{\alpha\beta}$. Then:*

1. (Label rigidity.) *$\varrho_t \in \mathcal{K}_{\alpha\beta}$ for every finite t: finite-time dynamics fixes the vacuum-pair label (α, β), and the argument fixes the labels rather than merely preserving disjointness.*

2. (Charge ledger.) *For an event separated in the sense of Definition 6.8, with an explicit cut $c \in W$ and leg-subtracted channel charges,*

$$2s \cdot \delta x + \big(q_{\text{out}} - q_{\text{in}}\big) = 0. \tag{97}$$

Idea of proof. For rigidity, let α_t be the finite-range Heisenberg automorphism. Lieb–Robinson quasi-locality supplies, for each local pair D, O, each fixed t and each $\varepsilon > 0$, *local* approximants $D_\varepsilon, O_\varepsilon$ — independent of the translation index n — with $\|\alpha_t(D) - D_\varepsilon\| < \varepsilon$ and $\|\alpha_t(O) - O_\varepsilon\| < \varepsilon$. Translation covariance gives $\alpha_t(\tau_n(O)) = \tau_n(\alpha_t(O))$, and stationarity of the vacua gives $\omega_{\alpha/\beta}(\alpha_t(O)) = \omega_{\alpha/\beta}(O)$. The two-factor estimate

$$\big|\varrho\big(\alpha_t(D)\alpha_t(\tau_n(O))\big) - \varrho\big(D_\varepsilon \tau_n(O_\varepsilon)\big)\big| \le \varepsilon\|O\| + \big(\|D\| + \varepsilon\big)\varepsilon \tag{98}$$

then lets one take $n \to \mp\infty$ first, in the factorised local limit, and $\varepsilon \downarrow 0$ afterwards, reproducing the same left and right vacuum functionals. For the ledger, a kink at coordinate m contributes $2s(m-c)$ to the regularised total charge, so the in and out totals are $2s(m_i - c) + q_{\text{in}}$ and $2s(m_f - c) + q_{\text{out}}$; conservation of the regularised charge and subtraction give (122). Specialising, reflection has $q_{\text{out}} = q_{\text{in}} = -1$ and hence $\delta x = 0$, while transmission has $(q_{\text{in}}, q_{\text{out}}) = (-1, +1)$ and hence $\delta x = -1/s$. □

Scope 7.5 (What this theorem does not say, and one thing it refutes). The charges in (122) are *leg-subtracted* channel charges, defined relative to the vacuum on which each separated leg sits. They are not raw half-line charges. Indeed, the former raw half-line formula $\delta x = (2s)^{-1}\Delta\mathfrak{q}_c^R = -(2s)^{-1}\Delta\mathfrak{q}_c^L$ is [REFUTED]: in the transmitted spin-1/2 channel it predicts $+1$ (or -1), while the ledger (122) gives -2. Separately, the proposed reading of the kink coordinate as a $\mathbb{Z}$-torsor, with a flat kink band, depends on Conjecture 6.14 and is therefore a conjecture, not a consequence of this theorem. The vacuum-pair label is rigid and is read off, not shifted; δx records the leg-charge transfer within that fixed label.

campaign id:	B3
proved in:	`theory/corner-b-draft.md` §7, steps (1.1)–(1.3), with the N9 density leaf supplied
tested by:	—
reviewed in:	`theory/verdicts/corpus-r2.md` (adjudicated with the N9 sweep)

7.4 Conditional quantization for the spin-1/2 kink

Theorem 7.6 (Windowed memory is channel-quantized, conditional on the scattering hypothesis)**.** [PROVED, CONDITIONAL] *Let $\Delta > 1$ and take the spin-1/2 Hamiltonian of Definition 6.6, with unbroken $U(1)$ charge S^z and asymptotic densities $+1/2$ on the left and $-1/2$ on the right, selecting the kink sector $\mathcal{K}_{\uparrow\downarrow}$. Prepare one right-moving magnon on the $\uparrow$ leg and use the windowed wall coordinate (77).*

Assume (H-AD), i.e. all four clauses AD1–AD4 of Definition 6.8, for this scattering vector. *Then, in the limit order fixed by (AD4), the asymptotic two-channel displacement observable obeys*

$$\Delta X = -\frac{1}{s}\,N_T, \qquad \delta x = -\frac{\langle N_T\rangle}{s}, \tag{99}$$

and consequently

$$\operatorname{spec}(\Delta X) \subseteq \left\{-\frac{1}{s},\,0\right\}, \qquad \delta x \in \left[-\frac{1}{s},0\right], \qquad \operatorname{Var}_\Psi(\Delta X) = \frac{1}{s^2}\langle N_T\rangle\big(1-\langle N_T\rangle\big). \tag{100}$$

The channel statement is independent of k, of Δ, of the packet shape and of the scattering phase. For the concrete spin-1/2 application $s = 1/2$, so $\delta x = -2\langle N_T\rangle$ and the variance is $4\langle N_T\rangle(1-\langle N_T\rangle)$.

Idea of proof. Choose a cut c and regularise the total charge by subtracting $+s$ to the left of c and $-s$ to the right; a kink whose windowed coordinate is m then contributes $2s(m-c)$. The regularised charge is exactly conserved, because the bond currents telescope and the frozen endpoints carry no hopping current, and (AD1) with (AD4) carries that fixed label into the in and out channel limits. On the reflected branch the in and out totals are $2s(m_i - c) - 1$ and $2s(m_R - c) - 1$, so $m_R = m_i$; on the transmitted branch they are $2s(m_i - c) - 1$ and $2s(m_T - c) + 1$, so $2s(m_T - m_i) = -2$ and $m_T - m_i = -1/s$. Hence on the two outgoing channel spaces the displacement is $0 \cdot P_L - (1/s)P_T$, and conjugating by W_+ gives $\Delta X = -(1/s)N_T$ on $\mathcal{H}_{\mathrm{sc}}$. Since N_T is an orthogonal projection, its spectrum lies in $\{0, 1\}$ and $N_T^2 = N_T$, which gives (100) immediately. ☐

Scope 7.7 (Exactly what is conditional, and what the variance is a variance of). The hypothesis (H-AD) is used at precisely three places: to say that dressing inside a fixed-charge kink sector does not change the kink eigenvalue, to replace interacting late-time states by separated channels, and to package the channel weights into the projection N_T. The flux identity Theorem 7.2 and the charge arithmetic use none of it. (H-AD) for the unprojected chain remains open; it is proved only for the projected three-wall component (Corollary 7.14).

The abstract arithmetic behind (99) is only the ledger (122); it does not by itself establish the required channels for a higher-spin XXZ chain, nor for a general triple (compact group, injective MPS, finite-range Hamiltonian). That generality gap is closed only in the conditional form of Theorem 7.9.

It is the *outcomes* that are quantized, not every expectation value. The expectation $\delta x = -\langle N_T\rangle/s$ is the convex combination selected by $\langle N_T\rangle \in [0, 1]$. Likewise $\operatorname{Var}(\Delta X)$ is the variance of the two-channel displacement observable; identifying it with the one-time variance $\mathrm{Var}(\mathfrak{X}_W)$ would additionally require a sharp initial wall eigenstate and an explicit two-time measurement convention, and neither is implicit here.

Remark 7.8 (Finite-window error budget). At finite window and finite separation it is useful to keep the theorem and the measurement apart:

$$\left|\delta x_W + \frac{\langle N_T\rangle}{s}\right| \le \epsilon_{\rm AD} + C_{\tilde\lambda}\tilde\lambda^{d_W} + \frac{N_W(t_i)+N_W(t_f)}{s} + \epsilon_{\rm num}, \tag{101}$$

with $\tilde\lambda \in (\lambda_E, 1)$, d_W the minimum kink-core-to-edge distance, $\epsilon_{\rm AD}$ an empirical local charge-decomposition defect incurred when the exact channels are approximated numerically, N_W the residual leg content inside W, and $\epsilon_{\rm num}$ the estimator and evolution errors. All terms vanish in the exact conditional limits, and $\epsilon_{\rm AD}$ is not a return to the superseded norm-mixture formulation of Remark 6.10. A projection error of order Δ^{-2} is added only when the full $\langle N_T\rangle$ is replaced by the Fano value of Theorem 7.15; it is not an error in (99).

campaign id:	M-quant
proved in:	`theory/memory-quantization.md` §4 steps (1.1)–(1.7) and §5 Corollary Mq-quant; `theory/corner-b-draft.md` §6
tested by:	`theory/checks/mquant_check.py` tests the flux identity and the empirical scan only; (H-AD) is an assumption and is tested by no gate
reviewed in:	`theory/verdicts/corpus-r2.md` (proved conditional on Definition 6.8); full-chain (H-AD) tracked as open work

7.5 The compact-group generalisation

Theorem 7.9 (The memory law for compact G and injective MPS vacua)**.** [PROVED, CONDITIONAL] *The following is proved* only *as a conditional implication: the hypothesis package H-MQG(1)–(4) together with H-AD-G implies the conclusions* (103)–(104)*.*

Hypotheses H-MQG.

(1) G is compact and $\{A_\gamma\}_{\gamma\in\Omega_{\rm vac}}$ is a G-covariant family of injective canonical MPS tensors of one common, arbitrary finite bond dimension χ (Section 3). Fix α and $\beta = g\cdot\alpha \neq \alpha$, and use the kink sector $\mathcal{K}_{\alpha\beta}$ supplied by Section 5.

(2) A common unbroken compact abelian subgroup lies in $H_\alpha \cap H_\beta$. Choose one primitive circle direction ξ in its Lie algebra and fix, once for both tails, the Hermitian on-site charge $S^z := -i\,q(\xi)$. In this fixed convention $\omega_\alpha(S^z) = +s$ and $\omega_\beta(S^z) = -s$, with $s > 0$. For a higher-dimensional torus the theorem is applied componentwise after choosing ξ; a finite abelian group with no circle direction is not enough, since it supplies no local current.

(3) H is translation invariant, finite range and G-invariant in the exact sense of Section 4, and the two vacua are stationary. Its infinite-volume dynamics therefore obeys the cut-current continuity equation for the selected charge.

(4) The selected incoming state is a fixed *wave packet in the ℓ^1 class $\mathcal{K}^{(1)}_{\alpha\beta}$ of Definition 6.7, and its wall coordinate is exactly the windowed observable* (77) *with the charge and s of item (2). No plane wave is admitted in place of this packet.*

Hypothesis H-AD-G. *This is shorthand, not a replacement definition, for all of Definition 6.8(AD1)–(AD4) holding for that vector, with one incoming left leg, one reflected outgoing leg and one transmitted outgoing leg, whose charges — measured relative to the vacuum supporting each leg, in the* *single convention of item (2) — are*

$$q_{\text{in}} = -1, \qquad q_L = -1 \text{ (reflected)}, \qquad q_T = +1. \tag{102}$$

(AD2)'s transmitted-channel projection is N_T; (AD1) excludes any further propagating channel, and the selected vector has no bound-state component.

Conclusion. *Form the infinite-volume dynamics and the wave operators first; fix the packet; at each fixed window take the incoming and outgoing large-time limits; only then increase $W = [a, b]$ to $\mathbb{Z}$, with both kink-core distances tending to infinity. In this order,*

$$\Delta X = -\frac{1}{s} N_T, \qquad \delta x = -\frac{\langle N_T \rangle}{s}, \tag{103}$$

$$\operatorname{spec}(\Delta X) \subseteq \Big\{0, -\frac{1}{s}\Big\}, \qquad \operatorname{Var}(\Delta X) = \frac{1}{s^2} \langle N_T \rangle \big(1 - \langle N_T \rangle\big). \tag{104}$$

The outcome quantum is fixed by charge conservation alone: reflection transfers no charge between the two vacuum conventions, while transmission changes the separated leg charge by $q_T - q_{\text{in}} = 2$. More generally, if that difference is some definite ν, the same proof gives the quantum $-\nu/(2s)$; the displayed $-1/s$ therefore includes the primitive channel normalisation of H-AD-G, and is not invariant under relabelling a charge-ν particle as charge one.

Idea of proof. Four steps. First, the selected state has a fixed asymptotic label (α, β) which the dynamics does not move, by the rigidity clause of Theorem 7.4; so δx is motion *within* a fixed sector, not a change of vacuum-pair label. Second, an exact finite-window identity: for a cut $c \in W$,

$$\sum_{x=a}^{c} \big(S_x^z - s\big) + \sum_{x=c+1}^{b} \big(S_x^z + s\big) = 2s\big(\mathfrak{X}_W - c\big), \tag{105}$$

which is pure finite algebra, valid for arbitrary states and arbitrary χ, since both sides expand to $\sum_{x=a}^{b} S_x^z + s(a + b - 1 - 2c)$. Third, the two tail series converge absolutely on the ℓ^1 class, the Heisenberg derivative of the window charge is the difference of the two physical cut currents, and the ordered limits of (AD3)–(AD4) carry the conserved total — kink plus separated leg — into the channels. Fourth, equating in and out totals gives the ledger (122), which returns $\delta x_R = 0$ and $\delta x_T = -1/s$; packaging these by (AD1)–(AD2) gives $\Delta X = -(1/s) W_+ P_T W_+^*$ and hence (103)–(104), without replacing the coherent outgoing state by a mixture. ☐

Scope 7.10 (What the generalisation does and does not buy). The theorem is proved *only* as the implication stated. H-AD-G itself is not proved; neither is the sector reduction Theorem 7.12 in any general setting, nor (H-AD) for the unprojected chain. No soft zero is proved here: if a separate result shows $\langle N_T \rangle \to 0$, then (103) transfers that zero to the memory, but nothing in this theorem produces it. No uniformity is claimed — not over packets, models, soft limits, a continuous vacuum manifold, G, α, β, χ, tensors, Hamiltonians, times, core decorations or deformations. At the general ℓ^1 level the tail error is the state-specific omitted sum; it tends to zero as $W \uparrow \mathbb{Z}$ with no rate. Only if the asymptotic kink core is one fixed two-sided MPS decoration with modifications in a fixed bounded support does one recover an exponential tail $C_{\tilde{\lambda}} \tilde{\lambda}^{d_W}$, on exactly that fixed data.

The transitivity hypothesis (T) is *not* required for (103)–(104). If it is imposed, it supplies the double-coset label $\mathfrak{d}(\alpha, \beta) \in H_\alpha \backslash G / H_\alpha$ of the fixed vacuum pair, and the scattering does not change that label. Finally, the measured displacement is charge bookkeeping inside a fixed sector; it is not a moment map, a flat kink torsor, an SPT index, or a deformation-independent absolute charge.

Two convention warnings. Rescaling the generator rescales both s and all leg charges, so only $-(q_{\text{out}} - q_{\text{in}})/(2s)$ is convention covariant. And a finite abelian symmetry alone is insufficient: without a circle generator there is only a modular charge label, not a local continuous current and not a real-valued displacement law.

Remark 7.11 (Consistency with the spin-1/2 case). Taking $G = U(1) \rtimes \mathbb{Z}_2$, $\alpha = \uparrow$, $\beta = \downarrow$ and $s = 1/2$ in Definition 6.6, the two tensors are injective product tensors with $\chi = 1$, so H-MQG(1)–(4) holds as the special case in which every transfer-tail error vanishes exactly. With $q_{\text{in}} = q_L = -1$ and $q_T = +1$, (103)–(104) reduce to $\Delta X = -2N_T$, $\delta x = -2\langle N_T\rangle$, $\operatorname{spec}(\Delta X) \subseteq \{0, -2\}$ and $\operatorname{Var}(\Delta X) = 4\langle N_T\rangle(1 - \langle N_T\rangle)$ — exactly the numbers of Theorem 7.6. A rank-two example needs a nonabelian ambient group: a bare abelian group, or a selected central factor, would force $\omega_\beta = \omega_\alpha$ on the selected charge and could not produce opposite vacuum densities.

campaign id:	M-quant-G
proved in:	`theory/memory-quantization-general.md` §2, steps (1.1)–(1.5), with the hypothesis package in §0 and the scope discipline in §4
tested by:	`theory/checks/mquant_general_check.py` (symbolic charge arithmetic and representation theory only; the red mode mutates q_T). It constructs no MPS tensors, no wave operators and no proof of H-AD-G
reviewed in:	`theory/verdicts/mquant-g-r2.md` (promoted as the conditional implication only)

7.6 The exactly solvable sector: the Fano graph

Theorem 7.12 (The projected three-wall component is a Fano graph)**.** [PROVED] *Take the spin-1/2 model of Definition 6.6 with $J > 0$, $\Delta > 1$, kink orientation $\uparrow$ on the left and $\downarrow$ on the right, and a fixed integer regularised charge label μ. Let P_3 project the fixed-S^z kink-plus-one-magnon space onto configurations with one or three domain walls, retain the connected component containing the three consecutive walls $(\mu - 1, \mu, \mu + 1)$, and put $H_3 = P_3 H_{XXZ} P_3$. Then, for every admissible frozen-boundary interval and at infinite volume, there is an explicit unitary U from that component onto $\ell^2(\mathbb{Z}) \oplus \mathbb{C}|d\rangle$ with*

$$UH_3U^* = H_0 + V, \qquad H_0 = \Big[E_c - \frac{J}{2}\big(\mathsf{T} + \mathsf{T}^*\big)\Big] \oplus E_d, \qquad V = -\frac{J}{2}\big(|0\rangle\langle d| + |d\rangle\langle 0|\big), \tag{106}$$

where T is the unit shift on $\ell^2(\mathbb{Z})$, $E_c = \frac{3}{2}J\Delta$ is the uniform channel on-site energy, and $E_d = E_c - J\Delta = \frac{1}{2}J\Delta$ is the energy of the side vertex. The channel structure is explicit:

- *the negative tail is the incoming and reflected leg, of charge -1, carrying a sharp kink at bond $m := \mu + 1$;*
- *the positive tail is the transmitted leg, of charge $+1$, carrying a sharp kink at bond $m-2 = \mu-1$;*
- *the side vertex $|d\rangle$ is the pure sharp kink at bond μ.*

Moreover every fixed-window observable maps to two exact channel-kink constants plus a finite-support graph remainder: for $O \in \mathfrak{A}_W$,

$$UP_COP_CU^* = o_L(O)\,\Pi_W^- + o_R(O)\,\Pi_W^+ + F_W, \tag{107}$$

with $o_L(O) = \langle K_{\mu+1}|O|K_{\mu+1}\rangle$, $o_R(O) = \langle K_{\mu-1}|O|K_{\mu-1}\rangle$, $\Pi_W^\mp$ the projections onto sufficiently remote negative and positive tails, and F_W supported on a finite graph set. For the wall observable itself, $o_L(\mathfrak{X}_W) = \mu + 1$, $o_R(\mathfrak{X}_W) = \mu - 1$ and $\langle K_\mu|\mathfrak{X}_W|K_\mu\rangle = \mu$: zero reflected displacement, and transmitted displacement exactly -2.

Idea of proof. In the S^z product basis, a configuration is labelled by its ordered wall bonds. In the fixed charge sector the unique one-wall word has its wall at bond μ, and every three-wall word is uniquely $(a, a+\ell, \mu+\ell)$ with $a \le \mu - 1$ and $\ell \ge 1$ — a rectangle of labels. The exchange term toggles the two wall indicators neighbouring the bond it acts on, so a three-wall vertex has a surviving edge only when two of its walls are adjacent; in rectangle coordinates that is exactly the row $\ell = 1$ and the column $a = \mu - 1$. Those two rays form a single two-sided path meeting at the junction $(\mu-1, \mu, \mu+1)$, the one-wall word K_μ is attached only to that junction, and every other rectangle vertex is isolated. Reading the row as negative labels and the column as positive labels gives the unitary, and the Ising energies ($3J\Delta/2$ per three-wall word, $J\Delta/2$ for the single wall) with uniform hopping $-J/2$ give (106). The finite maps are literally restrictions of the infinite one along every cofinal sequence, since each word is a finite deviation of a sharp kink. Identity (107) follows because a W-local operator has zero matrix element between product words differing outside W, so all but finitely many tail rows are diagonal with the two kink constants. □

Scope 7.13 (Projected, not full chain). This is a theorem about the compressed operator $H_3 = P_3 H_{XXZ} P_3$, not about the unprojected XXZ Hamiltonian. It does not remove the separate leakage problem $P_3 H_{XXZ}(1 - P_3) \neq 0$: five-wall and higher configurations are real, not absent. Their weight is *measured*, not bounded: full-sector exact diagonalisation at $N = 22$, $k = \pi/2$, packet width $\sigma = 2.6$, over $0 \le Jt \le 13$, gives $P(\ge 5 \text{ walls}) \approx 8 \times 10^{-3}$ at $\Delta = 8$, 3×10^{-2} at $\Delta = 4$ and 1×10^{-1} at $\Delta = 2$, consistent with $O(\Delta^{-2})$. That is evidence for a controlled lift, not a uniform-in-time scattering estimate.

campaign id:	Mq-E
proved in:	`theory/mq-e.md` §§1–7 (enumeration, component, finite unitary, infinite-volume compatibility, labels and charges, local-observable map)
tested by:	`theory/checks/mq_e_check.py` (finite-volume falsifier, safe under `python3 -O`; the red mode shifts the right wall pair and exits 1). Retained enumeration check: $N = 14$, wall at bond 6, component of size 12 with degree histogram {1:3, 2:8, 3:1}
reviewed in:	`theory/verdicts/blitz-mq-e-r1.md` (0 fatal, 0 major)

Corollary 7.14 (The projected component satisfies the scattering hypothesis)**.** [PROVED] (for the projected component) *The projected incoming at-most-three-wall component of Definition 6.6 satisfies all four clauses AD1–AD4 of Definition 6.8, for every incoming packet with smooth momentum amplitude compactly supported in $k \in (0, \pi)$. This does not prove Definition 6.8 for the unprojected full chain.*

Idea of proof. By Theorem 7.12 the compressed operator is (106), whose perturbation V has rank at most two and is therefore trace class. Kato–Rosenblum then gives the wave operators $W_\pm$ with $\operatorname{ran} W_\pm = \mathcal{H}_{\mathrm{ac}}(H_3)$, which is (AD1). Reflection about the junction splits off an odd free channel; the even channel together with $|d\rangle$ is a half-line Jacobi matrix differing from the constant-coefficient one in finitely many entries. With

$$m_0(z) = \langle 0 | \Big(z - E_c + \tfrac{J}{2}(\mathsf{T} + \mathsf{T}^*) \Big)^{-1} | 0 \rangle = \big[(z - E_c)^2 - J^2 \big]^{-1/2}, \tag{108}$$

every cyclic resolvent entry is rational in z and $m_0(z)$ with denominator $z - E_d - (J^2/4) m_0(z)$. Inside the band $[E_c - J, E_c + J]$ the boundary value $m_0(E + i0)$ has nonzero imaginary part, so that denominator has no real zero and the spectral measure is absolutely continuous there; outside the band its zeros are isolated and finite in number, and at the two thresholds the Jacobi solutions are constant or linear (up to a sign alternation) and are not square summable. Hence there is no singular continuous spectrum and $\mathcal{H} = \mathcal{H}_b \oplus \operatorname{ran} W_\pm$ with $\dim \mathcal{H}_b < \infty$. Finally, stationary phase for $E(k) = E_c - J \cos k$ shows a smooth packet supported away from $k = 0, \pi$ leaves every finite set as $t \to \pm\infty$; the wave operators transfer that local decay to H_3, and Theorem 7.12 labels the left and right tails as the reflected and transmitted channels with charges -1 and $+1$. Those are (AD2)–(AD3), and the strong limits impose (AD4). ☐

campaign id:	Mq-AD3
proved in:	`theory/memory-quantization.md` §3, steps (1.1)–(1.5); `theory/mq-e.md` §6
tested by:	`theory/checks/mq_e_check.py` tests the sector reduction, not the spectral theorems
reviewed in:	`theory/verdicts/blitz-mq-e-r1.md`

7.7 The transmission coefficient and its quadratic soft zero

Theorem 7.15 (Fano transmission and the quadratic zero)**.** [PROVED] (projected, unconditional) *On the projected incoming at-most-three-wall component of Definition 6.6, the kink–magnon scattering data of Definition 6.5 are*

$$t(k) = \Big[1 + \frac{iJ^2}{4\,\omega(k)\,v(k)}\Big]^{-1}, \qquad r(k) = -\frac{iJ^2}{4\,\omega(k)\,v(k)}\,t(k), \qquad T(k) = \Big[1 + \Big(\frac{J^2}{4\,\omega(k)\,v(k)}\Big)^2\Big]^{-1}, \tag{109}$$

with transmission phase $\delta_t(k) = -\arctan\big[J^2/(4\,\omega(k)\,v(k))\big]$ *and* ω, v *as in* (86)*. In the soft limit* $k \to 0$ *at fixed* Δ*, where* $v \to 0$ *and* $\omega \to J(\Delta - 1)$*,*

$$T(k) = 16(\Delta - 1)^2 k^2 + O(k^4), \qquad R(k) = 1 - 16(\Delta - 1)^2 k^2 + O(k^4). \tag{110}$$

A soft magnon is thus totally reflected, with a quadratic zero in the transmission. Combining with (99)*,* $\delta x(k) = -16(\Delta - 1)^2 k^2/s + O(k^4)$*.*

Idea of proof. By Theorem 7.12 the effective one-body problem is a uniform tight-binding chain with hopping $-J/2$ and a single side-coupled level — the pure kink — attached at the junction with amplitude $-J/2$ and detuning $\omega(k)$. Eliminating the side level leaves an energy-dependent single-site potential $U(k) = (J/2)^2/\omega(k)$ at the junction. Matching $\psi_n = e^{ikn} + re^{-ikn}$ for $n \le 0$ to $\psi_n = te^{ikn}$ for $n \ge 0$ across that impurity gives $-2i\tau r \sin k + Ut = 0$ and $t = 1 + r$; with $2\tau \sin k = J \sin k = v(k)$ this is (109). ☐

Remark 7.16 (Two useful limits)*.* For large Δ (the Ising regime), $J^2/(4\omega v) \approx 1/(4\Delta \sin k)$, so $R(k) \approx 1/(16\Delta^2 \sin^2 k)$: the wall is transparent up to $O(\Delta^{-2})$, the lattice counterpart of the reflectionless continuum domain wall. In the frequency variable, using $\omega - \omega_{\mathrm{gap}} = J(1 - \cos k) \approx Jk^2/2$, the soft law (110) reads $T \approx 32(\Delta - 1)^2(\omega - \omega_{\mathrm{gap}})/J$ — linear in the excess energy above the gap. The crossover momentum is $k_* = 1/(4(\Delta - 1))$.

Scope 7.17 (Projected and non-universal)*.* Equations (109)–(110) are exact for the Fano graph, and therefore unconditional on the projected component, because Theorem 7.12 identifies that graph with the all-volume incoming component. They are *not* exact for the unprojected H_{XXZ}, because of the measured $O(\Delta^{-2})$ leakage into five-wall and higher configurations. The unprojected formula, and the universality of both the quadratic zero and its coefficient $16(\Delta - 1)^2$, remain [CONJECTURE]. What a genuine soft theorem would have to supply is that this zero and its coefficient are functions of the kink's asymptotic data — vacuum pair, $U(1)$ charge, gap — and not of the microscopic tensor. That is the real obligation on the edge from the soft corner to the memory corner, and it is not discharged here.

A separate warning, worth repeating: $d\delta_t/dk$ is the Wigner–Eisenbud spatial shift of the *transmitted magnon*. It is smooth and non-quantised and it is not the wall displacement. The leakage affects $T(k)$; it does not affect the memory law (99), which is a conservation law.

campaign id:	M-tk
proved in:	`theory/corner-b-draft.md` §5.2 Eqs. (5.1)–(5.3); `theory/memory-quantization.md` §5 Eqs. (Mq.9)–(Mq.10); `theory/mq-e.md` §7 step (1.2)
tested by:	`theory/checks/mq_e_check.py`; `theory/checks/crosscheck_corner_b_tk.py` remains an empirical full-chain cross-check only
reviewed in:	`theory/verdicts/blitz-mq-e-r1.md`

7.8 What the numerics see

Observation 7.18 (The wavepacket scan reproduces the conditional law)**.** [SKETCH] An XXZ wavepacket scan matches the spin-1/2 prediction $\delta x = -2\langle N_T \rangle$ of Theorem 7.6 at the recorded precision. On the nine clean rows at $N = 160$, three-wall truncation, standoff 36, with trapped weight below 10^{-6}:

- the headline estimator satisfies $\max |\delta x_1 + 2T| = 0.004330$ sites, i.e. 0.004 at the reported precision;
- the integrated-magnetisation estimator δx_2, which is the direct proxy for (77), satisfies $\max |\delta x_2 + 2T| = 0.001233$ sites;
- the largest estimator spread is 0.005563 sites and the largest possible trapped-charge term is $2 \times (8.862 \times 10^{-7})$ sites, so both residuals lie inside the scan's own budget (101);
- norm and energy drifts are at most 7.75×10^{-13} and 6.08×10^{-10}, subleading to the measurement geometry.

This is [SKETCH] and stays there: it is numerical evidence for the conditional theorem, not a theorem, and in particular not a proof of Definition 6.8 for the full chain. Comparing three-wall against five-wall truncation shifts the robust integrated estimator by 0.0096 sites at $\Delta = 2$, $N = 56$, about half a percent, which together with the measured $O(\Delta^{-2})$ higher-wall probability supports but does not establish the full-chain lift.

campaign id:	N2
proved in:	no proof; empirical scan only, reported in `theory/memory-quantization.md` §6, named computation Mq-check N1
tested by:	`numerics/results/memory-scan-1.json`: $\delta x \approx -2T$ within 0.004 sites
reviewed in:	`theory/verdicts/corpus-r2.md` (held at sketch)

7.9 Are the two twos the same number?

Conjecture 7.19 (The soft slope and the memory quantum are one charge datum)**.** [CONJECTURE] *The soft phase slope of the two-magnon problem and the memory channel quantum of Theorem 7.6 are the same asymptotic-charge datum, namely $|q_{\mathrm{hard}}|/s$, where q_{hard} is the hard leg's $U(1)$ charge relative to its vacuum and s is the vacuum density of Definition 6.2.*

Where the conjecture comes from. In the isotropic ferromagnet the two-magnon phase has $d\delta_{\mathrm{phys}}/dk_s|_0 = 2$ on the half-zone (Section 10), i.e. a soft Wigner shift of two sites off a hard magnon; and the kink memory quantum of Theorem 7.6 is $1/s = 2$ sites at $s = 1/2$. Two independent-looking calculations return the same number. The conjecture is that this is not a coincidence.

Evidence, stated as evidence. A pre-registered falsifier was run and *survived*. Across site spins $s \in \{1/2, 1, 3/2, 2\}$, the measured soft slope is $1/s$ and the measured ratio is $\delta x/N_T = -1/s$, both inside pre-registered 8% criteria, on an ansatz-free ring plus dynamics computation with a Bethe residual below 1.1×10^{-14}. A second, sharper falsifier targeted the charge-two case: three-magnon exact diagonalisation at $N = 100$ measured 4.07001 ± 0.08981 against the predicted $|q_{\mathrm{hard}}|/s = 4$ at $s = 1/2$, inside the pre-registered absolute window of 0.35, with the dense-sector oracle green and the hopping mutation red.

Why this is not a proof. Both measurements are finite-packet evidence. There is no extrapolation in system size or packet width, and no charge-two kink-memory test at all — the memory half of the conjecture has been checked only at $|q_{\mathrm{hard}}| = 1$. The soft half is generalised to every site spin by the exact two-body slope theorem of Section 10, but the memory half rests on the conditional theorems Theorem 7.6 and Theorem 7.9, and the factor at $|q_{\mathrm{hard}}| > 1$ is open. No cross-reference between the two halves promotes the conjecture; it remains [CONJECTURE]. Should a spin-1 two-magnon phase ever return a slope of 2 rather than 1, the coincidence is numerology and must be dropped.

campaign id:	Bc
proved in:	no proof shard; statement recorded in `theory/corner-b-draft.md` §10 and §5.3
tested by:	`theory/checks/spin_s_slope_check.py`; `numerics/results/spin1-bc-falsifier.json` (84 ring, 16 dynamics and 22 memory rows); `numerics/results/spin1-bc-crosscheck.json`; `numerics/test/test_spins_twomagnon.jl` and `test_spins_memory.jl` (decision tests with fixed criteria); `theory/lanes/blitz-2026-08-29/bc-charge2/RESULT.md` with `test_bc_charge2.jl` and its green, red-stub and red-mutation logs
reviewed in:	falsifier survived 2026-08-26 and again on the charge-two lane 2026-08-29; status unchanged

8 The memory index: quantization without scattering theory

8.1 The inversion of the continuum logic

In the continuum infrared triangle the memory effect is the last of the three corners to be quantized, and it is quantized from the outside. One first builds a scattering theory: wave operators, an inventory of asymptotic channels, and enough completeness to say that the inventory is exhaustive. A soft theorem is then read off the S-matrix, and the memory — the permanent displacement left behind after everything has dispersed — inherits whatever discreteness the channel labels happen to carry. Every step of that chain is analytically expensive, and on a lattice model of any interest not one of the steps is available.

On the lattice the logic runs the other way. The quantum of memory is fixed *before* any dynamics is discussed, by two ingredients that a lattice model supplies for free:

1. *An integrality hypothesis on the on-site charge.* The selected charge generates a circle, so its 2π rotation acts on one site by a scalar. That single algebraic fact — hypothesis (INT), Definition 8.3 — pins the on-site spectrum into one coset of $\mathbb{Z}$.

2. *Window arithmetic.* The regularised charge of a finite window is a sum of on-site charges plus a real background number. Its spectrum therefore also lies in one coset of $\mathbb{Z}$; the coset is some fixed real offset κ_{W,c_0} that depends on the window and the cut, and on nothing else.

The measurement protocol does the rest. If one measures the *same* windowed observable twice — once early, once late — both readouts are drawn from that same coset, and their difference is an integer no matter how complicated the intervening evolution was, and no matter that the two Heisenberg observables fail to commute. The offset does not have to be computed; it has to cancel, and it cancels because it is the same offset twice. Time evolution enters only through the fact that it is a $*$-automorphism, so that it preserves spectra.

That is the whole mechanism, and it costs no wave operator, no channel inventory, and no completeness. What scattering theory buys, when it happens to be available, is exactly the residual question: *which* integer, and with what probability. Corollary 8.10 makes that precise — under the scattering hypotheses of Section 11 the law collapses to a two-point law and reproduces the channel result of Section 7 — but the support statement was already true without it.

Two disciplines are binding throughout and are worth stating before anything else, because most of the ways of misreading this section violate one of them.

Remark 8.1 (Only outcomes are quantized, never their mean). The wall displacement δx is an average over a probability law. It is never claimed to be quantized, and generically it is not. Quantization attaches to two objects only: the individual outcomes of the explicit two-measurement protocol below, and — separately, under the scattering hypotheses — the spectrum of the single displacement operator ΔX of Section 7. In particular, writing δx as an expectation over some coupling of two marginal laws is bookkeeping, not an outcome law, since no such joint law is constructed.

Remark 8.2 (Differences of lattice-valued observables are not lattice-valued). It is tempting to argue that both $\hat{Q}(t_-)$ and $\hat{Q}(t_+)$ have integer spectrum, hence so does their difference. That is false. On $\mathbb{C}^2$ take $Q_- = |0\rangle\langle 0|$ and $Q_+ = |+\rangle\langle +|$: both spectra are $\{0,1\} \subset \mathbb{Z}$, yet

$$\operatorname{spec}(Q_+ - Q_-) = \left\{-\tfrac{1}{\sqrt{2}}, +\tfrac{1}{\sqrt{2}}\right\}. \tag{111}$$

No argument below ever forms such a difference. Integrality enters only through the two *spectral resolutions* of the same window observable, at step (1.5).(2.3) of the proof shard — that is, through readouts, not through operator arithmetic.

campaign id:	register discipline L3; landmine exhibit L1
proved in:	`theory/memory-index.md` §0 (register paragraph) and §6 (L1, L3)
tested by:	`theory/checks/memory_index_check.py` IDX-C2 (the naive unmeasured difference must fail at $> 10^{-6}$: the dephasing defect is generically nonzero)
reviewed in:	`memory-index-r1.md` V3; `memory-index-r2.md`

8.2 The two definitions

Throughout this section the setting is that of Section 7: a compact on-site symmetry group with a covariant family of injective matrix-product vacua, a distinguished kink pair (α, β), one selected primitive circle direction ξ common to both unbroken subgroups, and the Hermitian on-site charge $S^z := -\mathrm{i}\, q(\xi)$ built from it. The vacuum densities are $\omega_\alpha(S^z) = +s$ and $\omega_\beta(S^z) = -s$ with $s > 0$, and $\mathfrak{X}_W$ is the windowed wall-position observable of the frozen definition recalled in Section 3.

Definition 8.3 (Circle-integral on-site charge, hypothesis (INT))**.** For the selected Hermitian circle charge $S^z = -\mathrm{i}\, q(\xi)$, hypothesis **(INT)** says that there is $c \in U(1)$ with

$$e^{2\pi \mathrm{i} S^z_x} = c\mathbb{1} \tag{112}$$

on one site. Equivalently, choosing the unique $\kappa \in [0, 1)$ with $c = e^{2\pi \mathrm{i}\kappa}$,

$$\operatorname{spec} S^z \subset \kappa + \mathbb{Z}. \tag{113}$$

This holds for every spin-S chain, with $c = e^{2\pi \mathrm{i} S}$.

No arithmetic condition on the real tail-density parameter ρ is part of (INT). In particular $\rho = 0$ is admitted — that is the AKLT cross-check — and the condition $s > 0$ that calibrates the wall coordinate is a separate hypothesis belonging to the kink setting, not to (INT). Under the additional smoothness hypothesis (S) at both tails the density is instead *derived* to satisfy $e^{2\pi \mathrm{i}\rho} = c = e^{-2\pi \mathrm{i}\rho}$, hence $2\rho \in \mathbb{Z}$; that is Theorem 8.15 below, a conclusion and not a hypothesis.

campaign id:	D26
proved in:	`definitions.md`, "Memory-index campaign — D26–D27" (merged from `theory/memory-index.md` §8.1)
tested by:	`theory/checks/memory_index_check.py` IDX-C1 (finite-window offset and TPM law); `theory/checks/memory_index_probe.py` P2
reviewed in:	`memory-index-r2.md` (objection 7 repair: $\rho = 0$ admitted); `memory-index-r3.md`

The second definition packages what has to be assumed about the long-time behaviour of the charge history. It is the only place in the section where a genuinely dynamical hypothesis is made, and it is assumed, never derived.

Definition 8.4 (Charge-history local relaxation, hypothesis (LR))**.** Fix a kink-class vector Ψ of the ℓ^1 class of Section 3, a cut c_0, and a padded exhaustion $W_m = [a_m, b_m] \uparrow \mathbb{Z}$ with $c_0 \in W_m$ for all m. Put

$$\hat{Q}_{W,c_0} := 2s\,(\mathfrak{X}_W - c_0) \in \mathfrak{A}_W, \tag{114}$$

and let $E_{W,t}$ be the spectral resolution of its Heisenberg translate $\hat{Q}_{W,c_0}(t) = \alpha_t(\hat{Q}_{W,c_0})$. Hypothesis **(LR)** consists of the following three clauses.

(LR1) Common-sequence Cesàro convergence. There is *one* sequence $T_n \to \infty$ such that, for every fixed window W, the two one-time Cesàro states

$$\omega^+_{W,n}(A) = \frac{1}{T_n}\int_{T_n}^{2T_n} \langle \Psi, \alpha_t(A)\Psi\rangle \, \mathrm{d}t, \qquad \omega^-_{W,n}(A) = \frac{1}{T_n}\int_{-2T_n}^{-T_n} \langle \Psi, \alpha_t(A)\Psi\rangle \, \mathrm{d}t, \qquad A \in \mathfrak{A}_W, \tag{115}$$

and the double-Cesàro two-measurement laws

$$p_{W,n}(\nu) = \frac{1}{T_n^2}\int_{T_n}^{2T_n} \mathrm{d}t_+ \int_{-2T_n}^{-T_n} \mathrm{d}t_- \sum_q \big\| E_{W,t_+}(\{q-\nu\})\, E_{W,t_-}(\{q\})\, \Psi \big\|^2 \tag{116}$$

converge. The sum runs over $q \in \operatorname{spec} \hat{Q}_{W,c_0}$, and spectral values that are absent contribute zero.

(LR2) Vanishing first-moment measurement back-action. With the dephasing (pinching) map $\mathcal{D}_{W,t_-}(A) = \sum_q E_{W,t_-}(\{q\})\, A\, E_{W,t_-}(\{q\})$, the double-Cesàro average of

$$\big\langle \Psi, \big[\mathcal{D}_{W,t_-}\big(\hat{Q}_{W,c_0}(t_+)\big) - \hat{Q}_{W,c_0}(t_+)\big]\Psi \big\rangle \tag{117}$$

tends to zero for every fixed W. This is a first-moment nondemolition condition only; asymptotic commutativity of the two Heisenberg observables is *not* assumed.

(LR3) First-moment tightness of the spatial family. Writing p_W for the fixed-window time limit supplied by (LR1), the family $\{p_{W_m}\}$ is first-moment tight:

$$\lim_{M\to\infty} \sup_m \sum_{|\nu|>M} (1+|\nu|)\, p_{W_m}(\nu) = 0. \tag{118}$$

(LR-conv) Convenience normalisation, optional. p_{W_m} converges weakly to a probability p. This clause buys only uniqueness of the *value* δx below; support quantization is subsequence-free without it, by Prokhorov's theorem on the closed set $\mathbb{Z}$.

The ordered wall expectation (a definition, not a hypothesis). δx is

$$\delta x := \lim_m \big[\omega^+_{W_m}(\mathfrak{X}_{W_m}) - \omega^-_{W_m}(\mathfrak{X}_{W_m})\big], \tag{119}$$

taken along the full sequence when (LR-conv) holds and along an (LR3) subsequence otherwise. Its existence is a corollary of (LR1)–(LR3): at every fixed m one has the exact ledger identity

$$\sum_\nu \nu\, p_{W_m}(\nu) = -2s\big[\omega^+_{W_m}(\mathfrak{X}_{W_m}) - \omega^-_{W_m}(\mathfrak{X}_{W_m})\big], \tag{120}$$

and (LR3) makes the left-hand side converge, hence the right-hand side. This δx is the ordered asymptotic value of the finite-time wall observable; the finite-time δx is unchanged. The order of limits is binding: the infinite-volume dynamics is formed first, the fixed-window time limits second, and the spatial exhaustion last. No plane-wave or soft-momentum interchange is included anywhere.

campaign id:	D27
proved in:	`definitions.md`, "Memory-index campaign — D26–D27"; the existence corollary for δx is `theory/memory-index.md` step (1.7).(2.2)
tested by:	`theory/checks/memory_index_check.py` IDX-C4 (a tight family with non-convergent first moment is rejected: bare tightness is not enough); `theory/checks/lr_d16_check.py` C1(a),(b)
reviewed in:	`memory-index-r1.md` (objection 1: the former clause LR4 was a theorem and is deleted; the interaction-range-collar clause was cited by no step and is deleted); `memory-index-r2.md`

Remark 8.5 (Why (LR3) carries the weight $(1+|\nu|)$). Bare tightness is not enough, and one-point expectation convergence can never supply the missing control. Two exhibits make this concrete. First, $\mu_W = (1-\frac{1}{W})\delta_0 + \frac{1}{2W}\delta_W + \frac{1}{2W}\delta_{-W}$ has pointwise limit δ_0 and exact signed first moment 0, yet $\sum_{|\nu|>R}|\nu|\mu_W(\nu) \geq 1$ for every R. Second, $p_m = (1-\frac{1}{m})\delta_0 + \frac{1}{m}\delta_m$ is tight with weak limit δ_0 but has first moment identically 1. The $(1+|\nu|)$ weight in (118) is exactly what prevents both loss of mass and loss of the first moment.

campaign id:	D27(LR3); landmine exhibit L2
proved in:	`theory/memory-index.md` §6 (L2)
tested by:	`theory/checks/memory_index_check.py` IDX-C4
reviewed in:	`memory-index-b-r1.md` objection 2; `memory-index-r1.md`

8.3 Finite windows: the protocol, and integrality by offset cancellation

The two-projective-measurement protocol

Everything downstream is a statement about one explicitly specified experiment, so it is worth stating the experiment before the theorem. Fix a finite window $W = [a, b]$, a cut $c_0 \in W$, and two times $t_- < t_+$. The observable is the single windowed regularised charge $\hat{Q}_{W,c_0}$ of (114); the same window, the same cut, and the same background are used at both times.

At time t_- one performs a projective measurement of $\hat{Q}_{W,c_0}(t_-)$, obtaining an eigenvalue $q_- = q$ with probability $\|E_{W,t_-}(\{q\})\Psi\|^2$ and leaving the (unnormalised) post-measurement vector $E_{W,t_-}(\{q\})\Psi$. At time t_+ one measures the same window observable again, now Heisenberg-translated to t_+, obtaining q_+. The joint law of the pair, indexed by the *escaped charge* $\nu := q_- - q_+$, is

$$p_{W;t_-,t_+}(\nu) = \sum_{q\in\kappa_{W,c_0}+\mathbb{Z}} \left\| E_{W,t_+}(\{q-\nu\})\, E_{W,t_-}(\{q\})\, \Psi \right\|^2. \tag{121}$$

This is the two-projective-measurement (TPM) law. Nothing in (121) presumes that the two Heisenberg observables commute; $p_{W;t_-,t_+}$ is a genuine outcome law of a stated procedure, not a spectral property of some difference operator (Remark 8.2).

The sign convention is fixed once. The measured window-charge increment is $\Delta Q_W := q_+ - q_-$, and the escaped charge is $\nu = -\Delta Q_W$; the ledger of Section 7 then reads

$$2s\,\delta x = \lim \mathbb{E}[\Delta Q_W] = -\sum_\nu \nu\, p_\nu, \qquad \text{i.e.} \qquad \delta x = -\frac{1}{2s}\sum_\nu \nu\, p_\nu. \tag{122}$$

The first moment of (121) has an exact closed form which is what clause (LR2) is designed to control:

$$\sum_\nu \nu\, p_{W;t_-,t_+}(\nu) = \big\langle \hat{Q}_{W,c_0}(t_-)\big\rangle - \big\langle \mathcal{D}_{W,t_-}\big(\hat{Q}_{W,c_0}(t_+)\big)\big\rangle. \tag{123}$$

The second term is the *dephased* final expectation, not the bare one; the difference between the two — the dephasing defect — is generically nonzero, and replacing (123) by the naive unmeasured difference $\langle\hat{Q}(t_-)\rangle - \langle\hat{Q}(t_+)\rangle$ is a registered checker mutation that fails.

The finite-window theorem

Theorem 8.6 (Windowed memory is quantized without scattering theory)**.** [PROVED] *Assume (INT) of Definition 8.3; assume the calibration of Section 7 fixing the wall-coordinate parameter $s > 0$ to be the tail density; and fix a finite window $W = [a, b]$ and a cut $c_0 \in W$. Then:*

(i) *the regularised wall charge has, expanded,*

$$\hat{Q}_{W,c_0} = \sum_{x=a}^{b} S_x^z + s\,(a+b-1-2c_0)\,\mathbb{1}, \tag{124}$$

and its spectrum lies in one coset of $\mathbb{Z}$, $\operatorname{spec}\hat{Q}_{W,c_0} \subset \kappa_{W,c_0} + \mathbb{Z}$, *with offset*

$$\kappa_{W,c_0} \equiv |W|\kappa + s\,(a+b-1-2c_0) \pmod{\mathbb{Z}}, \qquad |W| = b-a+1; \tag{125}$$

(ii) *the same coset occurs at every time, i.e.* $\operatorname{spec}\hat{Q}_{W,c_0}(t) \subset \kappa_{W,c_0} + \mathbb{Z}$ *for every* t*; and*

(iii) *the TPM law* (121) *is a probability law whose escaped increment satisfies* $\nu \in \mathbb{Z}$, *with no assumption that* $\hat{Q}_{W,c_0}(t_-)$ *and* $\hat{Q}_{W,c_0}(t_+)$ *commute.*

Scope 8.7. The hypotheses are exactly (INT), the density calibration $s > 0$, a finite window $W = [a, b]$, and a cut $c_0 \in W$; the offset is (125) and is time-independent *because* the dynamics acts by C^*-automorphisms, which preserve spectra — no further dynamical input is used or available. Integrality of the TPM increment is offset cancellation at fixed W, and it is *not* spectral arithmetic for a difference of noncommuting operators (Remark 8.2). The supporting numerical probe certifies $\operatorname{spec}\hat{Q}_W \subset \mathbb{Z}$ *by construction of the observable* in the exactly solvable instantiation of Section 6 — there the on-site charge and the vacuum density are both exactly $\pm\frac{1}{2}$, so every eigenvalue is an exact machine integer for any state, evolved or not. That is an arithmetic certificate, not a dynamical one, and it says nothing about the outcome law. This theorem is a finite additive statement about a 0-form charge: it implies no displacement law, no channel inventory, and no ordered limit.

Idea of proof. By (INT), $\operatorname{spec} S^z \subset \kappa + \mathbb{Z}$ on one site. The window charge (124) is a sum of $|W|$ commuting on-site charges plus a real multiple of the identity, so its spectrum is a sum of finite spectra shifted by that number; this gives (125). Spectra are automorphism-invariant, which gives (ii). For (iii), positivity of (121) is manifest, and summing over ν and q gives 1 because each of the two spectral resolutions sums to the identity (apply Parseval twice, sequentially). Finally both readouts q_- and q_+ are drawn from the *same* coset $\kappa_{W,c_0} + \mathbb{Z}$, because both measurements use the same W, the same c_0, and the same background; hence $\nu = q_- - q_+ \in \mathbb{Z}$. The offset is removed here, at fixed W, before any limit is taken — which is also why it cannot smear when the window is later enlarged.

campaign id:	M-INDEX-fin; D13(a), D26
proved in:	`theory/memory-index.md` steps (1.1) and (1.5); the offset cancellation is step (1.5).(2.3); the mean identity (123) is step (1.6)
tested by:	`theory/checks/memory_index_check.py` IDX-C1, IDX-C2 (ten registered rows; green exit 0, `--red` exit 1, RED-OK 10 of 10); `theory/checks/memory_index_probe.py` P2 (arithmetic certificate only)
reviewed in:	`memory-index-r2.md` §4; gates cleared in `memory-index-r3.md` §0 V3

8.4 The limit law

The finite-window theorem says nothing about what happens as the window grows and the times recede. That step is where a hypothesis has to be paid, and (LR) is that hypothesis. What follows is a conditional implication and is stated as one.

Theorem 8.8 (The ordered escaped-charge law is supported on the integers)**.** [PROVED, CONDITIONAL] *Assume the kink-and-calibration hypotheses (1)–(4) of Section 7 — in particular no scattering hypothesis and no channel inventory — together with (INT) of Definition 8.3 and (LR1)–(LR3)* *of Definition 8.4 in their tightness-only form. Then every subsequential ordered TPM escaped-charge law p is a probability supported on $\mathbb{Z}$, and along that subsequence*

$$\delta x = -\frac{1}{2s}\sum_{\nu\in\mathbb{Z}} \nu\, p_\nu. \tag{126}$$

Support quantization is subsequence-free; the optional clause (LR-conv) makes p and the value δx unique and does nothing else. Consequently the individual outcomes of this explicit history protocol — not their average — are multiples of $1/(2s)$ in wall-displacement units.

Scope 8.9. The theorem is proved only as the conditional implication just displayed: the kink hypotheses, (INT) and (LR1)–(LR3) in tightness-only form imply that every (LR3)-subsequential ordered TPM law is a probability on $\mathbb{Z}$ satisfying (126) *along that subsequence*. Support quantization is subsequence-free; (LR-conv) buys uniqueness of the value δx and nothing else. Hypothesis (LR) is assumed, never derived, and no implication from the scattering hypotheses of Section 11 to (LR) is claimed anywhere: the frozen words of the local-decay clause there fix no window-marginal topology and supply no uniform integrability, so a future bridge would have to add the reading of Corollary 8.10 explicitly and prove (LR) from it. No sector-wide total-charge operator is constructed here, and its unconditional existence is refuted — see Refuted claim 8.20. Bound states, absorption and additional channels are all permitted, precisely because none of them is named.

Idea of proof. Three moves, in the order the limits are taken. First, at fixed W, average the exact mean identity (123) doubly in Cesàro form and pass to the common sequence of (LR1); clause (LR2) says exactly that the dephasing defect disappears in that average, so what survives is

$$\sum_\nu \nu\, p_W(\nu) = \omega_W^-(\hat{Q}_{W,c_0}) - \omega_W^+(\hat{Q}_{W,c_0}). \tag{127}$$

This is the step that closes the two-time problem: it is (LR2), and not the false arithmetic of Remark 8.2, that does the work. Second, substitute the definition (114); the common scalar $-2sc_0$ cancels exactly and (127) becomes the ledger (120). Third, take the padded spatial exhaustion. Each p_{W_m} is a probability on the closed set $\mathbb{Z}$, so by Prokhorov every subsequence has a further subsequence converging weakly to a probability p', necessarily supported on $\mathbb{Z}$ — that part needs no more than tightness, which is why it is subsequence-free. The $(1+|\nu|)$ weight in (LR3) additionally passes the first moment to the limit, giving (126). The window-dependent offset κ_{W,c_0} cannot smear under this limit because it was already removed at fixed W, in Theorem 8.6(iii). No plane-wave or soft limit is taken: the normalisable packet is fixed before all limits.

campaign id:	M-INDEX-spec; depends on M-flux, M-quant-G, M-INDEX-fin, D13, D17, D26, D27
proved in:	`theory/memory-index.md` steps (1.4)–(1.8); the Prokhorov step is (1.7).(2.3)
tested by:	`theory/checks/memory_index_check.py` IDX-C1–IDX-C5 (ten registered rows; green exit 0, `--red` exit 1, RED-OK 10 of 10); `theory/checks/memory_index_probe.py` P1–P5 (PASS; `--red` exit 1 with 190 pre-registered violations, `--selftest` exit 0)
reviewed in:	`memory-index-r2.md` §4; gates cleared in `memory-index-r3.md`

Corollary 8.10 (Reduction to the channel picture, when a channel picture exists)**.** [PROVED, CONDITIONAL] *Assume in addition the scattering hypotheses of Section 11 in the form used in Section 7, with the local-decay clause read as* weak-$*$ convergence of the window restriction, per channel, to the corresponding kink charge eigenstate. *Then the limit law is the two-point law*

$$p = (1-\langle N_T\rangle)\,\delta_0 + \langle N_T\rangle\,\delta_2, \tag{128}$$

and (126) *together with its variance reproduces exactly the channel result of Section 7:*

$$\delta x = -\frac{\langle N_T\rangle}{s}, \qquad \frac{1}{(2s)^2}\,\mathrm{Var}_p(\nu) = \frac{1}{s^2}\langle N_T\rangle\big(1-\langle N_T\rangle\big). \tag{129}$$

Scope 8.11. The reading of the local-decay clause is a further hypothesis, named here because the frozen words of that clause fix no topology; it is the reading the channel argument of Section 7 needs for its own uses. Reflection changes the core charge by 0 and transmission by -2, so the escaped values are 0 and 2; the two-point form (128) holds only inside the two-channel inventory, which excludes any further propagating channel and any bound-state component of the selected vector. Coherence is retained throughout: no replacement of the state by a norm mixture is used.

Idea of proof. Two limits do the work. Incoming concentration: for the incoming kink-charge eigenvalue q_*, the window projection $E_W(\{q_*\})$ lies in $\mathfrak{A}_W$, and $\|E_W(\{q_*\})\Psi_t - \Psi_t\|^2 = 1 - \langle \Psi_t, E_W(\{q_*\})\Psi_t\rangle \to 0$ as $t \to -\infty$ by the named reading. Cross-term vanishing: for the outgoing superposition $r|R\rangle + t|T\rangle$, Cauchy–Schwarz gives $|\langle R|E_W(\{q_R\})|T\rangle| \leq \langle T, E_W(\{q_R\})T\rangle^{1/2} \to 0$ since the transmitted leg's window law concentrates on $q_T \neq q_R$. Hence the TPM Born weights receive no inter-channel cross terms and are $1 - \langle N_T\rangle$ and $\langle N_T\rangle$; the rest is a Bernoulli computation.

campaign id:	M-INDEX-spec (reduction clause)
proved in:	`theory/memory-index.md` step (1.8), with the local-decay topology lemma at (1.8).(2.2)
tested by:	`theory/checks/memory_index_check.py` IDX-C5 (mutating the transmitted escaped charge from 2 to 1 must fail both the mean and the variance)
reviewed in:	`memory-index-r1.md` objection 4; `memory-index-r2.md`; `memory-index-r3.md` objection 5 (indexing gloss withdrawn)

Remark 8.12 (An empirical fence for the channel-free register). The numerical probe of Section 17 finds, in the exactly solvable chain of Section 6, extra outcome mass at window-charge values -1 and -3, carried by near-threshold two-magnon channels with vanishing group velocity. Those channels lie outside the two-channel inventory of Corollary 8.10, yet the mass sits exactly on the integers. This is direct evidence for the channel-free register of Theorem 8.8, and a concrete reason why the two-point form (128) needs its inelastic-threshold fence.

campaign id:	M-INDEX-spec (fence remark)
proved in:	`theory/memory-index.md` §4 remark, §7 finding (i)
tested by:	`theory/checks/memory_index_probe.py` P3 (the outcome-support finding is P3's, not P2's)
reviewed in:	`memory-index-r2.md` objection 9; `memory-index-r3.md`

8.5 Clause (LR1) is a theorem

Of the three clauses of (LR), the first is not a hypothesis at all. It follows from soft-analysis inputs that any lattice model of the kind considered here supplies. This matters twice over: it shortens what has to be assumed, and it explains why the clause cannot simply be deleted.

Theorem 8.13 (Cesàro compactness of the charge history)**.** [PROVED] *Let $\mathfrak{A}$ be the quasi-local algebra of a quantum spin system on a countable lattice with uniformly finite local dimension, let α_t be a strongly continuous one-parameter group of $*$-automorphisms of $\mathfrak{A}$, let Ψ be a unit vector in* any *representation of $\mathfrak{A}$, let c_0 be a cut, and let $\hat{Q}_{W,c_0} \in \mathfrak{A}_W$ be self-adjoint with finite spectrum at each finite window $W \ni c_0$. Then* every *prescribed sequence $S_n \to \infty$ admits a subsequence $T_n = S_{n_j}$ along which,* simultaneously for every such W, *the two Cesàro states $\omega^{\pm}_{W,n}$ of* (115) *converge weak-$*$ on all of $\mathfrak{A}$ to states, and every double-Cesàro TPM weight $p_{W,n}(\nu)$ of* (116) *converges. Each limit p_W is a probability supported in the finite set* $\operatorname{spec}\hat{Q}_W - \operatorname{spec}\hat{Q}_W$*, which under (INT) lies in $\mathbb{Z}$.*

Scope 8.14. The proof consumes exactly three inputs: separability (and unitality) of $\mathfrak{A}$; strong continuity of α_t; and finiteness of $\operatorname{spec}\hat{Q}_{W,c_0}$ at each fixed W. It uses *no* spectral gap, no Lieb–Robinson velocity, no ergodicity, no scattering or completeness input, and no property of Ψ beyond $\|\Psi\| = 1$. What is *not* claimed: pointwise-in-t convergence, which (LR1) does not ask for; convergence along the full sequence; and anything whatsoever about (LR2) or (LR3). One further caveat is recorded: the ASSUME line of the proof step still over-lists two standing register hypotheses that the argument does not use, and is due a strengthening in a later round.

The clause is therefore a theorem rather than a restriction — but it is not removable. It existentially binds the *single* sequence over which clauses (LR2) and (LR3) are then quantified, and the substantive content of (LR) lies entirely in those two clauses.

Idea of proof. The quasi-local algebra is a C^*-inductive limit over a countable directed family of finite-dimensional matrix algebras, hence separable: rational-coefficient words in fixed finite generating sets of the local algebras form a countable norm-dense set. Therefore the state space $S(\mathfrak{A})$ is weak-$*$ compact *and metrizable*, hence sequentially compact, and two successive extractions produce one sequence along which both ω^+ and ω^- converge — one sequence, all windows at once, because convergence is on all of $\mathfrak{A}$. For the laws, fix W: since $\hat{Q}_{W,c_0}$ has finite spectrum σ_W, its spectral projections lie in $\mathfrak{A}_W$, the integrand of (116) is a finite sum of jointly continuous functions of (t_-, t_+) with values in $[0, 1]$, it vanishes unless $\nu \in \sigma_W - \sigma_W$, and it sums to 1. The index set of pairs (W, ν) is countable, so a Cantor diagonal extraction over Bolzano–Weierstrass in $[0, 1]$ yields one further sequence along which every $p_{W,n}(\nu)$ converges. Finally a pointwise limit of probability measures on a *fixed finite* set is a probability measure on that set: no mass can escape.

campaign id:	LR1-GEN; D26, D27
proved in:	`theory/lr-d16.md` steps (1.3).(2.1)–(2.8), restated at (1.6).(2.2); the scope inventory is step (1.3).(2.7)
tested by:	`theory/checks/lr_d16_check.py` C1(a) TPM normalisation (6.66×10^{-16}) and C1(b) raw integer spectrum and support; green exit 0, `--red c1-nonunitary` and `--red c1-noninteger` each exit 1 on exactly that row. These certify the finite-W TPM arithmetic only; *no* gate bears on the Cesàro convergence itself, which is not a finite-size testable statement and needs none — the proof is complete
reviewed in:	`lr-d16-r1.md`; `lr-d16-r2.md`

8.6 A by-product: the tail density is half-integer

The circle hypothesis (INT) was introduced to control window spectra. Applied to the two vacua themselves it yields something independent of any window, any cut and any dynamics: a rigidity statement in the Lieb–Schultz–Mattis family, proved here in the matrix-product register with no gap, no clustering and no dynamical input beyond injectivity.

Theorem 8.15 (Antisymmetric covariant tails have half-integer density)**.** [PROVED] *Assume: injective canonical matrix-product tensors A_α, A_β; covariance of the vacuum family with a common unbroken circle direction $\xi \in \mathfrak{h}_\alpha \cap \mathfrak{h}_\beta$ and on-site charge $S^z = -\mathrm{i}\, q(\xi)$; the uniqueness clause of the intertwining relation; the smoothness hypothesis (S) at both tails; (INT) of Definition 8.3 with scalar c; and the* ***antisymmetry*** *$\omega_\beta(S^z) = -\omega_\alpha(S^z) =: -\rho$. Then the intertwining phase $f_\alpha(\theta) := \theta_\alpha(\exp(\theta\xi))$ has slope $f'_\alpha(0) = \rho$, satisfies $f_\alpha(\theta) = \rho\theta \bmod 2\pi$ for all θ, and*

$$e^{2\pi \mathrm{i}\rho} = c = e^{-2\pi \mathrm{i}\rho}, \qquad \textit{hence} \qquad e^{4\pi \mathrm{i}\rho} = 1, \qquad 2\rho \in \mathbb{Z}, \qquad c \in \{\pm 1\}. \tag{130}$$

Scope 8.16*.* The antisymmetry is load-bearing. One tail alone gives only $\rho \in \kappa + \mathbb{Z}$; a general (non-antisymmetric) tail pair gives only $\rho_\alpha - \rho_\beta \in \mathbb{Z}$; the conclusion $2\rho \in \mathbb{Z}$ needs the antisymmetric pair. The density ρ is a free real parameter of the vacuum pair, and $\rho = 0$ is admitted — that is the AKLT cross-check. No step assumes any relation between ρ and the on-site dimension d: ρ is a fresh density symbol and is deliberately *not* the site-spin parameter whose standard gloss $d = 2s + 1$ appears in the symbol table; that gloss is the fully polarised special case and is not imported here. The result is derived, not assumed.

Idea of proof. Four steps. (i) *Slope equals density.* With $V_\theta := V_\alpha(\exp(\theta\xi))$ and $u(\theta) := e^{\mathrm{i}\theta S^z}$ on one site, the intertwining relation gives $\omega_\alpha(u_x(\theta)) = e^{\mathrm{i}f_\alpha(\theta)}\,\mathrm{tr}[V_\theta^{-1}E_\alpha(V_\theta r)]$. Differentiating at $\theta = 0$ — legitimate by (S) — the two virtual terms cancel because the transfer operator fixes the right environment and preserves traces, leaving $\frac{\mathrm{d}}{\mathrm{d}\theta}\omega_\alpha(u_x(\theta))|_0 = \mathrm{i}f'_\alpha(0)$, while the same derivative equals $\mathrm{i}\,\omega_\alpha(S^z) = \mathrm{i}\rho$. Hence $f'_\alpha(0) = \rho$. (ii) *Additivity and continuity.* f_α is additive mod 2π along the unbroken circle and is C^1 near the origin with $f_\alpha(0) = 0$; the additivity defect of a continuous lift is a continuous $2\pi\mathbb{Z}$-valued function vanishing at the origin, hence identically zero nearby, and Cauchy's functional equation gives $f_\alpha(\theta) = \rho\theta \bmod 2\pi$ globally. (iii) *Evaluate at $\theta = 2\pi$.* (INT) gives $u(2\pi) = c\mathbb{1}$, so the intertwining relation reads $cA^s_\alpha = e^{\mathrm{i}f_\alpha(2\pi)}V_{2\pi}^{-1}A^s_\alpha V_{2\pi}$; the uniqueness clause (applied with the trivial comparison gauge) forces $V_{2\pi}$ scalar and $e^{\mathrm{i}f_\alpha(2\pi)} = c$, i.e. $e^{2\pi\mathrm{i}\rho} = c$. (iv) *The other tail.* The same computation at β, whose density is $-\rho$ and whose on-site charge — hence whose scalar c — is the same, gives $e^{-2\pi\mathrm{i}\rho} = c$. Dividing yields (130).

campaign id:	M-IDX-density; D1, D2(a,b,e), D26
proved in:	`theory/memory-index.md` step (1.9); route recorded in `memory-index-b-r1.md` §1(b)
tested by:	`theory/checks/memory_index_check.py` IDX-C6, IDX-C7(i), IDX-C7(ii) (green exit 0, `--red` exit 1). The load-bearing two-tail step (1.9).(2.4) is separately red-certified as IDX-C7(ii), with twisted-transfer defects at the correct counterterm and $\lvert\varphi_W(2\pi)-1\rvert$ up to 1.9979 on the charge-shifted family, independently re-run in `memory-index-r3.md` §0 V3(i)–(iii)
reviewed in:	`memory-index-b-r1.md`; `memory-index-r2.md` §4 (objection 1 fence, note 10); `memory-index-r3.md`

Remark 8.17 (The offset vanishes identically on an antisymmetric pair). If in addition the wall-coordinate calibration parameter is identified with the tail density, $s = \rho$, then the coset offset (125) is not merely constant in time but identically zero: substituting $\kappa \equiv \rho \pmod{\mathbb{Z}}$ into (125) gives

$$\kappa_{W,c_0} \equiv \rho(b-a+1) + \rho(a+b-1-2c_0) = 2\rho\,(b-c_0) \equiv 0 \pmod{\mathbb{Z}}, \tag{131}$$

since $2\rho \in \mathbb{Z}$ and $b - c_0 \in \mathbb{Z}$. Hence $e^{2\pi\mathrm{i}\hat{Q}_{W,c_0}}$ is exactly the identity in $\mathfrak{A}_W$ and $\operatorname{spec}\hat{Q}_{W,c_0} \subseteq \mathbb{Z}$ for every window and cut. Nothing in Theorems 8.6–8.8 needs this: the TPM increment cancels the offset at fixed W regardless. It merely strengthens "one coset" to "the zero coset", and it is what the numerical probe's integrality findings instantiate at $s = \frac{1}{2}$.

campaign id:	M-IDX-density (corollary); D26, and the density calibration $s = \rho$
proved in:	`theory/memory-index.md` step (1.10)
tested by:	`theory/checks/memory_index_check.py` IDX-C7(i),(ii); `theory/checks/memory_index_probe.py` P1, P2
reviewed in:	`memory-index-r2.md` objection 4; `memory-index-r3.md`

8.7 An implementer on one kink folium

Theorem 8.6 is about finite windows. It is natural to ask whether the window charges assemble into a single self-adjoint charge operator in an infinite-volume representation. The next two results give the honest answer: on one fixed kink representation such an operator exists, but it is *not* obtained as a limit of the window charges, and the general existence statement is false.

Theorem 8.18 (A strongly continuous circle implementer on the fixed kink folium)**.** [PROVED] *Let ϱ_0 be the fixed bare kink state supplied by the corner-A construction of Section 5, with invertible junction insertion $V_\alpha(g)^{-1}$. Let the selected circle be common-unbroken, $h_\theta \in H_\alpha \cap H_\beta$, satisfy the smoothness hypothesis (S) at both tails, and obey (INT) of Definition 8.3. Then in the GNS representation of ϱ_0 the circle automorphisms are implemented by a strongly continuous unitary group*

$$U(\theta) = e^{\mathrm{i}\theta\hat{Q}}, \qquad U(\theta)\pi(A)U(\theta)^* = \pi(\gamma_\theta(A)), \quad A \in \mathfrak{A}, \tag{132}$$

whose Stone generator satisfies $e^{2\pi i\hat{Q}} = e^{2\pi i q_0}\mathbb{1}$ *for some real* q_0*; consequently* $\hat{Q}$ *is pure point with* $\operatorname{spec}\hat{Q} \subseteq q_0 + \mathbb{Z}$*, a single coset of* $\mathbb{Z}$*.*

Scope 8.19. This is a folium-equivalence and implementer theorem. It is *not* a strong-resolvent limit of the window charges $\hat{Q}_{W,c_0}$: for a nonscalar virtual circle action V_θ those bare window unitaries remain non-Cauchy, as Refuted claim 8.20 records, and there is no contradiction because an implementing unitary is determined only up to a scalar and is under no obligation to be the strong limit of any particular inner approximants. The scope is the fixed bare kink of Section 5, together with only those finite-core variants whose padded blocked tensors span the full matrix algebra $M_\chi(\mathbb{C})$; no assertion is made for arbitrary finite-core modifications. The common-unbroken recital is necessary: if the circle moved either tail label, the rotated state would lie in a disjoint sector and no implementer could exist in the fixed representation. The whole content is *existence*; once the implementer exists, the scalar-period clause is a one-line Schur argument.

Idea of proof. Injectivity localises the circle orbit. Since the left and right injectivity words each span $M_\chi(\mathbb{C})$ and the junction matrix is invertible, the blocked reference matrices $L_a M R_b$ span $M_\chi(\mathbb{C})$; hence a fixed padded block around the junction carries a local operator $B(M')$, affine and continuous in M' with $B(M) = \mathbb{1}$, such that $\pi(B(M'))\Omega$ realises exactly the state with junction M' replaced. Rotating by the circle telescopes all virtual gauges away except at the junction, where $M \mapsto M_\theta = V_\alpha(h_\theta)MV_\beta(h_\theta)^{-1}$; this is continuous and invertible near $\theta = 0$, so the rotated states are represented by a continuous family of unit vectors ξ_θ in the *same* Hilbert space. The fixed kink is pure — it is a vacuum composed with a half-string automorphism — so its GNS representation is irreducible, and the rule $W_\theta\pi(A)\Omega := \pi(\gamma_\theta(A))\xi_{-\theta}$ defines an isometry, extends to a unitary, and implements γ_θ. Uniqueness of implementers up to scalars makes $\theta \mapsto [W_\theta]$ a continuous projective homomorphism, which on a *one-dimensional* parameter group can be de-projectivised explicitly: the C^1 scalar two-cocycle $m(s,t) = W_s W_t W_{s+t}^*$ has argument $f(s,t) = a(s+t) - a(s) - a(t)$ for an explicit a, so $e^{ia(\theta)}W_\theta$ is an honest local group and extends to $\mathbb{R}$. Stone's theorem produces $\hat{Q}$; (INT) makes $\gamma_{2\pi}$ the identity automorphism, so $U(2\pi)$ lies in the scalar commutant, and spectral calculus confines the spectral measure to $q_0 + \mathbb{Z}$ and makes it pure point.

campaign id:	M-INDEX-LA-folium; A2, WI, D1(c,e′), D2(b,e), D26
proved in:	`theory/folium-implementer.md` steps (1.1)–(1.7); the compatibility fence with Refuted claim 8.20 is step (1.7); summarised at `theory/memory-index.md` step (1.3b)
tested by:	— (exact finite-dimensional and functional-analytic proof; no numerical claim and no checker required)
reviewed in:	`blitz-la-folium-r1.md` (0 fatal / 0 major)

8.8 What is refuted: there is no sector-wide charge operator

Refuted claim 8.20 (Unconditional existence of a regularised total charge)**.** [REFUTED] Withdrawn statement. *The kink-sector definition, the* ℓ^1 *decay class, and (INT) together imply the existence of a self-adjoint regularised total charge in every GNS representation of the kink sector* $\mathcal{K}_{\alpha\beta}$ *— the operator form of the finite-window theorem.*

This is false, by two distinct mechanisms.

Mechanism A (off-folium; the fluctuation counterexample). *This mechanism alone refutes the statement as written, and it needs neither the smoothness hypothesis (S) nor any identification of the calibration parameter with the density. Take one site with basis* $|m\rangle$*,* $m \in \{-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\}$ *and* $S^z|m\rangle = m|m\rangle$*; the symmetry group* $U(1) \rtimes \mathbb{Z}_2$ *with the circle generated by* S^z *and the reflection* $R|m\rangle = |-m\rangle$*; the product vacua* $\alpha = \bigotimes|\frac{1}{2}\rangle$ *and* $\beta = \bigotimes|-\frac{1}{2}\rangle$*; and* $H = 0$*. For* $n \geq 1$ *put*

$\varepsilon_n = (n+1)^{-1/2}$ *and*

$$|\psi_{-n}\rangle = \sqrt{1-\varepsilon_n^2}\,|\tfrac{1}{2}\rangle + \frac{\varepsilon_n}{\sqrt{2}}\big(|-\tfrac{1}{2}\rangle + |\tfrac{3}{2}\rangle\big), \tag{133}$$

and let ϱ be the product state with these vectors at the sites $-n$ and $|-\frac{1}{2}\rangle$ at every site $x \geq 0$. Then ϱ lies in the ℓ^1 kink class including *its optional spatial first moment, since $\langle\psi_{-n}|S^z|\psi_{-n}\rangle = \frac{1}{2}$* exactly *at every site: all the deviation sums vanish identically. Yet the window charge on $W_N = [-N, N]$ with $c_0 = -1$ has, in the cyclic vector, the distribution of $L_N = \sum_{n=1}^{N}(S^z_{-n} - \frac{1}{2})$, whose characteristic function is*

$$\varphi_N(t) = \prod_{n=1}^{N}\big[1 - \varepsilon_n^2(1-\cos t)\big]. \tag{134}$$

With $V_N = \sum_{n\leq N}\varepsilon_n^2 \sim \log N \to \infty$ and $1-\cos t \geq 2t^2/\pi^2$ on $[-\pi,\pi]$, one gets $|\varphi_N(t)| \leq e^{-2V_N t^2/\pi^2}$ and hence $\sup_k \mathbb{P}(L_N = k) \leq C\,V_N^{-1/2}$: the distribution spreads without bound. Splitting the resolvent matrix element at $|k| \leq K$ then gives $\langle\Omega_\varrho, (\hat{Q}_{W_N,-1} - \mathrm{i})^{-1}\Omega_\varrho\rangle \to 0$, which is impossible for a self-adjoint limit, since every self-adjoint $\hat{Q}$ has $\mathrm{Im}\langle\Omega, (\hat{Q}-\mathrm{i})^{-1}\Omega\rangle = \int(\lambda^2+1)^{-1}\mathrm{d}\mu(\lambda) > 0$. The moral is sharp: the ℓ^1 class controls means, *not charge fluctuations, and certainly not implementability. A second, purely definitional point reinforces it: the kink sector is a* set *of states, whereas a GNS representation belongs to one state, so "the GNS representation of $\mathcal{K}_{\alpha\beta}$" has no canonical meaning without an additional folium choice.*

Mechanism B (on-folium; a strengthening, not a second proof of the same statement). *This mechanism assumes strictly more — the smoothness hypothesis (S), the identification of the calibration parameter with the tail density, a two-sided decorated kink reference, the corner-A construction, and a nonscalar V_{θ_0} — and in exchange it kills the window-limit route* on *the kink folium itself, where fluctuations are perfectly tame. For nested windows differing by a shell I of length ℓ deep in the α tail, the on-site summands commute, so*

$$\big\|\big(e^{\mathrm{i}\theta\hat{Q}_{W',c_0}} - e^{\mathrm{i}\theta\hat{Q}_{W,c_0}}\big)\Omega\big\|^2 = 2 - 2\,\mathrm{Re}\langle\Omega, e^{-\mathrm{i}\theta Q_I^\alpha}\Omega\rangle, \tag{135}$$

where $Q_I^\alpha = \sum_{x\in I}(S_x^z - \rho)$ is the tail-normal-ordered shell charge — the identification of the two is exactly where $s = \rho$ is consumed. Pushing the shell deep into the tail and then letting $\ell \to \infty$ sends the right-hand side to $2\big(1 - |\mathrm{tr}(V_\theta r)|^2\big)$, and for a positive normalised environment $|\mathrm{tr}(Vr)| \leq 1$ with equality only for scalar V. Hence for nonscalar V_{θ_0}

$$\liminf\;\big\|\big(e^{\mathrm{i}\theta_0\hat{Q}_{W'}} - e^{\mathrm{i}\theta_0\hat{Q}_W}\big)\Omega\big\|^2 \;\geq\; 2\big(1 - |\mathrm{tr}(V_{\theta_0}r)|^2\big) > 0: \tag{136}$$

the window unitaries are not Cauchy, so the window charges have no strong-resolvent limit. (At $\theta \in 2\pi\mathbb{Z}$ there is no obstruction, exactly as Remark 8.17 predicts.)

What survives. *Mechanism A shows the operator form fails off every kink folium; mechanism B shows the* window route *fails on the folium as well. The surviving statements are: the finite-window theorem (Theorem 8.6) and the ordered limit law (Theorem 8.8), which never form such an operator; the implementer theorem (Theorem 8.18), which reaches a self-adjoint charge on one fixed folium by an entirely different route; and the arithmetic implication that* if *a self-adjoint $\hat{Q}$ is supplied and $e^{2\pi\mathrm{i}\hat{Q}}$ is scalar, then $\hat{Q}$ is pure point with spectrum in one coset of $\mathbb{Z}$ — which is a consequence of spectral calculus and proves neither premise.*

campaign id:	M-INDEX-LA-strong
proved in:	two mechanisms: `theory/memory-index.md` step (1.3) (the fluctuation counterexample, off-folium — this alone refutes the row as stated and needs neither (S) nor $s = \rho$) and step (1.12) (the nonscalar virtual-circle obstruction, on-folium — a *strengthening*, not a second proof of the same statement: it additionally assumes (S), $s = \rho$, D1(e′), A2, and nonscalar V_{θ_0}). The valid arithmetic implication is step (1.2)
tested by:	`theory/checks/memory_index_check.py` IDX-C3, IDX-C8 (green exit 0, `--red` exit 1)
reviewed in:	`memory-index-b-r1.md` objection 7(d) (nonexistence direction only); `memory-index-r2.md` objection 4; `blitz-la-folium-r1.md` (compatibility with Theorem 8.18)

8.9 Where memory can live: the compact-group scope

So far the symmetry has been one selected circle. For a general compact on-site group one can ask which directions of the symmetry algebra are capable of carrying a calibrated memory at all. The answer is sharp, and it is a genuine restriction: only the central directions of the unbroken subgroup.

Theorem 8.21 (Calibrated memory lives on the centre of the unbroken subgroup)**.** [PROVED] *Let G be a compact on-site symmetry group and α a vacuum of the covariant family, with unbroken subgroup H_α and Lie algebra $\mathfrak{h}_\alpha$. Then:*

- *(i) the vacuum density functional $\xi \mapsto \omega_\alpha(q(\xi))$ is $\mathrm{Ad}(H_\alpha)$-invariant and annihilates $[\mathfrak{h}_\alpha, \mathfrak{h}_\alpha]$; with an $\mathrm{Ad}(H_\alpha)$-invariant inner product its representing vector lies in $\mathfrak{h}_\alpha^{H_\alpha} = \mathrm{Lie}\big(Z(H_\alpha)^0\big)$, and every nonzero tail-density jump descends through the on-site scalar kernel to $\mathrm{Lie}\big(Z^0_{\alpha,\mathrm{eff}}\big)$;*
- *(ii) a Weyl representative n_w with $\beta = n_w \cdot \alpha \neq \alpha$ and $\mathrm{Ad}_{n_w^{-1}} \xi = -\xi$ supplies an opposite-density kink pair $(+s, -s)$ on a common unbroken circle, hence a kink sector of the corner-A type; and*
- *(iii) semisimple unbroken directions and finite groups supply no calibrated memory: in the first case the vacuum density vanishes, so no positive calibration s exists; in the second there is no Lie-algebra circle, hence no current to integrate and no real displacement calibration at all.*

Scope 8.22. The theorem classifies the only possible *location* and *lattice* of a memory effect; it does not assert that any given compact group actually has one. It does not claim that every G has an active centre, a Weyl element that flips a central direction, a kink, scattering channels, or relaxation. A Weyl element supplies a calibrated pair only on a direction in its -1 eigenspace and only when it moves α to a distinct vacuum; directions fixed by that element have equal tail densities and are spectators. The exclusion in (iii) concerns the semisimple part of the *unbroken* algebra, not of the ambient group: an ambient semisimple group may perfectly well exhibit memory after breaking to a subgroup with a central circle, as in $SU(2) \to U(1)$. Finite groups may still leave exact group-valued string endpoint operators, but that is a different observable and not an additive memory law. Finally, this construction supplies vacuum-pair hypotheses only; it supplies neither scattering hypotheses nor (LR).

Idea of proof. For (i), invariance of the vacuum under H_α gives $d_\alpha(\mathrm{Ad}_h \xi) = d_\alpha(\xi)$; differentiating along $h = \exp(t\eta)$ gives $d_\alpha([\eta, \xi]) = 0$, so the functional kills the commutator ideal. Compact reductivity splits $\mathfrak{h}_\alpha = \mathfrak{z}(\mathfrak{h}_\alpha) \oplus [\mathfrak{h}_\alpha, \mathfrak{h}_\alpha]$, and the invariance places the representing vector in the fixed-point set, i.e. in the Lie algebra of the connected centre. A direction acting by scalars on site has the same density in every state, so a nonzero *jump* descends to the effective centre. For (ii), covariance gives $d_\beta(\xi) = d_\alpha(\mathrm{Ad}_{n_w^{-1}} \xi) = -d_\alpha(\xi)$, which is the antisymmetric pair, and conjugation carries the same circle into $H_\beta = n_w H_\alpha n_w^{-1}$, so the circle is common-unbroken; the corner-A construction then supplies and superselects the kink sector. For (iii), the wall calibration carries an explicit factor $(2s)^{-1}$ with $s > 0$, which (i) makes unavailable on semisimple directions, while a finite group has trivial Lie algebra.

campaign id:	M-SCOPE-center; D2, D9, D10, A2, M-quant-G, S-IDX-fin-G, S-IDX-G-label
proved in:	`theory/compact-g-memory-scope.md` §§1, 3–4, i.e. steps (1.1), (1.3), (1.4), (1.5)
tested by:	`theory/checks/scope_g_check.py` SG-C1 (finite covariance only: 24 deterministic covariant tensors and 24 invariant density samples for a nonabelian $U(1) \times SU(2)$, checking zero density on all three semisimple generators and a permitted nonzero central density; the red mutation adds a highest-weight triplet component and breaks covariance)
reviewed in:	`blitz-scope-g-r1.md` (0 fatal / 0 major)

8.10 Higher rank: the torus generalization

When the effective centre has rank $r > 1$ the escaped charge is a vector, and the natural home for it is not $\mathbb{Z}^r$ but the weight lattice of the effective central torus; $\mathbb{Z}^r$ appears only after a choice of primitive coordinates.

Theorem 8.23 (The joint escaped charge lies in the central weight lattice)**.** [PROVED] *Let the vacuum pair share a connected torus of rank r inside $Z(H_\alpha)^0 \cap Z(H_\beta)^0$; quotient its on-site scalar kernel and choose r primitive cocharacters of the resulting effective torus, so that the free part of its character lattice is written $\mathbb{Z}^r$. Impose (INT) separately on each corresponding Hermitian generator $C_j = -\mathrm{i}q(\xi_j)$, and on every* active *direction impose the antisymmetric calibration $\omega_\alpha(C_j) = +s_j$, $\omega_\beta(C_j) = -s_j$ with $s_j > 0$ (directions with zero density jump are simply omitted from the calibrated list). Use one fixed window $W = [a, b]$, one cut $c_0 \in W$, and the same backgrounds at both measurement times. Then:*

(i) at finite times the joint TPM escaped-charge vector lies in the effective central weight lattice, hence in $\mathbb{Z}^r$ in the chosen primitive coordinates; and

(ii) under the joint-PVM strengthening of (LR) stated below, every ordered subsequential law p on that lattice satisfies, componentwise,

$$\delta x_j = -\frac{1}{2s_j} \sum_{\nu \in \mathbb{Z}^r} \nu_j \, p(\nu), \qquad j = 1, \ldots, r. \tag{137}$$

The joint-PVM strengthening is: the joint *TPM laws have the (LR1) limits on one common sequence; the* joint *initial dephasing has vanishing first-moment defect for every C_j, as in (LR2); and the laws are first-moment tight with $|\nu|$ replaced by $\|\nu\|_1$, as in (LR3). The optional full-sequence weak convergence plays exactly the convenience role it plays in Definition 8.4.*

Scope 8.24. Conclusion (i) is unconditional on the displayed finite hypotheses and needs no relaxation assumption of any kind. Conclusion (ii) needs the joint-PVM strengthening, and the distinction matters: applying the scalar hypothesis (LR) separately to each circle proves the r *marginal* statements of Theorem 8.8, but it does not by itself prove that the finer joint dephasing defect vanishes. An individual calibrated history outcome is the vector $\big(-\nu_1/(2s_1), \ldots, -\nu_r/(2s_r)\big)$; each δx_j is its mean and need not lie on the lattice — no mean quantization is claimed (Remark 8.1). No single scalar wall displacement is inferred unless the event separately makes the component values compatible. Under the smoothness hypothesis (S) at both tails, Theorem 8.15 additionally gives $2s_j \in \mathbb{Z}$ for every antisymmetric active circle, but that is a conclusion, not an added hypothesis.

Idea of proof. The generators C_j lie in one abelian torus, so they commute on site; hence all the window charges

$$\hat{Q}^{(j)}_{W,c_0} := 2s_j\big(\mathfrak{X}^{(j)}_W - c_0\big) = \sum_{x=a}^{b} C_{j,x} + s_j\,(a+b-1-2c_0)\nVdash \tag{138}$$

commute and admit a joint spectral resolution, and so do their Heisenberg translates at each fixed time. The joint on-site spectral labels form an affine torsor over the effective central character lattice, so every *difference* of labels evaluates on the chosen primitive cocharacters to an element of $\mathbb{Z}^r$; summing over the window and adding the scalar backgrounds changes only the common affine offset. Every nonzero atom of the joint TPM law then has the form

$$\big\|E_{W,t_+}(\{\lambda-\nu\})\,E_{W,t_-}(\{\lambda\})\,\Psi\big\|^2, \qquad \nu \in \mathbb{Z}^r, \tag{139}$$

and the affine offset cancels because both measurements use the same window, cut and backgrounds — the vector version of Theorem 8.6(iii), again with no cross-time commutation used. For (ii), the componentwise analogue of the mean identity (123) plus the joint (LR2) gives $\sum_\nu \nu_j p_W(\nu) = -2s_j[\omega^+_W(\mathfrak{X}^{(j)}_W) - \omega^-_W(\mathfrak{X}^{(j)}_W)]$, and vector first-moment tightness — $\|\nu\|_1$ dominates every $|\nu_j|$ — supplies the Prokhorov and first-moment steps of Theorem 8.8 in each component.

campaign id:	M-INDEX-torus; D13, D26, D27, M-INDEX-fin, M-INDEX-spec, M-IDX-density, S-IDX-G-label
proved in:	`theory/compact-g-memory-scope.md` §2, step (1.2), with the finite-window part at (1.2).(2.1)–(2.5) and the ordered-limit part at (1.2).(2.6)–(2.9)
tested by:	`theory/checks/scope_g_check.py` SG-C2 (finite arithmetic only; no gate bears on the relaxation hypothesis): three commuting on-site Cartan charges, two random same-time joint eigenbases with noncommuting cross-time observables, and the full 625-atom joint TPM law, checking normalisation and affine-offset cancellation into $\mathbb{Z}^3$; the red mutation shifts the final background by $\sqrt{2}/10$ and the lattice gate fails
reviewed in:	`blitz-scope-g-r1.md` (0 fatal / 0 major; the citation for the on-site joint spectrum was repaired there to D26 and Theorem 8.6)

8.11 What the section does and does not settle

Theorem 8.6 is the centrepiece and is unconditional: given the integrality hypothesis on the on-site charge and a fixed window, the outcomes of the stated two-measurement protocol are integers, for any dynamics that acts by automorphisms. Theorem 8.8 carries that statement to the ordered limit at the price of one explicitly named dynamical hypothesis, (LR), which is assumed and not derived; Theorem 8.13 removes its first clause from the price list, leaving the substantive content in the second and third. Theorem 8.15 is a by-product of independent interest, and Theorem 8.21 together with Theorem 8.23 says where in a general compact symmetry such a memory can live and what lattice it takes values in.

Two limits of the analysis should be kept in view. The operator form of the finite-window statement is false (Refuted claim 8.20), so "the total charge of the sector" is not an available object; only the law-level statements and the single-folium implementer (Theorem 8.18) survive. And no model instance of (LR2)–(LR3) has been exhibited: what is proved is a conditional whose hypothesis class is constrained from several sides in Section 9. The status of the triangle as a whole is assembled in Section 18.

9 Local relaxation, escape obstructions, and kink–magnon wave operators

The memory limit law of Section 8 is not unconditional. It delivers a quantized ordered wall displacement only on states for which the windowed charge history relaxes: the three clauses of local relaxation — convergence of the fixed-window Cesàro states and two-point-measurement laws, first-moment nondemolition, and first-moment tightness of the family of window laws — which we abbreviate (LR1), (LR2), (LR3), and collectively (LR). This section is the record of what has actually been proved about that hypothesis and about its enemies.

The material falls into three groups, and it is worth saying in advance which way each one cuts.

First, an exact algebraic identity: the regularised window charge is nothing but the windowed restriction of the conserved first-moment wall coordinate. This is unconditional, it costs nothing, and it is the sharpest single fact in this section, because it converts two apparently representation-theoretic questions — can a state have exponentially confined tails while a charged leg escapes? can such a state carry nonzero ordered memory? — into arithmetic. Both answers are negative.

Second, a package of theorems around a named exponential-tail hypothesis (K-TAIL). These say what the confined-charge corner looks like: on it one has a quantitative window-charge estimate with an explicit $\tilde{\lambda}^{d_W}$ rate, the tail vacua are forced to have sharp on-site charge, and — the sting — the ordered memory is *exactly zero*. The hypothesis class is inhabited (the exactly solvable kink model of Section 6 sits in it, with numerically measured decay rate), but it is disjoint from the escaping-leg class in which a nonzero memory could live.

Third, the complementary corner. An unconditional operator inequality bounds the *number* of phase boundaries in the easy-axis model uniformly in time, but says nothing about their *length*; the gap between the two is exactly the residual hypothesis (NR) from which clause (LR3) does follow. Against this, mean tail transport (M-ESC) is shown to destroy (LR3) outright. No state realising (M-ESC) is exhibited anywhere in this campaign, so that result constrains a hypothesis class rather than refuting anything; but together with the tightness transfer it yields the clean propositional statement that (M-ESC) and (NR) cannot both hold.

The section closes with the conditional construction of kink–magnon wave operators from exact ansatz band data, which is the machinery that would instantiate the channel structure the escape analysis presumes, and with the local-decay hypothesis it leaves open.

9.1 Standing notation

The following objects are defined in full in Section 7 and Section 8; they are recalled here so that this section can be read without turning back.

Definition 9.1 (Window observables and the relative window charge)**.** Fix a chain with two S^z-vacuum densities $s_\alpha = +s$ at $-\infty$ and $s_\beta = -s$ at $+\infty$, $s > 0$. For a finite window $W = [a, b] \subset \mathbb{Z}$ the *windowed wall-position observable* is

$$\mathfrak{X}_W \;:=\; a - 1 + \frac{1}{2s}\sum_{x=a}^{b}\big(S_x^z + s\big), \tag{140}$$

normalised so that $\mathfrak{X}_W = m$ on a sharp wall at the bond $(m \mid m{+}1)$ carrying no other content in W. For a cut $c_0 \in W$ the *relative window charge* is $\hat{Q}_{W,c_0} := 2s(\mathfrak{X}_W - c_0)$, a local self-adjoint element of $\mathfrak{A}_W$ with finite spectrum, and $E_W(\cdot)$ denotes its spectral measure. The *escaped wrong-phase content* outside W is

$$\mathcal{N}_W \;:=\; \sum_{x<a}\Big(\tfrac{1}{2} - S_x^z\Big) + \sum_{x>b}\Big(S_x^z + \tfrac{1}{2}\Big), \tag{141}$$

a sum of non-negative diagonal terms, so that $\mathcal{N}_{W'} \le \mathcal{N}_W$ whenever $W \subseteq W'$.

Definition 9.2 (Tail deviations and window padding)**.** For a finite core interval $K = [\ell, r] \subset \mathbb{Z}$ and a site $x \notin K$ put $\gamma(x) := \alpha$ if $x < \ell$ and $\gamma(x) := \beta$ if $x > r$, and define the *on-site tail deviation*

$$D_x \;:=\; S^z_x - s_{\gamma(x)}, \qquad s_\alpha = +s, \quad s_\beta = -s. \tag{142}$$

For a window $W = [a, b] \supseteq K$ the *core-to-edge padding* is $d_W := \operatorname{dist}(K, \mathbb{Z}\setminus W) = \min(\ell - a,\ b - r) + 1 \ge 1$. Where a cut c_0 is fixed rather than a core, we write instead the *step density* $s_{\gamma_{c_0}}(x) := +s$ for $x \le c_0$ and $-s$ for $x > c_0$.

Throughout, H-MQG(1)–(3) are the standing structural hypotheses of the general memory-quantization setting of Section 7: a compact group G with a covariant family of injective canonical matrix product state tensors of one common bond dimension and a selected kink sector $\mathcal{K}_{\alpha\beta}$; a common unbroken abelian subgroup with a fixed primitive on-site charge S^z calibrated by $\omega_\alpha(S^z) = +s = -\omega_\beta(S^z)$, $s > 0$; and a translation-invariant, finite-range, G-invariant Hamiltonian with stationary vacua. D26(INT) is the integrality condition on the on-site charge, and D17 is the ℓ^1 kink class.

9.2 The window charge is the conserved wall coordinate

The starting point costs nothing and pays for a great deal. The regularised window charge, which was introduced as a bookkeeping device for windowed two-point measurements, is literally the sum of the on-site tail deviations over the window — with no offset and no normalisation factor.

Theorem 9.3 (The window charge is the windowed sum of tail deviations)**.** [PROVED] *Assume H-MQG(1)–(2). For every finite window $W = [a, b]$ with $c_0 \in W$,*

$$\hat{Q}_{W,c_0} \;=\; \sum_{x \in W} \bigl(S^z_x - s_{\gamma_{c_0}}(x)\bigr) \qquad \textit{exactly.} \tag{143}$$

Consequently, on any state on which the increasing-window limit exists — in particular on every state satisfying the exponential-tail hypothesis (K-TAIL) of Definition 9.8, where the limit vector $\mathcal{Q}_{c_0}\varphi := \lim_n \hat{Q}_{W_n,c_0}\varphi$ exists and is independent of the exhaustion —

$$\mathcal{Q}_{c_0} \;=\; \sum_{x \in \mathbb{Z}} \bigl(S^z_x - s_{\gamma_{c_0}}(x)\bigr) \;=\; 2s\,(X_1 - c_0), \tag{144}$$

where X_1 is the first-moment wall coordinate of the dynamical dress (Section 7), the regularised total magnetisation, which is exactly conserved. The relative window charge is therefore not a new object but the vector-valued lift of X_1.

Scope 9.4. The identity (143) is unconditional under H-MQG(1)–(2). The identification (144) requires in addition that the increasing-window limit exist on the state in question. What is constructed is only the limit *vector* $\mathcal{Q}_{c_0}\varphi$ on a (K-TAIL) state, never a sector-wide charge operator: the refutation of the strong local-additivity claim (Section 16) is respected and not contradicted, its refuting state having polynomially decaying deviations and therefore lying outside the (K-TAIL) class.

Idea of proof. The frozen expansion of the windowed observable gives $\hat{Q}_{W,c_0} = \sum_{x\in W} S^z_x + s(a + b - 1 - 2c_0)$, and the step density sums to exactly the negative of that offset, $\sum_{x \in W} s_{\gamma_{c_0}}(x) = s\,[(c_0 - a + 1) - (b - c_0)] = -s(a + b - 1 - 2c_0)$. The two scalars cancel identically; there is nothing left to estimate. Summing the identity over an exhaustion and comparing with the definition of X_1 gives (144). □

campaign id:	LD-ID
proved in:	`theory/ace-ld.md` step (1.4).(2.7), THEOREM LD-ID, sub-step (3.1)–(3.2)
tested by:	No gate is required: the identity is exact algebra. Independently re-verified in `theory/verdicts/ace-ld-r3.md` §0(i) on all 2^{12} basis states of the exactly solvable battery over six windows (maximal discrepancy 0.000e+00), symbolically and on a 201-site lattice in `theory/verdicts/ace-ld-r2.md` §0(i), and reached independently by the parallel lane
reviewed in:	`ace-ld-r2.md`, `ace-ld-r3.md`, `ace-ld-r4.md`; parallel-lane confirmation `lr-d16-r1.md` §2 item 1

The identity has two corollaries, each on its own named hypotheses. The first closes off an entire class of loopholes.

Corollary 9.5 (No escaping charged leg has time-uniform confined tails)**.** [PROVED, CONDITIONAL] *Assume, in addition to the hypotheses of Theorem 9.3, that Ψ_t carries an outgoing free leg of charge $q_{\text{leg}} \neq 0$ which has left the window W in the sense of the asymptotic-decay clause D18(AD3), and that the state satisfies (K-TAIL) with* time-independent *constants $(K, C_K, \tilde{\lambda})$. Then*

$$\|\mathcal{Q}_{c_0}\Psi_t - \hat{Q}_{W,c_0}\Psi_t\| \;\geq\; \Big|\sum_{x \notin W} \omega_{\Psi_t}(D_x)\Big| \;\longrightarrow\; |q_{\text{leg}}| \quad (t \to \infty), \tag{145}$$

while (K-TAIL) bounds the same quantity by $2C_K\tilde{\lambda}^{d_W}/(1-\tilde{\lambda})$ uniformly in t. Choosing d_W so large that $2C_K\tilde{\lambda}^{d_W}/(1-\tilde{\lambda}) < |q_{\text{leg}}|$ is a contradiction. Hence no state in the escaping-leg class of D18(AD3) satisfies (K-TAIL) with time-independent constants, and distinct channel charges $q_L \neq q_T$ are unavailable within a single conserved-charge sector of D18.

Corollary 9.6 (On the confined class the ordered memory vanishes)**.** [PROVED, CONDITIONAL] *Assume in addition (K-Q) of Definition 9.9 at $\varepsilon_Q = 0$ together with time-uniform (K-TAIL) data, and adopt D27's definition of the ordered wall expectation δx. Then q_φ is a time-independent eigenvalue of the conserved $\mathcal{Q}_{c_0}$, and*

$$|\delta x| \;\leq\; s^{-1} \lim_m \frac{2C_K}{1-\tilde{\lambda}}\tilde{\lambda}^{d_{W_m}} \;=\; 0. \tag{146}$$

On that class the ordered memory is exactly zero — the conservation trap of the dynamical dress, now in windowed dress.

Scope 9.7. Neither corollary is claimed outside its stated hypotheses, and nothing is claimed about which regime any given model occupies. Corollary 9.5 additionally requires D18(AD3) and time-independent (K-TAIL) constants; Corollary 9.6 additionally requires (K-Q) at $\varepsilon_Q = 0$, time-uniform data, and D27's own definition of δx.

Idea of proof. For Corollary 9.5: by Theorem 9.3 the difference $\mathcal{Q}_{c_0}\Psi_t - \hat{Q}_{W,c_0}\Psi_t$ is exactly the sum of the tail deviations outside W, and Cauchy–Schwarz on unit Ψ_t bounds its norm below by the modulus of its expectation, which D18(AD3) drives to q_{leg}. The quantitative witness — one magnon over a polarised vacuum — has escaped tail sum saturating at exactly $2s = 1$ at every padding, with the minimal admissible C_K growing like $\tilde{\lambda}^{-vt}$ (numerically $10^{+79.8}$ at $t = 64$). For Corollary 9.6: the exponential estimate (148) applied on both time wings pins the window-charge expectation to q_φ at every time, and δx is by definition $(2s)^{-1}$ times the limit of the difference of those expectations. □

campaign id:	LD-ID
proved in:	`theory/ace-ld.md` step (1.4).(2.7), sub-steps (3.3) and (3.4)
tested by:	Corollary (i)'s quantitative witness re-measured in `theory/verdicts/ace-ld-r3.md` M1(b): $\|(\hat{Q}_{W'} - \hat{Q}_W)\varphi\| = 2s$ at every padding for an escaped leg inside the annulus
reviewed in:	`ace-ld-r3.md` (F1, M1, m1–m3), `ace-ld-r4.md`; mechanism reached independently in `lr-d16-r1.md` M2(b)2

9.3 Exponentially confined tails: the ε-theorem

We now name the confinement hypothesis precisely and state what it buys.

Definition 9.8 ((K-TAIL): confined-core exponential tail decay)**.** A unit vector φ satisfies **(K-TAIL)** if there are a finite interval $K = [\ell, r]$ and constants $C_K < \infty$, $\tilde{\lambda} \in (0, 1)$ such that for all $x, x' \notin K$ lying on the *same* side of K,

$$\left|\omega_\varphi(D_x D_{x'})\right| \;\le\; C_K^2\, \tilde{\lambda}^{\operatorname{dist}(x,K)}\, \tilde{\lambda}^{\operatorname{dist}(x',K)}. \tag{147}$$

Taking $x = x'$ gives the on-site bound $\|D_x\varphi\|^2 \le C_K^2 \tilde{\lambda}^{2\operatorname{dist}(x,K)}$. Conversely, since each D_x is self-adjoint, Cauchy–Schwarz gives $|\omega_\varphi(D_x D_{x'})| = |\langle D_x\varphi, D_{x'}\varphi\rangle| \le \|D_x\varphi\|\,\|D_{x'}\varphi\|$, so the on-site bound implies the full same-side clause. **(K-TAIL) is therefore an on-site tail-decay hypothesis, not a clustering hypothesis**, and a site-by-site numerical profile certifies it in full.

Definition 9.9 ((K-Q): sharp relative charge)**.** A unit vector φ satisfies **(K-Q)** if there are $q_\varphi \in \mathbb{R}$ and $\varepsilon_Q \ge 0$ with $\|\mathcal{Q}_{c_0}\varphi - q_\varphi\varphi\| \le \varepsilon_Q$, where $\mathcal{Q}_{c_0}\varphi = \lim_n \hat{Q}_{W_n,c_0}\varphi$ is the relative-charge limit vector. That limit exists on every (K-TAIL) state and is exhaustion independent — this is a lemma, not part of the hypothesis — and by Theorem 9.3 it equals $2s(X_1 - c_0)$. So (K-Q) is a sharpness hypothesis on the *conserved* charge, and (K-TAIL) says, up to the uniform bound below, that no charge lies outside the core.

Theorem 9.10 (Exponential window-charge estimate on confined-tail states)**.** [PROVED, CONDITIONAL] *Proved only as the following conditional implication. Assume H-MQG(1)–(2); a finite core $K = [\ell, r]$; a finite window $W = [a, b] \supseteq K$ with cut $c_0 \in W$ and padding $d_W \ge 1$ as in Definition 9.2; and a unit vector φ satisfying (K-TAIL) with constants $C_K, \tilde{\lambda}$ and (K-Q) with data q_φ, ε_Q. Then*

$$\|(\hat{Q}_{W,c_0} - q_\varphi)\varphi\| \;\le\; \varepsilon_Q + \frac{2C_K}{1-\tilde{\lambda}}\,\tilde{\lambda}^{\,d_W}, \tag{148}$$

and consequently $\operatorname{dist}\big(q_\varphi,\ \kappa_{W,c_0} + \mathbb{Z}\big)$ *is bounded by the same quantity, where $\kappa_{W,c_0} + \mathbb{Z}$ is the coset containing the spectrum of the window charge (the finite memory-index theorem of Section 8).*

Two consequences of (K-TAIL) are forced, and each is itself proved.

Corollary 9.11 (Forced consequences of confined tails)**.** [PROVED, CONDITIONAL] *Under (K-TAIL):*

(i) In the kink class of D9(a), (K-TAIL) is equivalent to the statement that no charge lies outside the core, quantitatively

$$\|\mathcal{Q}_{c_0}\varphi - \hat{Q}_{W,c_0}\varphi\| \;\le\; \frac{2C_K}{1-\tilde{\lambda}}\tilde{\lambda}^{\,d_W} \qquad \textit{for every finite } W \supseteq K \textit{ with } c_0 \in W, \tag{149}$$

and it forces both tail vacua to have sharp on-site charge*: $\omega_\gamma\big((S^z_x - s_\gamma)^2\big) = 0$ at every site, equivalently $\omega_\gamma\big(\Pi_x^{(s_\gamma)}\big) = 1$ for the on-site spectral projection of S^z_x. Hence $\pm s \in \operatorname{spec} S^z$; in particular a spin-1 chain with calibration $s = 1/2$ admits no (K-TAIL) kink state. See Theorem 9.15.*

(ii) With time-independent data $(K, C_K, \tilde{\lambda})$, (K-TAIL) is jointly unsatisfiable *with the clause of D18(AD3) that the free leg charge leaves the window. No state carrying an escaping leg of nonzero charge satisfies it with time-independent constants; $q_L \ne q_T$ is unavailable within one conserved-charge sector of D18; and on states that do satisfy it with $\varepsilon_Q = 0$ and time-uniform data the ordered wall displacement is $\delta x = 0$.*

Scope 9.12*.* The following are *not* claimed. That the ε-weakened local-decay statement supplies the cross-term step of the wave-operator construction of Section 9.10 — it does not, because the hypothesis fails on that entire class; the earlier claim to the contrary is retracted. The exact fixed-window local-decay hypothesis (AD3-ex) of Conjecture 9.39. The weak-$*$ clause of D18(AD3) or the reduction clause of the spectral memory-index statement. The $\chi = 1$ product form of the tail vacua under H-MQG alone (see Refuted claim 9.18). Any statement about generic injective matrix product state vacua — sharp on-site charge excludes every vacuum with nonzero on-site charge variance, and that is the scope boundary of this theorem. Any infinite-volume proof of (K-TAIL) or (K-Q) on any model. The existence of a sector-wide conserved charge operator: only the limit vector on a (K-TAIL) state is constructed, the strong local-additivity claim is refuted and is not contradicted, and the infinite-volume route to the time-uniformity of (K-Q) remains a conjecture.

Idea of proof. Two lemmas. The first is a window-difference identity: for $W = [a, b] \subseteq W' = [a', b']$ both containing the cut,

$$\hat{Q}_{W',c_0} - \hat{Q}_{W,c_0} \;=\; \sum_{x=a'}^{a-1} \big(S^z_x - s\big) + \sum_{x=b+1}^{b'} \big(S^z_x + s\big), \tag{150}$$

so the difference is exactly the sum of the on-site deviations over the annulus, the cut dependence cancelling because c_0 enters both terms with the same coefficient. The second is a tail-sum estimate: for a finite one-sided set F at distance d from K, expanding the square and using (K-TAIL) on every pair gives $\|\sum_{x\in F} D_x\varphi\| \le C_K\tilde{\lambda}^d/(1-\tilde{\lambda})$, because the distinct sites of a one-sided F have pairwise distinct integer distances to K, all at least d, so the geometric series sums. Applying this to the two annuli in (150) shows that $\{\hat{Q}_{W_n,c_0}\varphi\}$ is Cauchy along any exhaustion, that interleaving two exhaustions identifies the limits, and that letting $W' \uparrow \mathbb{Z}$ gives (149). The theorem is then the triangle inequality on $(\hat{Q}_{W,c_0} - q_\varphi)\varphi = (\hat{Q}_{W,c_0}\varphi - \mathcal{Q}_{c_0}\varphi) + (\mathcal{Q}_{c_0}\varphi - q_\varphi\varphi)$, and the coset location follows from $\|(\hat{Q}_{W,c_0} - q_\varphi)\varphi\|^2 = \int |\lambda - q_\varphi|^2\, d\mu_\varphi(\lambda) \ge \operatorname{dist}(q_\varphi, \operatorname{spec}\hat{Q}_{W,c_0})^2$ together with spectral permanence. □

Remark 9.13 (Why the time-uniform hypothesis is honest)*.* (K-Q) constrains the *conserved* charge, and that is what makes the time-uniform version of Theorem 9.10 an assumption about the initial data rather than a disguised assumption of the conclusion. If the realization carries a self-adjoint $\widehat{\mathcal{Q}}$ commuting with the dynamics and agreeing with the limit vector on (K-TAIL) states, then $\|(\widehat{\mathcal{Q}} - q)\varphi_t\| = \|(\widehat{\mathcal{Q}} - q)\varphi_0\|$, so (K-Q) at $t = 0$ propagates with the same q and ε_Q. The window charge $\hat{Q}_{W,c_0}$ is *not* conserved — that is the whole content of the conservation trap — and no such propagation is claimed for it: the estimate converts conserved-charge sharpness plus instantaneous tail decay into a window statement separately at each time.

Remark 9.14 (The hypothesis class is inhabited)*.* The exactly solvable easy-axis kink model of Section 6 sits inside the class. Its exact zero-energy sector states satisfy (K-Q) exactly, and they satisfy (K-TAIL) at rate

$$\tilde{\lambda} \;=\; q \;=\; \Delta - \sqrt{\Delta^2 - 1}, \tag{151}$$

independently verified by full 2^L exact diagonalisation at $L = 12$: the measured on-site defects are $\|D_x\varphi\| = 4.356 \times 10^{-2},\ 9.092 \times 10^{-3},\ 1.898 \times 10^{-3},\ 3.960 \times 10^{-4},\ 8.266 \times 10^{-5}$, with per-site ratios 0.20871 on *both* sides against $q = 0.208712$, and $C_K = 0.2087$, in the $\mathbb{Z}_2$ image of the frozen orientation. The checker's conclusion gates independently certify the $\tilde{\lambda}^{d_W}$ rate of (148), with defects from 4.45×10^{-2} down to 4.05×10^{-4} and ratios 0.2089, 0.2087, 0.2083 against the same q.

campaign id:	ACE-LD-eps
proved in:	`theory/ace-ld.md` step (1.4) (THEOREM ACE-LD-ε), with lemmas at (1.4).(2.1)–(2.3) and the forced consequences at (1.4).(2.7), (2.9)
tested by:	`theory/checks/ace_ld_check.py` gate LD-C7 (green exit 0; `--red` exit 1, RED-OK 23/23). Gates (b)–(c) certify the conclusion's $\tilde{\lambda}^{d_W}$ rate; gate (e) certifies the hypothesis (K-TAIL) itself site by site, and by Cauchy–Schwarz on the self-adjoint D_x the on-site clause is equivalent to the full same-side clause, so (e) certifies (K-TAIL) in full. No gate bears on (K-Q), on the infinite-volume statement, or on the sharp-charge corollary, which is exact algebra
reviewed in:	`ace-ld-r3.md` §8(A) (adjudicated scoping), `ace-ld-r2.md`, `ace-ld-r4.md` §0(iv)

9.4 Sharp on-site charge, and the refuted product-vacuum reading

The forced consequence (i) above deserves its own statement, because it is the cleanest description of exactly which corner of the theory the ε-theorem occupies — and because the natural strengthening of it is false.

Theorem 9.15 (Confined tails force zero on-site charge variance)**.** [PROVED] *Assume H-MQG(1)–(2), and let φ be a unit vector whose state ω_φ lies in the kink sector $\mathcal{K}_{\alpha\beta}$ of D9(a) and satisfies (K-TAIL) with core K and constants $C_K, \tilde{\lambda}$. Then both tail vacua have* sharp on-site charge*: for $\gamma \in \{\alpha, \beta\}$ and every site x,*

$$\omega_\gamma\big((S^z_x - s_\gamma)^2\big) = 0, \qquad \textit{equivalently} \qquad \omega_\gamma\big(\Pi_x^{(s_\gamma)}\big) = 1, \tag{152}$$

where $\Pi_x^{(m)}$ is the on-site spectral projection of S^z_x at eigenvalue m. Hence the tail density is an on-site eigenvalue*: $s_\alpha = +s \in \operatorname{spec} S^z$ and $s_\beta = -s \in \operatorname{spec} S^z$.*

Corollary 9.16 (An arithmetic exclusion)**.** [PROVED] *A spin-*1 *chain with calibration $s = 1/2$ admits* no *(K-TAIL) kink state, since $\pm 1/2 \notin \operatorname{spec} S^z = \{-1, 0, 1\}$.*

Scope 9.17. Sharp on-site charge means zero on-site charge variance and therefore excludes every vacuum with nonzero on-site charge variance, in particular generic injective matrix product state vacua. It is *not* claimed that the tail vacua are fully polarised S^z-product states; see Refuted claim 9.18. That stronger form needs s_γ to be a *simple* eigenvalue of S^z — true in the standard spin-S register with on-site dimension $d = 2s+1$, and true in the exactly solvable model, but not implied by H-MQG(1)–(2), by covariance, or by D26(INT), all of which admit degenerate on-site charge eigenspaces. The gloss $d = 2s+1$ in the notation table is explicitly fenced as the fully polarised special case and is not a constraint.

Idea of proof. Every input is named. D9(a) itself supplies the weak-$*$ tail relaxation $\omega_\varphi(\tau_x(O)) \to \omega_\gamma(O)$ for local O, so no extra relaxation hypothesis is needed. (K-TAIL) at $x = x'$ gives $\omega_\varphi(D_x^2) \le C_K^2 \tilde{\lambda}^{2\,\mathrm{dist}(x,K)} \to 0$; translating along the tail and passing to the limit gives $\omega_\gamma\big((S^z_x - s_\gamma)^2\big) = 0$. Cauchy–Schwarz on the self-adjoint $S^z_x - s_\gamma$ then pins the on-site spectral weights, and translation invariance of ω_γ makes the conclusion site independent. □

Refuted claim 9.18 (The tail vacua are fully polarised product states)**.** [REFUTED] Withdrawn statement. *That Theorem 9.15 concludes further $\omega_\gamma = \bigotimes_x |s_\gamma\rangle\langle s_\gamma|$, a bond-dimension-one S^z-product state.*

Why it fails. *Sharp on-site charge names a unique on-site* vector *only if* $\dim\ker(S^z - s_\gamma) = 1$, *and no hypothesis in force supplies that simplicity. Explicit counterexample: take the on-site space $\mathbb{C}^2 \otimes \mathbb{C}^3$ with $S^z := \sigma^z/2 \otimes \mathbb{1}_3$, so that* $\operatorname{spec} S^z = \{\pm 1/2\}$ *with each eigenvalue of multiplicity* 3 *and $e^{2\pi i S^z} = -\mathbb{1}$, whence D26(INT) holds with $\kappa = 1/2$. Put*

$$\omega_\alpha = \bigotimes_x \big(|\uparrow\rangle\langle\uparrow| \otimes \varrho_{\mathrm{AKLT}}\big), \qquad \omega_\beta = \bigotimes_x \big(|\downarrow\rangle\langle\downarrow| \otimes \varrho_{\mathrm{AKLT}}\big). \tag{153}$$

These are injective matrix product states of bond dimension $\chi = 2$ (bond Schmidt spectrum $(1/2, 1/2)$, transfer gap $1/3$), covariant for $G = U(1) \rtimes \mathbb{Z}_2$, with $\omega_\alpha(S^z) = +1/2 = s = -\omega_\beta(S^z)$. The sharp domain wall in the $\mathbb{C}^2$ factor lies in $\mathcal{K}_{\alpha\beta}$ and has $D_x\varphi = 0$ at every *site, so it satisfies (K-TAIL) with $C_K = 0$ — while its tail vacua are not product states.*

Surviving weaker statement. *Theorem 9.15: zero on-site charge variance, hence $\pm s \in \operatorname{spec} S^z$. When s_γ* is *a simple eigenvalue of S^z — the standard spin-S register, and the exactly solvable model of Section 6 — the sharp-charge conclusion does give $\omega_\gamma = \bigotimes_x |s_\gamma\rangle\langle s_\gamma|$ and $\chi = 1$; that conditional clause survives as a corollary, not as the theorem.*

campaign id:	ACE-LD-sharp
proved in:	`theory/ace-ld.md` step (1.4).(2.9)(b)
tested by:	`theory/checks/ace_ld_check.py` gate LD-C7(e) certifies the *hypothesis* (K-TAIL) on the exactly solvable instance (per-site ratios 0.20871 on both sides, $C_K = 0.209$, red-armed by `--red-c7-orientation` at $C_K = 2525$); the conclusion is exact algebra and needs no gate. The counterexample to the product form is verified numerically
reviewed in:	`ace-ld-r3.md` F1 and F1(b) (the refutation, adopted), `ace-ld-r4.md` (restatement)

9.5 The abstract mechanism: spectral diagonality from first-moment escape

Underneath the model-specific analysis sits a short, entirely abstract lemma. It is worth isolating because it shows exactly what local decay of the channel charges buys: nothing about the model, but a clean identification of channel projections with spectral projections of the window charge, with a constant uniform in time.

Definition 9.19 (The abstract escape package (A1)–(A5))**.** (A1) A Hilbert space $\mathcal{H}$, a strongly continuous unitary group e^{-itH}, a unit vector Ψ, and $\Psi_t := e^{-itH}\Psi$.

(A2) A bounded self-adjoint operator $\hat{Q}$ on $\mathcal{H}$, time independent (Schrödinger picture), whose spectrum is a *finite* set contained in a single coset $\kappa + \mathbb{Z}$, $\kappa \in [0, 1)$. In the instance $\hat{Q} = \hat{Q}_{W,c_0}$ at a fixed finite window, and the coset containment is the finite memory-index theorem of Section 8, cited and not re-proved.

(A3) A finite family $\{P_{\mathrm{ch}}\}_{\mathrm{ch}=1}^n$ of mutually orthogonal projections, each commuting with e^{-itH}, with $\sum_{\mathrm{ch}} P_{\mathrm{ch}}\Psi = \Psi$.

(A4) Real *channel charges* $\{q_{\mathrm{ch}}\}_{\mathrm{ch}=1}^n$ and first-moment escape: $\varepsilon_{\mathrm{ch}}(t) := \|(\hat{Q} - q_{\mathrm{ch}})P_{\mathrm{ch}}\Psi_t\| \to 0$ as $t \to +\infty$ for every channel.

(A5) The q_{ch} are pairwise distinct.

Lemma 9.20 (Channel projections become spectral projections)**.** [PROVED] *Assume (A1)–(A5). With $E(\cdot)$ the spectral measure of $\hat{Q}$ and $d_{\mathrm{ch}} := \operatorname{dist}\big(q_{\mathrm{ch}}, \operatorname{spec}\hat{Q} \setminus \{q_{\mathrm{ch}}\}\big) > 0$,*

$$\|E(\{q_{\mathrm{ch}}\})\Psi_t - P_{\mathrm{ch}}\Psi_t\| \;\le\; \sum_{\mathrm{ch}'=1}^{n} d_{\mathrm{ch}'}^{-1}\varepsilon_{\mathrm{ch}'}(t) \;\longrightarrow\; 0 \qquad \textit{for every channel,} \tag{154}$$

the constant $\sum_{\mathrm{ch}'} d_{\mathrm{ch}'}^{-1}$ being independent of t. If moreover $q_{\mathrm{ch}} \in \kappa + \mathbb{Z}$ then $d_{\mathrm{ch}} \ge 1$, so under (A2) the constant is at most n for coset-valued charges. The lemma holds for N channels and a general on-site spin.

Lemma 9.21 (Distinct channel charges are necessary). [PROVED] *Assume (A1)–(A4) with $n = 2$ and $q_1 = q_2 =: q$, and $w_i := \liminf_{t\to+\infty}\|P_i\Psi_t\| > 0$ for* both *$i = 1, 2$. Then for* every *$q' \in \mathbb{R}$,* $\liminf_{t\to+\infty}\|E(\{q'\})\Psi_t - P_1\Psi_t\| > 0$*; the diagonality display fails for channel 1 for every charge assignment, and by symmetry for channel 2. Hypothesis (A5) is therefore not an artefact of the proof.*

Scope 9.22. Nothing is claimed about any concrete model, no instantiation of (A4) is supplied, and no sharpness of the constant beyond the single spectral gap is asserted.

Idea of proof. The engine is a one-line gap estimate: if $\hat{Q}$ has finite spectrum S and $d_q := \operatorname{dist}(q, S \setminus \{q\}) > 0$, then $\hat{Q}$ restricted to $\operatorname{ran}(1 - E(\{q\}))$ has spectrum $S \setminus \{q\}$, so $(\hat{Q} - q)$ is invertible there with inverse norm d_q^{-1}, giving

$$\|(1 - E(\{q\}))\phi\| \;\le\; d_q^{-1}\|(\hat{Q} - q)\phi\| \qquad \text{for every } \phi \in \mathcal{H}. \tag{155}$$

Because the projections resolve Ψ and commute with the dynamics, they resolve Ψ_t for all t, and $E(\{q_{\mathrm{ch}}\})\Psi_t - P_{\mathrm{ch}}\Psi_t = -(1 - E(\{q_{\mathrm{ch}}\}))P_{\mathrm{ch}}\Psi_t + \sum_{\mathrm{ch}'\neq\mathrm{ch}} E(\{q_{\mathrm{ch}}\})P_{\mathrm{ch}'}\Psi_t$. The diagonal term is bounded by (155) directly. For a cross term, distinctness of the charges makes the two spectral projections orthogonal, so $E(\{q_{\mathrm{ch}}\})P_{\mathrm{ch}'}\Psi_t = E(\{q_{\mathrm{ch}}\})(1-E(\{q_{\mathrm{ch}'}\}))P_{\mathrm{ch}'}\Psi_t$ and (155) applies again at $q_{\mathrm{ch}'}$. Time uniformity is free: $\operatorname{spec}\hat{Q}$ is a property of the fixed Schrödinger-picture operator, and only Ψ_t moves. Two distinct points of one coset of $\mathbb{Z}$ differ by a nonzero integer, which is where the coset containment enters. □

campaign id:	ACE-LD-abs (with ACE-LD-nec)
proved in:	`theory/ace-ld.md` steps (1.1)–(1.3): the gap estimate at (1.1), the assembly at (1.2), the necessity of distinct charges at (1.3)
tested by:	`theory/checks/ace_ld_check.py` gates LD-C1–C4 (green exit 0; `--red` exit 1, 23 mutation modes, exit paths logged; every LD-C1 sub-gate armed, including orthogonality and resolution)
reviewed in:	`ace-ld-r2.md` (M4/m4 fixed), `ace-ld-r3.md`, `ace-ld-r4.md`

9.6 Energy bounds the number of phase boundaries, not their length

We turn to the complementary corner: the easy-axis kink model of Section 6, where one would like to *prove* local relaxation rather than assume it. There is an unconditional operator inequality, and it is instructive precisely because of what it fails to give.

Theorem 9.23 (Energy dominates the domain-wall number). [PROVED] *In the easy-axis model, in the two-site basis $(|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle)$, the bond term $h^{\mathrm{XXZ}}_{x,x+1}$ annihilates $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ and equals $(J\Delta/2)I - (J/2)\sigma^x$ on the domain-wall block, with eigenvalues $(J/2)(\Delta \mp 1)$. Hence, with P^{DW}_x the projection onto* $\operatorname{span}\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}_{x,x+1}$ *and $N_{\mathrm{DW}} := \sum_x P^{\mathrm{DW}}_x$ the domain-wall* number*,*

$$h^{\mathrm{XXZ}}_{x,x+1} \;\succeq\; \tfrac{J}{2}(\Delta - 1)P^{\mathrm{DW}}_x, \qquad H_{\mathrm{XXZ}} \;\succeq\; \tfrac{J}{2}(\Delta - 1)N_{\mathrm{DW}} \;\succeq\; 0, \tag{156}$$

the sum being a quadratic-form inequality on the finite-deviation core. Consequently, for any state Ψ whose energy $E_0 := \langle\Psi, H_{\mathrm{XXZ}}\Psi\rangle$ is finite and is conserved by the dynamics*,*

$$\langle\Psi, \alpha_t(N_{\mathrm{DW}})\Psi\rangle \;\le\; \frac{2E_0}{J(\Delta - 1)} \qquad \textit{for every } t. \tag{157}$$

Scope 9.24. This bounds the **number** of phase boundaries uniformly in time and **nothing about their length**: in one dimension a wrong-phase block of any length carries the same $O(1)$ Ising cost. It therefore does *not* give clause (LR3), and it gives neither clause of the hypothesis (NR) of Definition 9.25. Finiteness and conservation of E_0 are stated as hypotheses on the selected state, not derived: the proof step asserts both without proof, and the operator-hygiene paragraph of the shard covers only the diagonal operators, not H_{XXZ}. Both are expected to hold on an ℓ^1 packet of D17 — $P_x^{\mathrm{DW}} \preceq n_x + n_{x+1}$ and $0 \preceq h_x \preceq (J/2)(\Delta+1)P_x^{\mathrm{DW}}$ give $E_0 < \infty$ from the ℓ^1 clause — but that derivation is not in the shard. Finally, the bond inequality and its sum are exact in infinite volume *and* on any finite open chain; a finite-chain certificate must use the H_{XXZ} propagator, because H_{kink} and H_{XXZ} generate the same derivation only in infinite volume and differ at finite N by the boundary term $\frac{J}{2}\sqrt{\Delta^2-1}\,(S_1^z - S_N^z)$.

Idea of proof. The bond term is block diagonal: it kills the two aligned configurations, and on the two-dimensional domain-wall block it is $(J\Delta/2)I - (J/2)\sigma^x$, whose eigenvalues are $(J/2)(\Delta \mp 1)$. For $\Delta > 1$ the smaller one is positive, so the bond term dominates $(J/2)(\Delta-1)$ times the domain-wall projection. Summing bondwise gives (156), and taking the expectation in a state of conserved finite energy gives (157). As a sanity check, the bare kink has $E_0 = C_K = \frac{J}{2}\sqrt{\Delta^2-1}$, so $\langle N_{\mathrm{DW}}\rangle \le \sqrt{(\Delta+1)/(\Delta-1)}$, which tends to 1 as $\Delta \to \infty$ (a single sharp wall) and diverges as $\Delta \to 1^+$ (the kink delocalises) — both as they must. $\square$

campaign id:	LR-D16-EDW
proved in:	`theory/lr-d16.md` step (1.5).(2.6), named computation LRD-EDW
tested by:	`theory/checks/lr_d16_check.py` gate C4: (a) the bond inequality, minimum eigenvalue -0.0; (b) H_{XXZ} energy drift 1.11e−14 and $\max\langle N_{\mathrm{DW}}\rangle = 3.053 \le 4.753$; (c) finite-chain calibration $\langle K\vert H_{\mathrm{XXZ}}\vert K\rangle = 1.1456439081$ against $C_K = 1.1456439237$. Green exit 0; the mutations `--red c4-wrong-gap`, `--red c4-kink-propagator` (caught by the energy-drift gate at 1.007) and `--red c4-sharp-calibration` each exit 1 on exactly that row
reviewed in:	`lr-d16-r1.md`, `lr-d16-r2.md`

9.7 Tightness transfer: from bounded escaped content to first-moment tightness

The gap left open by Theorem 9.23 — block length rather than block number — is named, once, as a hypothesis, and clause (LR3) is derived from it.

Definition 9.25 ((NR): no unbounded escaped content, two clauses)**.** Let W_1 be the smallest window of the padded exhaustion, Ψ the selected vector, $E_{W,t}$ the spectral resolution of the Heisenberg translate of $\hat{Q}_{W,c_0}$, and $\mathcal{N}_W$ as in (141). Then (NR) consists of

$$\sup_{t\in\mathbb{R}} \langle \Psi, \alpha_t(\mathcal{N}_{W_1})^2 \Psi\rangle < \infty, \tag{158}$$

$$\sup_m \sup_{t_- < t_+} \sum_q \|\alpha_{t_+}(\mathcal{N}_{W_1}) E_{W_m,t_-}(\{q\})\Psi\|^2 < \infty, \tag{159}$$

and S_{NR} denotes twice the supremum of the sum of the two quantities.

Theorem 9.26 (Bounded escaped content implies first-moment tightness)**.** [SKETCH] *In the standing register (H1)–(H6) of the easy-axis setting, with clause (LR1), the two-clause hypothesis (NR) implies clause (LR3) for the same padded exhaustion and the same fixed-window laws, with the explicit tail bound*

$$\sup_m \sum_{|\nu|>M} (1+|\nu|)\, p_{W_m}(\nu) \;\le\; \frac{2S_{\mathrm{NR}}}{M} \;\longrightarrow\; 0. \tag{160}$$

Scope 9.27. The status is that of an exact conditional proof with an independent critic re-derivation, whose promotion in the claims graph is still pending; the implication itself is what is claimed, not its hypothesis. Specifically, the following are *not* claimed: (NR) itself; clause (LR2); the optional weak-convergence clause of (LR3); and either clause of (NR) as a consequence of Theorem 9.23, which controls wall number, not block length.

What is proved, and how. Four exact steps and one hypothesis. (1) An exact second-moment identity: with $\Delta\hat{Q}_W := \hat{Q}_{W,c_0}(t_+) - \hat{Q}_{W,c_0}(t_-)$ and $\mathcal{D}_{W,t_-}$ the pinching by the spectral projections at t_-, $\sum_\nu \nu^2 p_{W;t_-,t_+}(\nu) = \langle\Psi, \mathcal{D}_{W,t_-}\big((\Delta\hat{Q}_W)^2\big)\Psi\rangle$. (2) Chebyshev: for any probability p on $\mathbb{Z}$ and $M \geq 1$, $\sum_{|\nu|>M}(1+|\nu|)p(\nu) \leq 2\sum_{|\nu|>M}|\nu|p(\nu) \leq (2/M)\sum_\nu \nu^2 p(\nu)$, so a uniform second moment is exactly first-moment tightness. (3) A splitting onto the two window edges: $\Delta\hat{Q}_W$ equals the difference of the two Heisenberg translates of the complementary charge (the scalar cancels), and $2X^2 + 2Y^2 - (X-Y)^2 \succeq 0$ with positivity of the pinching gives

$$\sum_\nu \nu^2 p_{W;t_-,t_+} \;\leq\; 2\langle\Psi, \alpha_{t_-}(\hat{Q}_{W^c})^2\Psi\rangle + 2\sum_q \|\alpha_{t_+}(\hat{Q}_{W^c})\varphi_q\|^2. \tag{161}$$

(4) A monotonicity lemma: $\mathcal{N}_{W'} \leq \mathcal{N}_W$ for $W \subseteq W'$ and $|\hat{Q}_{W^c,c_0}| \leq \mathcal{N}_W$, so both terms may be majorised with the *fixed* observable $\mathcal{N}_{W_1}$. That is (NR). The remaining m-dependence sits in the dephased state, not in the observable — positivity of a pinching does not make its expectation of a non-commuting positive observable monotone — which is exactly why the second clause of (NR) retains its supremum over m. Convexity of the double Cesàro average and Fatou's lemma transport the uniform bound to the fixed-window limit laws.

What is missing. (NR) itself. It is genuine and non-circular: it controls supremum-in-time second moments of the state and of the dephased states, whereas (LR3) controls only tails of double-Cesàro limit laws, and (LR3) does not recover either clause. The honest reading is that Theorem 9.23 bounds the number of wrong-phase blocks unconditionally, (NR) is the statement that their length also stays bounded, and this theorem is the bridge from the latter to (LR3). □

Remark 9.28 (Why (NR) is the right hypothesis here). In the exactly solvable model the closed span of the zero-energy kink family is annihilated by the kink Hamiltonian, so the dynamics acts trivially on it up to a constant phase: every state in that closed span is exactly stationary. This flatness is what makes an assumption about escaped content, rather than about transport, the natural residual input. Exhaustion of the kernel by that family is a separate conjecture and is not used here.

campaign id:	LR-D16-NR
proved in:	`theory/lr-d16.md` step (1.5).(2.1)–(2.7), THEOREM LRD-3; independently re-derived in `theory/verdicts/lr-d16-r2.md` §4 item 3
tested by:	`theory/checks/lr_d16_check.py` gates C3(a)–C3(c) test finite identities and majorants only; no finite checker bears on the asymptotic implication
reviewed in:	`blitz-m-esc-nr-r1.md` (0F/0M/0m), `lr-d16-r1.md`, `lr-d16-r2.md`

9.8 The escape dichotomy: mean transport destroys tightness

The opposite hypothesis — that the kink's mean position transits the window at a rate proportional to the window size — is incompatible with local relaxation, and the argument needs remarkably little.

Definition 9.29 ((M-ESC): mean tail transport)**.** In the setting of D27 (a D17 vector Ψ, a cut c_0, a padded exhaustion $W_m = [a_m, b_m] \uparrow \mathbb{Z}$ containing c_0) with the fixed-window Cesàro states $\omega^{\pm}_{W_m}$, the hypothesis **(M-ESC)** is

$$\theta_{\mathrm{tr}} \;:=\; \liminf_{m} \frac{\left|\omega^{+}_{W_m}(\mathfrak{X}_{W_m}) - \omega^{-}_{W_m}(\mathfrak{X}_{W_m})\right|}{|W_m|} \;>\; 0. \tag{162}$$

Theorem 9.30 (Mean escape is incompatible with first-moment tightness)**.** [PROVED, CONDITIONAL] *Assume H-MQG(1)–(3), D26(INT), and the D27 setting with clauses (LR1)–(LR2) but* not *(LR3); assume (M-ESC). Then*

(a) $\left|\sum_\nu \nu\, p_{W_m}(\nu)\right| \geq 2s\,\theta_{\mathrm{tr}}|W_m|(1 - o(1)) \to \infty$*;*

(b) clause (LR3) ***fails****:* $\sup_m \sum_{|\nu|>M}(1 + |\nu|)p_{W_m}(\nu) = \infty$ *for every M;*

(c) the ordered wall expectation δx of D27 is undefined along that exhaustion, its defining differences diverging linearly at rate at least θ_{tr} per window site.

Positive form*, which is not a contrapositive: the same step gives directly that on any state satisfying (LR) the mean wall transport is* uniformly bounded*,*

$$\sup_m \left|\omega^{+}_{W_m}(\mathfrak{X}_{W_m}) - \omega^{-}_{W_m}(\mathfrak{X}_{W_m})\right| \;\leq\; \frac{M_0 + 1}{2s}, \tag{163}$$

where M_0 is any threshold with $\sup_m \sum_{|\nu|>M_0}(1 + |\nu|)p_{W_m}(\nu) \leq 1$*; in particular* $\theta_{\mathrm{tr}} = 0$*.*

Scope 9.31. **No model or state realising (M-ESC) is exhibited anywhere in this corpus** — independently confirmed by the parallel lane. The theorem therefore constrains a hypothesis class, and until a witness exists it makes no clause of any class theorem necessary.

Disclosure: boundedness of the mean transport under (LR3) is the quantitative form of D27's own existence corollary, already frozen in D27's definition paragraph. The genuine addition is the M_0 tightness route, and it is a strict strengthening: D27's corollary delivers convergence along the full sequence only under D27's optional convenience clause, and otherwise only along an (LR3) subsequence, whereas (163) is a supremum over *all* m, with no subsequence, no convenience clause, and no passage through weak convergence.

Nothing is claimed about the exactly solvable model or any confined-kink class. There $\theta_{\mathrm{tr}} = 0$, but only on named conditionals: every state in the closed span of the zero-energy kink family is exactly stationary (proved), while exhaustion of the kernel by that family is a conjecture; and the kink–magnon bound $|\delta x| \leq \langle N_T \rangle / s$ is the conclusion of the general memory-quantization theorem, which is proved *conditional* on the full asymptotic-decay package D18(AD1)–(AD4) for that vector — open for this model, since (AD3-ex) is a conjecture (Conjecture 9.39). Note that D18(AD3) is jointly unsatisfiable with (K-TAIL) (Corollary 9.11(ii)), so this $\theta_{\mathrm{tr}} = 0$ fence and the ε-theorem of Section 9.3 concern *disjoint* state classes: a kink–magnon ℓ^1 packet under the asymptotic-decay package, versus a magnon-free confined-tail sector state. Also not claimed: any statement about the packet class of D28 as such, which contains $\theta_{\mathrm{tr}} = 0$ members; mass-defectiveness of weak limits (first-moment divergence only); and the two-atom law as a theorem — it is numerically exhibited only.

Idea of proof. Three ingredients, all cited rather than re-proved. First, each p_{W_m} is a probability on $\mathbb{Z}$: at fixed m the law is supported on finitely many ν, because $|\nu|$ is bounded by the diameter of the finite set spec $\hat{Q}_{W_m,c_0}$, so no mass escapes in the Cesàro limit. Second, the exact first-moment identity, valid at every fixed m,

$$\sum_{\nu} \nu\, p_{W_m}(\nu) \;=\; -2s\big[\omega^{+}_{W_m}(\mathfrak{X}_{W_m}) - \omega^{-}_{W_m}(\mathfrak{X}_{W_m})\big]. \tag{164}$$

Third, tightness bounds the first absolute moment: if (LR3) held, pick M_0 with $\sup_m \sum_{|\nu|>M_0}(1+|\nu|)p_{W_m}(\nu) \le 1$; splitting the sum at M_0 and using total mass 1 gives $\sum_\nu |\nu| p_{W_m}(\nu) \le M_0 + 1$ for every m. Combining (164) with (M-ESC) makes the left side diverge linearly in $|W_m|$, contradicting that uniform bound; the quantitative form $\sum_{|\nu|>M}(1+|\nu|)p_{W_m}(\nu) \ge \big|\sum_\nu \nu p_{W_m}(\nu)\big| - M$ then gives (b), and (c) is D27's definition of δx read against (a). Reading the third ingredient forwards instead of backwards gives (163).

Non-circularity matters here and is checked: the first-moment identity is consumed at a step whose own leaf justifications name only (LR1), (LR2), and the exact finite-time mean formula — not the (LR3)-conditional statement of the spectral memory-index theorem — so the proof by contradiction against (LR3) does not assume what it refutes.

Finally, the mechanism. If in the incoming Cesàro states the window lies entirely in one tail in the mean, while in the outgoing states the kink has transited beyond W_m with weight p_{tr} and stayed with weight $1 - p_{\mathrm{tr}}$, then the definition $\mathfrak{X}_W = a - 1 + (1/2s)\sum_{x\in W}(S^z_x + s)$ gives $\omega^{-}_{W_m}(\mathfrak{X}_{W_m}) = a_m - 1 + o(|W_m|)$ and $\omega^{+}_{W_m}(\mathfrak{X}_{W_m}) = a_m - 1 + p_{\mathrm{tr}}|W_m| + o(|W_m|)$, so $\theta_{\mathrm{tr}} = p_{\mathrm{tr}}$. Only the *mean* tail densities enter: no variance hypothesis, no channel projections, no wave operators, no sign-definite velocity classes, no incoming-concentration step. □

Remark 9.32 (The dichotomy, and what it costs the memory ledger)*.* Taken together, Section 9.3 and Theorem 9.30 partition the reach of the *unsubtracted* window charge. On confined-charge sharp-tail states, where (K-TAIL) and (K-Q) hold and $\theta_{\mathrm{tr}} = 0$, the ε-version is available and Corollary 9.6 makes the ordered memory exactly zero. On mean-transiting states, where $\theta_{\mathrm{tr}} > 0$, clause (LR3) fails and no ordered outcome measure exists. Neither side of the dichotomy carries a nonzero ordered memory for the unsubtracted charge. A two-atom ledger with $\delta x \neq 0$ therefore requires a *subtracted* or co-moving observable — the leg subtraction of the conservation-trap paragraph, a definition-level move which is proposed but not made here.

The two-atom shape itself is real as a mechanism, and is recorded as evidence for the arithmetic $\theta_{\mathrm{tr}} = p_{\mathrm{tr}}$ rather than re-claimed as a theorem: a critic reproduced it to four digits on a scattering model (weights 0.5567 and 0.4433, incoming-concentration defect 2.4×10^{-9}, tail $(1+513)\times 0.4433 = 227.86$ against a measured 227.867), and a checker gate re-certifies the configuration on a barrier model.

campaign id:	ACE-LD-obst-prime
proved in:	`theory/ace-ld.md` step (1.5) (PROPOSITION ACE-LD-obst′), with the mechanism remark at (1.5).(2.5) and the fences at (1.5).(2.6)
tested by:	`theory/checks/ace_ld_check.py` gate LD-C5 (incoming concentration $\le 3.2\times10^{-6}$ at every window, and the exact two-atom support $\{\nu = 0,\ \nu = -\lvert W\rvert\}$ including the $\nu = 0$ atom, with weights matching independently measured $\lvert r\rvert^2 = 0.2704$, $\lvert t\rvert^2 = 0.7296$); gate LD-C6 (bounded-transport contrast at protocol time $t_+ = 4$: mass at $\lvert\nu\rvert > 3$ below 9.1×10^{-13}, first moment ≤ 0.453, with the bound 1.2 held across a recorded sweep $t_+ \in \{4, 20, 40, 200\}$ whose maximum is 1.0712; red-armed by `--red-c6-moving` at $12.4024 > 1.2$ and `--red-c6-weaktransit` at mass 3.742×10^{-2}). **Neither certificate exhibits an (M-ESC) state, and no gate bears on claims (a)–(c), which are proved, not tested**; the residual caveat is that the sweep gate itself is reached by no registered mutation and is green-side evidence only
reviewed in:	`ace-ld-r3.md` (M2 adopted verbatim, M3), `ace-ld-r4.md` §0(ii),(iv) (M3 repaired, M1 residual caveat), `ace-ld-r2.md`; no-witness confirmed independently in `lr-d16-r1.md` M2(b)1 and `lr-d16-r2.md` M4(b)3

9.9 Composition: mean escape excludes bounded escaped content

The two conditional theorems above point in opposite directions, and composing them is pure propositional logic.

Proposition 9.33 (Mean escape and bounded escaped content are incompatible)**.** [SKETCH] *Under the joint hypotheses of Theorem 9.30 and Theorem 9.26 — the same vector Ψ, cut, D27 sequence, fixed-window laws, and an arbitrary padded exhaustion — (M-ESC) implies $\neg$(NR): at least one of the two bounds* (158)*,* (159) *is infinite. This holds exhaustion by exhaustion.*

Scope 9.34*.* The status is a sketch that is a candidate for promotion: the argument is a pure composition of two exact conditional statements, and review and adjudication are pending. Not claimed: *which* clause of (NR) fails; any model or state realising (M-ESC); a counterexample in the exactly solvable model; or necessity of (NR) outside this named route.

Idea of proof. Theorem 9.30 gives (M-ESC) $\Rightarrow$ $\neg$(LR3), while Theorem 9.26 gives (NR) $\Rightarrow$ (LR3). Assuming both (M-ESC) and (NR) yields (LR3) and $\neg$(LR3). No new computation is involved, and neither source checker exhibits an (M-ESC) witness. $\square$

campaign id:	M-ESC-NR
proved in:	`theory/lr-d16.md` step (1.5).(2.9), THEOREM LRD-MESC-NR; composed from `theory/ace-ld.md` (1.5) and `theory/lr-d16.md` (1.5).(2.1)–(2.7)
tested by:	No new computation: propositional composition of two exact conditional rows; neither source checker exhibits an (M-ESC) witness
reviewed in:	`blitz-m-esc-nr-r1.md` (0F/0M/0m)

9.10 Kink–magnon wave operators from exact ansatz band data

The escape analysis above presumes a channel decomposition. This subsection records the one construction of such a decomposition that the campaign has proved — conditionally, on an explicitly displayed and unverified localization hypothesis. The general two-magnon theory is in Section 10 and the abstract wave-operator layer in Section 11; here we state the kink–magnon case and its hypothesis package in full.

Definition 9.35 (Exact fixed-packet kink–magnon band data)**.** Assume H-MQG(1)–(3). In addition:

1. *(Covariant kink realization.)* The kink folium has a covariant positive-energy Hilbert realization $(\mathcal{H}_{\alpha\beta}, \pi_{\alpha\beta}, U_{\alpha\beta})$ implementing the commuting infinite-volume time and lattice translations. This is extra structure beyond the state-set definition. The vacuum GNS triples are $(\mathcal{H}_\gamma, \pi_\gamma, \Omega_\gamma)$ for $\gamma \in \{\alpha, \beta\}$, with $H\Omega_\gamma = 0$; there is *no* vacuum vector in $\mathcal{H}_{\alpha\beta}$.

2. *(Exact kink band.)* Its Hamiltonian has an isolated finite-multiplicity kink band with dispersion $E_K \in C^2(\mathbb{T})$, matrix valued on the finite frame index. A fixed Gram-normalised, translation-covariant finite-core ansatz frame gives an **exact** band map Γ_K satisfying $H_{\alpha\beta}\Gamma_K = \Gamma_K E_K$; it is not merely a Rayleigh–Ritz approximation.

3. *(Exact magnon bands with momentum-filtered creators.)* Each relevant vacuum tail γ has an isolated selected magnon band ω_γ (written ω when symmetry relates the tails), gap $\Delta_M := \min_k \omega(k) > 0$, and Gram-normalised **exact** band map $\Gamma_{M,\gamma}$. A spacetime-Schwartz energy–momentum filter of a local observable gives almost-local creators $a_{\gamma,b}(n)$ with strictly positive energy transfer and with the *momentum-filtered normalisation*

$$a_{\gamma,b}(n)\Omega_\gamma \;=\; \Gamma_{M,\gamma}\big(\chi_\gamma e_n \otimes e_b\big), \qquad e_n(k) := e^{-ikn}, \tag{165}$$

for a fixed filter $\chi_\gamma \in C_c^\infty(\mathbb{T})$; translation covariance makes χ_γ independent of n. This *equality*, not merely nonzero overlap with the band, is part of the exactness hypothesis. A position-delta normalisation, the case $\chi_\gamma \equiv 1$, is not assumed and is in fact incompatible with a compactly supported energy–momentum filter.

4. *(Fixed packets, separated velocities, filter compatibility.)* Packet amplitudes are finite sums of C_c^∞ products and the resulting physical packet states lie in the ℓ^1 class D17. The dispersions are C^∞ on open neighbourhoods U_K, U_γ of their packet momentum supports (global C^2 is retained), their kink and magnon velocity supports have distance $\varepsilon_v > 0$, and their signs are: incoming-left $v_M - v_K \geq \varepsilon_v$ at $t \to -\infty$; outgoing-left $v_M - v_K \leq -\varepsilon_v$; outgoing-right $v_M - v_K \geq \varepsilon_v$ at $t \to +\infty$. This is a fixed-packet condition and excludes equal velocities and the soft endpoint. The filters of clause 3 and the kink filter $\chi_K \in C_c^\infty(\mathbb{T})$ satisfy $\chi_\bullet \equiv 1$ on the corresponding packet momentum support with $\operatorname{supp}\chi_\bullet \subset U_\bullet$, and companions $\widetilde{\chi}_\bullet \in C_c^\infty(\mathbb{T})$ satisfy $\widetilde{\chi}_\bullet \equiv 1$ on $\operatorname{supp}\chi_\bullet$ with $\operatorname{supp}\widetilde{\chi}_\bullet \subset U_\bullet$. The channel Hamiltonian symbols $h_K := E_K\widetilde{\chi}_K$ and $h_\gamma := \omega_\gamma\widetilde{\chi}_\gamma$ are then $C^\infty(\mathbb{T})$ and agree with the dispersions on the packet supports.

5. *(Two-cluster factorization through the vacuum tails.)* Write $\kappa_a^{(0)}(x) := \Gamma_K(e_x \otimes e_a)$ for the unfiltered exact kink Wannier vectors. There are $C_{\mathrm{cl}} < \infty$ and $\tilde{\lambda} \in \big(\max(\lambda_{E_\alpha}, \lambda_{E_\beta}), 1\big)$ such that for every $\gamma \in \{\alpha, \beta\}$, all frame indices a, a', all $x, x' \in \mathbb{Z}$, every $r \geq 1$, and every pair $A \in \mathfrak{A}_{\Lambda_\alpha}$, $B \in \mathfrak{A}_{\Lambda_\beta}$ with $\Lambda_\alpha \subset (-\infty, \min(x,x')-r]$ on the α side and $\Lambda_\beta \subset [\max(x,x')+r, \infty)$ on the β side (either factor may be $\mathbb{1}$),
$$\Big|\big\langle \kappa_a^{(0)}(x),\, AB\,\kappa_{a'}^{(0)}(x')\big\rangle - \omega_\alpha(A)\,\omega_\beta(B)\,\big\langle \kappa_a^{(0)}(x), \kappa_{a'}^{(0)}(x')\big\rangle\Big| \;\leq\; C_{\mathrm{cl}}\|A\|\,\|B\|\,\tilde{\lambda}^{\,r}. \tag{166}$$
The one-tail case is $B = \mathbb{1}$ (or $A = \mathbb{1}$); the genuinely two-sided case is used at exactly one step. The constants depend on the model and the frame only — *not* on $A, B, \Lambda_\alpha, \Lambda_\beta, r, x, x', a, a'$, and in particular not on any time parameter, the positions being packet-driven and time dependent downstream. This is the localization input that turns exact band equations into a two-cluster estimate; isolated eigenvalues alone do not imply it. It is the sole replacement for the vacuum clustering apparatus of the source literature, and no four-point analogue is assumed.

The asymptotic channel Hamiltonian is multiplication by $E_K(p) + \omega_\gamma(k)$. Smooth packet domains with the three velocity signs are denoted D_-^L, D_+^L, D_+^T, and P_L, P_T are the two outgoing direct-sum projections. We refer to (166) as the *two-cluster inequality.*

Remark 9.36 (A regime fence, which is not a hypothesis). Put $K_- := \min E_K$, $K_+ := \max E_K$, $W_K := K_+ - K_-$, and let I_{in} be the range of $E_K(p) + \omega(k)$ on the incoming packet support. Same-band kink absorption is excluded by $d_{\mathrm{abs}} := \operatorname{dist}(I_{\mathrm{in}}, [K_-, K_+]) > 0$, for which the packet-independent sufficient condition is $\Delta_M > W_K$. With $\Delta_r := \min_k \omega_r(k)$ for every other known one-particle vacuum band and $K_* := \inf_{j\geq 2,p} E_{K,j}(p)$ for every other known kink band, one asks $\sup I_{\mathrm{in}} + \eta_{\mathrm{inel}} < \Theta_{\mathrm{inel}}$ where $\Theta_{\mathrm{inel}} := \min\{K_- + 2\Delta_M,\ K_*,\ \inf_r(K_- + \Delta_r)\}$ and $\eta_{\mathrm{inel}} > 0$, absent entries being $+\infty$; the three entries respectively exclude an additional magnon, a higher-kink absorption channel, and a different one-particle channel. An uncomputed wall–magnon bound band is not excluded. **These inequalities are used nowhere in the proof.** Their role is to make the *exactness* hypotheses plausible — an isolated exact band is not to be expected when a competing channel is open — so they belong to the justification of Definition 9.35, not to its inferential content. Adding or deleting them changes no step.

Theorem 9.37 (Fixed-packet kink–magnon wave operators exist and are isometric)**.** [PROVED, CONDITIONAL] *Proved only as the following conditional implication. Assume H-MQG(1)–(3) and the whole of Definition 9.35 — exact kink and magnon band maps, covariant kink-sector realization, momentum-filtered creator normalisation, velocity separation $\varepsilon_v > 0$, and the displayed* two-sided *two-cluster inequality* (166)*. Then, on the fixed packet domains:*

(ACE.1) the Cook limits W_-^L, W_+^L and W_+^T exist;

(ACE.2) W_-^L and $W_+ := W_+^L \oplus W_+^T$ are isometries, and $N_T^{\mathrm{ex}} := W_+ P_T W_+^*$ is an orthogonal projection on the constructed out-space $\mathcal{H}_{\mathrm{out}}^{\mathrm{ex}} := \operatorname{ran} W_+$ — the transmitted-channel number of D18 restricted to that range, with nothing asserted on its orthogonal complement.

A third conclusion is ***triply conditional****, on* all three *of: the charge assignment $q_{\mathrm{in}} = q_L = -1$, $q_T = +1$; the named local-decay hypothesis (AD3-ex) of Conjecture 9.39, which this construction does not prove; and the existence of the ordered-limit outcome measure for the vector in question, which Definition 9.35 does not supply. Under all three:*

(ACE.3) for a normalized event vector in $\operatorname{ran} W_-^L \cap \mathcal{H}_{\mathrm{out}}^{\mathrm{ex}}$ for which that outcome measure exists*, the constructed-channel part of the measure has support* contained in $\{0, 2\}$ *(either weight may vanish),*

$$p_\nu^{\mathrm{ex}} = \|P_L W_+^* \Psi\|^2 \delta_{\nu,0} + \|P_T W_+^* \Psi\|^2 \delta_{\nu,2}, \qquad p_2^{\mathrm{ex}} = \langle \Psi, N_T^{\mathrm{ex}} \Psi \rangle, \tag{167}$$

and the ledger gives $\delta x_{\mathrm{ex}} = -p_2^{\mathrm{ex}}/s$.

The order of limits is frozen: infinite-volume dynamics first, then the wave-operator time limit, then any fixed-window limit, and the spatial exhaustion last. No statement is made at zero momentum or after a packet-soft limit.

Scope 9.38. **The two-cluster inequality** (166) **is the load-bearing hypothesis and is unverified on any model.** The threshold inequalities of Remark 9.36 are used in no step. No completeness, no implication from raw band data, no bound-state exclusion, no threshold-inequality use, and no soft limit is claimed. Nothing is asserted about the orthogonal complement of the constructed out-space.

Idea of proof. The construction is a Cook argument with the vacuum clustering apparatus replaced by the single displayed cut (166). One fixes a smooth monotone profile θ with θ' supported in $[-w, w]$ and $\|\theta'\|_\infty \le C/w$, sets $\theta_T := \theta$ and $\theta_L := 1 - \theta$, and defines the kink–magnon precursor identification

$$I_c F \;:=\; \sum_{x,y,a,b} \theta_c(y - x)\, F^{ab}(x, y)\, a_{c,b}(y)\, \kappa_a(x), \tag{168}$$

with the filtered Wannier frame $\kappa_a(x) = \Gamma_K(\chi_K e_x \otimes e_a)$. Both profiles are fixed, independent of t and of the packet. The free channel Hamiltonian is multiplication by $E_K(p) + \omega_\gamma(k)$, which on the packet domain coincides with convolution by the filtered kernels, because the packet momentum supports lie where the filters are identically one. Commuting the creator past the Hamiltonian produces a defect that splits into an exact band-equation term and two kernel-tail terms; velocity separation makes the supports of the kink and magnon packets separate ballistically, so the defect is integrable in t and the Cook limits exist.

Isometry is where the two-cluster inequality is consumed. One needs a four-cluster factorization,

$$\big|\langle a_{\gamma,b}(y)\kappa_a(x),\, a_{\gamma,b'}(y')\kappa_{a'}(x')\rangle - \langle \kappa_a(x), \kappa_{a'}(x')\rangle\, G_\gamma^{bb'}(y, y')\big| \;\le\; C_N \langle r \rangle^{-N}, \tag{169}$$

with G_γ the exact magnon Gram form. This is *derived*, not assumed: the product $a_{\gamma,b}(y)^\dagger a_{\gamma,b'}(y')$ is almost local about $\{y, y'\}$, so it may be truncated to a one-sided region at distance $\gtrsim r$ with rapidly decaying error, whereupon the one-tail case of (166) factors it and the vacuum evaluation is exactly the magnon Gram form by the filtered-creator normalisation (165). The genuinely two-sided case of (166) is used at one step only. The isometry and the projection property of N_T^{ex} follow on the constructed range, with no completeness statement anywhere. □

campaign id:	AC-EX; D28
proved in:	`theory/ansatz-scattering.md`: the theorem at §0, the almost-locality and defect estimates at step (1.3), velocity separation at (1.4), Cook integrability at (1.5), isometry and channel number at (1.6), and the conditional ledger at (1.7)
tested by:	`theory/checks/ansatz_scattering_check.py`, gates ACE-C1a/C1b/C2a/C2b (green exit 0; `--red`, `--red-slow-kernel`, `--red-equal-velocity`, `--red-absorption` each exit 1; all five re-run independently). Kernel decay is red-certified load-bearing on the majorant of step (1.5).(2.4)(i), not on the literal ℓ^2 defect, which is insensitive to it and whose mutation is registered dead; `--red-absorption` is a fence regression guard and is *not* a red test of this theorem. **Neither certificate touches the two-cluster inequality** (166)
reviewed in:	`memory-index-r3.md` §4 (amended sentence; the earlier hold on this row is lifted) and §0 V6, §1

9.11 The open local-decay hypothesis

What separates the conditional ledger (ACE.3) from a theorem is one named hypothesis, and it is worth displaying exactly as it stands.

Conjecture 9.39 (Local decay on the constructed channels)**.** [CONJECTURE] *Let $\Psi = W_+(F_L, F_T)$ and $\Psi_t := e^{-itH}\Psi$. For every fixed window W* ***containing the kink core region****, with $E_W(\cdot)$ the spectral measure of the regularised window charge and P_{ch}, $\mathrm{ch} \in \{L, T\}$, the constructed channel projections,*

$$\lim_{t\to+\infty} \|E_W(\{q_{\mathrm{ch}}\})\Psi_t - P_{\mathrm{ch}}\Psi_t\| = 0, \tag{170}$$

the limit being taken before *$W \uparrow \mathbb{Z}$.*

Scope 9.40. The order of limits is part of the statement: time first at fixed W, then the window exhaustion. The hypothesis is the corresponding clause of the general asymptotic-decay package D18, restricted to the constructed channels; Definition 9.35 deliberately does not contain it. No part of it is proved. What *is* proved from it is the vanishing of the inter-channel cross terms, at step (1.7).(2.2) of the construction. The gloss "the outgoing boundary-straddling charge on ∂W vanishes in the ordered limit" describes the *missing mechanism*, not the hypothesis: the free leg and its dressing do leave any fixed window at rate $O(|t|^{-N})$, but converting that into convergence of the window-charge *spectral projection* requires control of the boundary-straddling charge that no step of the construction supplies. Lemma 9.20 is the abstract statement of what such control would buy, and Theorem 9.10 is the ε-weakened version available on the confined-tail class — which, by Corollary 9.5, is disjoint from the class this hypothesis is about.

campaign id:	AD3-ex
proved in:	Displayed in `theory/ansatz-scattering.md` step (1.7), in the ASSUME block; consumed at (1.7).(2.2)
tested by:	—
reviewed in:	—

10 The two-magnon sector: Bethe oracle and the exact soft slope

Everything else in this labbook is an attempt to say something general. This section is the opposite: it is the one place where the campaign has a model it can solve exactly, and where a proposed soft law can therefore be checked against arithmetic rather than against intuition. The model is the isotropic Heisenberg ferromagnet on a chain. Its fully polarised state is an exact ground state, its one-magnon sector is a free hopping problem, and its two-magnon sector reduces — fibre by fibre in the total momentum — to a half-line Jacobi matrix with a single boundary defect. Nothing in this section needs integrability as a hypothesis: every statement below is obtained by writing down the lattice eigenvalue equations and solving them.

Three features of this referee should be kept in mind while reading.

First, the magnon dispersion is quadratic at small momentum, $\omega(k) = J(1 - \cos k)$, so "soft" here means small *momentum*, and the energy variable is a bad coordinate: the phase is analytic in k_s and only Puiseux (half-integer powers of ω_s) in the energy. Statements are therefore always organised in momentum.

Second, what is exactly true in the soft limit is weaker than the field-theory slogan. The two-magnon scattering ratio tends to one as the soft momentum vanishes — Dyson's decoupling of a zero-momentum magnon — but this is a plain limit, not an Adler zero: no theorem here says that a soft insertion is annihilated, only that the connected amplitude becomes the identity, with a computable first-order departure.

Third, that first-order departure is the campaign's actual number. On the physical outgoing/incoming channel the phase behaves as $\delta_{\text{phys}} = 2\,\text{sgn}(v_h - v_s)\,k_s + O(k_s^2)$ for spin 1/2, and, in the exact generalisation to every site spin S, as $\text{sgn}(v_h - v_s)\,k_s/S + O(k_s^2)$. The slope $1/S$ is the sharpest exactly proved statement the campaign owns.

10.1 The oracle model and its conventions

The next three definitions are frozen: they are the shared vocabulary of every later two-magnon statement, and they are reproduced here in full so that this document can be read without opening the definitions file.

Definition 10.1 (The Heisenberg-ferromagnet oracle model and its bases)**.** On the infinite chain — or on a periodic ring of N sites, whenever finite volume is stated — put $\boldsymbol{S}_x = \boldsymbol{\sigma}_x/2$, take $J > 0$, and let

$$H = \sum_x h_{x,x+1} = -J \sum_x \Big(\boldsymbol{S}_x \cdot \boldsymbol{S}_{x+1} - \tfrac{1}{4}\Big), \qquad h_{x,x+1} = \tfrac{J}{2}\big(1 - P_{x,x+1}\big), \tag{171}$$

where $P_{x,x+1}$ swaps the two spins. The vacuum is the fully polarised state $|\Omega\rangle = |\uparrow\uparrow\cdots\rangle$. The coordinate bases are $|x\rangle = S_x^-|\Omega\rangle$ and $|x,y\rangle = S_x^- S_y^-|\Omega\rangle$ with $x < y$. Lattice momenta lie in $(-\pi, \pi]$, we write $z(k) = e^{ik}$, and

$$|k\rangle_N = N^{-1/2} \sum_{x=0}^{N-1} e^{ikx}|x\rangle, \qquad k = 2\pi n/N. \tag{172}$$

On the infinite chain the same sum denotes the delta-normalised generalised state. The symbols $\omega(k)$ and $v(k) = d\omega/dk$ denote the one-magnon energy and group velocity derived from this H.

campaign id: D6
proved in: `definitions.md` D6 (FROZEN — not editable without reopening the oracle review loop)
tested by: `theory/checks/oracle_bethe_check.py`
reviewed in: `theory/verdicts/oracle-bethe-r2.md`

Definition 10.2 (Ordered-coordinate Bethe wave and the scattering convention)**.** In the chamber $x < y$ the two-magnon coordinate Bethe wave is

$$\psi(x, y) = A_{12}\, e^{i(k_1 x + k_2 y)} + A_{21}\, e^{i(k_2 x + k_1 y)}, \tag{173}$$

and one sets

$$S_{12}(k_1, k_2) := \frac{A_{12}}{A_{21}}, \qquad S_{21} := \frac{A_{21}}{A_{12}}, \tag{174}$$

while $E(k_1, k_2)$ denotes the corresponding scattering energy. For real wave packets with $v(k_2) > v(k_1)$ and the k_2 packet initially on the left, A_{21} is the *incoming* coefficient and A_{12} the *outgoing* one; reversing the channel convention gives $S_{21} = S_{12}^{-1}$. The rapidity is

$$\lambda(k) := \tfrac{1}{2} \cot(k/2). \tag{175}$$

Whenever $S_{12} = e^{i\delta}$ is expanded at a point where $S_{12} = 1$, the symbol δ means the unique continuous real branch vanishing at that point. For a two-magnon bound state we write $K := k_1 + k_2$ for the total momentum, $r := y - x$ for the separation, f_r for the relative-coordinate wave, t for its geometric decay factor, $\eta := -\log|t|$, and $E_b(K)$ for its energy.

campaign id:	D7
proved in:	`definitions.md` D7 (FROZEN)
tested by:	`theory/checks/oracle_bethe_check.py`
reviewed in:	`theory/verdicts/oracle-bethe-r2.md`

Definition 10.3 (The soft limit of the Bethe oracle)**.** The hard momentum $k_h \in (0, \pi)$ is held fixed while the *signed* soft momentum $k_s \to 0$. Write $\omega_h := \omega(k_h)$, $v_h := v(k_h)$, $\omega_s := \omega(k_s)$, $v_s := v(k_s)$, and $\sigma := \mathrm{sgn}(k_s)$ when a one-sided energy expansion is used. On this half-zone $v_h > v_s$ for all sufficiently small $|k_s|$, so the ratio S_{12} of Definition 10.2 is the physical outgoing/incoming ratio there. A simultaneous limit of k_h and k_s, or a hard momentum outside this half-zone, is *not* included unless explicitly stated.

campaign id:	D8
proved in:	`definitions.md` D8, FROZEN-AS-AMENDED: the naming of ω_h, v_h, v_s and the remark that $v_h > v_s$ for small $\lvert k_s\rvert$ were added in oracle round r2 and break no consumer
tested by:	`theory/checks/oracle_bethe_check.py`
reviewed in:	`theory/verdicts/oracle-bethe-r2.md` (residue 3)

The exact solution of the model, in this vocabulary, is the following block of facts. They are used repeatedly below and are stated once here.

Oracle fact 10.4 (Exact one- and two-magnon data of the ferromagnet)**.** [PROVED] Under Definitions 10.1 and 10.2:

$$\omega(k) = J(1 - \cos k), \qquad v(k) = J \sin k, \tag{176}$$

$$E(k_1, k_2) = \omega(k_1) + \omega(k_2) = J(2 - \cos k_1 - \cos k_2), \tag{177}$$

the exact coefficient ratio in the ordered chamber is

$$S_{12}(k_1, k_2) = -\frac{e^{i(k_1+k_2)} - 2e^{ik_1} + 1}{e^{i(k_1+k_2)} - 2e^{ik_2} + 1} = \frac{\lambda_1 - \lambda_2 + i}{\lambda_1 - \lambda_2 - i}, \tag{178}$$

with $\lambda_j = \lambda(k_j)$, and it has unit modulus for real momenta; and there is, up to scale, exactly one two-magnon bound state per total momentum,

$$f_r = t^{\,r-1}, \qquad t = \cos(K/2), \qquad E_b(K) = J \sin^2(K/2), \tag{179}$$

for $0 < |K| \le \pi$, degenerating at $K = 0$ into a non-normalisable threshold resonance.

campaign id:	oracle facts O1–O5 of `theory/oracle-bethe.md` (these have no separate row in `claims/CLAIMS.md`; the campaign rows that consume them are OR1 and OR2)
proved in:	`theory/oracle-bethe.md` steps (1.1)–(1.2), collected at (1.4)
tested by:	`theory/checks/oracle_bethe_check.py`
reviewed in:	`theory/verdicts/oracle-bethe-r2.md`

Idea of proof. For two spin halves $P = 2\boldsymbol{S}_1 \cdot \boldsymbol{S}_2 + 1/2$, so each bond term is $\frac{J}{2}(1-P)$, has eigenvalues 0 and J, and annihilates any parallel pair; hence $H|\Omega\rangle = 0$ and $H \geq 0$. Acting on one overturned spin leaves a nearest-neighbour hopping problem, which the Fourier states diagonalise. For two overturned spins the bond algebra gives a free equation away from contact — whence energy additivity — and a single modified equation at $y = x+1$; substituting the two-plane-wave form into that one contact equation gives a linear relation between A_{12} and A_{21}, and hence Equation (178). In a fixed total-momentum fibre the relative equation is a constant-coefficient second-order recursion whose two characteristic roots multiply to one; square summability therefore selects the single root inside the unit disc, and matching the $r = 1$ equation fixes both the decay factor and the energy in Equation (179).

10.2 Soft decoupling: what is actually true in the limit

Oracle fact 10.5 (A zero-momentum magnon decouples)**.** [PROVED] On the half-zone of Definition 10.3 — fixed $k_h \in (0, \pi)$, signed $k_s \to 0$ —

$$\lim_{k_s \to 0^+} S_{12}(k_s, k_h) = \lim_{k_s \to 0^-} S_{12}(k_s, k_h) = 1. \tag{180}$$

Scope 10.6*.* This is a plain limit and nothing more. It says that the connected two-magnon amplitude degenerates to the identity as the soft leg becomes a zero mode; it is explicitly *weaker* than an Adler-zero theorem, since it asserts no vanishing of a soft insertion, no factorisation, and no statement about any process other than exact two-magnon scattering in this model. The rate at which the limit is approached — the content of Theorems 10.9 and 10.14 — is a separate statement.

campaign id:	OR2 (oracle fact O6)
proved in:	`theory/oracle-bethe.md` step (1.3).(2).2
tested by:	`theory/checks/oracle_bethe_check.py`; the recorded run gives $\lvert S_{12} - 1\rvert = 2.966 \cdot 10^{-16}$ at the sampled soft points
reviewed in:	`theory/verdicts/oracle-bethe-r2.md`

Idea of proof. Put $k_s = 0$ in Equation (178). The numerator becomes $e^{ik_h} - 1$ and the denominator $1 - e^{ik_h}$; the outer minus sign turns their ratio into 1, and the denominator is nonzero because k_h is fixed inside $(0, \pi)$. The statement is thus simply $S_{12} - 1 \to 0$, obtained by subtracting the identity amplitude, with no low-energy theorem involved.

10.3 Completeness of the two-magnon sector

The soft expansions below are statements about a distinguished channel inside a Hilbert space. They are only meaningful once one knows what else that space contains. That is the content of the next theorem, which is proved by direct diagonalisation and, deliberately, without assuming that Bethe states are complete.

Theorem 10.7 (Complete two-magnon resolution, without assumed Bethe completeness)**.** [PROVED] *Assume Definitions 10.1 to 10.3, $J > 0$, and, in finite volume, $N > 3$.*

1. Finite ring. *Every nonzero two-magnon Bethe vector on the N-site ring is one of the real-pair, complex-pair (two-string), or $SU(2)$-descendant vectors classified in the proof. For even N*

the singular completion supplies exactly one further vector, the normalised contact state

$$|\chi_\pi\rangle = N^{-1/2}\sum_x(-1)^x|\{x,x+1\}\rangle, \qquad H|\chi_\pi\rangle = J|\chi_\pi\rangle, \tag{181}$$

of total momentum π, which carries the compactified rapidity pair $\{+i/2,-i/2\}$ and has no finite two-plane-wave normalisation. The formal coincident solutions are identically zero vectors and supply nothing. The physical vectors number exactly $N(N-1)/2$ and can be normalised to an orthonormal eigenbasis.

2. Infinite chain. *The two-magnon Hilbert space is the orthogonal sum of the scattering representation of Definition 10.2 and one bound-magnon band. In the centre-momentum chart, with $c=\cos(K/2)$, the fibre is the half-line Jacobi operator*

$$(h_K f)_1 = Jf_1 - Jcf_2, \qquad (h_K f)_r = 2Jf_r - Jc(f_{r-1}+f_{r+1}) \quad (r\geq 2), \tag{182}$$

the generalised eigenfunctions and energies are

$$\Psi^{\mathrm{sc}}_{Kq}(x,x+r) = \frac{e^{iK(x+r/2)}}{2\pi}\big[S(K,q)e^{iqr}+e^{-iqr}\big], \qquad 0<q<\pi, \tag{183}$$

$$S(K,q) = \frac{c-e^{-iq}}{e^{iq}-c}, \qquad E(K,q) = J(2-2c\cos q), \tag{184}$$

$$\Psi^{\mathrm{b}}_{K}(x,x+r) = \frac{e^{iK(x+r/2)}}{\sqrt{2\pi}}\sqrt{1-c^2}\; c^{r-1}, \qquad E_b(K) = J\sin^2(K/2), \tag{185}$$

and for every chamber vector ψ the resolution of the identity holds:

$$\|\psi\|^2 = \int_{-\pi}^{\pi} dK\int_0^{\pi} dq\;|\langle\Psi^{\mathrm{sc}}_{Kq},\psi\rangle|^2 + \int_{-\pi}^{\pi} dK\;|\langle\Psi^{\mathrm{b}}_{K},\psi\rangle|^2. \tag{186}$$

The Plancherel measures are $dK\,dq/(2\pi)^2$ and $dK/(2\pi)$ for the unnormalised kernels. The spectrum on this sector is purely absolutely continuous; the bound state of Equation (179) *becomes a band, present for every $K\neq 0$ and degenerating at $K=0$ into a threshold resonance.*

3. Consequence. *The finite-volume expansion of the charge-created state in the Bethe family is therefore* unconditional*, with the singular class separated explicitly: for the charge $Q_{k_s} = \sum_x e^{ik_s x}S_x^-$,*

$$Q_{k_s}|k_h\rangle_N = \sum_B c_B|B\rangle + \mathbf{1}\{2\mid N,\ k_s+k_h\equiv\pi\}\;2i\cos\!\big(\tfrac{k_h-k_s}{2}\big)|\chi_\pi\rangle, \tag{187}$$

the sum running over the normalised real, string and descendant states of the fibre. At $k_s=0$ the expansion collapses onto the single root-at-infinity descendant $Q_0|k_h\rangle_N/\sqrt{N-2}$.

Scope 10.8. This is a generalised-eigenfunction (Plancherel) statement. It does *not* construct Møller operators and does *not* prove asymptotic completeness in the scattering-theoretic sense; that is the separate content of Theorem 11.1 in Section 11. Nothing here is claimed for three or more magnons.

campaign id:	ML2
proved in:	`theory/ml2-completeness.md` steps (1.1)–(1.7); the finite-ring count is (1.3)–(1.4), the infinite-chain resolution (1.5), the charge-expansion consequence (1.6)
tested by:	`theory/checks/ml2_completeness_check.py`: eleven ring sizes, maximum spectral mismatch $8.882\cdot10^{-15}$, eigenvector residual $4.044\cdot10^{-14}$, projector error $4.640\cdot10^{-14}$, singular-overlap error $6.280\cdot10^{-15}$
reviewed in:	`theory/verdicts/ml2-r2.md` (PASS, 10/10 mutants killed)

Idea of proof. Translation invariance fibres the two-magnon sector by total momentum. On the ring one counts the relative basis vectors in each fibre — allowing for the identification of a pair with its reflection and for the special self-opposite separation at even N — and the dimensions sum to $N(N-1)/2$. In each fibre the Hamiltonian is a real Jacobi matrix whose interior recursion is the free one and whose first row carries the contact defect; its eigenvalue problem is solved through the Chebyshev-like polynomials of the recursion, and the resulting characteristic equation is matched, root by root, with the Bethe quantisation condition. The formal coincident roots are shown to produce identically vanishing coordinate waves, and at even N the $K = \pi$ fibre is diagonal, producing the singular contact vector Equation (181) explicitly rather than as a limit of two-plane-wave states. On the infinite chain the same fibration gives the half-line Jacobi operator Equation (182); a direct unit-circle integration of the scattering waves against each other yields the fibre completeness identity $\delta_{rs} = \int_0^\pi \frac{dq}{2\pi}\overline{w_q(r)}w_q(s) + (1-c^2)c^{r+s-2}$, in which the bound term is exactly the residue at the pole $z = c$; combining this with Fourier–Plancherel in the centre coordinate gives Equation (186). Absolute continuity follows because the energy maps $E(K,q)$ and $E_b(K)$ have nonvanishing gradient off a Lebesgue-null set. Finally, Parseval in the resulting orthonormal fibre basis, together with the explicit overlap of the charge-created state with $|\chi_\pi\rangle$, gives Equation (187).

10.4 The two-body soft theorem

We can now state the soft law of the spin-1/2 ferromagnet in the form the campaign uses. The point of the derivation is that it does *not* go through the closed Bethe ratio: it uses only the local current and the single contact equation, so the closed formula of Oracle fact 10.4 is available afterwards as an independent check rather than as an input.

Theorem 10.9 (Two-body soft expansion for the spin-half ferromagnet)**.** [PROVED] *Assume Definitions 10.1 to 10.3 with k_h fixed, $0 < |k_h| < \pi$, and signed $k_s \to 0$. Let S_{phys} be the physical outgoing/incoming ratio — that is, S_{12} when $\operatorname{sgn}(v_h - v_s) = +1$ and $S_{21} = S_{12}^{-1}$ when $\operatorname{sgn}(v_h - v_s) = -1$ — and let δ_{phys} be its continuous phase vanishing at $k_s = 0$. Then*

$$\delta_{\text{phys}} = 2\,\operatorname{sgn}(v_h - v_s)\,k_s + \frac{|v_h|}{\omega_h}k_s^2 + R_\delta, \tag{188}$$

$$S_{\text{phys}} = 1 + 2i\,\operatorname{sgn}(v_h - v_s)\,k_s + \left[i\frac{|v_h|}{\omega_h} - 2\right]k_s^2 + R_S, \tag{189}$$

where the quadratic coefficient has the equivalent invariant forms

$$\frac{|v_h|}{\omega_h} = \frac{2J-\omega_h}{|v_h|} = \sqrt{\frac{2J-\omega_h}{\omega_h}}, \qquad \textit{and } = \cot(k_h/2) \textit{ on the half-zone of Definition 10.3.} \tag{190}$$

The remainders are uniform on compact hard sets: for $0 < a < b < \pi$ there are finite explicit constants $C_\delta(a,b)$, $C_S(a,b)$, of size $O(a^{-2})$ as $a \downarrow 0$ at fixed b, with $|R_\delta| \le C_\delta(a,b)|k_s|^3$ and $|R_S| \le C_S(a,b)|k_s|^3$ on the compactum $\{|k_s| \le \varepsilon_{ab},\ a \le |k_h| \le b\}$.

Scope 10.10. The theorem is about the exact connected two-magnon multiplier of this model. It does *not* promote a process-independent soft theorem, and asserts nothing about amplitudes generated by an arbitrary quasi-local hard process: that would be the universality statement discussed in Section 13 and it is not established here. It is generalised — not replaced — by Theorem 10.14.

campaign id:	S2-2body
proved in:	`theory/soft-current-recon.md` steps (1.1)–(1.5) together with `theory/oracle-bethe.md` step (1.3)
tested by:	`theory/checks/soft_current_recon_check.py` (maximum operator and form-factor residual $1.560 \cdot 10^{-14}$, quadratic-phase fit error $2.167 \cdot 10^{-10}$), `oracle_bethe_check.py`, `ml2_completeness_check.py`, and the numerics scan `fm-displacement-scan`
reviewed in:	`theory/verdicts/corpus-r2.md` adjudication

Idea of proof. Write the complexified broken charge and its current,

$$Q_k = \sum_x e^{ikx} S_x^-, \qquad j^-_{x,x+1} = -[h_{x,x+1}, S_x^-] = \tfrac{J}{2}(S^-_{x+1} - S^-_x)P_{x,x+1}, \qquad J_k^- = \sum_x e^{ikx} j^-_{x,x+1}, \tag{191}$$

so that $[H, Q_k] = (e^{ik} - 1)J_k^-$. Acting with Q_{k_s} on a hard magnon produces a two-magnon state whose two ordered-chamber coefficients are *equal*; an on-shell scattering eigenstate has instead the ratio S_{12}. The exact mismatch is a single branch term,

$$Q_{k_s}|k_h\rangle_N - |B^{\rm in}\rangle = (1 - S_{12})|P_{12}\rangle, \tag{192}$$

with $|P_{12}\rangle = N^{-1/2}\sum_{x<y} e^{i(k_s x + k_h y)}|x, y\rangle$. Now the soft expansion is obtained without any closed formula: normalise the incoming coefficient to one, call the unknown outgoing coefficient $s(k_s, k_h)$, and impose only that the contact bond residual vanish. With $z_s = e^{ik_s}$ and $z_h = e^{ik_h}$ this is one linear equation,

$$(2z_h - z_s z_h - 1)\, s + (2z_s - z_s z_h - 1) = 0, \tag{193}$$

which for fixed k_h has a unique solution analytic at $k_s = 0$; expanding it gives $s = 1 + 2ik_s + [i\cot(k_h/2) - 2]k_s^2 + O(k_s^3)$, and taking the continuous logarithm gives the phase. All hard-momentum dependence cancels from the linear coefficient. Compactness of the hard set keeps $|z_h - 1|$ bounded below, which is what makes Taylor's remainder uniform. Finally the current identity supplies the mechanism behind the number 2: at zero soft momentum

$$\langle k_h|Q_0^\dagger J_0^-|k_h\rangle_N = 2iJ\sin k_h = 2i\, v_h, \tag{194}$$

so a hard external-pole reduction, which divides by the linear energy shift $v_h k_s$, cancels the hard velocity and leaves the coefficient 2.

Remark 10.11 (The number 2 is a displacement, not a scattering length). The magnitude of the linear coefficient in Equation (188) is the two-site Wigner phase-slope, i.e. a spatial-displacement coefficient of the soft packet, and its sign is the channel label $\mathrm{sgn}(v_h - v_s)$; the number is not hard-momentum-independent in an invariant sense, and only the half-zone of Definition 10.3 reduces it to $+2$. It is also not a scattering length: in the joint soft limit $k_1 = \varepsilon x$, $k_2 = \varepsilon y$ with $x \neq y$, the phase obeys $\delta_{12}/\varepsilon \to 2xy/(y - x)$, which vanishes but is not a constant multiple of $k_1 - k_2$, so no nonzero relative-momentum scattering length exists. Two further boundary facts are worth recording: as $k_h \downarrow 0$ the quadratic coefficient diverges like $2/k_h$, so truncation requires $|k_s| \ll k_h$; and as $k_h \to \pi$ the algebraic ratio stays regular but the kinematic ordering $v_h > v_s$ degenerates, requiring $k_s \ll \pi - k_h$.

Remark 10.12 (Momentum is the good coordinate). In the energy variable the same expansion is only Puiseux and remembers the sign of the soft momentum through $\sigma = \mathrm{sgn}(k_s)$:

$$\delta_{12} = 2\sigma\sqrt{2\omega_s/J} + \frac{2v_h}{J\omega_h}\,\omega_s + O(\omega_s^{3/2}), \tag{195}$$

with an explicit compact remainder bound. Both signed momentum limits give one, but the energy coordinate forgets σ and the connected term is $O(\sqrt{\omega_s})$ rather than a Taylor series.

Refuted claim 10.13 (The old label "$S2$" for the minimal two-body core). [REFUTED] *The identifier S2 was withdrawn — purely as a duplicate-status guard, so that one statement does not carry two rows — and its content is Theorem 10.9, which is the row that is proved.*

campaign id:	S2 (superseded label; use S2-2body)
proved in:	—
tested by:	—
reviewed in:	superseded by the S2-2body row of `claims/CLAIMS.md`

10.5 The exact soft slope at every site spin

The spin-1/2 result above turns out not to be an accident of $SU(2)$ spin-half algebra. The next theorem is the campaign's sharpest exact statement.

Theorem 10.14 (Exact two-magnon ratio and soft slope at every site spin). [PROVED] *Let $S \in \{1/2, 1, 3/2, \dots\}$, $J > 0$, and take on the infinite chain the fully polarised ferromagnet*

$$H_S = -J \sum_x \big(\boldsymbol{S}_x \cdot \boldsymbol{S}_{x+1} - S^2\big). \tag{196}$$

Use the ordered chamber, Fourier sign, and coefficient ratio $S_{12} = A_{12}/A_{21}$ of Definition 10.2. Put $z_j = e^{ik_j}$ and

$$a := 1 + z_1 z_2, \qquad b := z_1 + z_2, \qquad \mu := (2S-1)a + b. \tag{197}$$

On the regular scattering domain — real $k_1, k_2 \in (-\pi, \pi]$ with $k_1 \neq k_2 \pmod{2\pi}$, $ab \neq 0$, and $z_2\mu - Sab \neq 0$ — the two-magnon eigenvalue equation by itself *fixes*

$$\boxed{S_{12}(k_1,k_2) = \frac{S\,ab - z_1\mu}{z_2\mu - S\,ab}} \tag{198}$$

with $|S_{12}| = 1$ and $S_{12}(k_2,k_1) = S_{12}(k_1,k_2)^{-1}$. Setting $k_1 = k_s$, $k_2 = k_h$ with $0 < |k_h| < \pi$ fixed and signed $k_s \to 0$, and using the dispersion $\omega_S(k) = 2JS(1-\cos k)$, $v_S(k) = 2JS\sin k$ and the physical channel rule of Definition 10.2, the physical phase obeys, locally uniformly on compact hard sets on which the channel is fixed,

$$\delta_{\text{phys}}(k_s,k_h) = \frac{\operatorname{sgn}(v_h - v_s)}{S}\,k_s + O(k_s^2), \qquad S_{\text{phys}} = 1 + i\,\frac{\operatorname{sgn}(v_h - v_s)}{S}\,k_s + O(k_s^2), \tag{199}$$

so that

$$\boxed{\left.\frac{\partial \delta_{\text{phys}}}{\partial k_s}\right|_{k_s=0} = \frac{\operatorname{sgn}(v_h - v_s)}{S}} \tag{200}$$

which on the half-zone of Definition 10.3 is exactly $1/S$. At $S = 1/2$ the ratio Equation (198) collapses to Equation (178) and the slope to $2\operatorname{sgn}(v_h - v_s)$, recovering Theorem 10.9.

Scope 10.15. No integrability hypothesis and no Bethe completeness are used: the two-body wave is validated by direct substitution into every configuration class. What is proved is the exact *unit-charge two-body* slope, and only that. The theorem does not prove endpoint ($k_h \to 0, \pi$) or equal-velocity limits; it does not prove Bethe completeness at spin S; it does not prove a process-independent soft theorem; it does not prove the memory half of the triangle (see Section 8); it says nothing about a hard excitation of charge larger than one; and it therefore does not promote the combined slope-and-memory conjecture, which remains a conjecture.

campaign id:	S2-2body-S
proved in:	`theory/spin-s-twomagnon.md`, steps (1.1)–(1.5)
tested by:	`theory/checks/spin_s_slope_check.py` (the `--red` mutation replaces $2S-1$ by $2S+1$ in μ and must fail); `numerics/results/spin1-bc-falsifier.json`; `numerics/results/spin1-bc-crosscheck.json`
reviewed in:	`theory/verdicts/spin-s-r1.md` (PASS)

Idea of proof. In occupation variables $n_x = S - S_x^z$ the Hamiltonian is $H_S = 2JS\sum_x n_x - J\sum_x n_x n_{x+1} - \frac{J}{2}\sum_x(S_x^+S_{x+1}^- + S_x^-S_{x+1}^+)$. For $S \geq 1$ the two-magnon sector has two independent amplitudes: the separated wave $\psi(x,y)$ on $x<y$ and a genuine double-occupancy amplitude $d(x)$. There are therefore three exact coordinate equations — separated, adjacent, and doubly occupied — and, with $g = \sqrt{S(2S-1)}$, they read

$$E\psi(x,y) = 4JS\psi(x,y) - JS\big[\psi(x-1,y) + \psi(x+1,y) + \psi(x,y-1) + \psi(x,y+1)\big], \tag{201}$$

$$E\psi(x,x+1) = J(4S-1)\psi(x,x+1) - JS\big[\psi(x-1,x+1) + \psi(x,x+2)\big] - Jg\big[d(x) + d(x+1)\big], \tag{202}$$

$$E\,d(x) = 4JS\,d(x) - Jg\big[\psi(x-1,x) + \psi(x,x+1)\big]. \tag{203}$$

One takes the unsymmetrised two-plane-wave extension $\Psi(x,y) = Az_1^x z_2^y + Bz_2^x z_1^y$ on all of $\mathbb{Z}^2$, together with $d(x) = \rho(z_1z_2)^x$. The separated equation fixes the energy additively; the double-occupancy equation determines ρ in terms of the amplitudes; and eliminating ρ from the adjacent equation leaves the single contact relation $W\mu = S\Sigma ab$ with $\Sigma = A+B$ and $W = Az_2 + Bz_1$, which is exactly Equation (198). Unitarity follows by writing the ratio in centre/relative variables, where it takes the form $n/(-\bar{n})$. Differentiating $\log S_{12}$ at $z_1 = 1$ gives i/S, whence the slope.

One trap deserves recording, because it is what makes the answer $1/S$ rather than 2: the free extension Ψ is *not* symmetric across the diagonal unless $A = B$, so one may not replace $\Psi(x+1,x)$ by $\Psi(x,x+1)$ when extending the separated equation to coincident arguments. That illicit replacement collapses the two contact conditions into the spin-half one and falsely yields slope 2 at every S. The valid argument keeps the extension unsymmetrised, introduces $d(x)$ as an independent physical amplitude, and imposes both contact equations.

10.6 Consistency with the oracle, and the pending numerical test

Proposition 10.16 (The reconstruction agrees with the passed oracle)**.** [PROVED] *On the half-zone of Definition 10.3, the expansion obtained from the local current and the single contact equation Equation* (193) *agrees term by term — including the sign convention and the value* 2 *of the linear coefficient there — with the independently derived closed-form expansions Equations* (188) *and* (189) *of the Bethe oracle.*

Scope 10.17. This establishes the equality of two formulas, obtained by two independent routes. It does not establish process universality: agreement of the contact-equation route with the closed Bethe route says nothing about amplitudes generated by other hard processes.

campaign id:	OR1
proved in:	`theory/soft-current-recon.md` step (1.5).(2).1
tested by:	`theory/checks/oracle_bethe_check.py`, `theory/checks/soft_current_recon_check.py`
reviewed in:	`theory/verdicts/corpus-r2.md`

Conjecture 10.18 (Excitation-ansatz amplitudes reproduce the Bethe phase)**.** [CONJECTURE] *In a variational matrix-product excitation-ansatz computation of two-magnon scattering in the model of Definition 10.1, the extracted magnon amplitudes reproduce the exact Bethe ratio $S(k)$ as $k \to 0$.*

Scope 10.19. This numerical test has not been run. It is stated so that the tensor-network side of the campaign (Section 17) has an unambiguous, falsifiable target inherited from Theorem 10.9; no evidence for it is claimed here.

campaign id:	N1
proved in:	—
tested by:	— (not yet run)
reviewed in:	—

11 Two-magnon wave operators, the Ward projection, and its erratum

The previous section resolved the two-magnon Hilbert space spectrally. That is not yet scattering theory: a Plancherel decomposition tells you what the generalised eigenfunctions are, not that a wave packet prepared as two free magnons in the far past actually converges to one of them. This section builds the missing time-dependent layer, and then reports honestly on what happens when one tries to turn a symmetry (Ward) identity into a soft theorem at finite volume. The centre of gravity of the section is an erratum: a projected Ward identity that looked exact turned out to hold at one overturned spin only. The repair is displayed, the mechanism is named, and the downstream audit is recorded. That kind of entry is what this labbook is for.

11.1 The canonical two-magnon wave operators

Theorem 11.1 (Canonical two-magnon wave operators for the ferromagnet)**.** [PROVED] *Assume Definitions 10.1 to 10.3 and $J > 0$. In the centre/relative chart put*

$$\mathcal{C} := (-\pi, \pi] \times (0, \pi), \qquad c(K) := \cos(K/2), \qquad k_1 := K/2 - q, \quad k_2 := K/2 + q, \tag{204}$$

understood on the momentum torus, with $E(K, q) = J(2 - 2c(K)\cos q)$ and $E_b(K) = J(1 - c(K)^2)$. Let $\Sigma_{\text{bad}} \subset \overline{\mathcal{C}}$ be the union of $q \in \{0, \pi\}$, of $K = \pi$, and of the curves on which either $k_i \in \{0, \pi\}$; call a regular packet *any finite sum of smooth compactly supported product packets whose support has positive distance from Σ_{bad}. Write $\mathcal{H}_{\text{ch}} := L^2(\mathcal{C}, dK\,dq)$ and $\mathcal{H}_{b,\text{ch}} := L^2((-\pi, \pi], dK)$. Then there are canonical maps $W_\pm : \mathcal{H}_{\text{ch}} \to \mathcal{H}_2$ with the following properties.*

1. *$W_\pm$ are isometries with the same range $\mathcal{H}_{\text{sc}}$, and the two-magnon space splits orthogonally as $\mathcal{H}_2 = \mathcal{H}_{\text{sc}} \oplus \mathcal{H}_b$ with $\mathcal{H}_b \simeq \mathcal{H}_{b,\text{ch}}$.*

2. *They intertwine time and lattice translations, and*

$$W_+^* W_- = M_{S_{\text{phys}}}, \tag{205}$$

multiplication by the physical outgoing/incoming ratio, which in this chart equals S_{12} away from $K = \pi$ and has $|S_{\text{phys}}| = 1$. Concretely $W_- = \mathcal{U}_{\text{sc}}^*$ and $W_+ = \mathcal{U}_{\text{sc}}^* M_{S_{\text{phys}}^{-1}}$, where $\mathcal{U}_{\text{sc}}$ is the coefficient map against the delta-normalised scattering kernels Equation (183)*.*

3. *On every regular packet they are the strong time-dependent Møller limits*

$$W_\pm F = \operatorname*{s-lim}_{t \to \pm\infty} e^{itH}\, I\, e^{-itE} F, \tag{206}$$

for the canonical two-lowering identification I; so the limits exist and are isometric on precisely the packet class the campaign asked for.

4. *In each relative-coordinate fibre with $K \neq 0$ there is exactly one normalised bound eigenvector, $b_K(r) = \sqrt{1 - c(K)^2}\, c(K)^{r-1}$ for $r \geq 1$, of energy $E_b(K)$; at $K = 0$ it degenerates into a threshold resonance and is not an ℓ^2 vector. No bound vector lies in either wave-operator range.*

5. *The one-magnon scattering space is $\mathcal{H}_1 = \ell^2(\mathbb{Z}) \simeq L^2(\mathbb{T}, dk/2\pi)$; its incoming and outgoing maps coincide with the Fourier spectral map and are unitary.*

The union of regular packet spaces is dense in $\mathcal{H}_{\rm ch}$, so the compatible local isometries glue canonically to the global maps.

Scope 11.2. The time-dependent assertion is made only on regular packets. The extension across $\Sigma_{\rm bad}$ — equal velocities, one-particle endpoints, the $K = \pi$ line — is abstract, obtained by density from the isometries; no endpoint or equal-velocity time-dependent statement is claimed, and no incoming bound-pair channel is constructed. The bound band is a separate stable composite channel, not a two-free-magnon channel.

campaign id:	ML1
proved in:	`theory/ml1-wave-operators.md` steps (1.1)–(1.7): canonical spaces (1.2), stationary isometries and multiplier (1.3), identification with the Møller limits and dense gluing (1.4), bound isolation and one-magnon channel (1.5)
tested by:	`theory/checks/ml1_ml6_check.py` — which checks only the explicit fibre arithmetic, *not* wave-operator existence
reviewed in:	`theory/verdicts/blitz-ml1-ml6-r1.md` (no fatal or major objection on this row)

Idea of proof. The centre Fourier transform carries the chamber $\{x < y\}$ to a direct integral of the half-line Jacobi fibres Equation (182), which differ from the free constant-coefficient half-line operator only in the first diagonal entry. The Plancherel resolution Equation (186) of Theorem 10.7 makes the pair of coefficient maps against the scattering and bound kernels a unitary onto $\mathcal{H}_{\rm ch} \oplus \mathcal{H}_{b,\rm ch}$ conjugating H to multiplication by E and E_b; the two summands are therefore closed, orthogonal and reducing, and exhaust $\mathcal{H}_2$. In the chart the velocity difference is $v(k_2) - v(k_1) = 2J \cos(K/2) \sin q$, positive off the null line $K = \pi$, so the incoming/outgoing labelling of the two relative branches is fixed almost everywhere, giving the stationary maps and Equation (205). To identify these stationary maps with the time-dependent limits Equation (206), cover the compact support of a regular packet by finitely many product rectangles, on each of which the two velocity supports have a positive separation and all symbols are smooth; on each rectangle the fixed-packet Cook construction of Theorem 11.18 applies and its limit agrees with the stationary map. Overlapping rectangles therefore agree, the finite sum reproduces the packet, and density of the regular class extends everything uniquely.

11.2 Generalised one-magnon kernels beyond the product vacuum

The wave operators above live on the exactly solvable ferromagnet. To carry the same language to a general matrix-product vacuum one needs an abstract hypothesis package, which the campaign calls $D31$. It is used twice in this section, so it is stated here in full.

Definition 11.3 (Exact fixed-packet two-magnon data over one vacuum)**.** Fix one translation-invariant injective canonical matrix-product tensor A and its state and GNS triple $(\omega_A, \mathcal{H}_A, \Omega_A)$. All constructions use this one vacuum representation; no covariant family of vacua is required. Assume:

1. *Dynamics and grading.* A finite-range translation-invariant Hamiltonian generates the dynamics, is implemented in $\mathcal{H}_A$ by commuting positive-energy time and lattice translations, and is normalised by $H\Omega_A = 0$, $\ker H = \mathbb{C}\Omega_A$. A conserved circle charge grades the selected magnon by one unit, and the construction lies in its charge-two sector.
2. *An exact band.* There is a finite-multiplicity scalar magnon band $\omega \in C^2(\mathbb{T})$, isolated within the charge-one sector on the packet neighbourhoods of clause 4, and a Gram-normalised translation-covariant *exact* map $\Gamma_M : L^2(\mathbb{T}; \mathbb{C}^m) \to \mathcal{H}_A$ with $H\Gamma_M = \Gamma_M(\omega \otimes 1_m)$. The multiplicity is fixed; matrix-valued band crossings are outside the definition. Exactness is a hypothesis, not a Rayleigh–Ritz conclusion.

3. *Filtered creators.* For each packet window, a spacetime-Schwartz filter of compact energy and momentum support, applied to a charge-one local observable, gives uniformly almost-local creators of positive energy transfer, translation-covariant as $a_{i,b}(n) = \tau_n(a_{i,b}(0))$, and acting on the vacuum as $a_{i,b}(n)\Omega_A = \Gamma_M(\chi_i e_n \otimes e_b)$ with $e_n(k) = e^{-ikn}$.

4. *Packets.* Packet amplitudes are finite sums of C_c^∞ products. The dispersion is C^∞ on neighbourhoods U_i of the compact supports K_i; the velocity sets $V_i = \omega'(K_i)$ obey $\operatorname{dist}(V_1, V_2) \geq \varepsilon_v > 0$ and $\operatorname{dist}(V_i, \{0\}) \geq \varepsilon_0 > 0$. This last nonzero-velocity clause belongs to the campaign's packet register but enters no estimate of the theorem below; only ε_v does. Filters satisfy $\chi_i = 1$ on K_i and $\operatorname{supp}\chi_i \subset U_i$; companion filters satisfy $\tilde{\chi}_i = 1$ on $\operatorname{supp}\chi_i$ and $\operatorname{supp}\tilde{\chi}_i \subset U_i$; and $h_i = \omega\tilde{\chi}_i \in C^\infty(\mathbb{T})$.

5. *A spectral inventory with margins.* Supply an inventory $\mathcal{R}_{\text{inel}}$ of every alternative propagating charge-two band tuple $r = (r_1, \ldots, r_n)$ expressible from the stated exact band data, and set $\theta_r(K) := \inf\{\sum_j \omega_{r_j}(p_j) : \sum_j p_j = K,\ \sum_j q_{r_j} = 2\}$. With $I_2 = \{(\omega(k_1) + \omega(k_2), k_1 + k_2) : k_i \in K_i\}$, require for every r that $\omega(k_1) + \omega(k_2) + \eta_{\text{inel}} < \theta_r(k_1 + k_2)$ for one $\eta_{\text{inel}} > 0$ and all $k_i \in K_i$; absent channels count as $+\infty$. Every charge-two bound band in the supplied exact data is isolated, $d_B := \inf_{k_i \in K_i, j} |\omega(k_1) + \omega(k_2) - E_{B,j}(k_1 + k_2)| > 0$, and P_B denotes the sum of its joint spectral projections; bound bands are present but are not incoming or outgoing two-magnon channels.

No clustering inequality is assumed. Injectivity of A already implies, for every $\tilde{\lambda}$ between the transfer-spectrum gap and 1 and for local C, D separated by d_{sep},

$$|\omega_A(CD) - \omega_A(C)\,\omega_A(D)| \leq C_{A,\tilde{\lambda}} \|C\| \|D\|\, \tilde{\lambda}^{d_{\text{sep}}}, \tag{207}$$

and, after uniformly almost-local truncation, the four-operator version used below. Both are consequences, not hypotheses. The package is fixed-packet only: a soft scale and its limit are no part of it.

campaign id:	D31 (H-ACE2M)
proved in:	`definitions.md` D31, appended from `theory/ansatz-scattering-2m.md` §7 with no clause edited
tested by:	`theory/checks/ansatz_scattering_2m_check.py` corroborates only the clustering identity Equation (207) on AKLT
reviewed in:	`theory/verdicts/ansatz-scattering-2m-r6.md` (the merge that gave the clause set a numbered home)

Theorem 11.4 (Rigged generalised one-magnon kernels)**.** [SKETCH] *Assume Definition 11.3 with multiplicity m, and rig the one-magnon space as $C^\infty(\mathbb{T}; \mathbb{C}^m) \subset L^2(\mathbb{T}, dk/2\pi; \mathbb{C}^m) \subset \mathcal{D}'(\mathbb{T}; \mathbb{C}^m)$. Transporting this rigging through the Gram-normalised exact band isometry Γ_M is intended to produce generalised kernels $|k, b\rangle$, $1 \leq b \leq m$, unique up to a measurable unitary change of multiplicity frame, with*

$$\Gamma_M f = \sum_b \int \frac{dk}{2\pi} f_b(k) |k, b\rangle, \qquad \langle k, b \,|\, k', b'\rangle = 2\pi\delta(k - k')\delta_{bb'}, \qquad H|k, b\rangle = \omega(k)|k, b\rangle, \tag{208}$$

and with packets built from these kernels being exactly the one-particle legs carried by the two-magnon wave operators, at the unit external-leg weight the campaign's amplitude convention stipulates.

Scope 11.5. Two named gaps stand between this and a theorem, and both were raised by the critic.

- *Hypothesis binding.* The step identifying the kernels with the filtered creator legs uses clauses 3 and 4 of Definition 11.3, whereas the argument's own ASSUME binds only clauses 1 and 2. Either the full package must be assumed, or the theorem must be split into a clause-1-and-2 part and a creator-leg part.
- *The rigging is the wrong one.* The global C^∞ rigging is not preserved by multiplication by the dispersion, which Definition 11.3 only requires to be C^2. The fix is to rig by C^2 instead, or to use a local rigging.

Either way the statement is conditional on Definition 11.3, and it asserts nothing whatsoever about a merely variational excitation band.

campaign id:	ML1-D31-kernel
proved in:	`theory/ml1-wave-operators.md` step (1.6); the defective leaves are the creator-leg identification and the choice of rigging
tested by:	—
reviewed in:	`theory/verdicts/blitz-ml1-ml6-r1.md` (two major objections, both named above)

11.3 The finite-sector Ward projection, and its erratum

We now turn to the symmetry side. On a ring, in the sector $\mathcal{H}_{n,N}$ with n overturned spins, write $S^\pm$ for the global spin raising and lowering operators, J_0^a for the three zero-momentum current components, $D_{n,N} := Q_0|_{\mathcal{H}_{n,N}}$ for the restricted charge, and $A_n := D_{n,N}^\dagger D_{n,N}$. On a subspace where A_n is invertible put

$$P_{n,N} := D_{n,N} A_n^{-1} D_{n,N}^\dagger, \qquad R_{n,N} := (1 - P_{n,N}) J_0^-, \tag{209}$$

so that $P_{n,N}$ is the orthogonal projection onto the whole descendant subspace and $R_{n,N}$ is the part of the current orthogonal to it.

Theorem 11.6 (Exact Ward projection in every finite sector)**.** [PROVED] *On $\mathcal{H}_{n,N}$,*

$$D_{n,N}^\dagger J_0^- = 2J_0^z + J_0^- S^+. \tag{210}$$

On the highest-weight subspace $\ker S^+$ *with $n < N/2$ the norm identity*

$$D_{n,N}^\dagger D_{n,N} = (N - 2n)\,\mathbf{1} \qquad \textit{holds on } \ker S^+, \tag{211}$$

and on any subspace where A_n is invertible the general polar formula is

$$P_{n,N} J_0^- = D_{n,N} A_n^{-1} \big(2J_0^z + J_0^- S^+\big). \tag{212}$$

For one hard magnon of nonzero ring momentum h this gives

$$P_{1,N} J_0^- |h\rangle_N = \frac{2i\,v(h)}{N-2}\, Q_0 |h\rangle_N, \tag{213}$$

and the same identity holds diagonally for every hard packet supported away from $h = 0$.

Scope 11.7. These four statements are exact and were independently re-verified after the erratum below. What they do *not* give is a soft theorem: the complementary vector $(1 - P_{1,N}) J_0^- |h\rangle_N$ is nonzero for generic h, so the zero-momentum current is not velocity times total charge, and the Ward identity computes the leading projection only.

campaign id:	ML4-Ward
proved in:	`theory/ml4-ward-reduction.md` step (1.3) together with its erratum note
tested by:	`theory/checks/ml4_check.py` (Ward and projection residuals at $n = 1$: maxima $8.899 \cdot 10^{-16}$ and $3.473 \cdot 10^{-15}$); `theory/checks/ml4_ward_n2_check.py` (the $n \geq 2$ refutation and the corrected form, red-capable)
reviewed in:	`theory/verdicts/corpus-r2.md` adjudication, scoped per `theory/verdicts/soft-index-b-r1.md` F1 and the erratum work item; audit `theory/verdicts/ml4-ward-n2-audit.md`

Idea of proof. The current is an $SU(2)$ vector, so $[S^+, J_0^-] = 2J_0^z$; expanding $S^+J_0^-$ with this commutator gives Equation (210). If $S^+\psi = 0$ in the n-magnon sector then $S^+S^-\psi = [S^+, S^-]\psi = 2S^z\psi = (N-2n)\psi$, which is Equation (211). Substituting Equation (210) into the definition Equation (209) of the projector gives Equation (212). For $n = 1$ a single Fourier state is highest weight and satisfies $J_0^z|h\rangle_N = iv(h)|h\rangle_N$ by direct evaluation, which yields Equation (213); translation covariance promotes it to packets.

Refuted claim 11.8 (The scalar projected Ward identity, refuted for two or more magnons)**.** [REFUTED] *The withdrawn statement is the scalar display*

$$P_{n,N}J_0^- \overset{\text{false for } n\geq 2}{=} \frac{2}{N-2n}\,Q_0 J_0^z \qquad \text{on } \ker S^+. \tag{214}$$

It is exact at $n = 1$ and false for every $n \geq 2$.

Mechanism. *The derivation passes from the correct polar formula Equation* (212) *to Equation* (214) *by applying A_n^{-1} as the scalar $(N-2n)^{-1}$ to the vector $2J_0^z\psi$. That is legitimate only if $2J_0^z\psi$ lies in $\ker S^+$, which is the subspace on which Equation* (211) *makes A_n scalar. But $[S^+, J_0^z] = -J_0^+$, so $S^+J_0^z\psi = -J_0^+\psi$, which vanishes at $n = 1$ by momentum conservation and is* nonzero *for $n \geq 2$: numerically $\|J_0^+\psi\| = 0$, 0.66, 1.36 at $n = 1, 2, 3$ for $N = 8$. The scalar display then fails by 0.26 and 1.11 at $n = 2, 3$ against left-hand sides of norm 0.91 and 1.39, whereas the corrected form below is satisfied to $2.4 \cdot 10^{-15}$.*

Surviving statement, and it is register-dependent. *Exactly at every $n < N/2$, on $\ker S^+$,*

$$\boxed{P_{n,N}J_0^- = 2\,D_{n,N}A_n^{-1}J_0^z} \tag{215}$$

— that is, Equation (212) *with the $J_0^-S^+$ term dropped, because it dies on $\ker S^+$, but with A_n^{-1} kept as an* operator *inverse, A_n being taken on the* full *sector $\mathcal{H}_{n,N}$, where it is not scalar. The register matters and must always be named: in the highest-weight-restricted register $D_\lambda := Q_0|_{\ker S^+}$ one has $A_\lambda = (N-2n)\mathbf{1}$, so the same string $2D_\lambda A_\lambda^{-1}J_0^z$ collapses straight back to the refuted display. In that register the correct form is instead*

$$P_{n,N}J_0^- = \frac{1}{m_\lambda}\,Q_0\,\Pi_{\text{hw}}\,J_0^z, \qquad m_\lambda = \frac{N-2n}{2}, \tag{216}$$

with Π_{hw} the projection onto the highest-weight subspace.

A second casualty. *The two-hard specialisation, formerly written $P_{2,N}J_0^- = \frac{2}{N-4}Q_0J_0^z$, is refuted by* 100% *of its left-hand side on the singular contact vector Equation* (181) *of Theorem 10.7 — which is itself highest weight, since applying S^+ to the alternating contact sum gives $(-1)^x + (-1)^{x-1} = 0$ at each site. At $N = 8$ the true norm is* 0.775, *the old right-hand side has norm* 1.414, *and the error is* 0.775 *(relative error* 1.000*); at $N = 10$ the error is* 0.504 *(relative* 0.730*). The corrected full-sector form Equation* (215) *is exact there to $2.4 \cdot 10^{-16}$. No scalar $(N-4)^{-1}$ version of the two-hard Ward part is available.*

Downstream audit. *The audit is complete:* 44 *sites were examined, of which* 21 *were damaged,* 2 *repairable by substitution, and* 21 *clean, with* 0 *left unclear.* ***No headline claim was reached***

by the damage. *Equations Equations* (210) *to* (213) *were independently re-verified exact. The damaged statement sites in the two earlier soft-index shards fold into a unified rewrite rather than being patched twice.*

campaign id:	ML4-Ward (erratum clause)
proved in:	`theory/ml4-ward-reduction.md`, the erratum notes at step (1.3) and at step (1.5).(2).1
tested by:	`theory/checks/ml4_ward_n2_check.py` (green exit 0, `--red` exit 1); `theory/checks/ml4_check.py` at $n = 1$
reviewed in:	`theory/verdicts/ml4-ward-n2-audit.md` (the complete 44-site audit); first found in `theory/verdicts/soft-index-b-r1.md` F1

11.4 What actually kills the linear term

The erratum has a constructive moral, isolated in the following abstract lemma: the vanishing of the unwanted first-order contribution is not supplied by the Ward identity at all. The Ward identity identifies the projected part of the current. The zero comes from channel matching plus one derivative of regularity.

Lemma 11.9 (Matching plus C^1 regularity supplies the second soft zero)**.** [PROVED] *Let $U : \mathcal{H} \to \mathcal{K}$ be an isometry, $P = UU^\dagger$, and let $\Gamma : (-\epsilon, \epsilon) \to B(\mathcal{H}, \mathcal{K})$ satisfy*

$$\Gamma(0) = U, \qquad \sup_{|k|<\epsilon} \|\partial_k \Gamma(k)\| \le C_\Gamma. \tag{217}$$

Let $J \in B(\mathcal{H}, \mathcal{K})$, put $R = (1 - P)J$, and let $b \in C^1$ satisfy $b(0) = 0$ and $|b(k)| \le C_b|k|$. Then

$$\|b(k)\Gamma(k)^\dagger R\| \le C_b C_\Gamma \|J\|\, k^2, \tag{218}$$

and for a normalised rescaled soft packet $f_\varepsilon(k) = \varepsilon^{-1/2} f(k/\varepsilon)$,

$$\|b(k)\Gamma(k)^\dagger Rg\, f_\varepsilon(k)\|_{L^2(dk;\mathcal{H})} \le C_b C_\Gamma \|J\|\, \varepsilon^2 \|u^2 f(u)\|_{L^2(du)} \|g\|. \tag{219}$$

The same statement is allowed for channel traces rather than bounded maps, provided the derivative bound is read in the trace norm on the chosen packet class.

campaign id:	ML4-A
proved in:	`theory/ml4-ward-reduction.md` step (1.2)
tested by:	`theory/checks/ml4_check.py` fixed-volume probes
reviewed in:	`theory/verdicts/corpus-r2.md` adjudication

Idea of proof. Because U is an isometry, $U^\dagger R = U^\dagger(1-UU^\dagger)J = 0$, so $\Gamma(k)^\dagger R = [\Gamma(k)-\Gamma(0)]^\dagger R$. The fundamental theorem of calculus bounds the difference by $C_\Gamma|k|$, the projection bounds $\|R\|$ by $\|J\|$, and multiplying by $|b(k)| \le C_b|k|$ gives Equation (218); substituting $k = \varepsilon u$ and using that the normalisation factor cancels the Jacobian gives Equation (219).

Remark 11.10 (Where the zero comes from). The pair of properties that kills the orthogonal linear contribution is *energy-shell channel matching*, $\Gamma(0) = U$, together with C^1 *on-shell trace regularity.* Energy conservation and multiplicity one imply the first in a genuine two-body channel. The velocity sign $\operatorname{sgn}(v_h - v_s)$ only labels which continuous wave is called outgoing; it supplies no zero at all.

11.5 Off-shell interpolation at fixed volume — and why it is not a ring soft limit

Theorem 11.11 (Fixed-volume off-shell interpolation for one hard magnon)**.** [SKETCH] *Fix $I = [a,b] \Subset (0,\pi)$ and let $\Gamma_N(k)$ be the analytic interpolation obtained by transporting the coordinate scattering formula from total momentum $h+k$ to the $k=0$ fibre, normalised by $\Gamma_N(k)|h\rangle := B_N(k,h)/\sqrt{N-2}$, where on $0 \le x < y < N$*

$$B_N(k,h;x,y) = \frac{e^{ih(x+y)/2}}{\sqrt{N}} \big\{ s(k,h) e^{iq(y-x)} + e^{-iq(y-x)} \big\}, \qquad q = \tfrac{h-k}{2}, \tag{220}$$

with $s(k,h)$ the solution of the contact equation Equation (193) analytic at zero. Let

$$\mathsf{A}_{\perp,N}(k;f,g) := (e^{ik}-1)\,\langle \Gamma_N(k) f,\ R_{n,N}\, g\rangle \tag{221}$$

be the interpolated orthogonal contribution to the Ward-reduced connected numerator. At zero soft momentum the interpolation matches the descendant, $\Gamma_N(0)|h\rangle = Q_0|h\rangle_N/\sqrt{N-2}$, since $s(0,h) = 1$. For every fixed N there is then a constant $C'_{I,N} < \infty$ with

$$|\mathsf{A}_{\perp,N}(k;f,g)| \le C'_{I,N}\, k^2\, \|f\|_{I,N} \|g\|_{I,N} \tag{222}$$

for arbitrary smooth compactly supported hard packets; the corresponding soft-packet norm is accordingly $O_N(\varepsilon^2)$, and consequently

$$\lim_{N\to\infty} \lim_{k\to 0} \frac{\mathsf{A}_{\perp,N}(k;f,g)}{k} = 0, \tag{223}$$

where the inner limit is the off-shell analytic *limit.*

Scope 11.12. This is emphatically not an on-shell ring soft limit, and the distinction is not pedantic. At fixed periodic N the nonzero momenta are discrete, so $k \to 0$ at fixed N never runs through physical ring states; the interpolated $\Gamma_N(k)$ is an off-shell analytic vector. Worse, the first *physical* sequence refutes volume uniformity outright. Take $h = 2\pi/5$, ring sizes divisible by five, and $k_N = 2\pi/N$. The low-frequency relative-coordinate sum obeys

$$\lim_{N\to\infty} N^{-1} \langle B_N(c/N,h), B_N(0,h)\rangle = \frac{8\big[1-\cos(c/2)\big]}{c^2}, \tag{224}$$

which at $c = 2\pi$ equals $4/\pi^2$ rather than the value 1 at $c = 0$. Hence the transported trace tends to $2|v(h)|(1-4/\pi^2) > 0$ along that sequence, so its ratio to $|k_N|$ grows linearly in N and the normalised version grows like $\sqrt{N}$: no N-independent version of the estimate holds. What remains open, therefore, is exactly the statement the campaign wants — a packet-smeared infinite-volume on-shell trace bound controlling the regime $k = \Theta(1/N)$ — together with the two-hard, three-magnon channel discussed next.

campaign id:	ML4
proved in:	`theory/ml4-ward-reduction.md` step (1.4) together with step (1.5).(2).2; narrowed after the erratum, since the earlier step (1.5).(2).1 consumed the refuted scalar display
tested by:	`theory/checks/ml4_check.py`: fixed-volume fitted exponents 0.99700–0.99988 for the raw orthogonal trace and 1.99700–1.99988 after multiplication by $e^{ik}-1$, with the omitted projection as built-in red mutation; the joint-scaling probe at $k = 2\pi/N$ and its `--red-uniform` mutation reinstating the retracted uniform claim, which exits nonzero
reviewed in:	`theory/verdicts/corpus-r2.md` (held at sketch)

Remark 11.13 (The sharp obstruction in the two-hard channel). To infer a quadratic bound for the orthogonal part in the two-hard sector one would need a three-magnon outgoing channel satisfying the two hypotheses of Remark 11.10 in a packet trace norm uniform in volume. Theorem 10.7 resolves the *input* two-magnon state and says nothing about the three-magnon target. Unlike the two-body half-line fibre, a three-body fixed-energy shell can have several open channels; energy conservation alone then does not force every zero-soft channel into the range of the charge, and an orthogonal degenerate channel can contribute at first order after the single lattice difference. Excluding that multiplicity is the sharp missing hypothesis, and it needs three-body wave operators together with a limiting-absorption or trace estimate. Assuming closed Bethe factorisation would hide the issue rather than settle it, and is not permitted here.

Conjecture 11.14 (Uniform current form-factor regularity). [CONJECTURE] *The finite-volume regularity of the current form factors upgrades, after wave-packet smearing, to a bound uniform in volume and in the soft limit. Any proof must control the regime $k = \Theta(1/N)$ and must exclude a physical $1/k_s$ current pole, showing that the delta and principal-value pieces appearing in the coefficient formulas are distributional normalisation artefacts.*

campaign id:	ML3
proved in:	—
tested by:	—
reviewed in:	— (future work)

Conjecture 11.15 (Order of limits and channel contamination in general). [CONJECTURE] *The order of the infinite-volume limit, the wave-packet width, and the soft limit can be controlled in general, and the bound state and off-shell coefficients do not contaminate the leading real scattering channel.*

campaign id:	ML6
proved in:	—
tested by:	—
reviewed in:	— (future work)

11.6 The order of limits on a separated packet class

The general conjecture above is open. On a restricted but honest class of preparation protocols it is a theorem, and the following is what the campaign actually owns about limit order.

Theorem 11.16 (Ordered limits remove bound and string weight on the separated class). [PROVED] *Assume Definitions 10.1 to 10.3, $J > 0$, and a fixed hard interval $I = [a, b] \Subset (0, \pi)$. Take a one-sided normalised soft packet $f_\varepsilon(k) = \varepsilon^{-1/2} f(k/\varepsilon)$ with $\operatorname{supp} f \subset [c_1, c_2] \subset (0, \infty)$ and a normalised hard packet g_σ supported in I. For each fixed $\varepsilon > 0$ use the separated preparation sequence: translate the incoming hard packet by $R_j \to \infty$, take a settling time T_j with $d_\varepsilon T_j - R_j \to \infty$, impose the separation condition, and take the ring size large enough that the thermodynamic and sampling error tends to zero. The order is*

$$N \to \infty \ \prec \ (R, T, \sigma)_j \to (\infty, \infty, 0) \ \prec \ \varepsilon \downarrow 0. \tag{225}$$

Then:

1. *at each fixed ε, the total squared coefficient of the prepared state outside the selected incoming real-scattering packet tends to zero — this includes the two-string and bound band and all real-scattering coefficients off the selected support — and the analogous outgoing statement holds after the settling time;*

2. *along every actual row-measure limit subsequence the normalised connected readout is exactly*

$$\mathsf{A}_*(\varepsilon) = \int \big[S_{\mathrm{phys}}(k,h) - 1\big]\, d\mu_{*,\varepsilon}(k,h); \tag{226}$$

3. *with* $\bar{k}_*(\varepsilon) := \int k\, d\mu_{*,\varepsilon}$,

$$\mathsf{A}_*(\varepsilon) = 2i\, \bar{k}_*(\varepsilon) + O_I(\varepsilon^2), \tag{227}$$

   the remainder being uniform in the hard packet and in the actual limit measure, so neither bound nor off-shell coefficients contaminate the leading real scattering channel;

4. *no soft-uniform Cook estimate is needed, because the fixed-*ε *channel limit is completed first.*

Scope 11.17. The theorem proves no uniform pointwise limit of individual coefficients — only that their packet-smeared contaminating sum is controlled. It applies to the separated primitive packet class, not to an arbitrary preparation protocol. It constructs no incoming bound-pair wave operator, makes no endpoint or equal-velocity claim, and has no extension to $a \downarrow 0$: that is the same non-uniformity as $k_h \to 0$ recorded in Remark 10.11, and it is excluded rather than overcome.

campaign id:	ML6-HS-SEP
proved in:	`theory/ml6-hs-sep.md` steps (1.1)–(1.9): spectral fence (1.2), collective coefficient estimate (1.3), ordered limit (1.4), soft expansion in the purified channel (1.5), endpoint fence (1.6)
tested by:	`theory/checks/ml1_ml6_check.py`: green exit 0, with `--red-bound` and `--red-unitarity` exiting 1; arithmetic only
reviewed in:	`theory/verdicts/blitz-ml1-ml6-r1.md`

Idea of proof. At the same total momentum the continuum-to-bound separation is $E(K,q) - E_b(K) = J|e^{iq} - c|^2$, which at $k = 0$ equals $J\sin^2(h/2)$, so compactness of I gives a uniform positive fence. The overlap of the unseparated charge-created relative wave with the bound fibre is exactly

$$\beta(K,q) = \frac{2\sqrt{1-c^2}\,(\cos q - c)}{1 - 2c\cos q + c^2}, \qquad \beta\big(h+k, \tfrac{h-k}{2}\big) = 2k + O_I(k^2), \tag{228}$$

so the raw bound amplitude of a soft packet is $O_I(\varepsilon)$ — the same order as the connected law itself, hence *not* by itself enough to exclude contamination. The decisive input is the ordered limit. At fixed ε, the separated preparation makes the incoming Cook tail decay like a high inverse power of the separation, so the projection onto the bound band and onto real roots off the selected window vanishes in norm *before* the soft limit is taken; Parseval on the ring, together with strong convergence of the finite fibre matrices to the half-line matrices, converts this into a collective statement about smeared coefficients, with no root-by-root labelling and no principal-value cancellation required. Only then is $\varepsilon \downarrow 0$ taken, at which point the expansion Equation (189) of Theorem 10.9 is integrated against the limit measure. Because the contaminating projection is already exactly zero, it cannot contribute a competing term of order ε.

11.7 Cook limits over a general matrix-product vacuum

Theorem 11.18 (Fixed-packet two-magnon Cook limits from exact band data)**.** [PROVED, CONDITIONAL] *Assume Definition 11.3. Put* $\mathcal{H}_{0,12} := L^2(K_1;\mathbb{C}^m) \otimes L^2(K_2;\mathbb{C}^m)$ *with free Hamiltonian* $H_{0,12} = \omega \otimes 1 + 1 \otimes \omega$ *on the smooth product core* $\mathcal{D}_{12}$*, the two slots labelling the two disjoint packet windows rather than distinguishable species, and let* I *be the two-creator identification map. Then,* as a conditional implication from Definition 11.3 and nothing else*:*

(i) the fixed-packet limits $W_\pm F := \lim_{t\to\pm\infty} e^{itH} I e^{-itH_{0,12}} F$ *exist for* $F \in \mathcal{D}_{12}$*;*

(ii) $W_\pm$ are isometries, extend uniquely to the packet-domain closure, and intertwine joint time and space translations; their ranges are annihilated by the sum P_B of the listed fibrewise-isolated charge-two bound-band projections, and are orthogonal to every alternative propagating channel in the inventory, whose joint energies exceed $\sup_{I_2} E + \eta_{\mathrm{inel}}$.

Specialised to the ferromagnet of Definition 10.1, a fixed packet range lies in the matching part of the scattering summand of Theorem 10.7 and is orthogonal to its two-string summand, and $W_+^* W_-$ is exactly multiplication by the physical ratio S_{phys} of Definition 10.2 (the inverse ratio under the opposite velocity ordering), of modulus one.

The clustering estimates Equation (207) *and its four-operator version are* derived *from injectivity of the vacuum tensor, not assumed; no independent clustering hypothesis enters.*

The implication is not vacuous: the ferromagnet of Definition 10.1 satisfies all five clauses of Definition 11.3. Its all-up state is the injective bond-dimension-one product tensor, its shifted Hamiltonian is positive with $\ker H = \mathbb{C}\Omega$, the total magnetisation supplies the conserved circle grading, the Fourier map $g \mapsto \int \frac{dk}{2\pi} g(k) \sum_x e^{ikx}|x\rangle$ is an exact isometric band map, a spacetime-Schwartz filter of S_n^- gives the covariant creators, and Theorem 10.7 supplies exactly one bound band with a positive gap and an empty inelastic inventory.

Scope 11.19. Clauses 1–5 of Definition 11.3 — the exact band, the creators, the covariance and the threshold data — are *hypotheses*, not consequences of a variational ansatz; only the clustering estimates are derived. Nothing here is uniform in a soft scale: every estimate is at fixed packets, and the Cook and Gram constants carry the packet Schwartz seminorms $s_N(F)$. Along a soft rescaling $f_\varepsilon(k) = \varepsilon^{-1/2} f(k/\varepsilon)$ one has $\|\partial^j f_\varepsilon\|_\infty = \varepsilon^{-j-1/2}\|f^{(j)}\|_\infty$, so the N integrations by parts carry a factor ε^{-N} and the bound degrades without limit as $\varepsilon \downarrow 0$. The nonzero-velocity constant ε_0 of clause 4 enters no estimate in this theorem; only the velocity separation ε_v does. The theorem asserts nothing about the fixed-time interface of Theorem 11.20, which no step of its proof consumes. No endpoint or equal-velocity construction, no soft-uniform Cook bound, no bound-state wave operator, no compatibility or range exhaustion across packet windows, and no asymptotic completeness is claimed.

campaign id:	AC-EX-2M (parts A2M.1–A2M.2)
proved in:	`theory/ansatz-scattering-2m.md` steps (1.2)–(1.7′): fixed-packet fence (1.2), clustering upgrade (1.3), filtered one-particle equations (1.4), Cook integrability (1.5), isometry (1.6), intertwining and the exact match (1.7), nonvacuity on the ferromagnet (1.7′)
tested by:	`theory/checks/ansatz_scattering_2m_check.py`: green exit 0 and seven registered red modes exiting 1 under optimised Python, all nine gates evaluated before exit. The certificate corroborates *only* the clustering identity Equation (207) on AKLT, reproduced independently in a second tensor basis; it tests no Cook limit, no exact band, no spectral separation, and no identification with the ferromagnet's scattering summand. Two gate caveats are on record: the rescaled agreement gate does not certify the absence of a two-sided support-length factor (support-length independence is proved analytically, not numerically), and one guard gate is identically zero by algebra, so it is a code-shape guard rather than a numerical certificate.
reviewed in:	`theory/verdicts/ansatz-scattering-2m-r5.md` §9

Idea of proof. Exactness of the band map means the filtered creator defect $D_{i,b}(n) = [H, a_{i,b}(n)] - \sum_m h_i(m-n) a_{i,b}(m)$ annihilates the vacuum and is almost local uniformly in n. Expanding H on the two-creator vector and cancelling both filtered free convolutions therefore leaves only a commutator term, $(HI - IH_{0,12})F = \sum_{x,y} F(x,y)[D_1(x), a_2(y)]\Omega_A$, whose norm decays faster than any power of $x-y$. Outside its velocity cone each free packet is rapidly decreasing in $|x|+|t|$, while on the product of the two main cones the velocity separation forces $|x-y| \geq \varepsilon_v|t|/2$; splitting the sum accordingly gives an integrable majorant of order $|t|^{-3}$, and the fundamental theorem of calculus then gives both strong limits. Isometry follows by applying the derived clustering estimate to the four-creator scalar product on the product cones, where the Gram-normalised exact band maps factorise into the free tensor Gram form, with the complement again power-suppressed. Intertwining is the usual replacement of t by $t+s$; the joint spectral calculus then converts the fibrewise bound gap and the inelastic margin into exact annihilation of the corresponding projections. Finally, on the ferromagnet, stationary phase puts the two relative branches of Equation (183) in the incoming and outgoing regions exactly as the coefficient convention of Definition 10.2 prescribes, so the constructed labels are that convention's and not a new one.

11.8 The fixed-time interface: what is and is not established

Theorem 11.20 (The fixed-time protocol interface)**.** [SKETCH] *For the adjudicated fixed-time interface, two* separate *statements are established, and they are not composed.*

1. Diagonal compactness, conditional on assumed bounds. *Write the amputated datum at an allowed full index tuple α as $\mathsf{A}_\alpha(h) = \mathsf{N}_\alpha(h)/\mathsf{D}_\alpha(h)$ for $h \in I$. If the selected-hard-packet bounds*
$$\operatorname*{ess\,inf}_{\alpha} \operatorname*{ess\,inf}_{h\in I} |\mathsf{D}_\alpha(h)| \geq d_I > 0, \qquad \sup_{\alpha} \|\mathsf{N}_\alpha\|_{L^2(I)} \leq C_I < \infty \tag{229}$$
hold for every allowed tuple, then $\sup_\alpha \|\mathsf{A}_\alpha\|_{L^2(I)} \leq C_I/d_I$, so every full-index sequence respecting the precedence
$$N \to \infty \ \prec\ t \to \pm\infty \ \prec\ (W \uparrow \mathbb{Z}, \sigma \downarrow 0)_j \ \prec\ \varepsilon \downarrow 0 \tag{230}$$
with $\varepsilon \downarrow 0$ has a weakly convergent subsequence in $L^2(I)$. This is Banach–Alaoglu applied to assumed bounds: both displayed inequalities in Equation (229) *are assumptions, not derivations, and no iterated-order datum is asserted to exist.*

2. Creator-choice independence — a theorem about Haag–Ruelle families. *At fixed $\varepsilon > 0$, a soft creator family that is admissibly Haag–Ruelle, satisfies the one-particle on-shell normalisation, and has velocity support disjoint from the hard packet, may be replaced by the filtered creator of Definition 11.3 without changing any connected on-shell pairing in the limit of large times.*

Scope 11.21. **The second statement does not apply to the adjudicated interface.** Both protocols on record specify a *fixed-time* insertion $Q[f_\varepsilon]\psi$ of the charge on an already prepared hard vector, and that is not a Haag–Ruelle creator family: it never free-evolves its profile. The failure is precisely of Haag–Ruelle admissibility and not of the one-particle normalisation, which *is* proved on the ferromagnet for the family built from that same charge acting on the vacuum.

For the adjudicated datum the exact mismatch is visible at the soft law's own linear order. By Theorem 10.9 and Equation (192),
$$Q_{k_s}|k_h\rangle - |B^{\text{in}}\rangle = (1 - S_{12})|P_{12}\rangle = -2ik_s|P_{12}\rangle + O(k_s^2), \tag{231}$$
a relative size $\sqrt{2}\,k_s(1 + O(k_s))$ — nonzero exactly where the soft law lives. The identification of the fixed-time datum with the constructed channel of Theorem 11.18 is therefore **open**, and two steps are named as missing:

(i) prove, at order k_s and with a displayed uniform remainder, that the large-time readout's connected packet-amputated on-shell pairing equals the constructed-channel one *despite* that mismatch;

(ii) exhibit an instance of the soft-regularity condition for the fixed-time family, including existence of its volume and time limits.

The soft-regularity clauses have been verified for exactly one datum: the *constructed-channel* family on the infinite chain, which has a single member and involves no exhaustion,

$$\mathsf{A}(\varepsilon)(h) = \int d\mu_f(u)\, S_{\mathrm{phys}}(\varepsilon u, h) \tag{232}$$

on the ferromagnet, for which every uniformity clause is trivially satisfied because the family is a singleton and the limit-existence clause is bypassed rather than met. It therefore backs no claim about the interface. The window-and-width-uniform form of the soft-regularity condition is open on every model, the ferromagnet included; it remains a hypothesis with no verified instance. No fixed-time charge-to-channel identification, no fixed-time charge-to-scattering-vector equality, and no first-jet theorem for the interface is claimed.

campaign id:	AC-EX-2M-D29
proved in:	`theory/ansatz-scattering-2m.md` steps (1.8)–(1.9), excluding step (1.9).(2).4, which belongs to Theorem 11.18; the ported creator-choice theorem is taken from `refs/arxiv-1412.2970` (final clause of its Haag–Ruelle theorem)
tested by:	No certificate exists for this row: `theory/checks/ansatz_scattering_2m_check.py` tests none of the assumed bounds Equation (229), Haag–Ruelle admissibility, the missing step (i), the volume and time limits, or the soft-regularity condition
reviewed in:	`theory/verdicts/ansatz-scattering-2m-r5.md` §9 and `ansatz-scattering-2m-r6.md`

12 The soft index: Ward-identity constraints on soft amplitudes

12.1 Why an index, and not a soft factor

The programme that opened this part of the campaign was the obvious one: look for a lattice analogue of a universal soft factor — a single multiplier, built out of the charges and velocities of the hard legs alone, that multiplies an n-leg amplitude when one extra soft quantum of momentum k is emitted. That route failed, and it failed for a stated reason: on the lattice there is no proof that the soft multiplier is independent of the microscopic source, and there is an explicit local source for which it is not. The refutation is recorded in Section 13; what survives there is a criterion and a conditional shape, not a number.

The work in this section replaces that route by a different one. Rather than asking what the soft emission *is*, we ask what the symmetry algebra *forces* it to be. The symmetry supplies an exact operator identity in finite volume — a Ward identity between a charge, its associated current zero mode, and the projector onto the range of the charge — and that identity constrains a normalised ratio of matrix elements to be exactly one. We call that ratio the *soft index*. It is an index in the honest sense: it is an integer (or a one), it is insensitive to the couplings of the Hamiltonian, to the system size, and to the hard state inside the relevant sector, and its proof is finite-dimensional linear algebra plus representation theory.

The strategy therefore splits cleanly into two halves, and the split is the organising idea of this whole section.

- **Structure.** What the symmetry alone forces. This half is unconditional in finite volume (Theorem 12.4, Theorem 12.6, Theorem 12.8), and it becomes a conditional statement about limits of a protocol readout once one asks about the soft limit (Theorem 12.11). Structure never produces a number: it produces a shape with an open constant in it.
- **Value.** What pins the constant. Values come from on-shell two-body data — the exact spin-S two-magnon phase slope of Section 10 — and they enter only after a separately stated matching hypothesis that identifies the protocol readout with the on-shell multiplier. Where that matching is a hypothesis, the value statement is a conditional; where the matching is a theorem, the value statement is a theorem.

The one substantive advance of the campaign in this area is that on a restricted but explicitly displayed class of preparations — the *separated-preparation* class of Definition 12.15 — the matching hypothesis is no longer a hypothesis. It is proved, with zero remainder (Theorem 12.16), and with it the value follows unconditionally on that class (Theorem 12.18) and the factorisation takes an exact LSZ shape (Theorem 12.20). What remains open there is not the aggregate identity but its decomposition into named components; that residual gap is isolated as a single lemma (Scope 12.22).

12.2 The protocol datum: a proposed readout, not yet a definition

Everything in the "limit" half of this section is a statement about one concrete finite-volume number and its subsequential limits. That number, and the regularity package attached to it, are written down here in full. They are flagged [PROPOSED, NOT MERGED] throughout: they live in the merge-proposal block of the soft-index shard and have deliberately *not* been installed as numbered definitions in the campaign's definition file, because no adjudicated round has yet promoted them. Every theorem below that mentions them says so.

Definition 12.1 (The windowed, packet-smeared, charge-created ratio readout). [PROPOSED, NOT MERGED]

Fix a site spin S with $2S \in \mathbb{N}$ and the fully polarised bilinear ferromagnet $H_S = -J\sum_x(\mathbf{S}_x \cdot \mathbf{S}_{x+1} - S^2)$ on an N-site periodic ring, and fix a hard window $I = [a, b] \Subset (0, \pi)$ inside one physical velocity channel. The readout has six parts.

(1) Hard register. Let $\psi_{g,\sigma}$ be a normalised one-magnon hard packet drawn from $g_\sigma \in C_c^\infty(I)$ on the ring momenta. Write $m_{\lambda,N}$ for the weight of its sector; on the primitive one-magnon band $m_{\lambda,N} = NS - 1$, while an abstract use carries only $m_{\lambda,N}/N \to \rho > 0$. On the *full* weight sector put $D_{\lambda,N} = Q_0$, $A_{\lambda,N} = D_{\lambda,N}^\dagger D_{\lambda,N}$ and $P_{\lambda,N} = D_{\lambda,N} A_{\lambda,N}^{-1} D_{\lambda,N}^\dagger$, and for a scalar hard band with $v_h \neq 0$ record the *measured Ward register*

$$\ell_{\lambda,N}(h) := \frac{\langle h|Q_0^\dagger P_{\lambda,N} J_0^- |h\rangle}{2iv_h}. \tag{233}$$

(2) One scale-tied soft packet. Fix $f \in C_c^\infty((c_1, c_2))$ with $0 < c_1 < c_2 < 1$ and $\|f\|_2 = 1$ (or its reflection in the opposite channel), put $f_\epsilon(k) = \epsilon^{-1/2} f(k/\epsilon)$ and $\Lambda_N(\epsilon) = (2\pi\mathbb{Z}/N) \cap \operatorname{supp} f_\epsilon$. The *only* soft insertion is $Q[f_\epsilon] = \sum_{k\in\Lambda_N(\epsilon)} f_\epsilon(k) Q_k$, and the finite state is $\Phi_{N,\sigma}(0;\epsilon) = Q[f_\epsilon]\psi_{g,\sigma}$. There is no independent carrier or width limit and no arbitrary local source.

(3) Coordinate-kernel register. A normalised two-excitation vector is mapped to a symmetric $N \times N$ matrix by sending an occupation coefficient c_{xy}, $x < y$, to $c_{xy}/\sqrt{2}$ at the entries (x, y) and (y, x) and, when $S \geq 1$, a double-occupancy coefficient c_{xx} to (x, x); the momentum kernel is the discrete Fourier transform $\widehat{\Phi}(k, h) = N^{-1} \sum_{x,y} e^{-i(kx+hy)} \Phi(x, y)$.

(4) Interacting/free readout. For a settling time T let $\widehat{\Phi}_N(T)$ be the kernel of $e^{-iH_S T}\Phi_{N,\sigma}(0;\epsilon)$ and let $\widehat{\Phi}_N^{\text{free}}(T)(k, h) = \widehat{\Phi}_N(0)(k, h)\, e^{-i[\omega_S(k)+\omega_S(h)]T}$ be the freely propagated reference. With a hard-column set $K_h \subset I \cap (2\pi\mathbb{Z}/N)$ and an initial ordering matched to the physical out/in labelling, put

$$d_N(k) = \sum_{h\in K_h} |\widehat{\Phi}_N^{\text{free}}(T)(k, h)|^2, \qquad n_N(k) = \sum_{h\in K_h} \widehat{\Phi}_N(T)(k, h)\, \overline{\widehat{\Phi}_N^{\text{free}}(T)(k, h)}, \tag{234}$$

and, on the domain where $D_N(\epsilon) := \sum_k d_N(k) > 0$,

$$d\mu_{N,\epsilon}(k, h) = \frac{|\widehat{\Phi}_N^{\text{free}}(T)(k, h)|^2}{D_N(\epsilon)}, \qquad \mathcal{A}_{N,T,W,\sigma}(\epsilon) = \frac{\sum_k n_N(k)}{D_N(\epsilon)} - 1, \qquad \bar{k}(\epsilon) = \int k\, d\mu_{N,\epsilon}. \tag{235}$$

The aggregate is a norm projection, not a pointwise division by a hard amplitude; every finite datum is an explicit finite matrix quantity.

(5) Exact finite diagnostics. Record separately the full-sector split $J_k^- = P_{\lambda,N} J_k^- + (1 - P_{\lambda,N}) J_k^-$ and both terms of the current/contact identity, identifying neither with the other; and for an exact hard vector define the contact defect $\mathfrak{D}_N(k, h) = (H_S - \omega_S(k) - \omega_S(h)) Q_k |h\rangle_N$, whose time integral is the exact interacting-minus-free evolution by finite-dimensional Duhamel. No LSZ exhaustiveness, boundary decay, on-shell matching, value of the amputation constant, wave operator, or completeness statement is definitional here.

(6) Admissible indices and their order. An index is admissible only if $N\epsilon(c_2 - c_1) > 2\pi$, its initial geometry matches the physical channel, and its settling time lies in a declared settling/recollision sandwich $[T_{\min}, c_{\text{rec}} N/v_{\max}]$. Outer limits use admissible sequences $j \mapsto (N_j, T_j, W_j, \sigma_j)$ at *fixed* ϵ; only after an outer limit point has been selected is $\epsilon \downarrow 0$ taken. At fixed N the sample set $\Lambda_N(\epsilon)$ is empty once $c_2\epsilon < 2\pi/N$, so the joint sequence is not a limit of this family.

campaign id:	proposed D29 (PROTO)
proved in:	`theory/soft-index-r2.md` §7, step (1.14).(2.1) — quarantined in that shard's merge-proposal block; *not* installed in `definitions.md`
tested by:	`theory/checks/soft_index_r2_check.py` gates SIDXR2-C1–C3, C7–C8 (finite diagnostics and negative controls)
reviewed in:	`theory/verdicts/soft-index-r2-critic.md`

Definition 12.2 (The regularity target attached to the readout)**.** [PROPOSED, NOT MERGED]

This is a closure property for Definition 12.1 and asserts no existence of an outer limit point. Along a stated admissible outer subsequence, whenever the scalar datum has an actual cluster family on soft scales $E \subset (0, \epsilon_0]$ with $0 \in \overline{E}$, require: *(a) denominator nondegeneracy* — the aggregate free mass is nonzero, the normalised row measures are tight on the fixed hard window and the scale-tied soft support, and rows of zero free mass carry zero measure; *(b) component compactness* — the row measures, $m_{\lambda,N}/N$, $\ell_{\lambda,N}$, and every component named by any separately assumed decomposition have a further cluster point, with limits written $\mu_{*,\epsilon}$, $\rho > 0$, ℓ_h, and in particular $c_1\epsilon \leq |\bar{k}_*(\epsilon)| \leq c_2\epsilon$ in a one-sided channel; *(c) regularity only* — each cluster family admits the continuous or C^1 extension to $\epsilon = 0$ required at its point of use, with locally uniform bounds and equicontinuous first difference quotients; *(d) no hidden matching* — there is no equality with a chamber coefficient, no on-shell eigenvector, no physical multiplier, no LSZ exhaustiveness, no value or reality condition for the amputation constant, and no formula for ℓ_h in terms of a composite charge; *(e) optional convenience* — full-family convergence may be assumed, buying uniqueness of the selected limit function only.

Scope 12.3 (Regularity cannot supply a value). Clause (e) is stated with its own limitation because the limitation is the point. For any regular family with $1 + \mathcal{A}(\epsilon) \neq 0$ the transformation

$$1 + \mathcal{A}(\epsilon) \;\longmapsto\; [1 + \mathcal{A}(\epsilon)]\, e^{ic\epsilon} \tag{236}$$

preserves every clause (a)–(d) and shifts the phase jet $\partial_\epsilon \arg(1+\mathcal{A})|_0$ by the arbitrary real constant c. Any argument that reads a numerical coefficient out of the regularity package alone is therefore invalid: a separate value supplier must appear at the exact leaf where the coefficient is fixed. In this section that supplier is always the exact two-magnon phase slope of Section 10, and never a regularity clause.

campaign id:	proposed D30 (TGT)
proved in:	`theory/soft-index-r2.md` §7, step (1.14).(2.2); the value-shift obstruction is step (1.7)
tested by:	`theory/checks/soft_index_r2_check.py` gate SIDXR2-C6 (TGT-MOBIUS)
reviewed in:	`theory/verdicts/soft-index-r2-critic.md`

12.3 The finite soft-index theorem on the $SU(2)$ ring

The first theorem is the anchor of the whole section: it is exact, finite, and free of every limiting hypothesis. Its content is that the charge, its current zero mode, and the projector onto the range of the charge are locked together by the algebra, and that the resulting normalised ratio is one.

The one thing the reader must carry away is that the identity comes in *two registers* which look similar and are not interchangeable. In the first, the Gram operator $A = D^\dagger D$ is formed on the *whole* weight sector and inverted there. In the second, the charge is restricted to the highest-weight subspace first, and only then is the Gram operator formed; it is then a scalar. Substituting the scalar of the second register into the first is false as soon as the sector contains two or more magnons — a mistake the campaign made once and now guards against with a dedicated red mode.

Theorem 12.4 (Finite soft index on an $SU(2)$ ring)**.** [PROVED]

Let a finite periodic ring carry an on-site $SU(2)$ action and an $SU(2)$-invariant finite-range Hamiltonian. Let $Q_0 = S^-$, let (J_0^+, J_0^z, J_0^-) be its vector-current zero modes, and let $\mathcal{H}_{\lambda,N}$ be the full *weight sector on which $S^z = m_{\lambda,N} > 0$. Put*

$$D_{\lambda,N} := Q_0|_{\mathcal{H}_{\lambda,N}}, \qquad A_{\lambda,N} := D_{\lambda,N}^\dagger D_{\lambda,N} \quad \textit{on all of } \mathcal{H}_{\lambda,N},$$

let $P_{\lambda,N}$ project onto $\operatorname{ran} D_{\lambda,N}$, and let $\Pi_{\rm hw}$ project $\mathcal{H}_{\lambda,N}$ onto $K_{\lambda,N} := \ker S^+ \cap \mathcal{H}_{\lambda,N}$. Then $A_{\lambda,N}$ is strictly positive, and for every $\psi \in K_{\lambda,N}$:

(i) Full-sector Gram-inverse register.

$$P_{\lambda,N} J_0^- \psi = 2\, D_{\lambda,N} A_{\lambda,N}^{-1} J_0^z \psi. \tag{237}$$

(ii) Separately restricted highest-weight register. *With $D_{\rm hw}$ the restriction of $D_{\lambda,N}$ to $K_{\lambda,N}$ and $P_{\rm hw}$ the projector onto its range,*

$$D_{\rm hw}^\dagger D_{\rm hw} = 2m_{\lambda,N}\mathbb{1}, \qquad P_{\rm hw} J_0^- \psi = \frac{1}{m_{\lambda,N}}\, Q_0\, \Pi_{\rm hw} J_0^z \psi. \tag{238}$$

(iii) Primitive one-magnon residue. *For a nonzero-momentum primitive one-magnon vector $|h\rangle_N$ in the spin-S ferromagnetic band, where $m_{\lambda,N} = NS - 1$, the projector in Equation (238) may be removed and*

$$J_0^z |h\rangle_N = i\, v_S(h) |h\rangle_N, \qquad \langle h| Q_0^\dagger P_{\lambda,N} J_0^- |h\rangle_N = 2i\, v_S(h). \tag{239}$$

Equivalently, the normalised Ward register Equation (233) equals $\ell_{\lambda,N}(h) = 1$.

(iv) Pure charge-created anchor. *If ψ_g is a one-magnon packet of exact band vectors and $\Phi_0 := Q_0 \psi_g$ is the* pure *zero-mode-created state, then its interacting/free row ratio in the readout of Definition 12.1 at soft row $k = 0$, with nonzero free denominator, is exactly one for every finite N, T, W, σ and every site spin S:*

$$\mathcal{R}_{\Phi_0}(0) = 1. \tag{240}$$

Scope 12.5 (What the two registers do and do not say)*.* The two registers are not interchangeable, and the difference is not cosmetic. In Equation (237) the inverse Gram operator stays on the full source sector, because $J_0^z\psi$ need not remain highest weight once the sector contains two or more magnons; the scalar $2m_{\lambda,N}$ of Equation (238) may not be pulled through $A_{\lambda,N}^{-1}$. In particular the string $m_{\lambda,N}^{-1} Q_0 J_0^z$, without the highest-weight projector, is *false* for two or more magnons, where the cross-register error is $O(1)$ while Equation (237) is exact; that string appears nowhere as a claim.

The anchor Equation (240) is an anchor for the object just named and nothing else. It does not assert that the adjoined $k = 0$ row of the running family $Q[f_\epsilon]\psi_g$ equals one at $S \geq 1$ — a direct computation gives $2.1633 + 0.1527i$ for that row at $S = 1$, $N = 7$, against one to 9.5×10^{-16} for the pure charge-created row — and it implies nothing about $\epsilon \downarrow 0$, since the admissible family contains no fixed-N continuous path to that row. At $S = 1/2$ the same row identity holds for arbitrary two-magnon input, so there it is a readout tautology rather than independent evidence for a soft zero.

The theorem makes no assumption of the protocol readout, its regularity package, wave operators, completeness, integrability, or any soft limit.

Idea of proof. Three short steps. First, positivity: decomposing the finite $SU(2)$ representation into spin-j irreducibles, a weight-m vector satisfies $S^+S^- = (j+m)(j-m+1) \geq 2m > 0$, so $A_{\lambda,N}$ is invertible on the entire sector and the polar formula $P_{\lambda,N} = D_{\lambda,N}A_{\lambda,N}^{-1}D_{\lambda,N}^\dagger$ holds. Second, the Ward step: covariance of the invariant finite-range current gives $[S^+, J_0^-] = 2J_0^z$, so on a highest-weight ψ, $D_{\lambda,N}^\dagger J_0^-\psi = S^+J_0^-\psi = 2J_0^z\psi$; substituting this into the polar formula is exactly Equation (237). Restricting the domain first inserts the source projector Π_{hw} into the adjoint and turns the Gram operator into the scalar $2m_{\lambda,N}$, giving Equation (238). Third, the residue: for $h \neq 0$ the raised current $J_0^+|h\rangle_N$ vanishes because the zero-mode current preserves momentum while its raised target is the momentum-zero vacuum, so $J_0^z|h\rangle_N$ is itself highest weight and the projector drops; the continuity equation $-[\omega_S(h+k) - \omega_S(h)] = (e^{ik}-1)\langle h+k|J_k^z|h\rangle_N$ has removable value $i\partial_h\omega_S(h) = iv_S(h)$ at $k=0$, and pairing with $Q_0|h\rangle_N$, whose squared norm is $2m_{\lambda,N}$, cancels the weight and leaves $2iv_S(h)$. The anchor Equation (240) is immediate from $[H, Q_0] = 0$ and $\omega_S(0) = 0$: the interacting and free kernels coincide row by row.

campaign id:	S-IDX-fin-r2
proved in:	`theory/soft-index-r2.md` steps (1.3)–(1.5); positivity (1.3).(2.1), full register (1.3).(2.4), restricted register (1.3).(2.5), residue (1.3).(2.7)–(2.8), anchor (1.4)
tested by:	`theory/checks/soft_index_r2_check.py` gates SIDXR2-C1–C3, green exit 0 with eight public red modes; the `--red-register-trap` mode executes the standing cross-register mutation and propagates its outcome
reviewed in:	`theory/verdicts/soft-index-r2-critic.md` (PROMOTABLE NOW; proof independently recomputed by the critic, promoted after the mandated checker repair)

12.4 The same theorem for every compact group

Nothing in the previous proof used $SU(2)$ beyond one root ladder. Written in that language the theorem generalises verbatim to any compact Lie group, on any finite graph or periodic lattice, one represented root at a time. This is worth stating in full because it makes visible exactly which structural ingredients are load-bearing — a root triple, an adjoint-covariant current, and an occupied weight sector — and therefore exactly where the statement has no content at all.

Theorem 12.6 (Root-wise finite soft index for a compact group). [PROVED]

Let a finite graph or a finite periodic lattice carry a finite-dimensional on-site unitary representation $U_N = u^{\otimes N}$ of a compact Lie group G and a G-invariant finite-range Hamiltonian, and put $Q(X) = \sum_x q_x(X)$ for $X \in \mathfrak{g}_{\mathbb{C}}$. Suppose given a displayed complex-linear current zero mode $J_0 : \mathfrak{g}_{\mathbb{C}} \to \mathrm{End}(\mathcal{H}_N)$ satisfying the adjoint covariance

$$[Q(X), J_0(Y)] = J_0([X,Y]); \tag{241}$$

on a periodic lattice J_0 may be taken to be the displayed directional zero-momentum flux. Let α be a represented root of the semisimple Lie algebra of the identity component G^0, with a compact-compatible triple $[E_\alpha, F_\alpha] = H_\alpha$, $[H_\alpha, E_\alpha] = 2E_\alpha$, $[H_\alpha, F_\alpha] = -2F_\alpha$, $Q(E_\alpha)^\dagger = Q(F_\alpha)$ and $Q(H_\alpha)$ Hermitian. For an occupied positive coroot weight λ put

$$\mathcal{H}^\alpha_{\lambda,N} := \ker(Q(H_\alpha) - \lambda), \qquad D_\alpha := Q(F_\alpha)|_{\mathcal{H}^\alpha_{\lambda,N}}, \qquad A_\alpha := D_\alpha^\dagger D_\alpha,$$

let P_α project onto $\operatorname{ran} D_\alpha$, let $K^\alpha_{\lambda,N} := \ker Q(E_\alpha) \cap \mathcal{H}^\alpha_{\lambda,N}$ with projector Π^α_{hw}, and let D_{hw}, P_{hw} be the correspondingly restricted objects. Then $A_\alpha > 0$, and for every $\psi \in K^\alpha_{\lambda,N}$:

$$P_\alpha J_0(F_\alpha)\psi = D_\alpha A_\alpha^{-1} J_0(H_\alpha)\psi, \tag{242}$$

$$D_{\mathrm{hw}}^\dagger D_{\mathrm{hw}} = \lambda\mathbb{1}, \qquad P_{\mathrm{hw}} J_0(F_\alpha)\psi = \frac{1}{\lambda}\, Q(F_\alpha)\, \Pi^\alpha_{\mathrm{hw}} J_0(H_\alpha)\psi. \tag{243}$$

Whenever $\langle \psi, J_0(H_\alpha)\psi\rangle$ is nonzero, the normalised Ward index is exactly one:

$$\frac{\langle D_\alpha\psi,\ P_\alpha J_0(F_\alpha)\psi\rangle}{\langle \psi,\ J_0(H_\alpha)\psi\rangle} = 1. \tag{244}$$

If a possibly disconnected compact subgroup $K \le G$ is used for a Schur or sector statement, the required global covariance $U_N(k)J_0(Y)U_N(k)^\dagger = J_0(\mathrm{Ad}_k Y)$, $k \in K$, is displayed as a hypothesis; infinitesimal covariance alone is not used for disconnected elements.

Scope 12.7 (Three fences: where this theorem has no rows). The theorem has rows only for *represented roots* of the semisimple Lie algebra of G^0, and this is a genuine restriction with three separate consequences. A central-torus direction has no root ladder, hence no row at all. A finite group has no nonzero Lie-algebra current component, hence no Lie-current row. And an arbitrary finite graph supplies no canonical direction and no zero mode: on a periodic lattice one may use the displayed directional flux, but on a general graph the current map is an input, not something the geometry provides. A chosen finite-ring periodisation of a termwise invariant interaction is an *example* supplying Equation (241); it is not part of the abstract hypotheses, and no canonical half-line cut on a circle is asserted.

As in the $SU(2)$ case, the full-sector Gram-inverse register Equation (242) and the root-highest-restricted scalar register Equation (243) are not interchangeable: $J_0(H_\alpha)\psi$ need not remain root-highest, so λ cannot replace A_α in Equation (242).

Idea of proof. Restrict the finite unitary representation to the root $\mathfrak{sl}_2$ and use complete reducibility, $\mathcal{H}_N \cong \bigoplus_{n\ge 0} V_n \otimes M_{n,\alpha}$. On the weight-$\lambda$ line of V_n,

$$Q(E_\alpha)Q(F_\alpha) = \tfrac{1}{4}(n+\lambda)(n-\lambda+2)\mathbb{1}\ \ge\ \lambda > 0,$$

so A_α is strictly positive on the *entire* source sector and $P_\alpha = D_\alpha A_\alpha^{-1} D_\alpha^\dagger$. Adjoint covariance gives $D_\alpha^\dagger J_0(F_\alpha)\psi = Q(E_\alpha)J_0(F_\alpha)\psi = J_0(H_\alpha)\psi$ for root-highest ψ, and substituting into the polar formula yields Equation (242). For the restricted register, $Q(E_\alpha)Q(F_\alpha)\phi = (Q(F_\alpha)Q(E_\alpha) + Q(H_\alpha))\phi = \lambda\phi$ on $K^\alpha_{\lambda,N}$, and the adjoint of a restricted-domain map carries the source projector. The index Equation (244) needs no further work at all: since $D_\alpha\psi$ lies in $\operatorname{ran} D_\alpha$, the projector is invisible in the pairing and $\langle D_\alpha\psi, P_\alpha J_0(F_\alpha)\psi\rangle = \langle \psi, D_\alpha^\dagger J_0(F_\alpha)\psi\rangle = \langle \psi, J_0(H_\alpha)\psi\rangle$; divide. No injectivity, rank-one property, or scalarity of a reduced operator is used beyond the positivity already proved.

campaign id:	S-IDX-fin-G
proved in:	`theory/soft-index-general.md` §§1–2 (statement §1.2, positivity and full register, restricted register, normalised index); `theory/soft-index-2d.md` §§1–2
tested by:	`theory/checks/soft_index_2d_check.py` gates SIDX2D-C0–C3 and `--red-scalar-full`; the general-G and $SU(2)$ reference gates; fences computed as G5-C1-ABELIAN, G5-C2-FINITE, G5-C3-GLOBAL-FORM
reviewed in:	`theory/verdicts/soft-index-r2-critic.md` (reduction to Theorem 12.4 verified)

12.5 The soft index as a difference of superselection labels

There is a second, complementary statement in the general compact setting, and it is the one that deserves the name "index" most literally. Instead of asking about operator matrix elements it asks about *labels*: the charge-created and current-created vectors sit in a central character sector, and the soft datum is the difference between the target label and the source label. That difference is a character, it takes integer values on cocharacters, and it does not depend on the ring size, the hard vector, the Hamiltonian couplings, or any inventory of scattering channels.

Theorem 12.8 (Sector-label integrality of the soft index)**.** [PROVED]

Let a finite periodic ring carry a compact, possibly disconnected, on-site Lie symmetry G, let α be a vacuum label with unbroken subgroup H_α, write $Z_\alpha := Z(H_\alpha)$, and define the effective unbroken centre

$$K_u := \{z \in Z_\alpha : u(z) \in U(1)\not{\mathbb{1}}\}, \qquad Z_{\alpha,\mathrm{eff}} := Z_\alpha / K_u, \qquad \Lambda^*_\alpha := \mathrm{Hom}(Z_{\alpha,\mathrm{eff}}, U(1)). \tag{245}$$

All occupied Z_α characters have the same restriction to K_u, so they form an affine torsor whose differences lie in Λ^*_α; this character group carries the free lattice of the connected central torus and may in addition carry a finite torsion part contributed by disconnected components. Let a nonzero represented Lie component $X \in \mathfrak{g}_{\mathbb{C}}$ span a Z_α-character line of character $\chi_X \in \Lambda^*_\alpha$, put $Q_X = Q(X)$, let J_X be the displayed finite-ring current component, and assume the displayed global covariance

$$U_N(z) Q_X U_N(z)^\dagger = \chi_X(z) Q_X, \qquad U_N(z) J_X U_N(z)^\dagger = \chi_X(z) J_X, \qquad z \in Z_\alpha. \tag{246}$$

Then for every occupied central character γ, with Π_γ its projector,

$$\Pi_\delta Q_X \Pi_\gamma = \delta_{\delta,\gamma\chi_X}\, Q_X \Pi_\gamma, \qquad \Pi_\delta J_X \Pi_\gamma = \delta_{\delta,\gamma\chi_X}\, J_X \Pi_\gamma, \tag{247}$$

so that the finite sector-label index is the character difference

$$\mathrm{ind}_{Z_\alpha}(X;\gamma) := (\gamma\chi_X)\gamma^{-1} = \chi_X \in \Lambda^*_\alpha, \tag{248}$$

and for every cocharacter $\eta : U(1) \to Z^0_{\alpha,\mathrm{eff}}$,

$$\chi_X(\eta(e^{i\theta})) = e^{i n_\eta(\chi_X)\theta}, \qquad \Delta q_\eta = n_\eta(\chi_X) \in \mathbb{Z}. \tag{249}$$

A disconnected effective centre may additionally contribute the finite torsion label retained in Equation (248)*; it does not create an integer circle coordinate. Finally, for an exact hard eigenvector ψ_h with $Q_X\psi_h \neq 0$, the pure charge-created readout of Equation* (240) *is one whenever its denominator is nonzero.*

Scope 12.9 (What the label theorem does *not* imply)*.* The most important negative statement here is that Equation (247) does *not* imply the projected-current operator identity. Writing $D_\gamma := Q_X \Pi_\gamma$ and $P_{D,\gamma}$ for the projector onto its range, the theorem gives only the inclusion

$$\mathrm{ran}\, D_\gamma \subseteq \mathrm{ran}\, \Pi_{\gamma\chi_X}, \qquad P_{D,\gamma} \leq \Pi_{\gamma\chi_X}, \tag{250}$$

and the inclusion is strict in an explicit instance. In the four-qutrit fundamental $SU(3)$ ferromagnet one finds $\dim \mathcal{H}_{5/3} = 8$, $\dim \mathcal{H}_{2/3} = 24$, $\mathrm{rank}\, D = 8$, and the live replacement defect $\|(\Pi_{2/3} - P_D) J_X |h;2\rangle\| = \sqrt{8/3}$: replacing the descendant-range projector by the central-sector projector changes the current vector. The underdetermination behind this is abstract, not accidental: for a one-dimensional source line and a two-dimensional target of the same character, $D\psi = e_1$ and $J\psi = ae_1 + be_2$ have the same source and target labels for every a, b, while $P_D J\psi = ae_1$ varies. Sector labels alone do not determine an operator coefficient.

Two further degeneracies. If X is central its adjoint character is trivial and the label Equation (248) is zero — which is still distinct from a root row, since a central direction has no ladder. If G is finite there is no nonzero X and the theorem has no Lie-current instance at all; an independently supplied finite torsion label, or a finite-string endpoint identity, is a different statement.

Idea of proof. If $\psi = \Pi_\gamma \psi$ then Equation (246) gives $U_N(z) Q_X \psi = \gamma(z)\chi_X(z) Q_X \psi$ and likewise for J_X, so the image lies entirely in the $\gamma\chi_X$ block; Haar orthogonality of distinct characters — equivalently $\Pi_\delta = \int_{Z_\alpha} \overline{\delta(z)}\, U_N(z)\, dz$ — kills every other block and proves Equation (247). Dividing the target label by the source label proves Equation (248), the common affine restriction on K_u cancelling; and every continuous character of $U(1)$ is $e^{i\theta} \mapsto e^{in\theta}$ for a unique integer n, which is Equation (249). Since H_N commutes with the central action it preserves each Π_γ, so the label difference cannot change under finite-volume dynamics. This is compact representation covariance, not a statement about quantisation of an expectation value.

campaign id:	S-IDX-G-label
proved in:	`theory/soft-index-g4.md` steps (1.1)–(1.7); headline statement at (1.3), the pure zero-mode anchor at (1.4), the exact route boundary at (1.5), the $SU(2)$ reduction at (1.6), the four-site $SU(3)$ instance at (1.7); restated in `theory/soft-index-general.md` §4
tested by:	`theory/checks/soft_index_g4_check.py` gates G4-C0–C4; `theory/checks/soft_index_g_boundary_check.py` gates G5-C1–C3; `theory/checks/soft_index_r2_check.py` gates SIDXR2-C2–C3
reviewed in:	`theory/verdicts/soft-index-r2-critic.md`

12.6 The structural limit law, and the one remaining gap in one dimension

We now leave exact finite volume. The question is what the symmetry forces about *limits* of the readout of Definition 12.1 as the ring, the settling time and the window are taken out and the soft scale ϵ is then sent to zero. The honest register for this is the *actual limit point*: a common outer subsequence along which, for every ϵ in a set $E \downarrow 0$, the scalar datum, the row measures, the hard Ward registers, and every component named below all converge in the topologies required. The phrase asserts nothing about existence of such a subsequence. Along it the ordered soft variable is the packet mean $\bar{k}_*(\epsilon)$, not the carrier of an unsmeared plane wave, and it obeys $c_1\epsilon \le |\bar{k}_*(\epsilon)| \le c_2\epsilon$, with sign fixed by the chosen one-sided profile.

The whole content of the structural theorem rests on one hypothesis, and that hypothesis is the single remaining gap of the one-dimensional programme. It is displayed here as a named object so that it can never be smuggled into a definition.

Definition 12.10 (Hypothesis (PROTO-LSZ): an exhaustive component decomposition)**.** In the packet and amputation normalisation of Definition 13.2 (Section 13), after the outer limit and uniformly for $h \in I$, the limit datum decomposes exhaustively as

$$\mathcal{A}_* = \mathcal{A}_*^{\mathrm{desc}} + \mathcal{A}_*^{\perp} + \mathcal{A}_*^{\mathrm{dir}} + \mathcal{A}_*^{\partial}, \tag{251}$$

naming respectively the descendant, orthogonal-current, direct-contact, and two window-boundary-gradient contributions, where the descendant term has the external-flux form

$$\mathcal{A}_*^{\mathrm{desc}}(\epsilon) = \int (e^{ik} - 1)\, L(k,h)\, [\, 2iv_h \ell_h \,]\, d\mu_{*,\epsilon}(k,h), \tag{252}$$

with L uniformly C^1 and carrying exactly the h-profile $L(0,h) = \mathfrak{a}_{\mathrm{leg}}(\rho)\,(-i\,\mathrm{sgn}(v_h - v_s)/v_h)$ of Definition 13.5(3b), and

$$\mathcal{A}_*^{\perp} = O(\epsilon^2), \qquad \mathcal{A}_*^{\mathrm{dir}} = O(\epsilon^2), \qquad \mathcal{A}_*^{\partial} = o(\epsilon). \tag{253}$$

The hypothesis also carries the antecedents that the relevant class $\mathcal{S}_W(\rho)$ is nonempty and that the asymptotic one-magnon kernel exists — solely so that $\mathfrak{a}_{\mathrm{leg}}(\rho)$ is defined — and that the measured hard Ward register converges to a finite ℓ_h, with $\ell_h = 1$ only for the primitive one-magnon band proved in Theorem 12.4.

This hypothesis is uninstantiated. It carries the exhaustiveness, the kinematic normalisation, the reduced-channel regularity, the no-contact condition, and the finite-window boundary limit as *explicit assumptions*; it does not make exhaustiveness true by definition, and no current claim proves it for the adjudicated fixed-time readout.

Theorem 12.11 (Structural law for actual limit points)**.** [SKETCH]
Assume an actual limit point of Definition 12.1 in the sense above, the regularity package Definition 12.2, and hypothesis (PROTO-LSZ) *of Definition 12.10. Then every such limit point obeys*

$$\mathcal{A}_*(\epsilon) = 2i\,\mathfrak{a}_{\mathrm{leg}}(\rho)\ \mathrm{sgn}(v_h - v_s)\,\ell_h\,\bar{k}_*(\epsilon) + o(\epsilon), \tag{254}$$

and hence has the continuous Adler extension $\mathcal{A}_*(0) = 0$. Before any on-shell matching its phase jet is only

$$\frac{\arg(1 + \mathcal{A}_*(\epsilon))}{\bar{k}_*(\epsilon)} \longrightarrow 2\,\mathrm{Re}\,\mathfrak{a}_{\mathrm{leg}}(\rho)\ \mathrm{sgn}(v_h - v_s)\,\ell_h, \tag{255}$$

which is not a numerical law.

Scope 12.12 (Exactly what is proved and what is missing). What is proved is a complete conditional implication: given the displayed hypotheses, the algebra of Equation (254) is exact. What is missing is any instance. Hypothesis (PROTO-LSZ) is not proved for the adjudicated fixed-time readout; in particular the two window-boundary gradients and the identification of the fixed-time charge insertion with a constructed scattering channel are not controlled by the wave-operator results of Section 11, which explicitly exclude that interface. The soft-leg amputation constant $\mathfrak{a}_{\mathrm{leg}}(\rho)$ appearing in Equation (254) is the *open* class constant of Definition 13.5(3b): no value, and in particular no $1/\rho$ law, is claimed here at any density including $\rho = 1/2$. The regularity package cannot repair this by Scope 12.3. Finally, existence of an actual limit point is not asserted.

What the argument does. Uniformly on the packet support, $e^{ik} - 1 = ik + O(\epsilon^2)$ and $L(k, h) = \mathfrak{a}_{\mathrm{leg}}(\rho)(-i\,\mathrm{sgn}(v_h - v_s)/v_h) + O(\epsilon)$; multiplying the three leading factors in Equation (252) gives $(ik)\cdot\mathfrak{a}_{\mathrm{leg}}(-i\,\mathrm{sgn}\,/v_h)\cdot 2iv_h\ell_h = 2i\mathfrak{a}_{\mathrm{leg}}\,\mathrm{sgn}\ \ell_h\,k$, and integration replaces k by the packet mean. Every cross term is $O(\epsilon^2)$ uniformly because $I \Subset (0, \pi)$ keeps v_h away from zero in the chosen channel. Adding the three bounds of Equation (253) gives Equation (254); the Adler extension follows since the right-hand side is $O(\epsilon)$, and $\arg(1 + z) = \mathrm{Im}\,z + O(|z|^2)$ with the nonzero lower bound on $|\bar{k}_*|$ gives Equation (255).

campaign id:	S-IDX-spec-struct-r2
proved in:	`theory/soft-index-r2.md` §3, statement and proof at step (1.8); the limit-point register at (1.7)
tested by:	no certificate for the hypothesis; the gates test the finite inputs only (`theory/checks/soft_index_r2_check.py` gate SIDXR2-C5, arithmetic)
reviewed in:	`theory/verdicts/soft-index-r2-critic.md`

12.7 Adding on-shell data: the matched value

The structural law contains a shape and an open constant. To get a number one must connect the protocol readout to something on shell. That connection is a separate hypothesis, and it is displayed rather than assumed silently.

Theorem 12.13 (Matched spectral law under the matching hypothesis)**.** [SKETCH]
Assume the hypotheses of Theorem 12.11; a fully polarised spin-S vacuum, so that the tail density is $\rho = S$; a primitive one-magnon hard leg, so that $\ell_h = 1$; the class-membership antecedent

(β_S) *of Theorem 13.15 — existence of a source* $O \in \mathcal{S}_W(\rho)|_{\rho=S}$ *with* $M_1^O \neq 0$ *— and the following separate matching hypothesis:*

$$(\text{MATCH-S}): \qquad \mathcal{A}_*(\epsilon) - \int [\, S_{\text{phys}}(k,h) - 1\,]\, d\mu_{*,\epsilon}(k,h) = o(\epsilon), \tag{256}$$

a first-order identification of the fixed-time protocol readout with the on-shell multiplier, containing no numerical slope of its own. Then every actual limit point has phase slope

$$\frac{\arg(1+\mathcal{A}_*(\epsilon))}{\bar{k}_*(\epsilon)} \longrightarrow \frac{\operatorname{sgn}(v_h - v_s)}{S}, \tag{257}$$

and comparison with Equation (254) *supplies the conditional matched value of the amputation constant,*

$$\mathfrak{a}_{\text{leg}}(S) = \frac{1}{2S} = \frac{1}{Z_\rho}. \tag{258}$$

More generally, any datum at the same density that shares this class constant and satisfies (PROTO-LSZ) *obeys the structural slope* $\operatorname{sgn}(v_h - v_s)\ell_h/\rho$.

Scope 12.14 (Where the number comes from, and what is not claimed). The only numerical supplier is the exact two-body expansion $S_{\text{phys}}(k,h) - 1 = i\operatorname{sgn}(v_h - v_s)k/S + O(k^2)$ of Section 10, and it enters strictly *after* (MATCH-S), never as a clause of the readout and never as a regularity condition. Hypothesis (MATCH-S) is exactly the open first-order bridge that the wave-operator results of Section 11 decline to supply at this interface, and by Scope 12.3 the regularity package cannot substitute for it; the theorem is therefore a complete conditional implication with no instance in general.

The value Equation (258) at $S \in \{1/2, 1, 3/2, 2\}$ is precisely the existing conditional row of Theorem 13.15; at other half-integer spins it is the same one-line consequence, but it belongs to this sketch rather than being silently attributed to that row. Nothing here exhibits a member of $\mathcal{S}_W(\rho)$, and nothing here licenses a bridge-free conclusion about the amputation constant — that would be the conjecture of Conjecture 13.17. The final displayed generalisation carries ℓ_h as the *measured* Ward register; no composite-charge formula $\ell_h = |q|$ is asserted beyond the primitive $|q| = 1$ case, since the two-body slope supplier is scoped to unit charge. No leg-normalisation argument is used anywhere in the derivation of Equation (258): the exact charge-created/asymptotic leg conversion supplies only $Z_\rho^{-1/2}$, and never Z_ρ^{-1}.

campaign id:	S-IDX-spec-r2
proved in:	`theory/soft-index-r2.md` §3, steps (1.9).(2.1)–(2.5); the value interface and its fences at (1.11)
tested by:	`theory/checks/soft_index_r2_check.py --red-s2-value` tests the finite matching arithmetic only; no gate tests (MATCH-S)
reviewed in:	`theory/verdicts/soft-index-r2-critic.md`

12.8 The separated-preparation breakthrough: matching becomes a theorem

The results so far leave one hypothesis carrying all the weight. The advance recorded here removes it — not in general, but on an explicitly displayed subclass of preparations, and there completely.

The idea is physical and simple. If the soft and hard packets are prepared far apart, allowed to approach, collide once, and separate, and if the separations and settling times are taken out in the right order at fixed soft scale, then the fixed-time charge-created state can be compared *in norm* with a genuine incoming scattering channel, that channel can be pushed through the two-body wave operators, and the readout can be evaluated on the outgoing side. The comparison is a chain of Cook-type estimates; none of them needs to be uniform in the soft scale, because the soft scale is held fixed until the very end.

Definition 12.15 (The separated-preparation subclass)**.** Specialise the indices of Definition 12.1 as follows. At fixed $\epsilon > 0$ take hard widths $\sigma_j \downarrow 0$, initial separations $R_j \to \infty$, and settling times T_j with $L_j := d_\epsilon T_j - R_j \to \infty$, where d_ϵ is the velocity separation of the two packets. With $B_{M,\epsilon}$ the largest sup-norm of the derivatives of S_{phys} of total order at most M on the soft support times I, and $K_{M,\epsilon} := C_{M,S,J}(1 + B_{M,\epsilon})/d_\epsilon$, require for one integer $M \geq 8$

$$K_{M,\epsilon}\, s_M(f_\epsilon)\, s_M(g_{\sigma_j}) \left[(1 + R_j)^{3-M} + (1 + L_j)^{3-M}\right] \longrightarrow 0, \tag{259}$$

where s_M is the fixed-packet Schwartz seminorm evaluated on the *untranslated* envelopes, the unit-modulus translation phase being kept inside the stationary phase rather than differentiated. The hard-column window is the labelled rectangle formed by the soft support and $\operatorname{supp} g_{\sigma_j}$, and zero-mass rows carry zero measure. After (σ_j, R_j, T_j) are fixed, choose a ring N_j so large that $N_j\epsilon(c_2 - c_1) > 2\pi$, $T_j \leq c_{\mathrm{rec}} N_j / v_{\max}$, and the total norm error η_j between the periodised finite-ring kernels and their infinite-chain counterparts is at most $1/j$.

This class is nonempty and directly verifiable, not a renaming of the matching hypothesis: compactly supported smooth packets satisfy $s_M(g_\sigma) = O(\sigma^{-M-1/2})$, so taking, for instance, $R_j = \lceil \sigma_j^{-2} \rceil$ and $T_j = 2R_j/d_\epsilon$ makes $L_j = R_j$ and the left-hand side of Equation (259) equal to $O_\epsilon(\sigma_j^{M-13/2})$, which vanishes for $M \geq 8$. Suitable rings exist by Plancherel sampling and strong convergence of finite-range dynamics at fixed time, chosen diagonally.

Theorem 12.16 (On the separated class, matching holds with zero remainder)**.** [PROVED]
Let $H_S = -J\sum_x (\mathbf{S}_x \cdot \mathbf{S}_{x+1} - S^2)$ with $2S \in \mathbb{N}$, work on the primitive regular two-magnon channel, and assume the separated-preparation subclass of Definition 12.15. Then along every actual row-measure limit subsequence the fixed-time charge-created finite-ring readout converges and satisfies, for that same limiting row measure,

$$\mathcal{A}_*(\epsilon) = \int [\, S_{\mathrm{phys}}(k,h) - 1\,]\, d\mu_{*,\epsilon}(k,h). \tag{260}$$

In particular the matching hypothesis (MATCH-S) *of Equation* (256) *holds on this subclass with* zero *remainder, hence a fortiori through $o(\epsilon)$.*

Scope 12.17 (The boundary of the positive result)*.* The theorem is a whole-datum scattering identity. It does *not* identify the finite-separation fixed-time vector with a Haag–Ruelle creator or with a scattering vector; it does not require a soft-uniform Cook estimate; and it proves no component decomposition. In particular it does not prove (PROTO-LSZ): nothing above splits the datum into descendant, orthogonal-current, direct-contact and two boundary-gradient pieces, exhibits a microscopic member of $\mathcal{S}_W(\rho)$, or evaluates the amputation constant. Compare Equation (260) with Equations (251) and (253): the latter is strictly stronger content.

The result is primitive two-body only: no endpoint, equal-velocity, composite-charge, bound-state-channel, many-body, or asymptotic-completeness statement is asserted outside the two-magnon sector. The proof uses no scalar replacement of the full-sector Gram operator, respecting the register erratum of Scope 12.5.

Idea of proof: a norm bridge. Four movements. *(i) Wave operators.* Fourier transform in the centre coordinate decomposes the two-magnon Hamiltonian into fibres over the total momentum; on each fibre the relative motion is a constant-coefficient Jacobi recurrence away from contact, with only the adjacent row $r = 1$ and, for $S \geq 1$, the single normalised double-occupancy row $r = 0$ modified. A finite-rank boundary perturbation of a scalar Jacobi operator has isometric incoming and outgoing generalised transforms with equal range, and their composition is multiplication by the physical ratio, $(W_+^S)^* W_-^S = M_{S_{\mathrm{phys}}}$, with S_{phys} the exact two-body multiplier of Section 10. *(ii) Cook estimates.* The identification defect $(H_S I_S - I_S H_0) U_0(t) F_R$ is supported only on the collision rows $r \leq 1$; outside the two separated velocity cones repeated integration by parts gives decay $(1 + R + d_\epsilon |t|)^{2-M}$, and integrating in t gives the two displayed bounds — one comparing the fixed-time state with the incoming wave operator, one comparing the outgoing channel with free propagation. *(iii) The bridge.* Equality of ranges gives $W_-^S F_R = W_+^S M_{S_{\mathrm{phys}}} F_R$, and combining the two Cook bounds yields a single norm estimate that starts at the fixed-time charge state and ends at the freely propagated outgoing channel. No equality between those two states was assumed anywhere. *(iv) Through the readout.* The coordinate-kernel map and the row-rectangle selection are contractions; the free-reference error is the freely propagated initial collision rows plus a finite-ring sampling error, both $o(1)$ under Equation (259). Disjoint momentum supports make the ideal labelled rectangles orthogonal of equal mass 1/2, which bounds the denominators away from zero, so the row measures converge in total variation and the normalised pairing converges. In the ideal rectangle the free propagation phase and the translation phase occur on both sides of the pairing and cancel, leaving exactly $\int S_{\mathrm{phys}}\, d\mu$. Since the difference in (MATCH-S) is identically zero at each fixed ϵ after the outer limit, it is trivially $o(\epsilon)$ when $\epsilon \downarrow 0$ is finally taken — which is precisely why the fixed-packet divergence of the Cook constants does not obstruct this subclass.

campaign id:	S-IDX-MATCH-HS-SEP
proved in:	`theory/proto-lsz-match.md` steps (1.1)–(1.6): the subclass (1.2), wave operators (1.3), the fixed-time-to-incoming-channel bound (1.4), passage through the readout (1.5), the theorem (1.6); honest boundary at (1.11)
tested by:	`theory/checks/proto_lsz_match_check.py` gates PMLM-C1–C5; finite corroboration only, with no wave-operator certificate
reviewed in:	`theory/verdicts/blitz-proto-lsz-r1.md`

Theorem 12.18 (The value on the separated class)**.** [PROVED]

Under the separated-preparation subclass of Definition 12.15, along every actual row-measure limit subsequence the primitive fixed-time scalar datum obeys

$$\mathcal{A}_*(\epsilon) = \frac{i\,\mathrm{sgn}(v_h - v_s)\,\bar{k}_*(\epsilon)}{S} + O(\epsilon^2), \qquad \frac{\arg(1 + \mathcal{A}_*(\epsilon))}{\bar{k}_*(\epsilon)} \longrightarrow \frac{\mathrm{sgn}(v_h - v_s)}{S}. \tag{261}$$

Scope 12.19 (This row is about the readout, not about the class constant)*.* The value enters only after Theorem 12.16 has already identified the readout with the multiplier, and then solely through the exact two-body slope of Section 10. The row makes no claim about the amputation constant $\mathfrak{a}_{\mathrm{leg}}$, about membership in the no-contact class of Definition 13.5, about hypothesis (PROTO-LSZ), or about endpoints, equal velocities, composite charges, or any other protocol class. In particular, no value of $\mathfrak{a}_{\mathrm{leg}}$ follows from this corollary alone; if (PROTO-LSZ), class membership, and the other antecedents of Theorem 12.13 were separately supplied, the comparison there would give the conditional value $\mathfrak{a}_{\mathrm{leg}}(S) = 1/(2S)$ — this theorem neither supplies nor replaces those antecedents.

Idea of proof. Insert $S_{\mathrm{phys}}(k,h) - 1 = i\,\mathrm{sgn}(v_h - v_s) k/S + O(k^2)$ into Equation (260) and integrate against the same measure; the scale-tied support $|k| \leq c_2 \epsilon$ makes the integrated remainder $O(\epsilon^2)$. Then $\arg(1+z) = \mathrm{Im}\, z + O(|z|^2)$ together with the one-sided lower bound $|\bar{k}_*| \geq c_1 \epsilon$ gives the slope.

campaign id:	S-IDX-D29-value-HS-SEP
proved in:	`theory/proto-lsz-match.md` step (1.10) — the only place in that shard where the number $1/S$ is invoked
tested by:	`theory/checks/proto_lsz_match_check.py` gate PMLM-C4 plus the frozen partial gate C5
reviewed in:	`theory/verdicts/blitz-proto-lsz-r1.md`

12.9 An exact LSZ-shaped scalar factorisation

The matched identity can be rewritten so that it has the *shape* the universality analysis of Section 13 predicts — an external flux factor times a Ward residue, integrated against the row measure — with a reduced multiplier that extends continuously to zero soft momentum. This is worth isolating, because it shows that the LSZ shape itself is available on the separated class without any of the component hypotheses.

Theorem 12.20 (Exact scalar factorisation of every separated limit point)**.** [PROVED]
On the fully polarised bilinear ferromagnet H_S with $2S \in \mathbb{N}$, and on the separated-preparation subclass of Definition 12.15, define for $k \neq 0$ the scalar scattering witness

$$L_S^{\mathrm{sc}}(k,h) := \frac{S_{\mathrm{phys}}(k,h) - 1}{(e^{ik}-1)\,2iv_h}. \tag{262}$$

Then L_S^{sc} has a unique uniformly C^1 extension to $k = 0$ on the compact channel, with

$$L_S^{\mathrm{sc}}(0,h) = \frac{-i\,\operatorname{sgn}(v_h - v_s)}{2S\,v_h}, \tag{263}$$

and every actual row-measure limit point has the exact factorisation

$$\mathcal{A}_*(\epsilon) = \int (e^{ik}-1)\,L_S^{\mathrm{sc}}(k,h)\,[\,2iv_h\,]\,d\mu_{*,\epsilon}(k,h), \qquad \mathcal{A}_*(\epsilon) = \frac{i\,\operatorname{sgn}(v_h - v_s)\,\bar{k}_*(\epsilon)}{S} + O(\epsilon^2). \tag{264}$$

Scope 12.21 (Same profile is not same object). Equation Equation (263) has the same functional profile as the no-contact class clause Definition 13.5(3b) with scalar coefficient $1/(2S)$. That observation does *not* identify L_S^{sc} with the descendant quotient L of that clause, and it does not prove a member of $\mathcal{S}_W(\rho)|_{\rho=S}$: the class's L is the quotient of an independently identified descendant external-leg component inside an exhaustive decomposition, and is defined only when the class is nonempty. The superscript is a deliberate warning label. Nor does the theorem prove the orthogonal-current, direct-contact, or window bounds of (PROTO-LSZ), which remains open. No step obtains $1/(2S)$ from the charge-created/asymptotic-leg conversion, which supplies only $Z_\rho^{-1/2}$.

Scope 12.22 (The exact remaining lemma (COMP-HS)). Aggregate matching cannot determine component estimates, and the obstruction is elementary: for any nonzero bounded smooth b on I and $B(k,h) = k\,b(h)$, the replacement $E_{\mathrm{desc}} \mapsto E_{\mathrm{desc}} + B$, $E_\perp \mapsto E_\perp - B$ leaves every whole-datum equality unchanged while moving both components. What is therefore needed, and all that is needed, is one model lemma.

(COMP-HS): on the separated class, using at finite volume the exact full-sector split $J_k^- = P_{\lambda,N}J_k^- + (1-P_{\lambda,N})J_k^-$ and the exact Duhamel contact defect — with no scalar replacement of the Gram operator — the finite readout functional admits four independently defined terms, from the descendant-range current projection, the complementary projection, the direct source/contact functional, and the two window gradients, such that on the ordered outer limit: *(1)* the descendant quotient converges in the packet norm to a uniformly C^1 multiplier L_S^{desc} with $L_S^{\mathrm{desc}}(0,h) = L_S^{\mathrm{sc}}(0,h)$, equivalently $L_S^{\mathrm{desc}} - L_S^{\mathrm{sc}} = O(k)$ uniformly; *(2)* the complementary-current and direct-contact terms are each $O(\epsilon^2)$ uniformly on I; *(3)* the two window-gradient terms sum to $o(\epsilon)$ under Equation (259); *(4)* the limiting descendant construction exhibits a nonzero microscopic member of $\mathcal{S}_W(\rho)|_{\rho=S}$.

Together with Theorem 12.20 this would prove full (PROTO-LSZ) on the separated class without changing any definition. No existing proved claim supplies it: the finite Ward theorem gives no volume-uniform reduced-channel regularity, and the matching theorem controls only the sum after the outer limit. A viable attack is a two-body fibre limiting-absorption or Feshbach estimate — insert the full-sector descendant projector before taking the Jacobi boundary resolvent, prove a uniform reduced-resolvent bound on the compact hard window, and push the four finite functionals through the already proved norm bridge — which tests items (1)–(3) directly rather than inferring them from their sum.

Idea of proof. The multiplier S_{phys} is analytic in k and smooth in h near the compact rectangle with $S_{\text{phys}}(0,h) = 1$ and $\partial_k S_{\text{phys}}(0,h) = i\,\text{sgn}(v_h - v_s)/S$. The analytic removable-factor theorem therefore gives $S_{\text{phys}} - 1 = (e^{ik} - 1)G_S(k,h)$ with G_S uniformly C^1 and $G_S(0,h) = \text{sgn}(v_h - v_s)/S$, since $\partial_k(e^{ik} - 1)|_0 = i$; dividing by $2iv_h$, which is bounded away from zero on I, preserves regularity and gives Equation (263). Substituting the resulting pointwise identity $S_{\text{phys}} - 1 = (e^{ik} - 1)L_S^{\text{sc}}\,[2iv_h]$ — valid at $k = 0$ by continuity — into the matched identity Equation (260) gives Equation (264). The number $1/(2S)$ entered only through the k-derivative of the physical multiplier, after its branch was fixed.

campaign id:	S-IDX-PROTO-SCALAR-HS-SEP
proved in:	`theory/proto-lsz-scalar.md` steps (1.1)–(1.7): removable quotient (1.2), exact factorisation (1.3), supplier separation (1.4), the aggregate/component obstruction (1.5), the lemma (COMP-HS) (1.6)
tested by:	`theory/checks/proto_lsz_check.py` gates PLSZ-C1–C4 (corroboration of the scalar algebra only, not of regularity)
reviewed in:	`theory/verdicts/blitz-proto-lsz-r1.md` (0 fatal, 0 major)

12.10 The Haag–Ruelle value instance

Finally, there is a second and logically independent place where the number $1/S$ appears: not in the fixed-time protocol readout at all, but in the constructed scattering channel of Section 11. It is recorded here to keep the two clearly apart.

Theorem 12.23 (Value in the constructed Haag–Ruelle channel)**.** [SKETCH]
Under the exact-band register on the ferromagnetic model, the fixed-packet constructed Haag–Ruelle channel of Section 11 has multiplier S_{phys}, and its scale-tied packet average

$$\mathcal{A}_{\text{HR}}(\epsilon) := \int [\, S_{\text{phys}}(\epsilon u, h) - 1 \,]\, d\mu_f(u,h) \tag{265}$$

has phase slope $\text{sgn}(v_h - v_s)/S$*, with* $\bar{k} = \epsilon \int u \, d\mu_f$*.*

Scope 12.24 (An on-shell instance, explicitly not a protocol instance)*.* Equation Equation (265) is a value instance for a *constructed* channel. It is not the fixed-time charge-created protocol datum of Definition 12.1: the Cook and Gram constants of the wave-operator construction diverge along the rescaled soft packet, and creator-choice independence does not apply to the fixed-time insertion $Q[f_\epsilon]\psi_g$. It is therefore no basis for upgrading either the structural or the matched spectral row. Consequently there is at present no proved instance of Theorem 12.11 or Theorem 12.13 for the general protocol, no proved microscopic member of $\mathcal{S}_W(\rho)$, and no permission to use the amputation conjecture of Conjecture 13.17 as a value supplier.

campaign id:	S-IDX-HR-value-r2
proved in:	`theory/soft-index-r2.md` §4, instances and non-instances at step (1.10); interfaces and fences at (1.11)
tested by:	—
reviewed in:	`theory/verdicts/soft-index-r2-critic.md`

12.11 Reading of the section

The symmetry algebra alone delivers a genuine index: exact, finite, integral where an index should be integral, and valid for every compact group one root at a time. That is Theorems 12.4, 12.6 and 12.8, and none of it is conditional.

Passing to the soft limit costs one hypothesis — an exhaustive component decomposition of the limiting readout — and passing from a shape to a number costs a second — the identification of that readout with the on-shell multiplier. On the separated-preparation class the second cost is paid in full: matching is a theorem there, with zero remainder, and the value and the LSZ shape follow. The first cost is not yet paid anywhere, and it is now isolated to the single lemma (COMP-HS) of Scope 12.22. That lemma is the one remaining gap of the one-dimensional soft-index programme.

13 Universality: the class, the refutation, and the two candidate constants

13.1 What "universal" could have meant, and how the naive version died

In the continuum, the appeal of a soft theorem is that the soft emission factorises off the hard process with a multiplier that knows nothing about the hard process — only the charges and velocities of the external legs. That is what makes it a theorem about symmetry rather than about a model. The natural lattice ambition was the same: a multiplier S, depending on the asymptotic leg data alone, such that adding one soft magnon of momentum k_s to an n-leg amplitude multiplies it by S up to a controlled remainder.

Stated over unrestricted local or quasi-local sources, that ambition is false. There is an explicit four-site source on the ferromagnet whose one-hard amplitude is completely insensitive to a deformation parameter while its hard-plus-soft amplitude is not; the deformation moves the soft first jet by an h-dependent amount that no leg-data multiplier can reproduce. The refutation is Refuted claim 13.9, and it is not a technicality: it says that the soft first jet on the lattice is *not* a function of the asymptotic data alone.

What survives is a three-part structure, and the rest of this section is that structure.

- A precisely delimited *class* of sources — the Ward-covariant no-contact class of Definition 13.5 — inside which factorisation can be asked about at all. The class is defined by five conditions, of which two (exhaustiveness of the LSZ decomposition, and absence of a direct soft contact) are exactly the content that the finite-volume Ward identity does not supply.
- An exact *criterion* for factorisation on any linear source class (Theorem 13.11), which is a genuine equivalence and is agnostic about the value of the constant, and a conditional *shape* theorem on the class (Theorem 13.13), whose conclusion contains a class constant and no number.
- Two candidate readings of that constant, carried side by side and neither adopted: a conditional matched value that is a theorem as an implication with unproved antecedents (Theorem 13.15), and a bridge-free amputation conjecture with a sharp quantitative fence against it (Conjecture 13.17).

One piece of history has to be told honestly, because it explains why the definition below has an unusual shape. The class was originally frozen with a clause that *stipulated* the amputation constant to be 1. Under the corpus's identification of the class's soft jet with the two-body physical phase jet, that stipulation forces a soft slope of 2 at every tail density — which contradicts the proved spin-S law $1/S$ of Section 10 at every density except $\rho = 1/2$. The frozen class was therefore provably *empty* away from $\rho = 1/2$: the definition had accidentally been made inconsistent rather than merely unpopulated. The repair was to split the offending clause into a hypothesis about the shape of the external flux factor and a separate, honest statement that its constant is open. That split is the $(3a)/(3b)$ structure below, and everything downstream is written to keep the constant visible as a name rather than a number.

13.2 The definition of the class

Definition 13.1 (Hard sources)**.** A *local source* is an element of the local algebra $\mathfrak{A}_{\mathrm{loc}}$. An *exponentially quasi-local source* is a norm-convergent sum $O = \sum_X O_X$ with

$$\|O\|_\mu := \sum_X e^{\mu\,\mathrm{diam}(X)}\|O_X\| < \infty \tag{266}$$

for some $\mu > 0$. Local sources have finite $\|\cdot\|_\mu$.

Definition 13.2 (Amputated amplitudes, and the normalisation they are measured in)**.** For a source O, write $M_1^O(h)$ for its amputated one-hard amplitude and $M_2^O(k,h)$ for its connected hard-plus-soft amplitude in the physical channel. Both are linear in O, and both are measured as multipliers in $L^2(I, dh)$ for a fixed hard window $I = [a,b] \Subset (0,\pi)$. Remainders are measured as $\|R_{S2}(k)\|_{L^2(I)} \le C_I|k|^2\|M_1^O\|_{L^2(I)}$, or, for the rescaled packet $f_\epsilon(k) = \epsilon^{-1/2}f(k/\epsilon)$, as $O(\epsilon^2)$ in the product packet norm. Plane waves are generalised kernels inside compactly supported smooth packets.

Amputation and leg normalisation. Both amplitudes are taken against asymptotic one-magnon kernels normalised by $\langle k|k'\rangle = 2\pi\delta(k-k')$, inside the campaign's packet discipline. That kernel is *constructed* in the fully polarised spin-S register, where it is $|k\rangle = \sum_x e^{ikx}|x\rangle$, an exact one-magnon eigenvector of H_S; in an exact-band register it is the target of a separate sketch row, and outside those registers its existence is open. Every statement about the class constant of clause (3b) below is therefore scoped to a register where the kernel exists. The *same* amputation is applied to the *same* hard leg in both amplitudes, and the soft leg enters M_2^O as *one additional δ-normalised asymptotic magnon of momentum k carrying unit leg weight* — not as a charge-created or current-created vector. This is strictly narrower than the analytic amputation admitted by the fixed-volume Ward reduction, which allows any bounded C^1 multiplier family: here every soft-leg envelope $\lambda(k)$ must satisfy $\lambda(0) = 1$.

Scope 13.3 (The convention is load-bearing, and it is only a convention). A change of normalisation $M_1^O \mapsto c\,M_1^O$, $M_2^O(k,\cdot) \mapsto c'(k)M_2^O(k,\cdot)$ with $c, c'(k) \neq 0$ — necessarily independent of O, by linearity — leaves clauses (1), (4), (5) of Definition 13.5 invariant, because clause (1) is homogeneous in one amplitude while clauses (4) and (5) are two-amplitude bounds whose existentially quantified constant absorbs c'/c. Clause (2), whose residue is quoted against the charge-created soft leg, transforms as c'/c; the external flux factor and the amputation constant transform as $L(k,h) \mapsto (c'(k)/c)L(k,h)$ and $\mathfrak{a}_{\mathrm{leg}} \mapsto (c'(0)/c)\mathfrak{a}_{\mathrm{leg}}$. Fixing $c'(0)/c = 1$ is exactly what pins $\mathfrak{a}_{\mathrm{leg}}$ against these rescalings, and this definition supplies that fixing alone. Clause (2) is *not* a second covariance anchor: read against the charge-created leg it is a cross-normalisation *membership* condition on the source, tying the descendant term to $2iv_hM_1^O$, and it is that — not a second fixing of the covariance freedom — which makes $\mathfrak{a}_{\mathrm{leg}}$ a number rather than a name. Without both, the assertions "$\mathfrak{a}_{\mathrm{leg}} = 1$" and "$\mathfrak{a}_{\mathrm{leg}} = 1/Z_\rho$" are statements about a convention and not about a model. What remains free does not move $\mathfrak{a}_{\mathrm{leg}}$: rescaling the source; any hard-leg amputation convention applied identically in both amplitudes, which cancels in the clause-(3a) quotient; and any soft-leg envelope with $\lambda(0) = 1$.

Two consequences are used repeatedly below. *(i)* In the fully polarised spin-S ferromagnet the charge-created soft leg is *not* this convention's soft leg: on normalised one-flip states $S_x^-|\Omega\rangle = \sqrt{Z_\rho}|x\rangle$, hence exactly

$$Q_k^-|\Omega\rangle = \sqrt{Z_\rho}\,|k\rangle \tag{267}$$

on the vacuum, the two legs coinciding only at $\rho = 1/2$; on a one-hard-magnon descendant the same conversion holds per site only up to $O(1/(Z_\rho N))$, since $\|Q_q^-|h\rangle\|^2 = Z_\rho N - 2$ exactly for

$q \neq h$, becoming exactly $\sqrt{Z_\rho}$ in the LSZ limit. *(ii)* Because clause (2)'s residue is quoted against the charge-created leg while the clause-(1) descendant summand is measured against this convention's asymptotic leg, a soft-leg normalisation argument *alone* contributes exactly that conversion mismatch to $\mathfrak{a}_{\mathrm{leg}}$ — $Z_\rho^{-1/2}$ in the LSZ limit and $(Z_\rho - 2/N)^{-1/2}$ at finite N on the descendant leg — and not Z_ρ^{-1}. This is the fence on Conjecture 13.17.

Definition 13.4 (The contact first jet)**.** With $\chi(h,k) := \operatorname{sgn}(v(h) - v(k))$ and $\mathfrak{a}_{\mathrm{leg}}$ the soft-leg amputation constant of Definition 13.5(3b) below — the forward reference is deliberate, since this quantity is *by definition* the obstruction to the factorisation that the class predicts —

$$\mathfrak{c}_h(O) := \partial_k M_2^O(k,h)\big|_{k=0} - 2i\,\mathfrak{a}_{\mathrm{leg}}\,\chi(h,0)\,M_1^O(h). \tag{268}$$

Definition 13.5 (The Ward-covariant no-contact class $\mathcal{S}_W(\rho)$)**.** Fix a broken-symmetry tail density $\rho := \omega_\alpha(S^z) > 0$ and write $Z_\rho := 2\rho$ for the order-parameter density. (In an $\mathfrak{su}(2)$ model with a fully polarised tail, $Z_\rho = \omega_\alpha([S_x^+, S_x^-])$ — a special case, not a membership condition. Under the tail hypotheses and the integrality hypothesis of Section 8, together with the antisymmetric tail pair $\omega_\beta(S^z) = -\omega_\alpha(S^z) = -\rho$, one has $2\rho \in \mathbb{Z}$, so Z_ρ is then a positive integer. The value $\rho = 0$ is excluded: there is then no broken order parameter and no soft leg of this type. In the fully polarised spin-S ferromagnet $\rho = S$.)

Amplitudes below — M_1^O, M_2^O, and the clause-(1) summand E^O_{desc} — are in the normalisation of Definition 13.2; clause (2)'s residue display is quoted in the charge-created normalisation of the Ward reduction, as that clause states; clause (3b) is empty without Definition 13.2. The class $\mathcal{S}_W(\rho)$ consists of the sources satisfying all five of the following (clause 3 has two parts; the *count* of conditions is deliberately unchanged).

(1) Exhaustive normed LSZ decomposition. In $L^2(I)$, M_2^O is exactly the sum of the descendant external-leg term E^O_{desc}, the orthogonal-current term, and the direct source/contact term named below; there is no additional reduced term, and the equality holds in the packet norm of Definition 13.2.

(2) Ward covariance. The descendant current residue — **quoted against the charge-created soft leg $Q_0|h\rangle$, and not against the asymptotic leg** — factorises as $2iv_h M_1^O(h)$. The two legs differ by $\sqrt{Z_\rho}$ *in the LSZ limit*; at finite N the descendant leg carries $\sqrt{Z_\rho - 2/N}$ instead, since $\|Q_q^-|h\rangle\|^2 = Z_\rho N - 2$ exactly for $q \neq h$. That mismatch is part of what clause (3b)'s constant records: against the asymptotic leg the same residue reads $2iv_h M_1^O(h)/\sqrt{Z_\rho}$ in that limit, the two readings coinciding only at $\rho = 1/2$. The clause is *consistent* at every density: its left-hand side is exactly Z_ρ-linear on the fully polarised spin-S tail, with $\langle h|Q_0^\dagger J_0^-|h\rangle = 2iv_S(h)$ and $v_S(h) = Z_\rho J \sin h$, the frozen reconstruction shard being its $Z_\rho = 1$ reading. Whether a given O also makes the right-hand side match is a membership question, which no computation decides. The alternative reading, against the asymptotic leg, would be unsatisfiable by an O-independent factor at every $\rho \neq 1/2$ — which is why this normalisation and not that one.

(3) Kinematic LSZ normalisation, in two parts.

(3a) Hypothesis. The external flux factor

$$L(k,h) := \frac{E^O_{\mathrm{desc}}(k,h)}{(e^{ik}-1)\,2iv_h\,M_1^O(h)} \tag{269}$$

is well defined by clauses (1)–(2) for $k \neq 0$ at every h with $M_1^O(h) \neq 0$; the hypothesis is that it extends to a *process-independent*, uniformly C^1 function on $\{|k| \leq \epsilon\} \times I$, and $L(0,h)$ denotes that extension's value at $k = 0$.

(3b) The profile of $L(0,h)$, and the open amputation constant. The quantity $v_h\, L(0,h)/\chi(h,0)$ does not depend on h on I; equivalently

$$L(0,h) = \mathfrak{a}_{\mathrm{leg}}(\rho) \cdot \frac{-i\, \chi(h,0)}{v_h} \tag{270}$$

for some nonzero constant $\mathfrak{a}_{\mathrm{leg}}(\rho)$. **What this clause asserts is that h-profile — the reciprocal LSZ energy denominator $[\omega(h+k)-\omega(h)]^{-1}k$, the campaign's fixed provenance for L — and no value.** By (3a), L is a single class-level function, so $\mathfrak{a}_{\mathrm{leg}}(\rho)$ is a datum of the class $\mathcal{S}_W(\rho)$ and not of any individual source; it is defined only when that class is nonempty, and only in a register where the asymptotic one-magnon kernel of Definition 13.2 is constructed. **This definition fixes no value of $\mathfrak{a}_{\mathrm{leg}}(\rho)$ at any density, $\rho = 1/2$ included.**

(4) Reduced-channel regularity. The orthogonal current channel is $O_{L^2(I)}(k^2)$ in the stated target limit, with the relative norm bound of Definition 13.2. The fixed-volume Ward reduction does not by itself supply a volume-uniform bound.

(5) No direct soft contact. The amputated source commutator/contact term is bounded by $C_I|k|^2\|M_1^O\|_{L^2(I)}$.

Scope 13.6 (Nonemptiness, the retired stipulation, and what the repair did not buy). Clauses (1) and (5) are the extra LSZ and contact content beyond the fixed-volume Ward reduction, and by Definition 13.4 the zero-intercept and zero-first-jet conditions are necessary, not cosmetic. Symmetry-generated external-leg insertions are the intended seed, but **no nontrivial microscopic class has been proved to satisfy all five conditions**; nonemptiness in the target scattering register is future work.

As frozen, clause (3) stipulated $\mathfrak{a}_{\mathrm{leg}} = 1$. Under the campaign's jet-identification bridge — the identification of this definition's soft multiplier jet with the two-body physical phase jet, an *unproved* bridge carried as antecedent (α) in Theorem 13.15 — that stipulation contradicts the proved spin-S two-body slope at every $\rho \neq 1/2$, so the frozen class was provably empty there. The stipulation is withdrawn. Two candidate readings are carried and neither is part of this definition: the conditional matched value $\mathfrak{a}_{\mathrm{leg}}(\rho) = 1/Z_\rho$ of Theorem 13.15, and the bridge-free amputation conjecture of Conjecture 13.17. The fence on the latter is Scope 13.3(ii): a leg normalisation alone supplies $Z_\rho^{-1/2}$, not Z_ρ^{-1}.

The repair removed a *proof* that $\mathcal{S}_W(\rho)$ is empty for $\rho \neq 1/2$; it supplies no member and fixes no value, so nonemptiness stays open at *every* ρ. At $\rho = 1/2$ the two candidate readings and the retired stipulation all coincide at 1, so displays elsewhere in the corpus of the soft coefficient 2 remain numerically consistent with this clause — but those displays rest on the contact/oracle route of Section 10, not on this clause.

Scope 13.7 (The counterexamples that shape the definition). Two explicit local sources on four consecutive sites are recorded as part of the definition, because each rules out a simpler alternative.

The *process-dependence* source is

$$D := S_0^- S_1^- - S_1^- S_2^- + S_2^- S_3^- - S_0^- S_3^-, \qquad O_\eta := S_0^- + \eta D, \tag{271}$$

discussed as Refuted claim 13.9 below.

The *intercept* source is

$$V_{\mathrm{int}} := \tfrac{9}{4} S_0^- S_2^- - \tfrac{3}{2} S_0^- S_3^- - \tfrac{3}{2} S_1^- S_2^- + S_1^- S_3^-. \tag{272}$$

It is local and has $M_1^{V_{\mathrm{int}}} = 0$ and $\partial_k M_2^{V_{\mathrm{int}}}(0,h) = 0$, but

$$M_2^{V_{\mathrm{int}}}(0,h) = \tfrac{3}{4} - \tfrac{1}{2}e^{-ih} + \tfrac{3}{4}e^{-2ih} - \tfrac{1}{2}e^{-3ih}, \tag{273}$$

which is not identically zero. A zero first jet therefore does *not* replace the zero-intercept condition of Definition 13.4: this is the in-model red guard against the superseded first-jet-only criterion.

campaign id:	D24 (a)–(e)
proved in:	`definitions.md` D24, promoted at the 2026-08-26 freeze from `theory/ml5-universality.md` steps (1.1)–(1.3) and repaired thereafter; `theory/verdicts/d24d3-adjudication-r5.md` §5 is the merged text of record for clause (d), superseding r4 §5 where marked (clause 2 scope, clause 3b hypotheses); the finite-N leg ratio is r5 §1.1 and the two-hypothesis fence is r5 §2
tested by:	`theory/checks/d24d3_normalization_check.py` named computations D24N-C3 (the residue in both normalisations and both scopes) and D24N-C8 (the exact leg-conversion constant, vacuum and descendant legs); `theory/checks/ml4_check.py` named computation ML5-I1 (the intercept counterexample)
reviewed in:	`theory/verdicts/d24d3-adjudication-r3.md`, `d24d3-adjudication-r4.md`, `d24d3-adjudication-r5.md`

Definition 13.8 (The corner-C soft multiplier)**.** In the n-leg conjecture of Conjecture 13.19, $\mathsf{S}(k_s; \{q_i, v_i\})$ denotes the proposed soft multiplier in

$$M_{n+1}(k_s; p_1, \ldots, p_n) = \mathsf{S}(k_s; \{q_i, v_i\})\, M_n(p_1, \ldots, p_n) + R_{S2}(k_s). \tag{274}$$

Its arguments are the asymptotic leg charges and velocities and, for kink legs, the vacuum-pair and collective-coordinate data. Here *universal* means independent of the microscopic hard tensors within the Ward-covariant no-contact class $\mathcal{S}_W$ of Definition 13.5. The conjectured behaviour is $\mathsf{S} \to 0$, with leading order linear in k_s and $R_{S2} = o(\mathsf{S})$ in the packet norm of Definition 13.2. This definition names the object and does not strengthen its status: existence, factorisation, and universality all remain the conjecture Conjecture 13.19. The symbol is distinct from the symmetry-protected endpoint operator and from the Bethe amplitudes S_{12}, S_{21}, S_{phys} of Section 10.

campaign id:	D25
proved in:	`definitions.md` D25
tested by:	—
reviewed in:	—

13.3 The refutation

Refuted claim 13.9 (Universality over unrestricted local sources)**.** [REFUTED]

Withdrawn statement. *For every local or exponentially quasi-local source O, the hard-plus-soft amplitude factorises as $M_2^O(k,h) = 2i\mathfrak{a}_{\text{leg}}\chi\, k\, M_1^O(h) + O_{L^2(I)}(k^2)$, with a multiplier depending only on the asymptotic leg data.*

The mechanism that kills it. *Take the four-site family O_η of Equation (271). Every O_η is local, and it is deliberately not a single $SU(2)$ tensor source: it mixes one- and two-lowering components. The one-hard amplitude is independent of η, while*

$$M_2^{O_\eta}(k,h) = \eta\big\{\, 2i(1 - e^{-3ih})\, k + O_I(k^2)\,\big\}, \qquad \mathfrak{c}_h(\eta D) = 2i\eta(1 - e^{-3ih}) \neq 0 \tag{275}$$

on any hard packet supported where $1 - e^{-3ih} \neq 0$. The contact first jet therefore fails to vanish while the one-hard amplitude is unchanged, so no multiplier built from the asymptotic leg data can reproduce $\partial_k M_2$. Unrestricted universality is false.

What survives. *The equivalence Theorem 13.11, which converts the question into two testable conditions on any linear source class, and the conditional shape theorem Theorem 13.13 on the restricted class $\mathcal{S}_W(\rho)$. This negative row asserts nothing about the repaired conditional implication.*

Scope 13.10 (The counterexamples are blind to the constant). Both sources of Scope 13.7 have $M_1^O = 0$, so the $\mathfrak{a}_{\mathrm{leg}}$-dependent term in Equation (268) vanishes identically for every value of $\mathfrak{a}_{\mathrm{leg}}$. The refutation and its checker are therefore untouched by the clause-(3b) repair; neither needed an edit when the stipulation $\mathfrak{a}_{\mathrm{leg}} = 1$ was withdrawn.

campaign id:	ML5
proved in:	`theory/ml5-universality.md` step (1.4)
tested by:	`theory/checks/ml4_check.py`, the O_η obstruction: soft zero 3.14×10^{-16}, coefficient error 9.96×10^{-6}
reviewed in:	`theory/verdicts/corpus-r2.md`

13.4 The repaired criterion

Theorem 13.11 (Intercept-and-first-jet criterion for factorisation)**.** [PROVED]

Let $\mathcal{S}$ be a linear source class in the norm of Definition 13.1 such that for every $O \in \mathcal{S}$ one has $M_1^O \in L^2(I)$ and $M_2^O(k,\cdot)$ is C^2 for $|k| \le \epsilon$ as an $L^2(I)$-valued map, and put

$$K_O(\epsilon) := \tfrac{1}{2} \sup_{|q|\le\epsilon} \|\partial_q^2 M_2^O(q,\cdot)\|_{L^2(I)} < \infty. \tag{276}$$

Then the factorisation

$$M_2^O(k,h) = 2i\,\mathfrak{a}_{\mathrm{leg}}\,\chi\,k\,M_1^O(h) + O_{L^2(I)}(k^2) \tag{277}$$

holds on $\mathcal{S}$ if and only if both

$$M_2^O(0,\cdot) = 0 \qquad \textit{and} \qquad \mathfrak{c}_h(O) = 0 \tag{278}$$

in $L^2(I)$ for every $O \in \mathcal{S}$. Under Equation (278) *the remainder obeys the exact control*

$$\|M_2^O(k,\cdot) - 2i\mathfrak{a}_{\mathrm{leg}}\chi\,kM_1^O\|_{L^2(I)} \le K_O(\epsilon)|k|^2. \tag{279}$$

On an affine class $O_0 + \mathcal{V}$ both conditions must hold on every source difference $V \in \mathcal{V}$, and the base source must obey Equation (277) *with the same stated norm control.*

Scope 13.12 (Two qualifications). First, the *relative* bound used in Definition 13.2 requires the additional uniform hypothesis $K_O(\epsilon) \le C_I \|M_1^O\|_{L^2(I)}$ on the class; it does not follow from C^2 regularity alone. Second, and more importantly, **the criterion is constant-agnostic**: its proof evaluates at $k = 0$ and differentiates once, and never uses the value of $\mathfrak{a}_{\mathrm{leg}}$; only $\mathfrak{a}_{\mathrm{leg}} \neq 0$ is needed. The theorem is a criterion, not an assertion that any particular quasi-local source obeys it.

Idea of proof. Necessity: evaluate Equation (277) at $k = 0$ to get the intercept condition, then differentiate at $k = 0$ — the $O(k^2)$ remainder has vanishing first derivative — to get $\mathfrak{c}_h(O) = 0$. Sufficiency: with both conditions, the Banach-space Taylor theorem with integral remainder gives Equation (279) and hence Equation (277). The affine statement follows because both amplitudes, and therefore both $M_2(0,\cdot)$ and $\mathfrak{c}_h$, are linear in the source.

campaign id:	ML5-A
proved in:	`theory/ml5-universality.md` step (1.2), necessity (2.1), sufficiency (2.2), affine case (2.3); stated with the constant $2i\mathfrak{a}_{\mathrm{leg}}\chi$ since the clause-3 repair
tested by:	`theory/checks/ml4_check.py` named computation ML5-I1 (nonzero intercept with zero numerical first jet)
reviewed in:	`theory/verdicts/corpus-r2.md` (promotion adjudication); `theory/verdicts/d24d3-adjudication-r3.md` (constant-agnostic restatement)

13.5 The conditional shape theorem

Theorem 13.13 (Factorisation shape on the no-contact class)**.** [PROVED, CONDITIONAL]
Let $\mathcal{S}_W(\rho)$ be a source class satisfying all five conditions of Definition 13.5 — including, as an explicit hypothesis, the exhaustive normed LSZ decomposition. Then every $O \in \mathcal{S}_W(\rho)$ obeys

$$M_2^O(k,h) = 2i\,\mathfrak{a}_{\mathrm{leg}}(\rho)\,\chi\,k\,M_1^O(h) + O_{L^2(I)}(k^2), \tag{280}$$

where $\mathfrak{a}_{\mathrm{leg}}(\rho)$ is the soft-leg amputation constant of clause (3b). The coefficient is the signed $2\mathfrak{a}_{\mathrm{leg}}$-site Wigner displacement, and it is independent of the hard process within the class.

Scope 13.14 (A shape with a class constant, not a number)**. The conclusion Equation (280) contains no number at any density, $\rho = 1/2$ included.** Clause (3b) fixes no value of $\mathfrak{a}_{\mathrm{leg}}$, so this theorem predicts a factorisation *shape* — linear in k, proportional to M_1^O, with a class constant — and not a coefficient. The number 2 was never a consequence of this theorem; it was a consequence of this theorem *plus* the frozen clause-3 stipulation $\mathfrak{a}_{\mathrm{leg}} = 1$, which is withdrawn. Corpus displays of the soft coefficient 2 rest on the contact/oracle route of Section 10, which does not use Definition 13.5 at all. The “two-site” Wigner reading is the $\mathfrak{a}_{\mathrm{leg}} = 1$, $\rho = 1/2$ register.

Hypotheses (1), (4), (5) are explicit assumptions and not consequences of the finite-volume Ward identity: the velocity-domain condition fixes the sign χ and the external flux orientation, but it does not suppress the orthogonal current and does not constrain a direct contact source. What remains for future work is a nonempty class containing more than symmetry-generated external-leg insertions; one needs an LSZ Ward identity for the quasi-local source showing that its amputated commutator has zero first jet, not merely that it is regular. For two hard magnons one additionally needs the two-hard Ward reduction, and a sum of external Wigner displacements is conditional on the absence of degenerate three-body channels. Nonemptiness and microscopic membership remain open at every ρ.

On the promotion history, one sentence: the clause-3 split reparametrised this theorem rather than reopening it — the proof is unchanged apart from carrying $\mathfrak{a}_{\mathrm{leg}}$ as a nonzero constant factor, the relevant step needing only $\mathfrak{a}_{\mathrm{leg}} \neq 0$ and v_h bounded below on I — so the earlier corpus promotion survives verbatim with the constant carried.

Idea of proof. By clause (2) the descendant external-leg contribution is $(e^{ik}-1)L(k,h)\,[2iv_h M_1^O(h)]$. Expanding $e^{ik} - 1 = ik + O(k^2)$ and, by clauses (3a)–(3b), $L(k,h) = \mathfrak{a}_{\mathrm{leg}}(-i\chi/v_h) + O_I(k)$, and using that v_h is bounded away from zero on the compact window and that $\mathfrak{a}_{\mathrm{leg}} \neq 0$, this equals $2i\mathfrak{a}_{\mathrm{leg}}\chi\,kM_1^O + O_{L^2(I)}(k^2)$. By clauses (4) and (5) the orthogonal current and direct source terms are each $O(k^2)$, and by clause (1) these are *all* the remaining terms. Adding gives Equation (280), whose phase slope is $2\mathfrak{a}_{\mathrm{leg}}\chi$.

campaign id:	ML5-B
proved in:	`theory/ml5-universality.md` step (1.3), descendant leg (2.1), exhaustive remainder (2.2), sum (2.3); re-scoped at the clause-3 repair, `theory/verdicts/d24d3-adjudication-r3.md`; superseded in part by `d24d3-adjudication-r4.md` §0.2 and `d24d3-adjudication-r5.md` §0.2, with r5 §5 the merged text of record
tested by:	`theory/checks/ml4_check.py` (local obstructions only, and blind to $\mathfrak{a}_{\mathrm{leg}}$ since both counterexamples have $M_1^O = 0$); `theory/checks/d24d3_normalization_check.py` D24N-C1–C3 test the kinematics of the conclusion, not its constant
reviewed in:	`theory/verdicts/corpus-r2.md` (promotion as a conditional implication; the r3 annotation records that the certified multiplication survives verbatim with $\mathfrak{a}_{\mathrm{leg}}$ carried)

13.6 The two candidate constants

The shape theorem leaves exactly one number undetermined. Two independent routes to it are carried side by side. Neither is adopted, and the difference between them is precisely one unproved bridge.

Theorem 13.15 (Conditional matched value of the amputation constant). [PROVED, CONDITIONAL]
Work in the normalisation of Definition 13.2, and assume:

(α) the jet-identification bridge — *the soft multiplier jet of Definition 13.5 equals the two-body physical phase jet of Section 10 (this is the corpus identification, and it is* not *a proved statement); and*

(β_S) class membership — *existence of a source $O \in \mathcal{S}_W(\rho)|_{\rho=S}$ with $M_1^O \neq 0$ in the fully polarised spin-S register. (The notation $\mathcal{S}_W(\rho)|_{\rho=S}$, rather than $\mathcal{S}_W(S)$, keeps visible that the density here is the tail density of Definition 13.5 and not a spin label, and in particular not the file-local double-occupancy coefficient of the two-magnon shard.)*

Then, for each such S separately,

$$\mathfrak{a}_{\text{leg}}(S) = \frac{1}{2S} = \frac{1}{Z_\rho}, \qquad \textit{i.e.} \qquad L(0,h) = \frac{-i\chi}{2\rho\, v_h}, \qquad \textit{with shape-theorem jet } \frac{\chi}{\rho}, \tag{281}$$

at $S \in \{1/2, 1, 3/2, 2\}$. Within the one-parameter family $\mathfrak{a}_{\text{leg}} = Z_\rho^{-p}$, the exact law

$$(2S)^p = 2S \tag{282}$$

forces $p = 1$ at any single $2S \neq 1$.

Scope 13.16 (The implication is a theorem; its antecedents are not). This row is currently VACUOUS-OR-UNKNOWN, and that qualifier is content, not modesty. Antecedent (β_S) is open at every S — no member of the class has ever been exhibited — and antecedent (α) is unproved. The row therefore *constrains future work* — any member ever exhibited must have this value — rather than supplying evidence now. It cannot be upgraded past that reading until a member is exhibited or the conjecture Conjecture 13.17 closes.

Two further cautions. The point $S = 1/2$ is degenerate in Equation (282), and the four densities at which the row is stated are four points of one analytic curve, not four independent measurements. Extension to non-half-integer ρ is hypothesis, not consequence.

Idea of proof. Under (β_S) the class constant is defined and the shape theorem Theorem 13.13 applies, giving a soft phase jet $2\mathfrak{a}_{\text{leg}}(\rho)\chi$. Under ($\alpha$) that jet is the two-body physical phase jet, which the proved spin-S two-body result of Section 10 fixes at χ/S. Equating gives $2\mathfrak{a}_{\text{leg}}(S) = 1/S$, i.e. Equation (281), since $Z_\rho = 2\rho = 2S$. The power law Equation (282) is then the observation that $Z_\rho^{-p} = Z_\rho^{-1}$ at a single density with $Z_\rho \neq 1$ already forces the exponent.

campaign id:	D24-VAL
proved in:	`theory/verdicts/d24d3-adjudication-r3.md` §2, step (1.3); superseded in part by `d24d3-adjudication-r4.md` §0.2 (preamble, clause-2 normalisation, invariance list, conversion step) and `d24d3-adjudication-r5.md` §0.2 (clause-2 LSZ scope, clause-3b hypotheses); r5 §5 is the merged text of record
tested by:	`theory/checks/d24d3_normalization_check.py` D24N-C6 tests the numerical consistency of the *consequent* against the ansatz-free two-magnon slopes at a band derived from the data's own quoted error; neither antecedent is tested by any gate, and no gate bears on L, E^O_{desc}, or membership
reviewed in:	`theory/verdicts/d24d3-adjudication-r3.md`, `d24d3-adjudication-r4.md`, `d24d3-adjudication-r5.md`

Conjecture 13.17 (Soft-leg amputation supplies the order-parameter density). [CONJECTURE]

For every ρ at which $\mathcal{S}_W(\rho) \neq \varnothing$ and *the asymptotic one-magnon kernel of Definition 13.2 exists — currently the fully polarised spin-S family, kernel existence off that family being itself a conjecture — charge-created soft-leg amputation contributes the per-site order-parameter density $Z_\rho = 2\rho$ to the external flux, i.e.*

$$\mathfrak{a}_{\text{leg}}(\rho) = \frac{1}{Z_\rho} \qquad \text{without } \textit{the jet-identification bridge } (\alpha). \tag{283}$$

Scope 13.18 (The quantitative fence: a leg normalisation gives only half the power). Dropping the bridge (α) is precisely what distinguishes this conjecture from Theorem 13.15, so the two do not double-count in the argument DAG. Like that theorem, this row is currently VACUOUS-OR-UNKNOWN: nonemptiness is open at every ρ, so it constrains future work — any member ever exhibited must have this value if the conjecture holds — rather than supplying evidence now. It is statable only relative to the normalisation of Definition 13.2: a change of normalisation rescales $\mathfrak{a}_{\text{leg}}$, so before that convention the statement had no truth value.

The substantive fence is quantitative. The leg-conversion factor is computed *exactly*: $Q_k^-|\Omega\rangle = \sqrt{Z_\rho}|k\rangle$ as in Equation (267), and this is the exact mismatch, in the LSZ limit, between clause (2)'s charge-created residue normalisation and the asymptotic leg of Definition 13.2 (at finite N the descendant-leg ratio is $\sqrt{Z_\rho - 2/N}$). A leg normalisation *alone* therefore supplies only $Z_\rho^{-1/2}$.

That half-power value is refuted against the ansatz-free two-magnon data, but **only under two hypotheses, neither of them proved**: (i) the charge-created reading of the descendant term — that the clause-(1) descendant summand is exactly the propagated charge-created leg with no further dressing — and (ii) the jet-identification bridge (α), without which a two-magnon phase slope and an amputation constant are not comparable at all. Under both, the refutation is at the pre-registered band 0.08, with deviations 0.42, 0.73, 1.00 at $S = 1, 3/2, 2$ and margins 5.2–12.5×; the relevant red mode's own message quotes the tighter runtime-derived band 0.0479, and the refutation needs neither band to be tight.

The conclusion is therefore sharp and useful: the conjecture can be true only if a second factor $Z_\rho^{-1/2}$ arises from a mechanism that is *not* a leg normalisation. Any proposed proof must be checked against the exact leg conversion, or it double-counts the leg.

campaign id:	AMP
proved in:	—, originally named at `theory/verdicts/soft-index-r1.md` F1(c)(i); the quantitative fence is `theory/verdicts/d24d3-adjudication-r3.md` step (1.4), in one-step form `d24d3-adjudication-r4.md` §1.3, with the two hypotheses shipped explicitly per `d24d3-adjudication-r5.md` §2
tested by:	`theory/checks/d24d3_normalization_check.py` D24N-C8 (the leg-conversion constant on the vacuum and descendant legs) and D24N-C6 `--red-halfpower` (the leg-normalisation-only value refuted)
reviewed in:	`theory/verdicts/d24d3-adjudication-r3.md`, `d24d3-adjudication-r4.md`, `d24d3-adjudication-r5.md`

13.7 The surviving n-leg conjecture

Conjecture 13.19 (Lattice soft theorem with n hard legs). [CONJECTURE]

With S *as in Definition 13.8, the* $(n+1)$*-leg amplitude factorises as*

$$M_{n+1}(k_s) = \mathsf{S}(k_s; \{q_i, v_i\})\, M_n + R_{S2}, \tag{284}$$

with S *universal — independent of the microscopic hard tensors — only on the repaired class $\mathcal{S}_W$ of Definition 13.5. Over unrestricted local sources, universality is refuted (Refuted claim 13.9).*

Scope 13.20 (The live obligations). The conjecture is stated with its outstanding debts attached, and they are substantial: existence of the asymptotic one-magnon kernel outside the fully polarised register; the reduced-channel estimates; the packet-smeared infinite-volume Ward reduction, and the same reduction at two or more hard legs; the exhaustive LSZ decomposition; and microscopic membership in $\mathcal{S}_W$. Every one of these is a separate open item, and the last is the one that also blocks both candidate constants above. The status of the conjecture within the wider argument, including how it interacts with the memory and asymptotic-symmetry corners, is summarised in Section 18; the negative results that bound the search space are collected in Section 16.

campaign id: S-general
proved in: —, statement carried in `claims/CLAIMS.md`; definition in `definitions.md` D25; depends on the two-body row of Section 10 and on Theorem 13.13
tested by: —
reviewed in: —

13.8 Reading of the section

The honest summary is short. Universality in the naive sense is false on the lattice, and the counterexample is explicit and checked. What replaces it is a criterion that is an exact equivalence, and a conditional theorem that delivers a factorisation shape on a class defined by five conditions — two of which encode exactly the content the Ward identity does not give. The single constant in that shape is open. Two candidate values for it are carried, one a theorem with unproved antecedents and one a conjecture with a sharp quantitative fence, and both are worth precisely what their antecedents are worth: they tell us what any future class member must satisfy, and nothing about whether such a member exists.

The one place in the campaign where a constant of exactly this profile has been *derived* rather than assumed is the separated-preparation class of Section 12, where the reduced multiplier is proved to extend with value $-i\chi/(2Sv_h)$. That is the same profile with the same coefficient — and, as recorded there, that agreement is not yet an identification.

14 Symmetry-protected topology: edge residues, twists, and the rigidity dichotomy

A one-dimensional gapped chain with an on-site symmetry group G carries a discrete label that no local symmetric deformation can change: the class $[\omega] \in H^2(G, U(1))$ of the multiplier by which the symmetry acts on the virtual bond space of its matrix-product description (Section 5). This section reports what happened when we asked whether that label leaves a fingerprint in *soft* data — the coefficients that survive when a symmetry insertion is smeared over a window whose width is then sent to infinity. The hope was a novel prediction: a soft amplitude that is quantised by the symmetry-protected class in the same way that a soft graviton amplitude is fixed by an asymptotic charge.

The hope, in the two forms in which we first wrote it down, is false, and the falsity is instructive. The first form said that a closed bulk symmetry insertion cannot see $[\omega]$ at all, because its two endpoint multipliers cancel. The cancellation is real, but the inference is not: what is left after the multipliers cancel is the honest adjoint representation $\mathrm{Ad}(V)$, and adjoint representations of projective and linear lifts are genuinely different objects. The second form said, in the opposite direction, that $[\omega]$ can never appear in any soft coefficient, at any order. That argument leaned on Whitehead's lemma, which trivialises only the *infinitesimal* central cocycle of a compact semisimple Lie algebra; it says nothing about the dimension of an endpoint module, its integrality weights, or the behaviour of finite group elements. Both statements are recorded below as refutations, because knowing exactly how each died is what fixed the correct statement.

What survives is a dichotomy between the bulk and an unpaired endpoint. Closed bulk insertions have no projective multiplication anomaly and their normalised coefficients are ordinary continuous transfer data: along a path of symmetric injective tensors with a common transfer gap, they move continuously, and continuity is not quantisation. An explicit computation on the anisotropic Affleck–Kennedy–Lieb–Tasaki (AKLT) family makes this concrete — a natural bulk soft-charge coefficient runs from $1/8$ to $0.2401\ldots$ inside a single nontrivial phase. An unpaired half-chain endpoint behaves quite differently. Its registered residue is an operator on a fixed finite-dimensional module that carries the projective action, its spectrum lies in a lattice shifted by the class, and on the same AKLT family where the bulk coefficient runs, the residue is exactly $-Z/2$ throughout. That rigidity is the honest topological content.

The last part of the section reports the memory statement. It is deliberately modest and deliberately honest. A two-projective-measurement (TPM) protocol on a boundary window records an integer at every finite window length, before any limit, for a reason that is generic circle-charge arithmetic: the same window and the same normal-ordering offset are used at both readouts, so the offset cancels in the difference. The symmetry-protected class does not enter that arithmetic. It enters the *interpretation*: it is what guarantees that the boundary hosting the protocol carries a projective module of dimension at least d_ω, and hence has the capacity to store a nonzero outcome. Whether an actual dynamics writes a nonzero outcome into that capacity is a separate, still open, question, and is stated below as a conjecture with its missing steps named.

14.1 The registered objects

Everything in this section is stated first on a finite-dimensional register built from the transfer operator, and only afterwards — and only under an explicitly displayed hypothesis — transported to a physical half-chain Hilbert space. That discipline is what the five definitions below install. It matters because the object one is tempted to call "the edge" has at least three inequivalent meanings (a Schmidt register, a padded-window matrix algebra, and a physical low-energy subspace), and conflating them was the source of the first round of errors.

Definition 14.1 (Soft profiles, the Schmidt/edge register, and the order of limits)**.** Let C be an injective left-canonical tensor in the sense of Section 3, with strictly positive right fixed point $r > 0$. For an operator O supported on the first n sites of a half-chain, write $\mathbb{E}_O$ for its transfer contraction through those sites and define the *normalised boundary transfer compression*

$$\mathcal{C}_C^{(n)}(O) \;:=\; r^{-1/2}\,\mathbb{E}_O(r)\,r^{-1/2} \;\in\; \mathrm{End}(E_C), \qquad E_C := \mathbb{C}^\chi, \tag{285}$$

so that $\mathcal{C}_C^{(n)}(I) = I$. We call E_C the *Schmidt/edge register*. It is a fixed finite-dimensional space, defined *before* any scalar matrix element and before any limit. It is not the padded-window matrix module of Definition 14.3, and its identification with a physical edge Hilbert space is the separate hypothesis (H-split) stated there.

Fix once and for all the bump function $\varphi(y) = c_\varphi\, e^{-1/(1-y^2)}$ for $|y| < 1$ and $\varphi(y) = 0$ otherwise, with $c_\varphi > 0$ chosen so that $\int|\varphi|^2 = 1$. The three profile families used below are

$$f^{\mathrm{bulk}}_{L,\kappa}(x) = Z_{L,\kappa}^{-1/2}\,\varphi(x/L)\,e^{i\kappa x/L}, \qquad\qquad Z_{L,\kappa} = \sum_x |\varphi(x/L)|^2, \tag{286}$$

$$f^{\mathrm{edge}}_L = \mathbf{1}_{[0,L-1]}, \tag{287}$$

and, for a finite group, the endpoint profile $g^{\mathrm{edge}}_L(x) = g$ for $0 \le x < L$ and $g^{\mathrm{edge}}_L(x) = e$ otherwise, applied through the normal-ordered on-site operator $\check{u}_C(g) = e^{-i\theta_C(g)}u(g)$ so that no extensive phase survives. Every object at finite L is strictly local.

The *order of limits* is fixed and is part of every statement below: take $N \to \infty$ at fixed L with $N - L \to \infty$; normalise the packet states at that fixed L; take scattering or time limits if any are present; and only then let $L \to \infty$, so that the associated momentum scale $k_L \sim L^{-1}$ tends to zero. For a finite group G the last operation is a large-window endpoint/string limit only. It is not an infinitesimal group limit and not a momentum-soft limit, and it must not be described as one.

campaign id:	D19
proved in:	`definitions.md` D19; `theory/spt-rebuild.md` step (1.2).(2.2)–(2.3)
tested by:	`theory/checks/spt_rebuild_check.py` S-C2, S-C4
reviewed in:	`theory/verdicts/corpus-r2.md` (defect S1); `corpus-r4.md`

Definition 14.2 (Operator-valued soft insertions and their scalar form factors)**.** Fix normalised finite-dimensional bulk channel registers $K_{\mathrm{in}}, K_{\mathrm{out}}$ and packet embeddings $W^{N,L}_{\mathrm{in/out}}$ into a finite chain, and define the operator-valued bulk insertion

$$\mathfrak{F}^{\mathrm{bulk}}_{N,L}(\xi) := \left(W^{N,L}_{\mathrm{out}}\right)^\dagger Q[f^{\mathrm{bulk}}_{L,\kappa};\xi]\,W^{N,L}_{\mathrm{in}} \;\in\; \mathrm{Hom}(K_{\mathrm{in}}, K_{\mathrm{out}}). \tag{288}$$

Along a path of tensors, every external tensor, channel embedding, tangent-space gauge fixing, Gram normalisation, and differentiated profile entering (288) is *required* to be continuous in the path parameter, and C^p whenever a coefficient through order p is asserted. These are hypotheses on the external data, not consequences of the tensor path.

The modulated charge $Q[f;\xi]$ of Section 4 is anti-Hermitian, so put $Q^H[f;\xi] := -iQ[f;\xi]$. On the edge register define the Hermitian residue and the compensated group residue

$$\mathfrak{R}_{C,L}(\xi) := \mathcal{C}_C\big(Q^H[f^{\mathrm{edge}}_L;\xi]\big), \qquad \mathbb{S}^{\mathrm{comp}}_{C,L}(g), \tag{289}$$

where $\mathbb{S}^{\mathrm{comp}}_{C,L}(g)$ is the decorated contraction of the normal-ordered string $\check{U}_{C,[0,L-1]}(g)$ with $V_C(g)^{-1}$ inserted at the remote cut, viewed on the dual left-edge register. That insertion cancels the far endpoint left by the Ward identity of Section 4; the *uncompensated* string overlap is a different object and may decay to zero. Soft residues are operator-norm limits taken in the order of Definition 14.1. Scalar form factors are only matrix elements of those operators, for instance

$$F^{\mathrm{edge}}_{C,L}(e', e; \xi) = \langle e'|\mathfrak{R}_{C,L}(\xi)|e\rangle, \tag{290}$$

and a scalar is never declared to be algebra-valued.

Thermodynamic limits are uniform only on compact tensor paths carrying a common transfer bound. A coefficient through order p carries the hypothesis (H-soft-p): the soft limit is uniform through p derivatives. Any statement about scattering carries in addition the wave-operator and asymptotic-completeness hypotheses required by Section 8; transfer continuity supplies neither.

campaign id:	D20
proved in:	`definitions.md` D20; `theory/spt-rebuild.md` steps (1.2).(2.4)–(2.6)
tested by:	`theory/checks/spt_rebuild_check.py` S-C3, S-C6
reviewed in:	`theory/verdicts/corpus-r2.md` (defect S1)

Definition 14.3 (The endpoint module, the shifted charge lattice, and the physical-edge hypothesis)**.** The unconditional object is the Schmidt/edge register $E_C = \mathbb{C}^\chi$, carrying the projective action $V_C(g)$ with multiplier ω_C. It is distinct from the padded-window matrix module of Section 5,

$$\mathcal{M}_\chi(C) \;\cong\; M_\chi(\mathbb{C}) \;\cong\; E_C \otimes E_C^*. \tag{291}$$

Under left multiplication $V_C(g)$ acts on the first tensor factor with the dual factor a spectator, so $\mathcal{M}_\chi(C)$ is exactly χ copies of the E_C projective module. This multiplicity distinction changes none of the dimension bounds below. Write d_ω for the minimal dimension of an ω-projective irreducible representation; a nontrivial class has $d_\omega > 1$, since a one-dimensional multiplier is a coboundary.

Hypothesis (H-split). The half-chain ground or low-energy subspace admits a normal split realisation together with an isometry $J_C : E_C \to H_{\mathrm{edge}}$ intertwining the Schmidt/edge action V_C with the physical half-chain symmetry, in such a way that registered local contractions converge to the corresponding compressed physical operators. (H-split) is a statement about E_C and not about all χ^2 padded-window matrices. It is the explicit form of the missing normality/infinite-volume Schmidt step flagged in Section 5.

Suppose the identity component of the compact group G contains a circle, and let the universal cover of that component have kernel Z. Assume that a lift normalised compatibly with the full group and with the physical vacuum charge fixes a class-invariant central character $\nu_\omega : Z \to U(1)$. If the lifted 2π path of a circle generator ξ ends at $z_\xi \in Z$, the raw lift offset $q_\omega(\xi) \in \mathbb{R}/\mathbb{Z}$ is defined by $e^{2\pi i q_\omega(\xi)} = \nu_\omega(z_\xi)$. Set

$$X_C^\circ(\xi) := X_C(\xi) - \mathrm{Tr}\big(rX_C(\xi)\big)I, \qquad \bar{q}_C(\xi) := -i\ \mathrm{Tr}\big(rX_C(\xi)\big), \tag{292}$$

and $q^\circ_{\omega,C}(\xi) := q_\omega(\xi) - \bar{q}_C(\xi) \bmod \mathbb{Z}$. The Hermitian registered endpoint charge is

$$Q_{\mathrm{edge}}(\xi) := -i\, X_C^\circ(\xi), \qquad \text{with} \qquad \operatorname{spec} Q_{\mathrm{edge}}(\xi) \subset q^\circ_{\omega,C}(\xi) + \mathbb{Z}. \tag{293}$$

Under the phase-gauge change $V_C(\exp(\varepsilon\xi)) \mapsto e^{ia\varepsilon}V_C(\exp(\varepsilon\xi))$, both q_ω and $\bar{q}_C$ shift by a, while X_C°, Q_{edge}, and $q^\circ_{\omega,C}$ are unchanged. In the centered phase gauge $\mathrm{Tr}(rX_C) = 0$ the centered and raw offsets coincide. For $SO(3)$ that gauge gives the two offsets $1/2$ and $0 \bmod \mathbb{Z}$; for the explicit $O(2)$ comparison pair, the reflection pins the same two alternatives, AKLT having $V(2\pi) = -I$ and centered offset $1/2$, the large-D product having $V(2\pi) = I$ and centered offset 0. The full global group and the 2π loop are part of this statement: $U(1)$ alone does not protect the offset.

campaign id:	D21
proved in:	`definitions.md` D21; `theory/spt-rebuild.md` steps (1.2).(2.7)–(2.8)
tested by:	`theory/checks/spt_rebuild_check.py` S-C4, S-C5, and the phase-gauge red mutant
reviewed in:	`theory/verdicts/corpus-r2.md` (defects S1, S5); `corpus-r4.md`

Definition 14.4 (Twists, ordered endpoint products, and boundary-window edge memory)**.** A finite g-twist is a string on an interval and therefore has *two* endpoints. A single registered endpoint carries $V(g)$ and the other carries the inverse/dual multiplier, so that all gauge phases cancel globally. A physical endpoint observable requires both (H-split) and the dressing hypothesis

Hypothesis (H-dress). the endpoint operators have nonzero transfer overlap with the relevant virtual sectors, and the endpoint separation is sent to infinity before any other limit.

For commuting g, h, conjugation at one endpoint gives

$$V(h)V(g)V(h)^{-1} = e^{\,i[\omega(h,g)-\omega(g,h)]}\,V(g), \tag{294}$$

with the other endpoint compensating the phase, and ordered endpoint-soft operators obey $\mathbb{S}(h)\mathbb{S}(g) = e^{i\omega(h,g)}\mathbb{S}(hg)$ in the registered module.

For the channel-free operational register, use the boundary window $f_L^{\mathrm{edge}} = \mathbf{1}_{[0,L-1]}$ of Definition 14.1 and the local Hermitian charge $Q_L^{\partial} := Q^H[f_L^{\mathrm{edge}};\xi]$ of Definition 14.2. Let $E_{L,t}$ be the spectral resolution of $\alpha_t^+(Q_L^{\partial})$. The boundary two-projective-measurement (TPM) law records the escaped charge $\nu = q_- - q_+$ and the edge-memory change $m = q_+ - q_- = -\nu$ through

$$p_{L;t_-,t_+}(\nu) := \sum_q \|E_{L,t_+}(\{q-\nu\})\,E_{L,t_-}(\{q\})\,\Psi\|^2. \tag{295}$$

The hypothesis (E-LR) consists of three clauses. *(E-LR1)*: one common sequence $T_n \to \infty$ gives, for every fixed L, weak-$*$ limits of the positive- and negative-time Cesàro states and limits of all double-Cesàro TPM weights. *(E-LR2)*: writing $\mathcal{D}_{L,t_-}(A) := \sum_q E_{L,t_-}(\{q\})\,A\,E_{L,t_-}(\{q\})$, the double-Cesàro average of $\langle\Psi, [\mathcal{D}_{L,t_-}(Q_L^{\partial}(t_+)) - Q_L^{\partial}(t_+)]\Psi\rangle$ tends to zero for every fixed L. *(E-LR3)*: writing p_L for the fixed-window time limit,

$$\lim_{M\to\infty}\ \sup_L \sum_{|\nu|>M} (1+|\nu|)\,p_L(\nu) = 0. \tag{296}$$

Optionally one may require $p_L \Rightarrow p$; that clause selects a unique limit law and a unique first moment. Dynamics is formed first, fixed-L time limits second, and $L \to \infty$ last. This operational law is distinct from the channel operator below and assumes no wave operator and no channel inventory.

Finally, under the separate hypothesis (H-AD-edge) — asymptotic wave operators exist, bulk packets separate from the edge, and the total charge decomposes as $Q_{\mathrm{tot}} = Q_{\mathrm{edge}} + Q_{\mathrm{bulk}}$ on each channel — define the channel edge-memory operator

$$\Delta Q_{\mathrm{edge}} := W_+^{\dagger} Q_{\mathrm{edge}} W_+ - W_-^{\dagger} Q_{\mathrm{edge}} W_-. \tag{297}$$

Three words are used with fixed meanings. *Protected* refers to the projective edge module and its degeneracy; *quantised* refers to channel eigenvalues of (297), not to arbitrary expectation values; *permanent* additionally requires isolated charge-preserving edge dynamics after the packet has departed. Protection does not force any particular reflection matrix element to be nonzero. Neither (H-AD-edge) nor the operator (297) is part of the operational TPM theorem of Theorem 14.20.

campaign id:	D22
proved in:	`definitions.md` D22; `theory/spt-rebuild.md` steps (1.2).(2.9)–(2.10); `theory/spt-tpm.md` hypothesis set (T1)–(T4)
tested by:	`theory/checks/spt_rebuild_check.py` S-C5; the finite TPM arithmetic is inherited from the red-capable memory-index checker
reviewed in:	`theory/verdicts/corpus-r3.md`; `theory/verdicts/blitz-spt-tpm-r1.md`

Definition 14.5 (Exact comparison tensors and their boundary Hamiltonians)**.** In the Cartesian spin-one basis the AKLT tensor is $A^a = \sigma_a/\sqrt{3}$, equivalently $A^0 = Z/\sqrt{3}$ and $A^{\pm 1} = \sqrt{2/3}\,\sigma_\pm$. Its D_2 virtual operators are Pauli matrices and carry the nontrivial class. The symmetric same-phase deformation path is

$$A^x_b = a_b X, \quad A^y_b = a_b Y, \quad A^z_b = bZ, \qquad a_b = \sqrt{\tfrac{1-b^2}{2}}, \quad 0 < b < 1, \tag{298}$$

with the isotropic AKLT point at $b = 1/\sqrt{3}$.

The exact normalised injective trivial tensor at bond dimension $\chi = 2$ is

$$T^x_t = \begin{pmatrix} 0 & t\sqrt{3}/2 \\ t/2 & 0 \end{pmatrix}, \qquad T^y_t = 0, \qquad T^z_t = \operatorname{diag}\Big(\sqrt{1-\tfrac{t^2}{4}},\ \sqrt{1-\tfrac{3t^2}{4}}\Big), \qquad 0 < t \le 1, \tag{299}$$

with $\theta = (1, -1, -1, 1)$ and $V_T = (I, Z, I, Z)$ on (e, R_x, R_y, R_z); here V_T is an honest linear representation. At $t = 1$ the right fixed point is $r = \operatorname{diag}(3/4, 1/4)$, the transfer spectrum is $\{1, \sqrt{3}/2, 0, 0\}$, and the length-two words have rank four; as $t \downarrow 0$ the family tends symmetrically to $|z\rangle^{\otimes\infty}$. The $O(2)$ Lie-charge trivial comparator is instead the $\chi = 1$ product tensor $P^z = 1$; it is not an $SO(3)$ comparator, and no fictitious $SO(3)$-symmetric trivial $\chi = 2$ spin-one tensor is used.

For any such tensor C set

$$\Gamma^C_2(|i\rangle \otimes |j\rangle) = \sum_{s,t} (C^s C^t)_{ij} |s, t\rangle, \qquad W^C_2 = \Gamma^C_2 \big[(\Gamma^C_2)^\dagger \Gamma^C_2\big]^{-1/2}, \qquad h_C = I - W^C_2 (W^C_2)^\dagger, \tag{300}$$

and let the half-chain parent Hamiltonian be $H_{C,+} = h_{\partial,C} + \sum_{x \ge 0} \tau_x(h_C)$. For AKLT one has $h_C = P^{(S=2)}$ and $h_{\partial,A} = 0$. For $T = T_1$ one takes $h_{\partial,T} = W^T_2\big(\mu|1\rangle\langle 1| \otimes I\big)(W^T_2)^\dagger$ with $\mu > 0$; it is D_2-symmetric and selects the one-dimensional trivial-character edge. For the $\chi = 1$ $O(2)$ product comparator, $h_P = I - |zz\rangle\langle zz|$ and $h_{\partial,P} = 0$.

The model referred to by the dynamical conjecture of Conjecture 14.24 is narrowed to the Hamiltonian already supplied here, with no additional unspecified boundary term:

$$H^{\mathrm{dyn}}_{A,+} := H_{A,+} = h_{\text{mag-edge}} + \sum_{x \ge 1} P^{(S=2)}_{x,x+1}, \qquad h_{\text{mag-edge}} := P^{(S=2)}_{0,1} = \tfrac{1}{6}\big(\mathbf{S}_0{\cdot}\mathbf{S}_1 + 2\big)\big(\mathbf{S}_0{\cdot}\mathbf{S}_1 + 1\big). \tag{301}$$

The conjectured reflection matrix is thus the one for this specific open AKLT half-chain, not for an unnamed extra coupling.

campaign id:	D23
proved in:	`definitions.md` D23; `theory/spt-rebuild.md` step (1.2).(2.1)
tested by:	`theory/checks/spt_rebuild_check.py` S-C1, S-C5
reviewed in:	`theory/verdicts/corpus-r2.md` (defects S4, S5)

14.2 Two claims that died

Refuted claim 14.6 (Pointwise blindness of closed bulk insertions)**.** [REFUTED] ***Withdrawn statement.*** *Closed bulk contractions built only from the adjoint action* $\mathrm{Ad}(V)$ *are pointwise blind to the class* $[\omega]$*, so no closed bulk soft coefficient can distinguish a projective from a linear lift.*

Why it is false. *The multiplier does cancel in* $\mathrm{Ad}(V)$ *(Theorem 14.8), but the surviving adjoint* representation *is not the same object for the two lifts. For the finite group* D_2*, conjugation by the Pauli projective representation decomposes* $M_2(\mathbb{C})$ *into all four one-dimensional characters exactly once, with multiplicities* $(1,1,1,1)$*; conjugation by the scalar trivial representation gives four copies of the trivial character,* $(4,0,0,0)$*. The closed scalar* $\mathrm{Tr}\,\mathrm{Ad}(V(R_x))$ *is accordingly* 0 *in the first case and* 4 *in the second. An explicit* $\chi = 2$ *trivial comparator has multiplicities* $(2,0,2,0)$ *with* $\mathrm{Tr}\,\mathrm{Ad}(V(R_y)) = 4$ *against* 0 *for the projective lift. These are closed* $\mathrm{Ad}(V)$*-only numbers that separate the classes, which is precisely what the withdrawn statement forbade.*

What survives. *The correct residue of the argument is Theorem 14.8: a closed insertion has no projective* failure of multiplication*, and closed bulk amplitudes are invariant under rephasing the lift. Nothing follows about the isomorphism type of* $\mathrm{Ad}(V)$*, and Theorem 14.10 states exactly what the surviving coefficients are: continuous, deformable transfer data.*

campaign id:	SPT-B-r1
proved in:	disproved in `theory/spt-rebuild.md` step (1.3).(2.3)
tested by:	`theory/checks/spt_rebuild_check.py` S-C2
reviewed in:	`theory/verdicts/corpus-r2.md` (objection S2, conceded)

Refuted claim 14.7 (The all-orders edge no-go)**.** [REFUTED] ***Withdrawn statement.*** *The class* $[\omega]$ *cannot appear in any soft coefficient at any order, an edge residue included, because Whitehead's lemma trivialises the relevant cocycle.*

Why it is false. *Whitehead's lemma supports only the narrow statement that an infinitesimal central cocycle for a compact semisimple Lie algebra is cohomologically trivial and can be gauged away in a chosen phase section. It removes a coboundary; it does not erase the dimension, the weights, the Casimir data, or the integrability class of a representation, and it says nothing at all about finite group elements or about a global* 2π *loop. The counterexample is explicit and exact: on the anisotropic AKLT family the registered edge residue is* $-\frac{1}{2}[1-(2b^2-1)^L]Z$ *at every finite window length, with limit* $-Z/2$ *and residue spectrum* $\{-\frac{1}{2}, +\frac{1}{2}\}$*, while the trivial* $O(2)$ *product edge has residue* 0*. A half-integral quantised soft coefficient therefore exists, and it is correlated with the class.*

What survives. *The bulk no-go concerns a bulk projective multiplier and its topological interpretation, not all coefficient data. The edge residue is itself a quantised coefficient; see Theorems 14.12 and 14.14.*

campaign id:	SPT-nogo
proved in:	`theory/spt-rebuild.md` step (1.4)
tested by:	`theory/checks/spt_rebuild_check.py` S-C4
reviewed in:	`theory/verdicts/corpus-r2.md` (objections S3, T8)

14.3 What survives in the bulk

Theorem 14.8 (Closed symmetry insertions carry no projective multiplier)**.** [PROVED] *Consider ordered closed on-site symmetry insertions in the setting of Definitions 14.1 and 14.2, with every external leg in a fixed normalised bulk register and no string endpoint. Then the endpoint action appearing in a closed contraction is the doubled action* $V(g)\otimes\bar{V}(g)$*, equivalently* $\mathrm{Ad}(V(g))$*, and the projective multipliers cancel exactly:*

$$\big(V(h)\otimes\bar{V}(h)\big)\big(V(g)\otimes\bar{V}(g)\big) = \omega(h,g)\,\bar{\omega}(h,g)\;V(hg)\otimes\bar{V}(hg) = V(hg)\otimes\bar{V}(hg). \tag{302}$$

Consequently the projective failure of multiplication of a closed insertion is identically one, and closed bulk amplitudes are invariant under rephasing a fixed lift, $V(g)\mapsto\lambda(g)V(g)$*.*

Scope 14.9. The theorem says nothing about the isomorphism type of the honest representation $\mathrm{Ad}(V)$, and in particular does not imply that closed bulk data are blind to the class; see Refuted claim 14.6. It applies to insertions with no string endpoint: a hidden endpoint violates the register hypothesis and makes the object an edge or twist amplitude by definition.

Idea of proof. A finite on-site symmetry string on a bulk interval telescopes, by the Ward identity of Section 4, to exactly $V(g)^{-1}$ at one boundary bond and $V(g)$ at the other. In a closed contraction those two factors appear together as $V(g) \otimes \bar{V}(g)$, on which a rephasing acts trivially. Ordered multiplication then produces $\omega(h,g)\bar{\omega}(h,g) = |\omega(h,g)|^2 = 1$, which is (302). The same conclusion holds before any tensor contraction, because the truncated on-site physical operators themselves multiply honestly, with no projective scalar; for noncommuting elements only the ordinary group commutator can survive.

campaign id:	SPT-B-mult
proved in:	`theory/spt-rebuild.md` step (1.3).(2.1)
tested by:	`theory/checks/spt_rebuild_check.py` S-C2
reviewed in:	`theory/verdicts/corpus-r2.md` (promotion adjudicated)

Theorem 14.10 (Normalised bulk coefficients are continuous, not topological)**.** [PROVED] *Let $t \in I \mapsto A(t)$ be a compact path of injective canonical tensors, all covariant under the same on-site group G, with a common transfer bound $\sup_t \lambda_E(t) < \tilde{\lambda} < 1$. Assume every external leg sits in a fixed normalised bulk register with no string endpoint and uniformly positive Gram matrices, and that the external tensors, channel embeddings, tangent-space gauge fixings, Gram normalisations, and profiles are continuous in t — and C^p, together with every differentiated profile, whenever a coefficient of order p is asserted. Assume the profiles are the bulk packets of Definition 14.1 or any finite-support, no-net-jump profiles, and assume (H-soft-p) whenever a soft coefficient is asserted. Then every finite-window normalised scalar coefficient, and every thermodynamic or soft coefficient covered by those uniform hypotheses, is continuous in t, and is C^p when the data and (H-soft-p) are. Consequently a bulk scalar coefficient is a symmetry-protected invariant* only *if it is separately shown to be locally constant on every symmetric injective component; multiplier cancellation alone does not supply that proof, and bulk numbers may distinguish inequivalent* $\mathrm{Ad}(V)$ *representation types and may correlate with $[\omega]$ through deformable non-universal data.*

Scope 14.11. The theorem does not assert that AKLT and trivial bulk amplitudes are equal, and it does not say that complete bulk tomography cannot identify the phase. It also does not say that every coefficient varies: a locally constant bulk representation invariant may well exist, but it needs its own proof. Continuity is not local constancy, and the theorem must not be read as a no-go. Only compact subpaths with a common transfer bound are used; the trivial family of Definition 14.5 loses injectivity at its product endpoint, and no statement here crosses that endpoint.

Idea of proof. At finite chain length and finite window, an unnormalised contraction is a polynomial in the entries of $A(t)$, its conjugate, the external tensors, the embeddings, the gauge-fixed representatives, and the profiles, all of which are continuous (respectively C^p) by hypothesis. Normalisation multiplies by inverse square roots of finite Gram matrices, which is continuous by the functional calculus because the smallest Gram eigenvalue is bounded away from zero. On the compact path the isolated fixed-point projection P_t is continuous and the common spectral separation yields a uniform bound $\|\mathbb{E}_t^m - P_t\| \leq C_{\tilde{\lambda}} \tilde{\lambda}^m$, so the thermodynamic contractions converge uniformly in t, and a uniform limit of continuous functions is continuous. Under (H-soft-p) the same argument applies to the first p derivatives.

That continuity is not vacuous, and the campaign exhibits an actual deformation inside one fixed nontrivial phase. With the specified bulk packet, define

$$C_{\mathrm{bulk}}(b) := \lim_{L\to\infty} \frac{\langle Q[f^{\mathrm{bulk}}_{L,\kappa}; S^z]^\dagger Q[f^{\mathrm{bulk}}_{L,\kappa}; S^z]\rangle_c}{\sum_x |f^{\mathrm{bulk}}_{L,\kappa}(x+1) - f^{\mathrm{bulk}}_{L,\kappa}(x)|^2}, \tag{303}$$

equivalently, in terms of the kernel $S^{(b)}_{zz}(k) = \sum_{n\in\mathbb{Z}} e^{ikn}\langle S^z_0 S^z_n\rangle_c$, the limit $\lim_{k\to 0} S^{(b)}_{zz}(k)/[2(1-\cos k)]$. Transfer contraction on the family (298) gives $\langle (S^z_0)^2\rangle = 1-b^2$ and $\langle S^z_0 S^z_n\rangle_c = -(1-b^2)^2(2b^2-1)^{n-1}$ for $n \geq 1$, whence

$$C_{\mathrm{bulk}}(b) = \frac{b^2}{4(1-b^2)}. \tag{304}$$

Thus $C_{\mathrm{bulk}}(1/\sqrt{3}) = 1/8$ while $C_{\mathrm{bulk}}(0.7) = 0.240196078431\ldots$, even though $V(R_x)$ and $V(R_z)$ remain the same anticommuting Pauli pair and $[\omega]$ is fixed along the path. The coefficient (303) is therefore not topological, and no other raw coefficient is promoted merely because it happens to be written using $\mathrm{Ad}(V)$.

campaign id:	SPT-B'
proved in:	`theory/spt-rebuild.md` steps (1.3).(2.2)–(2.3)
tested by:	`theory/checks/spt_rebuild_check.py` S-C2, S-C3 (Fourier kernel at $k = 10^{-3}$ and the direct packet at $L = 2048$, $\kappa = 1.3$)
reviewed in:	`theory/verdicts/corpus-r3.md` (promotion adjudicated)

14.4 The endpoint is rigid

Theorem 14.12 (The exact AKLT-family edge residue)**.** [PROVED] *For the anisotropic AKLT family* (298)*, the Hermitian registered partial-charge compression of Definition 14.2, evaluated on the boundary window* f^{edge}_L *and the circle generator* S^z*, is exactly*

$$\mathfrak{R}_{A_b,L}(S^z) = -\tfrac{1}{2}\big[1-(2b^2-1)^L\big]\,Z \qquad \xrightarrow[L\to\infty]{} \qquad -\tfrac{Z}{2}. \tag{305}$$

The residue spectrum is $\{-\frac{1}{2}, +\frac{1}{2}\}$ *at every point of this nontrivial path.*

Scope 14.13. This is an exact statement about equation (305) and its AKLT-family limit, and nothing more. It is a registered transfer computation, not a scattering amplitude and not a statement about a physical edge Hilbert space; the latter needs (H-split). It does not assert that a magnon-to-edge transition matrix element is nonzero.

Idea of proof. On the family (298) the right fixed point is $r = I/2$, the boundary charge insertion contracts to $\mathbb{E}_{S^z}(r) = -(1-b^2)Z/2$, and the charge transfer channel acts as $\mathbb{E}_b(Z) = (2b^2-1)Z$. Summing sites $0,\ldots,L-1$ and applying the register normalisation $r^{-1/2}(\cdot)r^{-1/2} = 2(\cdot)$ turns the sum into the geometric series $-(1-b^2)\sum_{n=0}^{L-1}(2b^2-1)^n Z = -\frac{1}{2}[1-(2b^2-1)^L]Z$, which converges because $|2b^2-1| < 1$. At the isotropic point the convergence factor is $(-1/3)^L$. The comparison case is equally exact: the $O(2)$ product tensor has $S^z_x|z\rangle = 0$ at every site, hence zero residue.

campaign id:	SPT-E-AKLT
proved in:	`theory/spt-rebuild.md` step (1.4).(2.4), equation (4.1)
tested by:	`theory/checks/spt_rebuild_check.py` S-C4 (finite window formula and direct eigenvalues at $L = 1, 2, 4, 8, 24, 64$)
reviewed in:	`theory/verdicts/corpus-r2.md` (promotion adjudicated)

Theorem 14.14 (The registered endpoint residue and the shifted charge lattice)**.** [PROVED] *Work in the unbroken, normal-ordered setting of Definitions 14.1 to 14.3 and 14.5, and for the Lie-charge clauses assume the operator-norm convergence of the residues* (289)*. Then, in the fixed Schmidt/edge register* E_C *with the declared dual-left orientation:*

1. *the* compensated *endpoint group residue is* $\mathbb{S}_C(g) = V_C(g)$*, and the registered Lie residue is the compression of the Hermitian partial charge, equal to the centered operator* $\mathfrak{R}_C(\xi) = Q_{\mathrm{edge}}(\xi) = -iX_C^\circ(\xi)$*; scalar edge form factors are its matrix elements* (290)*;*

2. *every irreducible summand of the endpoint is* ω_C*-projective, so a protected summand has dimension at least* d_ω*, and* $d_\omega > 1$ *whenever* $[\omega_C] \neq 0$*;*

3. *at a fixed tensor,* $\operatorname{spec} Q_{\mathrm{edge}}(\xi) \subset q^\circ_{\omega,C}(\xi) + \mathbb{Z}$*, and both* Q_{edge} *and its centered offset are invariant under rephasing the virtual lift;*

4. *the padded-window module is* $\mathcal{M}_\chi(C) \cong E_C \otimes E_C^*$*, that is,* χ *copies of the* E_C *projective module, which changes none of the bounds above.*

A statement about a physical *half-chain edge Hilbert space is conditional on (H-split); under it,* $J_C\mathbb{S}_C(g)J_C^\dagger$ *and* $J_C\mathfrak{R}_C(\xi)J_C^\dagger$ *are the corresponding physical edge operators.*

Scope 14.15. No deformation constancy of the centered offset is claimed: clause (iii) is lift-gauge invariance at a fixed tensor, not local constancy along a path. The theorem protects the module and its degeneracy, not any chosen nonzero transition amplitude, and it supplies a quantised residue spectrum together with selection rules rather than dynamics. The offset is absolute only once the full global group and the 2π loop have been fixed; $U(1)$ alone does not protect it. The conclusion about the compensated string carries no implication for the uncompensated string overlap, which may decay to zero.

Idea of proof. Applying the normal-ordered string $\check{U}_{[0,L-1]}(g)$ pulls it through the tensors and leaves $V(g)^{-1}$ at the physical boundary and $V(g)$ at the remote cut; the prescribed $V(g)^{-1}$ insertion in $\mathbb{S}^{\mathrm{comp}}_{C,L}$ cancels the remote factor exactly, and the dual left-edge orientation fixes the surviving factor as $V_C(g)$. For the Lie residue the derivative at the identity is taken at each fixed L first; differentiating the Ward identity in the same orientation and applying the normalised compression yields $-i[X_C(\xi) - \mathrm{Tr}(rX_C(\xi))I] = -iX_C^\circ(\xi)$, where the subtraction is forced by $\mathcal{C}_C(I) = I$ and is exactly what makes the answer invariant under rephasing. Clause (ii) is the projective multiplication law $V(h)V(g) = e^{i\omega(h,g)}V(hg)$ on E_C together with the observation that a one-dimensional projective irreducible representation would exhibit its multiplier as a coboundary. Clause (iii) follows because the lifted 2π loop acts both as $e^{2\pi i q_{\mathrm{raw}}}$ and as $\nu_\omega(z_\xi)$, so each raw eigenvalue differs from $q_\omega(\xi)$ by an integer; centering subtracts $\bar{q}_C$ from every eigenvalue and shifts q_ω by the same amount under a rephasing. For $SO(3)$ the nontrivial lift to $SU(2)$ sends the nontrivial central element to -1, giving $q_\omega = 1/2 \bmod \mathbb{Z}$, and the trivial lift gives 0; the $O(2)$ pair reproduces the same alternatives directly from $V_A(2\pi) = -I$ and $V_P(2\pi) = I$. Clause (iv) is the padded-window injectivity statement of Section 5, in which $V_C(g)$ acts on the first tensor factor with the dual factor a spectator. The exact instance (305) of Theorem 14.12 shows the whole picture is nonvacuous: the residue is rigid where the bulk coefficient (304) runs.

campaign id:	SPT-E’
proved in:	`theory/spt-rebuild.md` step (1.4), clauses (i)–(iv)
tested by:	`theory/checks/spt_rebuild_check.py` S-C4, S-C5, and the phase-gauge red mutant (uncentered generator after a nontrivial rephasing must fail)
reviewed in:	`theory/verdicts/corpus-r4.md` (final promotion)

14.5 Twists and ordered endpoint products

Theorem 14.16 (Twist endpoints carry the slant character)**.** [PROVED] *Adopt the normalised cocycle convention* $V(h)V(g) = e^{i\omega(h,g)}V(hg)$ *and the twist setting of Definition 14.4. On one registered* *endpoint,*

$$V(h)V(g)V(h)^{-1} = e^{\,i[\omega(h,g)-\omega(hgh^{-1},h)]}\,V(hgh^{-1}), \tag{306}$$

and for h in the centraliser of g this reduces to the slant character (294)*, the H^2 commutator. The other endpoint carries the inverse phase.*

Scope 14.17. Only a relative two-endpoint charge, or a dressed string correlator, is gauge invariant; a single endpoint phase is not an observable. A *physical* observable is conditional on both (H-split) and (H-dress). The old unqualified one-endpoint formula for a general group is withdrawn: (294) is restricted to commuting elements, and the general case is (306).

Idea of proof. Write the projective law twice, as $V(h)V(g) = e^{i\omega(h,g)}V(hg)$ and $V(hgh^{-1})V(h) = e^{i\omega(hgh^{-1},h)}V(hg)$, and eliminate the common factor $V(hg)$; this is (306), and setting $hgh^{-1} = g$ gives (294). A finite physical twist has two endpoints and the combined action is honest, so the second endpoint necessarily carries the inverse multiplier. An endpoint matrix element is physical only when its dressing has nonzero transfer overlap, which is (H-dress).

campaign id:	SPT-T'
proved in:	`theory/spt-rebuild.md` step (1.5).(2.1), equations (5.1)–(5.2)
tested by:	—
reviewed in:	`theory/verdicts/corpus-r3.md` (promotion adjudicated; objection S7)

Theorem 14.18 (Ordered endpoint-soft products realise the cocycle)**.** [PROVED] *With the finite-group endpoint profiles and compensated endpoint contractions of Definitions 14.1 and 14.2, taking the thermodynamic limit and then $L \to \infty$ while retaining the fixed endpoint register, the ordered endpoint-soft operators satisfy*

$$\mathbb{S}_C(h)\,\mathbb{S}_C(g) = e^{\,i\omega_C(h,g)}\,\mathbb{S}_C(hg) \qquad \text{in } \operatorname{End}(E_C). \tag{307}$$

For a compact semisimple Lie group, differentiating (307) *near the identity yields an infinitesimal central cocycle that is cohomologically trivial and can be gauged away; in the phase convention where that coboundary has been removed, the ordinary Lie bracket is displayed. The torsion class survives in finite and large transformations and in global integrability data.*

Scope 14.19. The invariant information is the projective module, or a relative endpoint composition — not a gauge-invariant phase of one scalar matrix element. Physical use of (307) carries both (H-split) and (H-dress). Removing the infinitesimal coboundary in a chosen phase section does not identify half-integer and integer modules, and this statement is not the continuum double-soft scattering theorem of the gauge-theory literature; the mechanisms there (a massless current pole and its associated anomalous term) have no counterpart in these gapped registered form factors.

Idea of proof. Each single endpoint-soft limit equals $V_C(\cdot)$ by clause (i) of Theorem 14.14, and their ordered product is (307) by the defining projective multiplication law. Rephasing changes ω by a coboundary and conjugates no physical two-endpoint observable, so the class and the module are invariant. The Lie-algebra clause is Whitehead's lemma, used only in the narrow sense described in Refuted claim 14.7: the infinitesimal central cocycle is a coboundary, and a phase-section change removes it, without any claim that it vanishes in every section.

campaign id:	SPT-D'
proved in:	`theory/spt-rebuild.md` step (1.5).(2.2), equation (5.3)
tested by:	`theory/checks/spt_rebuild_check.py` S-C5
reviewed in:	`theory/verdicts/corpus-r3.md` (promotion adjudicated; objection S3)

14.6 Boundary-window memory

Theorem 14.20 (Boundary-window memory is integral without scattering theory)**.** [PROVED] *(the finite clause unconditionally; the ordered-limit clause as a conditional implication on (E-LR2)–(E-LR3)). Take the boundary windows of Definition 14.1 and the Hermitian normal-ordered charge* Q_L^∂ *of Definition 14.2, a strongly continuous half-chain automorphic dynamics, and a unit preparation* Ψ*. Assume the integral-circle condition of Section 8: the selected normal-ordered Hermitian one-site charge has spectrum in* $\kappa + \mathbb{Z}$ *for some* $\kappa \in [0, 1)$*. Then:*

1. (Finite, unconditional.) *At every finite* L *there is* $\kappa_L \in \mathbb{R}/\mathbb{Z}$ *with* $\operatorname{spec} Q_L^\partial(t) \subset \kappa_L + \mathbb{Z}$ *for all* t*; the TPM law* (295) *is a probability law; and every escaped-charge outcome* ν *and edge-memory outcome* $m = -\nu$ *lies in* $\mathbb{Z}$*. No commutativity of the two Heisenberg observables* $Q_L^\partial(t_-)$ *and* $Q_L^\partial(t_+)$ *is assumed.*
2. (Ordered limit, conditional on (E-LR2)–(E-LR3).) *With the common sequence supplied by the general common-subsequence theorem of Section 8, every ordered limit-point law is a probability on* $\mathbb{Z}$*, and along that subsequence its first moment is the ordered boundary-window charge change,*

$$\sum_m m\, r(m) = \lim_j \Big[\omega_{L_j}^+\big(Q_{L_j}^\partial\big) - \omega_{L_j}^-\big(Q_{L_j}^\partial\big)\Big], \qquad r(m) := p(-m). \tag{308}$$

   Under the optional full-sequence convergence clause, the limit law and the value in (308) *are unique.*

Adding (H-split) and Theorem 14.14 supplies the physical projective-edge and capacity interpretation, and nothing else.

Scope 14.21. No wave operator, channel inventory, definite channel charge, nonzero amplitude, or permanence is assumed or proved anywhere in this theorem; (H-AD-edge) and (H-dress) are not used. Charge conservation is needed only to read the first moment as a current flow, never for the quantisation. The support is all of $\mathbb{Z}$, not $\{-1, 0, +1\}$, unless a separate edge-doublet confinement theorem is supplied. The ordered mean (308) is an ordinary probability average and need not be an integer even though every outcome is. A projective measurement history is *not* the spectral measure of the operator difference $Q_L^\partial(t_+) - Q_L^\partial(t_-)$. The clauses (E-LR2) and (E-LR3) are assumed, not derived, and no nonzero model instance of them is exhibited.

One point of intellectual honesty is worth stating explicitly, because the critic made it and it is easy to lose. The support half of this theorem is generic circle-charge arithmetic: it holds for any integral circle charge and any automorphic dynamics, and by itself it diagnoses no symmetry-protected phase. The symmetry-protected content enters only through the interpretation supplied by (H-split) and Theorem 14.14 — that the boundary hosting the protocol carries an ω_C-projective module with a shifted charge lattice and a capacity of dimension at least d_ω.

Idea of proof. The finite clause is offset cancellation. Each on-site charge has spectrum in $\kappa+\mathbb{Z}$; the different-site terms commute, so $Q_L^\partial = \sum_{x=0}^{L-1} S_x^z$ has spectrum in $L\kappa + \mathbb{Z}$, and that coset is preserved at every time because a C^*-automorphism preserves spectrum. Every weight in (295) is a squared norm, hence nonnegative, and summing over the final and then the initial resolution gives total mass one by two applications of Parseval, with no interchange of the two noncommuting projections. Since both readouts use the *same* window, the same profile, the same normal ordering, and the same observable, both q_- and q_+ lie in the common coset $\kappa_L + \mathbb{Z}$, so their difference is an integer before any limit is taken. For the ordered clause, a lemma identifies the finite-window TPM mean as the pinched difference $\langle Q_L^{\partial}(t_-)\rangle - \langle \mathcal{D}_{L,t_-}(Q_L^{\partial}(t_+))\rangle$. Clause (E-LR1), supplied as a theorem by the general common-subsequence result, provides one time sequence good for all L; (E-LR2) drives the dephasing defect to zero, converting that mean into the difference of the two Cesàro states; and (E-LR3) supplies tightness and uniform integrability, so that Prokhorov extraction on the closed set $\mathbb{Z}$ retains total mass, integer support, and the first moment across the spatial limit.

campaign id:	SPT-M', D26, M-INDEX-fin, M-INDEX-spec, LR1-GEN
proved in:	`theory/spt-tpm.md` steps (1.1)–(1.7); the finite theorem at step (1.1), the ordered theorem at steps (1.3)–(1.4), the dependency audit at step (1.6)
tested by:	finite arithmetic inherited from the red-capable finite memory-index checker `theory/checks/memory_index_check.py` IDX-C1, IDX-C2; no gate bears on (E-LR2)–(E-LR3)
reviewed in:	`theory/verdicts/blitz-spt-tpm-r1.md` (0 fatal, 0 major; records that the support theorem is generic circle-charge arithmetic and not by itself a symmetry-protected diagnostic)

Corollary 14.22 (Channel bookkeeping for the edge charge)**.** [PROVED, CONDITIONAL] *Assume (H-split), (H-AD-edge), conservation of the selected charge, and asymptotic channels with definite bulk and edge charges. Then the channel edge-memory operator* (297) *obeys*

$$\Delta Q_{\mathrm{edge}} = -\big(Q_{\mathrm{bulk,out}} - Q_{\mathrm{bulk,in}}\big), \tag{309}$$

and fixed-system channel differences are integral. For the restricted AKLT edge doublet they lie in $\{-1, 0, +1\}$.

Scope 14.23. This is an *optional* corollary. It is the former channel form of the memory statement, retained unchanged, and it is not part of Theorem 14.20, whose arithmetic uses none of its four hypotheses. It quantises post-selected channel eigenvalues; superposed or mixed expectation values remain continuous. The restriction of the support to $\{-1, 0, +1\}$ holds only after restricting to the channel edge multiplet, and is not available for the operational protocol.

Idea of proof. Under (H-AD-edge) the asymptotic total charge splits as $Q_{\mathrm{tot}} = Q_{\mathrm{edge}} + Q_{\mathrm{bulk}}$ on each channel. Conservation of the total charge across the scattering process then rearranges into (309). Integrality of the fixed-system channel differences is the same shifted-lattice arithmetic as clause (iii) of Theorem 14.14: both channels sit in the same coset $q^{\circ}_{\omega,C} + \mathbb{Z}$, so the common offset cancels in the difference. For the AKLT edge doublet the two available edge eigenvalues are $\pm\frac{1}{2}$ by Theorem 14.12, and their differences are $\{-1, 0, +1\}$.

campaign id:	SPT-M'-ch
proved in:	`theory/spt-rebuild.md` step (1.6).(2.1), retained as an optional corollary
tested by:	— (the dynamical follow-on is tracked as `tns-cpq`)
reviewed in:	`theory/verdicts/corpus-r3.md` (conditional status inherited from the adjudication of the former channel claim)

Conjecture 14.24 (A nonzero edge-changing magnon reflection for the open AKLT half-chain)**.** [CONJECTURE] *For the specific open AKLT parent Hamiltonian* (301)*, with boundary-magnon coupling* $h_{\text{mag-edge}} = P_{0,1}^{(S=2)}$ *and no additional unspecified boundary term, at least one symmetry-allowed edge-changing magnon reflection amplitude is nonzero on an open interval of momenta, and the resulting boundary-window history leaves a nonzero post-selected charge memory in the sense of Theorem 14.20.*

Scope 14.25. The missing steps are named exactly: half-chain wave operators, the hypothesis (H-AD-edge) for the scattering-channel identification, the on-shell reflection matrix for (301), and nonvanishing of the relevant matrix element. None of these is supplied. The conjecture also does not claim nonzero edge-changing reflection at every momentum; it does not claim vanishing memory for every trivial boundary, since an accidental trivial edge mode can store memory perfectly well; it does not claim quantisation of an unconditioned expectation value; and it does not claim permanence without an isolated charge-preserving edge after the packet has departed. A zero edge-changing channel in the planned numerics would refute this model-specific conjecture and would refute nothing in Theorem 14.14; a nonzero channel for the trivial comparator would demonstrate accidental boundary memory and would refute only a revived "trivial implies zero" claim.

The evidence is structural rather than dynamical. Theorem 14.14 guarantees a protected edge module of dimension at least d_ω with the half-integral residue spectrum computed exactly in Theorem 14.12, so the capacity to record a nonzero outcome certainly exists; symmetric local edge terms cannot replace that module by a unique symmetric state. What topology does not do is force any particular reflection matrix element to be nonzero, and the campaign has been careful never to let the existence of a capacity be read as the existence of a signal. The pre-registered numerical protocol for deciding the conjecture — open chains of 48, 64 and 96 sites, incoming one-magnon packets centred at least twelve correlation lengths from the edge, with a probability-conservation defect below 5×10^{-3}, a per-channel edge charge budget below the same threshold, finite-size drift below 10% between $N = 64$ and $N = 96$, and a channel counted as nonzero only if its probability exceeds twenty-five times the combined truncation and finite-size error at three adjacent momenta — is recorded with the shard and is discussed in Section 17.

campaign id:	SPT-M'-dyn
proved in:	`theory/spt-rebuild.md` step (1.6).(2.2); model fixed by `definitions.md` D23, equation (301)
tested by:	— (follow-on `tns-cpq`; the pre-registered gates are in `theory/spt-rebuild.md` step (1.7).(2.4))
reviewed in:	`theory/verdicts/corpus-r2.md` (HOLD; objection S6)

15 The triangle in two space dimensions: anyon labels and 2D memory

The one-dimensional story of the preceding sections has three corners: a charge attached to an endpoint, a memory recorded by a windowed observable, and a soft theorem relating the two. In two space dimensions the first two corners change character in a way that is worth stating plainly before any theorem. An endpoint is no longer labelled by a group element or a projective class; it is labelled by an *anyon type*, an object of the Drinfeld centre of the input fusion category, and those labels do not in general subtract. A memory is no longer the displacement of a point-like kink; it is a ledger of charge crossing a closed curve, a disk or an annulus, whose boundary grows with the window instead of consisting of two endpoints. This section reports what the campaign proved about those two corners, and it is honest about the third: no two-dimensional soft theorem is claimed here.

That honesty is not modesty for its own sake. A survey of the continuum antecedents produced three warnings, and each of them is a live trap for a lattice calculation in $2+1$ dimensions.

First, memory means something different in $2+1$. Direct continuum computations show that permanent-displacement memory is an accident of $d = 4$. In $d = 3$ the retarded field carries a tail, the force decays only as $U^{-1/2}$, and the verdict is an "infinite momentum memory effect" with no gravitational memory at all. A lattice observable in $2+1$ dimensions should therefore be a momentum or velocity kick, stated as a rate or on a finite window, and never as a finite permanent displacement. If a lattice computation returns a finite permanent shift, the honest first hypothesis is that the lattice regulator produced it, not the physics. Everything below respects this: the memory statements of this section are *charge* ledgers on a finite window, and no displacement is defined.

Second, a confined or dressed charge sector silently trivialises the triangle. Logarithmic confinement in three-dimensional quantum electrodynamics forbids finite-energy asymptotic charged states, so the asymptotic symmetry acts trivially on the scattering matrix and the memory effect is abolished. The lattice analogue is immediate and dangerous: in a gapped $2+1$-dimensional topological phase there is no radiative sector at all, so a "soft theorem" derived there risks being the identity $0 = 0$ dressed up in notation. The pre-registered falsifier is that the sector carrying the charge must be shown to possess finite-energy asymptotic states — gapless at a boundary, or tuned to criticality — before any status label is allowed to move. The toric-code theorem below is stated as a *label* theorem precisely so that it is not mistaken for a radiative soft theorem: it is exactly and nontrivially true in a gapped phase, and it makes no soft claim.

Third, name the order of the fall-off. The continuum literature separates a radiative $1/r^{d/2-1}$ sector from a Coulombic $1/r^{d-3}$ sector, and ties only the null piece to flux and hence to the soft sector. That analysis is stated for $d \geq 4$ and does not cover $d = 3$, where the two exponents invert: the Coulombic term $1/r^0$ does not decay at all and dominates the radiative $1/r^{1/2}$. This structural oddity is flagged here rather than quietly extrapolated, and no result below depends on a radiative/Coulombic split.

What the antecedents *do* leave open is the holonomy route. Colour memory measures the change in a flat connection by parallel transport of test charges: an observable that needs no propagating radiation and no permanent displacement. Its lattice counterpart is the change in a string or matrix product operator holonomy encircling a region, whose labels are exactly the central idempotents of a tube algebra. That is the route taken by the first two theorems below.

15.1 The toric-code endpoint carries an exact anyon label

Definition 15.1 (The toric-code model, its boundary-circle measurement, and the same-circle protocol)**.** Let $\Lambda = C_{L_x} \square C_{L_y}$ with $L_x, L_y \geq 3$ be a finite square cellulation of the torus, with one qubit on each edge and Pauli operators X_j, Z_j. For each vertex v and plaquette p put

$$A_v := \prod_{j \ni v} X_j, \qquad B_p := \prod_{j \in \partial p} Z_j, \qquad H_{\mathrm{TC}} := -\sum_v A_v - \sum_p B_p. \tag{310}$$

A star and a plaquette share zero or two edges, so all these terms commute. Let Ω be any torus ground vector, $A_v\Omega = B_p\Omega = \Omega$. The two global relations $\prod_v A_v = \prod_p B_p = I$ leave a four-dimensional ground space, but no choice of logical ground state enters anything below.

The *anyon label set* is the pointed Drinfeld centre

$$\mathsf{A} := \mathrm{Irr}\, Z(\mathrm{Vec}_{\mathbb{Z}_2}) = \{1, e, m, \epsilon\} \;\cong\; \mathbb{Z}_2^e \times \mathbb{Z}_2^m, \tag{311}$$

written $x = (x_e, x_m)$ with $e = (1,0)$, $m = (0,1)$, $\epsilon = (1,1)$; addition in (311) is fusion. We write A rather than $\mathcal{A}$ to keep this label set distinct from the asymptotic symmetry group $\mathcal{A}$ of Section 5. The group structure of (311) is special to this pointed model, and it is exactly what makes a difference of labels meaningful.

Choose a proper contractible disk $D \subset \Lambda$ with compatible vertex and plaquette sets $V(D)$, $P(D)$, and define the closed boundary operators and their joint projection-valued measure

$$F_e(D) := \prod_{v \in V(D)} A_v, \qquad F_m(D) := \prod_{p \in P(D)} B_p, \qquad P_a(D) := \tfrac{1}{4}\big(I + (-1)^{a_e} F_e(D)\big)\big(I + (-1)^{a_m} F_m(D)\big). \tag{312}$$

Every interior edge occurs twice in the appropriate product, so F_e and F_m are closed lattice boundary operators; they commute, are self-adjoint, and square to the identity. Equivalently, the two commuting bits $\hat{q}_D = \big((I - F_e)/2,\, (I - F_m)/2\big)$ take values in (311), F_e reading the parity of enclosed electric charge and F_m that of enclosed magnetic charge.

Let γ be a direct-lattice path and γ^* a dual-lattice path, each with one endpoint inside D and one outside, so that each crosses ∂D an odd number of times. Put

$$Z(\gamma) = \prod_{j \in \gamma} Z_j, \qquad X(\gamma^*) = \prod_{j \pitchfork \gamma^*} X_j, \qquad W_x = Z(\gamma)^{x_e}\, X(\gamma^*)^{x_m}, \tag{313}$$

with an irrelevant Pauli phase chosen so that W_x is unitary. *Pure* means that the single x-component of (313) has been selected; a coherent sum of different W_x need not have one label and lies outside the definite-sector clause. For $x = \epsilon$ the direct and dual endpoints in one small neighbourhood form the composite endpoint. For $x = 1$, $W_1 = I$ is the empty ribbon and is not a physical endpoint.

The *same-circle protocol* is: for an arbitrary initial density matrix ρ, measure the projection-valued measure (312), apply W_x, and measure the *same* measure again. Its joint law and its sector increment are

$$p_x(a, b) := \mathrm{Tr}\big[P_b W_x P_a\, \rho\, P_a W_x^\dagger P_b\big], \qquad \Delta q_D := b - a \in \mathsf{A}. \tag{314}$$

Using the *same* finite boundary register twice is load-bearing, exactly as in the finite windowed memory theorem of Section 8: the two outcomes belong to one affine register before their difference is taken. Here the register has zero offset and is the fusion group itself, and (314) is not spectral arithmetic for a difference of noncommuting operators.

campaign id:	A-INDEX-TC-fin
proved in:	`theory/anyon-label-index.md` §§1.1–1.4
tested by:	`theory/checks/anyon_label_check.py` ANYON-C0–C4 on a 4×4 torus (exact rational weights)
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W2)

Theorem 15.2 (A toric-code ribbon endpoint has a definite anyon label and an exact same-circle shift)**.** [PROVED] *In the setting of Definition 15.1, for every $a, b, x \in \mathsf{A}$:*

1. (Definite endpoint sector.) *The ribbon has exact boundary covariance,*

$$F_e W_x = (-1)^{x_e} W_x F_e, \qquad F_m W_x = (-1)^{x_m} W_x F_m, \tag{315}$$

and hence the selection rule

$$P_b W_x P_a = \delta_{b,a+x}\, W_x P_a. \tag{316}$$

In particular $P_b W_x \Omega = \delta_{b,x} W_x \Omega$: the enclosed string endpoint sits in the definite Drinfeld-centre sector x.

2. (Quantised two-measurement law.) *The law* (314) *is a probability distribution supported on $\{(a, b) : b = a + x\}$, so every run has the group-valued outcome $\Delta q_D = x$. Starting from any torus ground state, $p_x(1, x) = 1$.*

3. (Braiding readout.) *A closed probe ribbon of type $y = (y_e, y_m)$ around the circle may be represented as $L_y(D) := F_m(D)^{y_e} F_e(D)^{y_m}$ and obeys*

$$L_y W_x = (-1)^{x_e y_m + x_m y_e}\, W_x L_y, \tag{317}$$

a nondegenerate pairing, so the two measured bits distinguish all four sectors.

The theorem is finite-volume and exact; it assumes neither a thermodynamic limit nor any tensor-network approximation.

Scope 15.3. The label shift is invariant under ribbon deformations that preserve the inside/outside endpoint partition, including arbitrary extra even crossings of the circle. It need *not* survive moving an endpoint across ∂D, and it need not survive changing the measured circle between the two measurements: both fences are part of the statement. A coherent superposition of different ribbon types is not a pure endpoint and is outside the theorem. Nothing here is a soft theorem, a current identity, a Ward identity, or a persistence statement; and nothing here alters or repackages any one-dimensional row of the campaign. The finite windowed memory theorem of Section 8 is used only as the protocol design pattern — the same register measured twice — and no current, wall coordinate, scattering limit, or Gram inverse is imported from it.

Idea of proof. A direct-lattice path enters and leaves every internal vertex, so it overlaps the corresponding star in zero or two edges, while at either endpoint the overlap is a single edge; the Pauli relation $XZ = -ZX$ then delivers exactly one minus sign per endpoint. The dual statement holds for $X(\gamma^*)$ against the plaquettes. Multiplying the star identities over $V(D)$ and the plaquette identities over $P(D)$ cancels every interior incidence modulo two and leaves exactly one endpoint sign in each active component, which is (315). Moving F_s through W_x in each spectral factor of (312) gives $P_b W_x = W_x P_{b-x}$, and right multiplication by P_a together with orthogonality of the measure gives (316). On a ground vector every factor of F_e and F_m has eigenvalue $+1$, so $P_a \Omega = \delta_{a,1} \Omega$ and the definite-sector clause follows. For the second clause, each weight in (314) is the trace of $K_{a,b} \rho K_{a,b}^{\dagger}$ with $K_{a,b} = P_b W_x P_a$, hence nonnegative; it vanishes off the support because the Kraus operator itself vanishes there, before any property of ρ is used; and summing over b first, using $\sum_b P_b = I$, cyclicity, and $W_x^\dagger W_x = I$, gives total mass $\operatorname{Tr}\rho = 1$ with no assumption that ρ commutes with the first measurement. For the third clause, the closed Z-type boundary string F_m has odd intersection with the active dual X-string exactly when $x_m = y_e = 1$, and the direct/dual exchange of that argument gives the other factor; every nontrivial x is detected by the probe $y = m$ when $x_e = 1$ and by $y = e$ otherwise.

There is also an exact non-abelian instance, which is worth recording because it shows precisely why the group-valued reading is special. For a Levin–Wen model with Ising input, whose bulk category is the doubled Ising theory, take the endpoint type and the enclosed source both equal to $(\sigma, \bar{1})$. The exact bulk fusion rule is $(\sigma, \bar{1}) \otimes (\sigma, \bar{1}) = (1, \bar{1}) \oplus (\psi, \bar{1})$, so a boundary-circle measurement may return either of two outcomes, each through a one-dimensional channel. The retained datum is the fusion event $(x, a \to b; \mu)$, not a quotient b/a; the two possible outputs are the explicit obstruction to subtracting Drinfeld-centre names. A group-valued shadow survives — the universal grading of Ising is $\mathbb{Z}_2$, giving $U(Z(\text{Ising})) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, in which both outcomes have the same grade and $|b| - |a| = |x|$ — but that shadow conserves only a coarse parity and chooses neither channel. This calculation is exact fusion arithmetic; its identification with a chosen microscopic endpoint tensor is *not* proved, and is not part of Theorem 15.2.

campaign id:	A-INDEX-TC-fin
proved in:	`theory/anyon-label-index.md` §2 (theorem) and §3 (proof, steps (1.1)–(1.4)); the doubled-Ising instance is §4 and is not part of the promoted row
tested by:	`theory/checks/anyon_label_check.py` ANYON-C0–C4, with the registered red mode `--red wrong-sector`; fusion arithmetic separately in `theory/checks/cat_hunt_check.py` CATH-C1–C2
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W2) and its promotion table

15.2 Two-dimensional memory is a charge ledger on a finite window

Definition 15.4 (Two-dimensional windows, the background-subtracted window charge, and its two-measurement law)**.** Let $\Gamma = \mathbb{Z}^2$, or any countable locally finite two-dimensional lattice, with uniformly finite on-site Hilbert spaces and quasi-local algebra $\mathfrak{A}$. Let a compact group act site by site, choose one closed one-parameter subgroup and a Hermitian on-site generator q_x, and impose the integral-circle condition of Section 8 in the form

$$e^{2\pi i q_x} = c\,I, \qquad c = e^{2\pi i\kappa}, \quad \kappa \in [0,1), \qquad \text{equivalently} \qquad \operatorname{spec} q_x \subset \kappa + \mathbb{Z}. \tag{318}$$

Compactness of the group alone is not silently turned into a charge theorem: a circle subgroup and (318) must be selected, and a finite or disconnected symmetry with no such generator is outside every statement below. Let α_t be the infinite-volume dynamics; the finite theorem needs only that it is a group of C^*-automorphisms.

The declared protocol windows are disks and annuli,

$$B_R(z) = \{x : \operatorname{dist}(x,z) \le R\}, \qquad A_{r,R}(z) = \{x : r < \operatorname{dist}(x,z) \le R\}, \tag{319}$$

both finite; a disk has one boundary component and an annulus two, each oriented outward from the annular region. The finite theorem in fact applies to every finite $W \subset \Gamma$. A disk exhaustion $B_{R_m}(z)$ exhausts the plane; an annular exhaustion $A_{r_0,R_m}(z)$ with fixed r_0 exhausts the exterior of a fixed core and not the plane, and nothing is called an exhaustion here unless the domain it exhausts is named.

The measured operator is a finite sum with a scalar counterterm,

$$\hat{Q}_W = \sum_{x \in W} q_x - b_W I. \tag{320}$$

Two preparations are intended. For a *symmetry-broken pair*, choose two zero-temperature vacua, a reference interface curve, and the phase profile on its two sides, and let b_W be the corresponding sum of vacuum densities; the curve fixes only the normal-ordering convention and no sharp interface-position operator is assumed. For a *symmetric vacuum*, take $b_W = |W|\rho_0$ with ρ_0 the vacuum density; there is then no wall coordinate at all, and the protocol measures charge emitted from or absorbed by the window. The scalar b_W may be non-integral and may change with W; this is harmless below precisely because both measurements use the same window and hence the same offset.

Let $E_{W,t}$ be the spectral resolution of $\hat{Q}_W(t) = \alpha_t(\hat{Q}_W)$. Measuring projectively at $t_- < t_+$, the recorded escaped charge and its law are

$$\nu := q_- - q_+, \qquad p_{W;t_-,t_+}(\nu) := \sum_q \|E_{W,t_+}(\{q-\nu\})\, E_{W,t_-}(\{q\})\,\Psi\|^2, \tag{321}$$

absent spectral values contributing zero. This is a sequential measurement law, not the spectral law of the generally noncommuting difference $\hat{Q}_W(t_-) - \hat{Q}_W(t_+)$.

campaign id:	M-INDEX-2D-fin, D26
proved in:	`theory/memory-index-2d.md` steps (1.1)–(1.4)
tested by:	`theory/checks/memory_index_2d_check.py` M2D-C1–C4
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W3)

Theorem 15.5 (Every finite disk or annulus records an integer)**.** [PROVED] *Assume* (318)*, a finite disk or annulus W, a real scalar background b_W, a unit vector Ψ in any representation of $\mathfrak{A}$, and finite times $t_- < t_+$. Put $\kappa_W := |W|\kappa - b_W \bmod \mathbb{Z}$. Then*

$$\operatorname{spec}\hat{Q}_W(t) \subset \kappa_W + \mathbb{Z} \qquad \textit{for every automorphic time } t, \tag{322}$$

and the law (321) *is a probability distribution whose escaped increment ν lies in $\mathbb{Z}$. No commutativity of the two Heisenberg observables is assumed.*

Scope 15.6. This is a finite, additive, 0-form charge theorem and nothing more. It implies no displacement: the one-dimensional conversion of escaped charge into a unique kink position does not transplant, because a two-dimensional interface has shape modes and a swept area rather than one canonical coordinate. It implies no 1-form or topological charge, no anyon content, no loop-operator statement, and no result about topological order — the symmetry here is an ordinary on-site 0-form symmetry. It supplies no channel inventory and no ordered limit; the limit statement is Theorem 15.13 and is conditional. It uses no spectral gap and no exponential clustering, so it holds verbatim in a gapless symmetry-broken phase with Goldstone modes: gaplessness can change probabilities, relaxation, and tightness, but it cannot move a same-window branch off $\mathbb{Z}$ while (318) holds. Finally, the *total* unnormalised increment is what is integral; dividing by the perimeter $|\partial W|$ would give a different, density-valued law supported in $|\partial W|^{-1}\mathbb{Z}$, and that must never be advertised as the integer-valued index.

Idea of proof. Each on-site eigenvalue differs from κ by an integer. The operators $\{q_x : x \in W\}$ commute because they act on different sites, so finite spectral addition on (320) puts every spectral value of $\hat{Q}_W$ in $|W|\kappa - b_W + \mathbb{Z}$, and α_t preserves spectrum, giving (322) at every time with the same offset. Every summand of (321) is a squared norm, and summing first over the final and then over the initial resolution gives total mass one by two applications of Parseval, with no interchange of the two noncommuting projections. Since the observable, the window, and the background are the same at both times, both q_- and q_+ lie in the single coset $\kappa_W + \mathbb{Z}$, so $\nu = q_- - q_+ \in \mathbb{Z}$ before any time or spatial limit. Every leaf of this argument sees only one fixed finite set and its cardinality: none sees spatial dimension, boundary length, a spectral gap, or a scattering channel, which is why the one-dimensional proof transplants without loss.

What does change with dimension is the size of the boundary term, and it is worth displaying because it is the obstacle to the limit theorem. For a finite-range interaction $H = \sum_Z \Phi(Z)$ with a termwise charge-conserving decomposition $[\Phi(Z), \sum_{x\in Z} q_x] = 0$, every term lying wholly inside or wholly outside the window cancels, so

$$\frac{d}{dt}\hat{Q}_W(t) = i\,\alpha_t\Big(\sum_{Z\,:\,Z\cap W\neq\emptyset,\ Z\cap W^c\neq\emptyset}\big[\Phi(Z), \textstyle\sum_{x\in W} q_x\big]\Big), \qquad \Big\|\tfrac{d}{dt}\hat{Q}_W(t)\Big\| \le C_{\Phi,q,r}\,|\partial_r W|. \tag{323}$$

On a disk of radius R the number of contributing terms is $O(R)$, and on an annulus $O(r+R)$ because both boundary components contribute, in place of the two-endpoint boundary of an interval. This perimeter growth is a real structural difference, catalogued alongside the other two-dimensional boundary findings in Section 16; a Lieb–Robinson estimate localises the terms but does not by itself cancel the perimeter factor.

campaign id:	M-INDEX-2D-fin, D26, M-INDEX-fin
proved in:	`theory/memory-index-2d.md` steps (1.5)–(1.6), with the leaf-by-leaf transplant ledger at step (1.6) and the perimeter audit at step (1.11)
tested by:	`theory/checks/memory_index_2d_check.py` M2D-C2 and the registered red mode `--red fractional-charge`; donor one-dimensional memory-index and common-subsequence gates
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W3): promotable at proved strength

15.3 Categorical selection: what a general anyon endpoint obeys

Definition 15.7 (The tensor-network typing hypotheses for an annular endpoint register)**.** Let $\mathcal{C}$ be a finite unitary fusion category and let a finite, matrix-product-operator-injective tensor-network lattice model supply all of the following *exact* data. These are hypotheses, not consequences.

(PT1) Tube measurement. An annular representation of the tube algebra $\mathrm{Tub}(\mathcal{C})$ whose minimal central idempotents $\{P_a\}_{a\in \mathrm{Irr}\,Z(\mathcal{C})}$ are mutually orthogonal and sum to the identity on the chosen boundary-circle register.

(PT2) Pure endpoint. A small-circle resolution of the endpoint tensor T_x with $P_y^{\mathrm{end}} T_x = \delta_{y,x} T_x$ for one simple $x \in \mathrm{Irr}\,Z(\mathcal{C})$. A sum over x is not called a pure endpoint.

(PT3) Pulling-through module action. Exact zipper, associator, and pulling-through equations identifying the large-circle action of T_x on a source sector a with the semisimple decomposition

$$x \otimes a \;\cong\; \bigoplus_b \mathrm{Hom}(b, x\otimes a)\otimes b. \tag{324}$$

(PT4) Protocol instrument. The physical charge-creation operation is supplied either as a normalised family of channel-resolved Kraus maps $T_{(x,\mu)}$, or else the experiment is explicitly conditioned on the success of one displayed Kraus map. Category data alone supply neither a normalisation nor probabilities, and (PT4) is the only normalisation input anywhere in this subsection.

For the momentum form, let $T_x(y; w)$ be a translated, windowed representative of the same pure endpoint data, with endpoint position y along a chosen direction and all microscopic smearing data collected in w. On a finite periodic boundary of length L set

$$T_x(k; w) = \sum_{y=0}^{L-1} e^{iky}\, T_x(y; w), \qquad k \in (2\pi/L)\mathbb{Z}. \tag{325}$$

In an infinite-volume protocol, "a limit" means a declared sequence of windows and momenta in a declared operator, weak, or matrix-element topology in which multiplication by the fixed finite-sector projectors is continuous.

campaign id:	A-INDEX-PEPS
proved in:	`theory/anyon-selection-hybrid.md` §§1–2; canonical typing in `theory/anyon-label-index.md` §5
tested by:	`theory/checks/cat_hunt_check.py` CATH-C0–C4; `theory/checks/soft_2d_hunt_check.py` S2DH gates
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W6) and §9 item 5

Theorem 15.8 (Fusion selection for an annular endpoint register)**.** [PROVED, CONDITIONAL] *Assume (PT1)–(PT3) of Definition 15.7, and for the momentum form the notation and declared topology stated there; assume (PT4) only for the instrument-normalisation clause. Then on the finite annular register*

$$P_b\, T_{(x,\mu)}\, P_a = 0 \qquad \textit{unless} \qquad \mu \in \mathrm{Hom}(b, x\otimes a) \;\; \textit{and} \;\; N_{xa}^{b} > 0, \tag{326}$$

and consequently the same-circle instrument weight

$$p_x(a,b,\mu) = \mathrm{Tr}\big(P_b T_{(x,\mu)} P_a\, \rho\, P_a T_{(x,\mu)}^{\dagger} P_b\big) \tag{327}$$

is supported only on the fusion events allowed by (326)*, normalised or postselected exactly as stipulated by (PT4). Moreover the windowed/momentum form obeys the same rule, $P_b T_x(k;w) P_a = 0$ unless $\mathrm{Hom}(b, x\otimes a) \neq 0$, at every finite momentum and window, and every existing limit point in the declared topology obeys the same zero-block constraint.*

The retained fine datum is the fusion event $(x, a \to b; \mu)$. If x is invertible, tensoring by x is an equivalence and the target is uniquely $b = x\otimes a$, so the supplied index is group-valued, $x \in \mathrm{Inv}(Z(\mathcal{C}))$; the toric-code theorem Theorem 15.2 is the case in which every simple object is invertible. For general x the only group-valued statement is the universal grading shadow: presenting the universal grading group by $[c] = [r][s]$ whenever $N_{rs}^{c} > 0$, every allowed event obeys

$$|b| = |x||a|, \qquad \textit{equivalently} \qquad |b||a|^{-1} = |x|. \tag{328}$$

Scope 15.9. Three fences are part of this statement. First, no microscopic instantiation beyond the toric code is claimed: supplying (PT2)–(PT4) for a particular lattice model is an open, model-specific obligation, and it is not asserted that every representative of a topological phase supplies them. Second, no Taylor coefficient inside an allowed channel is claimed. The theorem says a forbidden block vanishes; it says nothing about the value, the leading behaviour, or the softness of an allowed block, and it therefore supplies no categorical soft theorem and no universal soft coefficient. Third, this is one selection-rule statement and is not an omnibus packaging of the group-label, windowed-memory, and tensor-network results, which consume different hypotheses and remain separate rows.

Two further limitations are worth naming. The labels used are *bulk* Drinfeld-centre labels, because the boundary circle sits in the two-dimensional bulk; a physical edge or condensate requires its own module or boundary-excitation category, under which bulk anyons may condense, be identified, or split. And tube projectors select sectors only: they supply no current operator, no Ward identity, no Gram inverse, no transition amplitude, and no persistence theorem. The shadow (328) does not select b, does not fix a probability, and does not authorise subtraction in $\mathrm{Irr}\, Z(\mathcal{C})$; the doubled-Ising branching recorded after Theorem 15.2 is the explicit instance.

Idea of proof. By (PT2) the endpoint carries one simple type x, and by (PT3) its action on the source sector a is the semisimple decomposition (324), which has no summand labelled b when $\mathrm{Hom}(b, x \otimes a) = 0$. By (PT1) the P_b are mutually orthogonal central idempotents resolving the annular register, so left multiplication kills every component of $T_x P_a$ except its b summand; resolving the multiplicity space, whose dimension is N_{xa}^b, gives (326). A forbidden block makes the corresponding Kraus operator in (327) zero, so its weight vanishes by positivity alone; normalisation is used only in the form supplied by (PT4). For the momentum form, each summand $P_b T_x(y; w) P_a$ of (325) is zero on a forbidden pair, and Fourier summation and windowing are linear, so their finite linear combination vanishes. Finally, a net of identically zero blocks has zero limit whenever multiplication by the fixed projectors is continuous in the declared topology — which asserts neither existence nor uniqueness of any soft limit.

campaign id:	A-INDEX-PEPS
proved in:	`theory/anyon-selection-hybrid.md` step (1.1), leaves (2.1)–(2.6)
tested by:	`theory/checks/anyon_label_check.py` ANYON-C0–C4; `theory/checks/cat_hunt_check.py` CATH-C0–C4; `theory/checks/soft_2d_hunt_check.py` S2DH gates. These finite gates establish neither (PT2)–(PT4) for an arbitrary model nor the existence of an infinite-volume momentum limit
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W6) and §9 item 5 (promoted after the structured-proof repair, orchestrator-verified)

Theorem 15.10 (Contractible deformations do not change the endpoint morphism)**.** [PROVED, CONDITIONAL] *Assume an exact fixed-point string or matrix-product-operator representation together with its pulling-through and associator equations. Let γ and γ' have the same resolved endpoints and be related by a finite sequence of contractible local pulling-through moves inside a disk containing no other insertion. Then, after identifying the endpoint fusion spaces by the associator maps supplied by those moves, the code-projected endpoint morphisms for γ and γ' agree.*

Scope 15.11. The contractibility hypothesis is load-bearing and so is the fixed-point hypothesis. A move sequence winding around another insertion is not contractible in the complement of that insertion, so the coherence argument does not identify it with the trivial path; its composition may instead be the corresponding braid or monodromy action. No off-fixed-point statement is made: nothing is claimed about a dressed string in a generic phase, and nothing is claimed about a gapless on-shell soft factor.

Idea of proof. A single local pulling-through move is, by hypothesis, an equality of the two code-projected tensor networks across that move, with the endpoint fusion spaces identified by the associator map it supplies. Composing those equalities along the finite move sequence identifies the first network with the last, since equality and the supplied identifications are stable under finite composition. Two contractible move sequences with the same ends give the same categorical identification by pentagon coherence, which relates their two finite compositions of associators. The exception is exactly the non-contractible case, which the coherence argument does not reach.

campaign id:	SHAPE-FLAT
proved in:	`theory/anyon-selection-hybrid.md` step (1.2), leaves (2.1)–(2.5)
tested by:	`theory/checks/soft_2d_hunt_check.py` with the registered deformation mutation `--red wrong-path`; the toric-code patch supplies the nonzero witness (two paths give the same state, with an endpoint gap of four)
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W6) and §9 item 5 (promoted, orchestrator-verified)

15.4 The conditional two-dimensional limit law

Definition 15.12 (Two-dimensional local-relaxation clauses and the charge memory)**.** Fix a unit vector Ψ, a declared disk exhaustion $W_m = B_{R_m}(z)$ of the plane, or an annular exhaustion of a separately declared exterior domain, and the operators (320). Infinite-volume dynamics is formed first, fixed-window time limits second, and the spatial exhaustion last. The three clauses are as follows.

(LR1$_{\mathrm{2D}}$) Common Cesàro subsequence. There is one sequence $T_n \to \infty$ such that, for every fixed admissible window, the two Cesàro states $\omega^{\pm}_{W,n}$ formed by averaging $\langle \Psi, \alpha_t(A)\Psi\rangle$ over $[T_n, 2T_n]$ and $[-2T_n, -T_n]$ converge weak-$*$, and every double-Cesàro average $p_{W,n}(\nu)$ of (321) converges. By the general common-subsequence theorem of Section 8 this clause is itself a theorem — the argument consumes only separability of the quasi-local algebra, strong continuity of the dynamics, and finiteness of each local spectrum, none of which sees dimension — but it is retained as the binder that supplies one sequence for all clauses.

(LR2$_{\mathrm{2D}}$) First-moment nondemolition. With $\mathcal{D}_{W,t_-}(A) = \sum_q E_{W,t_-}(\{q\}) A E_{W,t_-}(\{q\})$, the double-Cesàro average of $\langle \Psi, [\mathcal{D}_{W,t_-}(\hat{Q}_W(t_+)) - \hat{Q}_W(t_+)]\Psi\rangle$ tends to zero for every fixed window along that same sequence. This is a scalar first-moment condition, not operator asymptotic commutativity.

(LR3$_{\mathrm{2D}}$) First-moment tightness on the declared exhaustion. Writing p_W for the fixed-window time limit,

$$\lim_{M\to\infty} \sup_m \sum_{|\nu|>M} (1+|\nu|)\, p_{W_m}(\nu) = 0. \tag{329}$$

One may optionally require p_{W_m} to converge weakly to a probability; that clause buys uniqueness of the limiting law and of its first moment, and is not needed for integer support.

The memory observable supplied by this definition is the *charge memory* $M_Q(p) := \sum_{\nu\in\mathbb{Z}} \nu\, p(\nu)$, defined per subsequence, or uniquely under the optional clause. No universal scalar displacement is defined.

campaign id:	M-INDEX-2D-spec, D27
proved in:	`theory/memory-index-2d.md` steps (1.7)–(1.8)
tested by:	`theory/checks/memory_index_2d_check.py` (finite arithmetic only; it proves nothing about (LR2$_{\mathrm{2D}}$)–(LR3$_{\mathrm{2D}}$))
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W3)

Theorem 15.13 (Conditional two-dimensional charge ledger)**.** [SKETCH] *Assume the integral-circle condition* (318) *together with (LR1$_{\mathrm{2D}}$)–(LR3$_{\mathrm{2D}}$) of Definition 15.12. Then every ordered spatial limit point p of the window laws is a probability supported on $\mathbb{Z}$, and along its subsequence its first moment obeys the charge ledger*

$$M_Q(p) = \lim_j \Big[\omega^{-}_{W_{m_j}}(\hat{Q}_{W_{m_j}}) - \omega^{+}_{W_{m_j}}(\hat{Q}_{W_{m_j}})\Big]. \tag{330}$$

Under the optional full-law convergence clause, the limit law and the value in (330) *are unique.*

Scope 15.14. The conditional implication above is complete and was independently verified; what is missing is an instance of its hypotheses. Clauses (LR2$_{\mathrm{2D}}$) and (LR3$_{\mathrm{2D}}$) are *assumed, not derived*, and are not implied by symmetry or by a Lieb–Robinson bound; the boundary terms they must control are perimeter-sized by (323). No scalar interface displacement, channel completeness, unconditional Goldstone instance, 1-form-symmetry result, or topological-order result is claimed.

The status is capped at sketch by the campaign's non-vacuity test rather than by any defect in the argument: no nonzero model instance of ($\mathrm{LR2}_{2\mathrm{D}}$)–($\mathrm{LR3}_{2\mathrm{D}}$) has been exhibited. In particular the exact 9×9 diagonalisation described below is *not* such an instance, and must not be presented as one. Promotion requires a model in which both clauses are proved and the limiting law is not the point mass at zero.

What is proved, and what is missing. The proved part is the implication. At finite times, expanding the two measured values identifies the finite-window mean as the pinched difference

$$\sum_{\nu} \nu\, p_{W;t_-,t_+}(\nu) = \langle \hat{Q}_W(t_-) \rangle - \langle \mathcal{D}_{W,t_-}\big(\hat{Q}_W(t_+)\big) \rangle, \tag{331}$$

an identity in which no geometry enters. Double-Cesàro averaging (331) and applying ($\mathrm{LR2}_{2\mathrm{D}}$) removes the dephasing defect and converts the mean into the difference of the two Cesàro states at each fixed window. Clause ($\mathrm{LR3}_{2\mathrm{D}}$) supplies tightness and uniform integrability, so Prokhorov extraction on the closed set $\mathbb{Z}$ yields a further weakly convergent subsequence with no loss of mass and with the first moment passing to the limit; integer support survives because every finite-window law already has it, the possibly drifting offset κ_W having been cancelled at each fixed window by Theorem 15.5 before the spatial limit is taken.

The missing part is any verified instance. The relevant numerical probe is an exact small diagonalisation of the spin-$\frac{1}{2}$ nearest-neighbour ferromagnet on a 9×9 open square, whose fully polarised state is an exact ground state and whose one-spin-flip sector is an exact magnon sector with the quadratic Goldstone dispersion $\varepsilon(k) = J(2 - \cos k_x - \cos k_y)$; the measured charge is the magnon number, with on-site spectrum $\{0, 1\}$. It confirms the finite theorem convincingly — on a 13-site disk the law is $p(0) = 0.176188$, $p(1) = 0.823812$, matching the integrated outward current to a residue of 9×10^{-16}; on a 24-site annulus, initially empty so that magnon entry gives a negative increment, $p(-1) = 0.691271$ with residue 5×10^{-15}; and a seeded state straddling the disk boundary, measured at two times with a common irrational offset $\sqrt{3}/11$ added to both readouts, reproduces exact integer support with the offset cancelling as predicted. All of that is finite-volume and finite-time. It establishes no infinite-volume time limit and no bound of the form (329), and the finite gap of the 9×9 system is not evidence for a thermodynamic gap. Those runs are described further in Section 17.

What replaces the kink picture. In one dimension the escaped charge converts into a wall displacement; in two dimensions it does not, and the honest replacement is spin-charge radiation through a closed curve. A local disturbance launches Goldstone wave packets, and the disk record counts their net selected charge across the circumference, an annulus recording the signed balance across its two boundary components. Angular profiles, multipoles, and neutral pairs are invisible to a single total-charge record. For a zero-temperature antiferromagnet the conserved observable is a selected *uniform* spin generator; the staggered magnetisation is an order parameter, not the conserved charge, and must not be substituted into the theorem. Finite-window integrality remains exact in every one of these cases, while the ordered limit remains open.

campaign id:	M-INDEX-2D-spec, M-INDEX-2D-fin, LR1-GEN
proved in:	`theory/memory-index-2d.md` steps (1.8)–(1.11), with the conditional theorem at step (1.9)
tested by:	`theory/checks/memory_index_2d_check.py` (finite arithmetic only; its green run proves nothing about ($\mathrm{LR2}_{2\mathrm{D}}$)–($\mathrm{LR3}_{2\mathrm{D}}$) and there is no gate for the limit law)
reviewed in:	`theory/verdicts/w21-joint-critic.md` §(W3): the implication is correct and the row is held at sketch by the non-vacuity test

16 Negative results: refutations, obstructions, and dead ends that shaped the campaign

The campaign's constitution treats a sharp refutation, delivered together with the weaker statement that survives it, as a result in its own right: a pile of suggestive analogies is worth less than one clean demonstration that a tempting statement is false. That rule has been applied often enough that the negative outcomes now form a body of knowledge of their own, and this chapter collects them in one place so that no reader — and no later session — walks a dead end twice.

Two kinds of material appear here. The first is a ledger of every withdrawn claim: the statement as it was once made, the mechanism that killed it, and what remains true. Each of those claims belongs to a positive chapter, where it is presented in full alongside the repaired result it forced; the entries below are two-to-four-sentence summaries with cross-references, so that this chapter can be read on its own as the campaign's "what is false" list. The second is the set of negative results owned by this chapter, which have no positive home: an obstruction theorem about the soft-leg amputation constant, a fully mapped and closed search for an equivariant-localisation mechanism, and a no-go fence separating what a fusion category can and cannot fix.

One procedural point must be stated once, because it governs the provenance blocks throughout. Under the campaign's process rules a negative result earns no adversarial review rounds: it is recorded, its checker (if any) is run, and the line is closed. Consequently the "reviewed in" field of every statement below is empty, and none of these statements carries a [PROVED]tag. The absence of a review artifact is deliberate and is not a defect; where a verdict file does exist for a neighbouring positive row it is named, and nowhere else.

16.1 The refutation ledger

The vacuum-pair orbit label. [REFUTED] The first form of the asymptotic-symmetry orbit statement asserted that the set of vacuum pairs is the orbit $\mathcal{A} = (G_L \times G_R)/G_{\text{diag}}$, labelled by the class $[g_L g_R^{-1}]$. This is false: $G_L \times G_R$ acts componentwise on $\Omega_{\text{vac}} \times \Omega_{\text{vac}}$, so the stabiliser of a pair (α', β') is $H_{\alpha'} \times H_{\beta'}$ and the space is $(G/H_\alpha) \times (G/H_\alpha)$, not a quotient of $G \times G$ by the diagonal; the proposed label is not invariant under the global diagonal symmetry and was withdrawn. What survives is stronger and cleaner: modulo the diagonal symmetry the complete invariant of a vacuum pair is the double coset $\mathfrak{d}(\alpha_L, \alpha_R) = H_\alpha\, g_L^{-1} g_R\, H_\alpha \in H_\alpha \backslash G / H_\alpha$, which for $G = SU(2)$, $H_\alpha = U(1)$ is the relative angle $\cos\theta = \hat{n}_L \cdot \hat{n}_R$; the classification additionally carries the transitivity hypothesis explicitly, since covariance of the vacuum family does not imply it. See Section 5.

campaign id:	A2-orbit-r1 (refuted); repaired statement A2(e)
proved in:	disproved in `theory/corner-a-kinks.md` step (1.9).(2.5).(3.3)
tested by:	`theory/checks/corner_a_check.py` C7
reviewed in:	—

The universal soft factor and its Adler zero. [REFUTED] The first form of the Goldstone-tensor corollary called the kinematic factor $e^{ik} - 1$ a *universal* soft factor, and inferred from it both an Adler zero and the hard-independent linear coefficient of the two-magnon oracles. Both inferences are withdrawn. In a matrix element the operator identity says only

$$\langle \text{out}|[H, Q_k]|\text{in}\rangle = (e^{ik} - 1)\, \langle \text{out}|J_k|\text{in}\rangle, \tag{332}$$

so all dependence on the Hamiltonian, on the symmetry direction, on the vacuum and on the hard legs sits inside $\langle \text{out}|J_k|\text{in}\rangle$: if that matrix element is $C_{\text{hard}} + O(k)$ then hard data enter at order

k, and if it carries a $1/k$ infrared singularity there is no zero at all. The factor is also convention-dependent — the kernel e^{-ikx} gives $e^{-ik} - 1$, and reversing the current's orientation flips it — so its independence of H is the tautology that a discrete difference was factored out *after* an H-dependent current had been defined. The exact finite-window summation-by-parts identity survives untouched, and what the argument does rederive is the one-magnon oracle only. See Section 5.

campaign id: G0-soft-r1 (refuted); surviving statement G0(c)–(e)
proved in: withdrawn in `theory/corner-a-goldstone.md` step (1.6).(2.7)
tested by: —
reviewed in: —

The superseded two-body label. [REFUTED] This entry is bookkeeping rather than mathematics, and is listed only because the claims DAG records it with the same tag as a genuine falsehood. The old label for the minimal two-magnon core was withdrawn so that a single mathematical statement could not carry two statuses at once; the content it named is alive and proved under its current label, and is generalised to every spin by the exact regular-channel ratio. Nothing was retracted here except a name. See Section 13.

campaign id: S2 (withdrawn label); live content S2-2body, S2-2body-S
proved in: —
tested by: —
reviewed in: —

Unrestricted soft universality. [REFUTED] The claim that the soft factorisation holds for every local or quasi-local source is false, and the counterexample is explicit and four sites wide. With

$$D := S_0^- S_1^- - S_1^- S_2^- + S_2^- S_3^- - S_0^- S_3^-, \qquad O_\eta := S_0^- + \eta D, \tag{333}$$

every O_η is local; because $D|\Omega\rangle$ lies in the two-magnon sector it leaves the one-magnon amplitude $M_1^{O_\eta}(h)$ completely independent of η, while the two-magnon amplitude acquires the first jet

$$M_2^{O_\eta}(k, h) = \eta\big\{2i\,(1 - e^{-3ih})\,k + O_I(k^2)\big\}. \tag{334}$$

On any hard packet supported where $1 - e^{-3ih} \neq 0$ the parameter η therefore moves the linear soft coefficient without moving the hard process at all, so no process-independent coefficient can exist on the unrestricted local class. The surviving statements are the repaired criterion — process independence holds exactly when the source's amputated commutator has vanishing first jet — and the conditional factorisation on the restricted class $\mathcal{S}_W(\rho)$, whose conclusion fixes a *shape* and not a number. See Section 13.

campaign id: ML5 (refuted); surviving statements ML5-A, ML5-B
proved in: `theory/ml5-universality.md` step (1.4)
tested by: `theory/checks/ml4_check.py`, the O_η obstruction
reviewed in: —

Memory as the direct-current limit of the soft factor. [REFUTED] The brief's memory conjecture, read literally — the wall displacement equals the zero-frequency limit of the soft factor, summed over the event — is false in the easy-axis chain. Two independent reasons kill it: conditional charge bookkeeping fixes the same displacement coefficient for every momentum, every anisotropy and every packet once the scattering channels exist, whereas any soft-factor expression varies with all three; and the displacement is entirely insensitive to the transmission phase, since a purely transmitting wall with vanishing phase still displaces by exactly $-1/s$. The soft factor is a phase, the memory quantum is a charge. What survives is the exact flux statement — the displacement is the finite-time direct-current weight of the *physical* boundary current — together with the conditional quantisation statement, into which soft data enter only through the transmission probability $T(k)$; no reading of virtual or bond data is claimed. See Section 7.

campaign id:	M (refuted); surviving statements M-flux, M-quant (conditional)
proved in:	`theory/memory-quantization.md` §1; `theory/corner-b-draft.md` §§2, 6, 10
tested by:	`theory/checks/mquant_check.py` (flux residues plus an empirical scan only)
reviewed in:	—

Pointwise blindness of closed adjoint contractions. [REFUTED] An early symmetry-protected-phase argument claimed that closed contractions built only from the adjoint action $\mathrm{Ad}(V)$ of the virtual symmetry are pointwise blind to the cohomology class $[\omega]$. They are not. For the two-element group with Pauli-projective virtual action the adjoint representation decomposes into four distinct characters, with multiplicities $(1,1,1,1)$, whereas the scalar-trivial lift gives four copies of the trivial character, $(4,0,0,0)$; the closed scalar $\mathrm{Tr}\,\mathrm{Ad}(V(R_x))$ is then 0 in the first case and 4 in the second. What survives is the exact multiplier cancellation for ordered closed insertions, where the endpoint action is $V(g) \otimes \bar{V}(g)$ and the projective phases cancel identically, together with the continuity statement for normalised coefficients — which becomes topological only after a separate local-constancy proof, since continuity does not imply local constancy. See Section 14.

campaign id:	SPT-B-r1 (refuted); surviving statements SPT-B-mult, SPT-B'
proved in:	disproved in `theory/spt-rebuild.md` step (1.3).(2.3)
tested by:	`theory/checks/spt_rebuild_check.py` S-C2
reviewed in:	—

The all-orders cohomology no-go. [REFUTED] The companion early claim was that the cohomology class can appear in no coefficient at all, an edge residue included. This over-read Whitehead's lemma, which trivialises only the *infinitesimal* central cocycle of a compact semisimple Lie algebra and only in a stated phase section; it never erases the dimension, the weights, the Casimir data or the integrability class of a representation, so global edge weights and module dimensions survive. What remains true is the narrow version: in the registered endpoint the infinitesimal cocycle is a coboundary and is gauged away in the stated phase convention, while the edge residue itself is the centred, phase-gauge-invariant charge whose spectrum lies in a fixed coset of $\mathbb{Z}$ and whose irreducible module has dimension at least d_ω. See Section 14.

campaign id:	SPT-nogo (refuted); surviving statements SPT-D', SPT-E'
proved in:	`theory/spt-rebuild.md` step (1.4)
tested by:	`theory/checks/spt_rebuild_check.py` S-C4
reviewed in:	—

A sector-wide regularised total charge. [REFUTED] It was claimed that the kink hypotheses together with integrality force the existence of a self-adjoint regularised total charge operator in every kink representation. They do not. The decisive counterexample is a product state in a kink sector whose on-site deviations decay only polynomially: with $\varepsilon_n = (n+1)^{-1/2}$ the accumulated variance $V_N = \sum_{n \le N} \varepsilon_n^2$ diverges, the window-charge distribution spreads with $\sup_k P(L_N = k) \le C V_N^{-1/2}$, and hence $\langle \Omega_\varrho, (\hat{Q}_{W_N,c_0} - i)^{-1} \Omega_\varrho \rangle \to 0$, which no self-adjoint limit can produce, since for any self-adjoint $\hat{Q}$ the imaginary part of that resolvent expectation is strictly positive. The mean-controlling hypothesis governs first moments, not charge fluctuations, and hence not implementability. A second, on-folium mechanism reaches the same conclusion for a nonscalar virtual circle action, but it is a strengthening under strictly more hypotheses, not an independent proof of the same statement.

What survives is that the relative window charge exists as a *limit vector* on states with exponentially clustering tails, where it coincides with the exactly conserved first-moment wall coordinate; no sector-wide operator is constructed anywhere in the corpus. See Section 8.

campaign id:	M-INDEX-LA-strong (refuted); surviving statements LD-ID, ACE-LD-eps
proved in:	`theory/memory-index.md` step (1.3) (fluctuation counterexample, off-folium; alone sufficient) and step (1.12) (nonscalar virtual circle, on-folium; a strengthening under additional hypotheses)
tested by:	`theory/checks/memory_index_check.py` IDX-C3, IDX-C8
reviewed in:	—

Remark 16.1 (Partial refutations that live inside proved rows). Three further negative findings are not listed above because they are corrections *within* statements that are otherwise proved, and are presented with those statements rather than here: the two-magnon Ward projection erratum, where a scalar-normalised projection formula holds at one magnon only and its register-dependent correction is non-scalar at two or more (Section 11); the withdrawal of the raw half-line displacement formula in favour of the leg-subtracted event identity $2s\,\delta x + (q_{\mathrm{out}} - q_{\mathrm{in}}) = 0$ (Section 7); and the refutation of the fully polarised product-state form of the tail vacua under the general kink hypotheses, which survives only when the tail density is a simple on-site eigenvalue (Section 9). Each is recorded as an erratum next to the theorem it scopes.

16.2 The amputation obstruction: the class fixes only a square root

The remainder of this chapter presents the negatives that have no positive home. The first is the sharpest of them, because it names precisely what a future proof must supply.

Definition 16.2 (Soft-leg amputation constant)**.** Work at a fixed tail density $\rho > 0$, put $Z_\rho := 2\rho$, and assume the restricted source class $\mathcal{S}_W(\rho)$ is nonempty and the asymptotic one-magnon kernel exists. Fix the normalisation convention in which the hard leg is amputated identically in the one-magnon amplitude M_1^O and the two-magnon amplitude M_2^O, while the extra soft leg of M_2^O is a unit-weight, delta-normalised asymptotic magnon. Decompose M_2^O into a descendant term E^O_{desc}, an orthogonal-current term and a direct/contact term, the last two bounded by $O(k^2)$; fix the descendant current residue in the charge-created normalisation to $2iv_h M_1^O(h)$; and for $k \neq 0$ define the descendant quotient

$$L(k,h) := \frac{E^O_{\mathrm{desc}}(k,h)}{(e^{ik}-1)\,2iv_h\,M_1^O(h)}, \tag{335}$$

assumed to have a process-independent, uniformly C^1 extension to $k = 0$. The *soft-leg amputation constant* $\mathfrak{a}_{\mathrm{leg}}(\rho) \neq 0$ is then defined by the value of that extension,

$$L(0,h) = \mathfrak{a}_{\mathrm{leg}}(\rho)\,\frac{-i\chi(h,0)}{v_h}, \tag{336}$$

with χ the channel sign. In this convention the conditional factorisation theorem reads $M_2^O(k,h) = 2i\,\mathfrak{a}_{\mathrm{leg}}(\rho)\,\chi\,k\,M_1^O(h) + O_{L^2(I)}(k^2)$, which contains no number at any density.

campaign id:	D24(b), D24(d)1–3, D24(d)3b; consumers ML5-B, D24-VAL, AMP
proved in:	`definitions.md` D24; `theory/amp.md` §1
tested by:	`theory/checks/d24d3_normalization_check.py` D24N-C1–C3 (kinematics of the conclusion, not its constant)
reviewed in:	`theory/verdicts/d24d3-adjudication-r5.md` §5 is the merged text of record for the clause repair

The natural hope was that this constant is fixed by the definition of the universality class, with the physically expected value $\mathfrak{a}_{\mathrm{leg}}(\rho) = 1/Z_\rho$ following from a careful accounting of how a charge-created soft leg is converted into an asymptotic one. That hope is refuted at the level of derivability.

Refuted claim 16.3 (The universality class fixes the amputation constant)**.** *Withdrawn statement: the normalisation conventions and analytic relations defining the class $\mathcal{S}_W(\rho)$ determine the value of $\mathfrak{a}_{\mathrm{leg}}(\rho)$, and in particular force $\mathfrak{a}_{\mathrm{leg}}(\rho) = 1/Z_\rho$. This is false. Those relations determine $\mathfrak{a}_{\mathrm{leg}}(\rho)$ only up to the constraint $\mathfrak{a}_{\mathrm{leg}}(\rho) \neq 0$ together with process independence; on the charge-created reading they supply exactly one factor $Z_\rho^{-1/2}$, and the remaining scalar is free. The statement is neither proved nor physically refuted — it is undecidable from the present hypotheses, and a completed proof requires a genuinely new dynamical input, displayed in* (342) *below.*

Mechanism. Three independent steps close the question. The first is an exact leg-conversion computation. On a fully polarised tail with $Z_\rho = 2S$, the charge-created soft leg on the vacuum and the descendant vector $u_N := Q_q^- |h\rangle$ built on a hard magnon at a commensurate momentum satisfy, on a finite ring,

$$Q_k^- |\Omega\rangle = \sqrt{Z_\rho}\, |k\rangle, \qquad \|u_N\|^2 = Z_\rho N - 2. \tag{337}$$

The rank-one orthogonal projection onto the descendant line must then be read in the normalised register:

$$P_N w_N = u_N \frac{\langle u_N, w_N\rangle}{Z_\rho N - 2} = \frac{u_N}{\sqrt{Z_\rho N - 2}}\, \frac{\langle u_N, w_N\rangle}{\sqrt{Z_\rho N - 2}}. \tag{338}$$

The first square root normalises the *vector*; only the second is the amplitude coefficient against that normalised vector. Counting both as amplitude factors changes register mid-equation, and that mutation is registered as an explicit failing mode of the checker. Removing the common finite-ring delta-normalisation leaves the exact descendant conversion factor $(Z_\rho - 2/N)^{-1/2}$, so that

$$\lim_{N\to\infty} \big(Z_\rho - 2/N\big)^{-1/2} = Z_\rho^{-1/2}, \qquad \big(Z_\rho - 2/N\big)^{-1/2} = Z_\rho^{-1/2}\Big[1 + \frac{1}{Z_\rho N} + O(N^{-2})\Big]. \tag{339}$$

The finite-size mismatch is therefore a vanishing correction, not a surviving second factor.

The second step checks the four operations that might have supplied a second factor, and finds that none does. The descendant projection is already exhausted by (338). Windowed-charge smearing is linear and supplies no density-dependent scalar, since the packet register fixes delta-normalised kernels and soft envelopes with value 1 at $k = 0$; a constant envelope $Z_\rho^{-1/2}$ would change that value and would be a forbidden second leg rescaling rather than a new mechanism. The two normalisation anchors remove a rescaling freedom and impose the Ward residue as a membership condition, but neither evaluates $L(0, h)$. And the finite-ring mismatch vanishes, as just displayed. The one remaining place where a new mechanism could live is the descendant propagation itself, which the class definition asserts to exist and to have a C^1 quotient without ever specifying a map whose normalised residue could be evaluated.

The third step makes the underdetermination explicit at the level of the relations. Fix a compact hard window on which $v_h \neq 0$ and the channel sign is constant, choose a nonzero profile $m \in L^2(I)$, and for any $a \in \mathbb{C} \setminus \{0\}$ consider the source-linear package

$$\begin{aligned} &M_1(h) = c\, m(h), && R(h) = c\, 2iv_h\, m(h), \quad L_a(k,h) = a\,\frac{-i\chi}{v_h}, \\ &E_{\mathrm{desc},a}(k,h) = (e^{ik} - 1)\, L_a(k,h)\, R(h), && M_{2,a} = E_{\mathrm{desc},a}, \qquad\qquad \text{other terms } = 0. \end{aligned} \tag{340}$$

Every clause of the class definition holds exactly: the decomposition is trivial, the remainder bounds hold with zero remainder, the Ward residue is $R = 2iv_h M_1$, and the descendant quotient is exactly L_a, with $iv_h L_a(0,h)/\chi = a$. Two distinct values of a leave M_1, the Ward residue, the packet measure and the leg convention unchanged while changing L and M_2, so they are not related by any normalisation covariance. Hence every nonzero constant is admitted by the displayed relations.

What a completed amputation proof must supply. Granting the charge-created reading of the descendant term, factor the class datum as

$$\mathfrak{a}_{\mathrm{leg}}(\rho) = Z_\rho^{-1/2}\left[\sqrt{Z_\rho}\,\mathfrak{a}_{\mathrm{leg}}(\rho)\right], \tag{341}$$

where the leg conversion (337) fixes the first factor and the class definition constrains the bracket only to be nonzero and process-independent. The desired value is therefore equivalent to the single additional equation

$$\sqrt{Z_\rho}\,\mathfrak{a}_{\mathrm{leg}}(\rho) = Z_\rho^{-1/2}. \tag{342}$$

Any proof of (342) must be a post-conversion descendant-propagation or LSZ residue theorem, stated after both sides are already expressed against the same delta-normalised soft leg; it is not a leg normalisation, and no clause of the present definition supplies it. Every candidate argument must be checked against the exact leg-conversion computation (337), or it double-counts the leg.

Scope 16.4 (What is and is not decided). The obstruction is at relation level: it exhibits packages satisfying the displayed algebraic and analytic relations for every nonzero constant, and it does not construct a microscopic local source, so it is not a physical counterexample and does not refute the value $1/Z_\rho$ in a future nonempty class. A separate, strictly conditional falsifier does bear on the leg-normalisation-only value: assuming both the charge-created reading of the descendant term and the jet-identification bridge that equates the class multiplier's jet with the two-body physical phase jet, a pure-leg proof predicts the jet $2/\sqrt{2S}$, whose scaled deviations against the ansatz-free two-magnon data are 0.0004, 0.4158, 0.7347 and 1.0033 at $S = 1/2, 1, 3/2, 2$, i.e. margins of 5.2, 9.2 and 12.5 times the pre-registered band of 0.08 at the three non-degenerate spins. Neither antecedent is proved, and without them a two-magnon phase slope and an amputation constant are not comparable at all; the relation-level obstruction above uses neither. Finally, both candidate value rows remain vacuous or unknown in a further sense: nonemptiness of the class is open at every density, so they constrain future work — any member ever exhibited must carry the stated value — rather than supplying evidence now.

campaign id:	AMP [CONJECTURE]; the obstruction is recorded as the proposed row AMP-OBS; inputs D24(b), D24(d)
proved in:	`theory/amp.md` steps (1.1)–(1.4); the honest disposition is step (1.4).(2.2)
tested by:	`theory/checks/amp_check.py` AMP-C1–C3 with the registered failing modes `--red-double-count` and `--red-pure-leg`; AMP-C4 is the conditional falsifier only; `theory/checks/d24d3_normalization_check.py` D24N-C8 (leg-conversion constant), D24N-C6 `--red-halfpower`
reviewed in:	— (no review round: negative results earn none by design; the claims DAG keeps AMP at conjecture)

16.3 The Duistermaat–Heckman hunt, fully mapped

The second negative is a closed search. Its target was attractive enough to be worth stating precisely: the memory law produced by the two-time measurement protocol is an atomic probability measure on $\mathbb{Z}$, and exact integrality statements of that kind are exactly what equivariant localisation delivers in symplectic geometry. If the Duistermaat–Heckman theorem — exactness of stationary phase for a Hamiltonian torus action, equivalently the statement that the pushforward of Liouville measure under the moment map is piecewise polynomial — could be applied to the outcome law, the campaign would gain both the support and the weights from geometry alone. It cannot, and this subsection records why in enough detail that the route need not be reopened.

Definition 16.5 (The two objects that were to be identified)**.** The *Duistermaat–Heckman law* of a compact Hamiltonian T-manifold (M, ϖ, μ) is the pushforward $\mu_*(\mathrm{vol}_\varpi)$ of Liouville measure by the moment map: it depends on the manifold, its symplectic form, the torus action and the moment map, and on nothing else. The *memory outcome law* is $p_W = \sum_{\nu \in \mathbb{Z}} p_W(\nu)\,\delta_\nu$, where each weight is a matrix element of a product of two spectral projections of the same windowed charge at two different times, followed by an ordered window and Cesàro limit; it depends on the state, on the dynamics, on the back-action of the first measurement, and on the limit order.

Refuted claim 16.6 (The memory outcome law is a Duistermaat–Heckman measure)**.** *Withdrawn statement: the escaped-charge law of the memory protocol is the Duistermaat–Heckman measure of a natural Hamiltonian torus space, so that its integer support and its weights follow from equivariant localisation. No candidate identification survives. The two objects have different types; the one construction that produces atoms does so only by inserting the desired atoms and their volumes by hand; and every route that reaches the soft coefficient consumes the representation-theoretic or scattering data it was meant to produce.*

Mechanism, candidate by candidate. Six natural routes were worked to their stopping points.

route	why it stops
the memory law as a localisation measure	a Duistermaat–Heckman measure on a positive-dimensional Hamiltonian torus manifold is a continuous, piecewise-polynomial pushforward, while the outcome law is atomic and state- and dynamics-dependent
weight-multiplicity asymptotics	the projective law takes the weight multiplicities as its Dirichlet parameters, so it consumes them rather than producing them
localisation for the soft coefficient	the internal circle acts on a one-magnon band by a scalar character, so the whole band is fixed and $k_s = 0$ is not an isolated fixed locus
wall crossing as phase rigidity	localisation walls are critical moment values inside one fixed space, whereas the phase-diagram sweep changes the Hamiltonian, the tensor and the symplectic data themselves
convexity and integral vertices	integral vertices do not make interior moment values integral: already on the one-qubit sphere the moment map fills $[-\frac{1}{2}, \frac{1}{2}]$
the kink Kähler cylinder	in two dimensions the moment-map identity alone gives the answer, so localisation performs no inference

The second route deserves its exact computation, because it kills the premise cleanly. Take two spin-$\frac{1}{2}$ sites, $V = (\mathbb{C}^2)^{\otimes 2}$, with circle generator $Q = S_1^z + S_2^z$, so that $V = V_{-1} \oplus V_0 \oplus V_{+1}$ with dimensions $(1, 2, 1)$; let $M = \mathbb{P}(V) = \mathbb{CP}^3$ carry normalised Fubini–Study measure and the moment map $\mu([v]) = \langle v, Qv\rangle / \langle v, v\rangle$. The normalised weight-counting law and the localisation law are then

$$\eta_{\mathrm{wt}} = \tfrac{1}{4}\delta_{-1} + \tfrac{1}{2}\delta_0 + \tfrac{1}{4}\delta_{+1}, \qquad \mathrm{d}\eta_{\mathrm{DH}}(x) = \tfrac{3}{2}\,(1 - |x|)^2\,\mathbf{1}_{[-1,1]}(x)\,\mathrm{d}x, \tag{343}$$

the second obtained by writing the block fractions of a Fubini–Study random ray as a Dirichlet vector with parameters $(1, 2, 1)$ and integrating out the neutral block. They are not the same measure, and not merely by normalisation: the first is supported on three points and the second is absolutely continuous, and their second moments are $\frac{1}{2}$ and $\frac{1}{10}$ respectively. The general statement is the same in every torus representation $V = \bigoplus_\lambda V_\lambda$: the block fractions (p_λ) are Dirichlet with parameters $\dim V_\lambda$ and $\mu_T = \sum_\lambda p_\lambda \lambda$, so the density is a spline whose inputs already include every multiplicity.

The third route fails for a structural reason worth stating on its own. The internal circle acts on a one-magnon state by its charge character, which is projectively trivial, so the entire one-magnon band is fixed rather than a point; on a finite ring the translation symmetry is the cyclic group and has no infinitesimal generator; and in infinite volume the Brillouin circle labels irreducible translation characters and is not a Hamiltonian circle action on a compact phase space. The two meanings of "stationary phase" also fail to meet: localisation needs a phase that is a moment map, while the scattering estimates use $t\omega(k) - kx$ plus an on-shell scattering phase, and no definition in the corpus turns a dispersion relation into an internal-charge moment map. Finally, the soft coefficient is not a symmetry-only integral at all: multiplying the Hamiltonian by a constant leaves the projective symmetry geometry unchanged while rescaling the current-velocity residue $2iv_S(h)$.

The sixth route is instructive because it *succeeds* and thereby shows what success would be worth. On a two-dimensional Hamiltonian circle manifold with coordinates (x_0, φ), invariant form $\varpi = \Omega(x_0)\,\mathrm{d}x_0 \wedge \mathrm{d}\varphi$ and moment map $\mu(x_0)$, the defining identity $\mathrm{d}\mu = \iota_{\partial_\varphi}\varpi$ gives $\mu'(x_0) = -\Omega(x_0)$, hence a constant pushforward density and

$$\int_{x_0}^{x_0+1}\int_0^{2\pi} \varpi = 2\pi\big[\mu(x_0) - \mu(x_0+1)\big]. \tag{344}$$

With the calibration $|\Delta\mu| = 2s$ the area per lattice translation is $4\pi s$ and the position quantum is $1/(2s)$ — the right answer, obtained by integrating the defining identity and nothing more. That is geometric interpretation, not inference, and it was pre-registered as the reproduction-only failure mode before the hunt began.

Observation 16.7 (The coadjoint-orbit soft-phase envelope)**.** One genuinely new result did survive the hunt, on the phase rather than on the weights. Write the classical spin phase space as a product of coadjoint orbits $P_S = \prod_x \mathcal{O}_S$ with $\mathcal{O}_S \cong S^2$ and form $\Omega_S = S\sum_x \omega_{\mathrm{KKS},x}$, so that the effective semiclassical parameter is $1/S$. Put $p = (k_s + k_h)/2$, $q = (k_s - k_h)/2$, $c_p = \cos p$, $c_q = \cos q$, $s_q = \sin q$, and in the regular chamber $c_p s_q \neq 0$ set

$$F = \frac{c_q\,(c_p - c_q)}{c_p\, s_q}, \qquad d = -1 + \frac{c_q}{c_p}. \tag{345}$$

Then the exact two-magnon contact phase, taken on the continuous branch with $\delta_S(0, k_h) = 0$, is

$$\delta_S(k_s, k_h) = 2\arctan\!\Big(\frac{F}{2S + d}\Big) = \frac{F}{S} + \frac{G}{S^2} + \frac{H}{S^3} + O(S^{-4}), \qquad G = -\frac{Fd}{2}, \quad H = \frac{Fd^2}{4} - \frac{F^3}{12}, \tag{346}$$

with every function fixed before any comparison and no fitted coefficient anywhere. At $k_s = 0$ one has $c_p = c_q$, hence $F(0, k_h) = 0$, $\partial_{k_s}F(0, k_h) = 1$ and $d(0, k_h) = 0$, so $G = O(k_s^2)$ and

$$\partial_{k_s}\delta_S(0, k_h) = \frac{1}{S} : \tag{347}$$

the proved finite-spin soft slope is exactly the leading semiclassical phase, and the first subleading term cannot renormalise the linear soft coefficient.

Status and evidence. This is a speculative result about the large-spin envelope of the two-magnon phase, not a second proof of the finite-spin slope: (346) reparameterises the exact contact ratio that the proved two-body theorem already supplies, and its new content is that the hierarchy is visible and testable. Eight frozen soft-window rows at $S \in \{1/2, 1\}$ and $k_s \leq 0.12$, previously compared only with the full exact answer, agree with the truncation $F/S + G/S^2$ to within 2.62% in the worst row — the softest-momentum row at $S = 1/2$ — inside the declared 3% descriptive gate, and adding the parameter-free G/S^2 term reduces the error of every one of them by at least 89.8%. An independent exact-diagonalisation sequence on an eighteen-site ring, running through $S = 8$ and computed from finite-volume level displacements rather than from the closed-form contact ratio, lies inside the derived pointwise remainder bound at every spin, with second-order residuals decreasing monotonically and $\max_S S^3|\text{error}| = 0.0089$. Identifying G with any particular fluctuation determinant is speculative and is not needed; the checked statement is only that G/S^2 is the first subleading contact correction of exactly that size.

Scope 16.8 (No localisation law for the outcome probabilities)*.* The envelope result does not extend to the memory weights, and the negative verdict there is sharp. The one-magnon protocol holds the escaped charge at order one rather than letting a highest weight grow, so the usual semiclassical bridge has the wrong limit: dividing the escaped integer by the spin or by the window size collapses the law to a point mass at zero, while declining to divide leaves an atomic law rather than a continuous density. The frozen memory data confirm the split. Thirteen memory rows at $S \in \{1/2, 1, 3/2\}$ have displacement-per-unit-time within 2.69% of $-1/S$, well inside the file's declared band — but that shared $1/S$ is the kinematic conversion from one unit of transmitted charge to wall displacement, not a geometric volume; the transmission weights themselves vary with the anisotropy, the packet momentum, the packet quality and the trapped mass, and neither the envelope nor any localisation volume predicts them. The independent probe puts at least 0.99601 of its final weight on a single outcome with zero off-lattice mass, which tests exact support and late-time channel concentration, again not a density. There is a real asymptotic statement in the neighbourhood — the normalised weight multiplicities of the line-bundle family $H^0(M, \mathcal{O}(n))$ do converge to the density in (343) — but that is a normalised-trace multiplicity family, and substituting it for the measured law would change the experiment; inserting the transmission probability as an equivariant weight would insert the answer. The absence of one further artifact should be noted explicitly: the hunt shard deliberately has *no* checker, since verification machinery is not built for a terminal negative side-result, whereas the surviving envelope does carry one.

campaign id:	no claims-DAG row (negative lane record, plus a speculative companion-paper result)
proved in:	`theory/dh-hunt.md` §§1–5 (the six verdicts and the exact projective computation); `theory/dh-semiclassical.md` §§1–4 (the envelope, the error model and the data comparisons)
tested by:	`theory/checks/dh_semiclassical_check.py` DHSC-C0–C4 with failing modes `--red wrong-leading` and `--red drop-fluctuation`; no checker exists for `theory/dh-hunt.md` by design
reviewed in:	—

16.4 The no-categorical-soft-theorem fence

The third negative fixes the boundary between what a fusion category can determine and what it cannot. It matters because the two-dimensional lane's positive selection theorem is genuinely categorical, and it would be easy to over-read that success as a categorical soft theorem.

Definition 16.9 (Endpoint register and endpoint form factor)**.** Let $\mathcal{C}$ be a finite unitary fusion category represented by exact matrix-product operators, and let $\mathcal{M}$ be the chosen indecomposable module category recording the boundary or phase register. The category of topological endpoint charges on that register is the dual category $\mathcal{D}_{\mathcal{M}} := \mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{M})$; a statement assigning endpoint labels from $\mathcal{C}$ alone is not invariantly typed. Let P_a project onto the simple sector $a \in \mathrm{Irr}(\mathcal{D}_{\mathcal{M}})$, and call an endpoint operator *pure of type* x when its intertwiner is resolved into a single simple x. Writing $T_x(k; w)$ for a windowed, momentum-resolved endpoint family, its allowed block in a one-dimensional fusion space factorises as

$$F^{x}_{ba}(k; w) = f^{x}_{ba}(k; w)\, I^{x}_{ba}, \tag{348}$$

where the intertwiner I^{x}_{ba} is categorical while the scalar form factor f^{x}_{ba} depends on the endpoint tensor, the window, the Hamiltonian, the external states and the normalisation.

The selection half is a theorem: a pure endpoint of type x satisfies $P_b T_x P_a = 0$ unless the fusion multiplicity N^{b}_{xa} is positive, and a nonzero block carries a channel vector $\mu \in \mathrm{Hom}(b, x \otimes a)$, so the fine categorical datum is the resolved fusion event $(x, a \to b; \mu)$ rather than any difference of labels; when x is invertible the target is unique and the index lives in $\mathrm{Inv}(\mathcal{D}_{\mathcal{M}})$, while for general x the maximal group-valued shadow is the universal grading, $|b||a|^{-1} = |x|$. The Ising algebra shows why the coarse statement cannot be improved: σ has quantum dimension $\sqrt{2}$ and $\sigma \otimes \sigma = 1 \oplus \psi$ has two fine targets, so there is no canonical subtraction, yet σ still carries a definite odd universal degree. None of this fixes a number inside an allowed channel.

Refuted claim 16.10 (A fusion category fixes a soft coefficient)**.** *Withdrawn statement: the categorical data of an endpoint — its type, its source and target sectors, its fusion channel, its support geometry and the exact homogeneous pulling-through and intertwiner equations — determine the soft behaviour of its form factor, in particular an Adler zero or a universal linear coefficient. This is false, and sharply so: fix all of that data together with the value $T_x(0) \neq 0$ of a nonzero analytic family near $k = 0$; then no positive-order Taylor coefficient of the allowed form factor follows. Categorical data give selection rules, not derivatives.*

Mechanism. Sector projection and every pulling-through or intertwiner equation are linear and homogeneous in the endpoint tensor, so for any analytic scalar c with $c(0) = 1$ the family

$$\widetilde{T}_x(k) = c(k)\, T_x(k) \tag{349}$$

has the same endpoint type, the same sectors, the same channel, the same support, the same zero-momentum operator and the same homogeneous equations. Choosing $c(k) = e^{i\alpha k^n}$ preserves the norm and every derivative below order n while changing the n-th derivative by $i\alpha\, n!\, T_x(0)$, and α is arbitrary; imposing norm preservation therefore still leaves an arbitrary phase jet. Without fixing $T_x(0)$ the conclusion is even simpler: I^{x}_{ba} and $k\, I^{x}_{ba}$ obey the same homogeneous categorical equations in an allowed one-dimensional channel. Fusion data force an exact zero only in the *forbidden* blocks of the selection rule; in an allowed block they force neither a zero nor a nonzero constant. An exact stabiliser-code computation makes the freedom concrete rather than formal: on a single plaquette, dressing a string operator with a slowly varying scalar phase profile of wavelength $1/k$ leaves the endpoint syndrome, the fusion channel and the amplitude magnitude untouched while giving the amplitude the arbitrary first jet

$$\partial_k A_{\mathrm{ph}}(0) = i\lambda\, (s_0 + s_1 + 2c). \tag{350}$$

Why gapless input is the missing ingredient. The failure is not an accident of bookkeeping; it identifies precisely what ordinary soft theorems use and this setting lacks. A massive anyon stays separated from the ground register by a positive energy as $k \to 0$, so the limit here is a long-wavelength limit of the chosen source rather than an energetic soft-particle limit. What gives a soft theorem its force is a gapless on-shell state together with a Ward identity or factorisation statement, and neither a finite tube algebra nor a non-invertible pulling-through relation contains a current zero mode. A fusion category with a gap can therefore be expected to supply selection rules and no coefficient; supplying a coefficient requires naming a gapless edge Hamiltonian and its current algebra as new input, which is a separate work order and must not be advertised as determined by the bulk category.

Scope 16.11 (Strength of the no-go). This is a data-insufficiency theorem, not a theorem about every possible normalisation convention: a convention can remove the scalar freedom, but the convention is then extra kinematic data and still fixes no measurable magnitude response. The witness varies the momentum-dependent probe family and is not a pair of fully specified Hamiltonian or tensor-network models with one fixed normalised protocol. No claim is made that every gapped response starts at order k: a constant term, an arbitrary analytic correction, or a convention-dependent phase is compatible with the categorical data, and the conclusion is the absence of a category-universal coefficient, not a universal order of vanishing. Under the negative-result rule this fence is deliberately kept out of the claims DAG; it is carried instead inside the "not claimed" list of the conditional selection theorem it fences, whose statement asserts no Taylor coefficient inside an allowed channel.

Observation 16.12 (The fusion datum is categorical, the Ward operator is Lie)**.** The companion finding of the same hunt is a clean split of the campaign's machinery by what each half actually consumes. On the categorical side stand the sector or tube projector and its selection rule, the universal grading shadow, and — for an *invertible* on-site string, where locality permits the conjugation $U_R^\dagger H U_R - H$ to be written as a sum of two endpoint defects — the finite string endpoint identity. On the Lie side stand the Ward relation $D^\dagger J\psi = R\psi$ and the invertibility and scalar ladder value of the Gram operator $D^\dagger D$, neither of which follows from fusion data. The witness is elementary and exact: in a two-dimensional register the operators D, $J_1 = D$ and $J_3 = 3D$ all carry the same odd categorical covariance, yet their projected vectors differ by $1 : 3$, and replacing D by $2D$ changes the Gram value by $1 : 4$ without changing any category datum. A non-invertible string has no conjugation at all, so its pulling-through equation continues the string-endpoint half and not the current-zero-mode half. Accordingly the sector-label theorem is the categorical half of the campaign, and the normalised Gram-inverse current theorem is genuinely Lie-algebraic under the hypotheses presently known.

Deliberately unpromoted. Neither this observation nor the categorical selection theorem behind it is proposed for the Letter. The hunt's own honest assessment is that any corpus row derived from it would have to enter at sketch status, because the artifact has not undergone the campaign's capped hostile review — and by the negative-results rule it does not get one. The material is therefore parked as companion-paper content, alongside the semiclassical envelope of Section 16.3, pending an explicit decision by the campaign owner; the joint critic's disposition is the same, recording the fence as a scope statement and a lane record rather than a claim. What did travel from this lane into the claims DAG is only the conditional selection theorem on the two-dimensional register, which carries this fence in its own scope.

campaign id:	no claims-DAG row (negative lane record and unpromoted companion material); the fence is carried inside A-INDEX-PEPS's not-claimed list
proved in:	`theory/soft-2d-hunt.md` §3 (the no-go and its probe-family fence) and §4 (the exact stabiliser patch); `theory/cat-hunt.md` §§2–3, §5.3 (the operator/categorical split) and §9 (honest status)
tested by:	`theory/checks/cat_hunt_check.py` CATH-C0–C4 with failing mode `--red missing-ising-channel`; `theory/checks/soft_2d_hunt_check.py` S2DH gates
reviewed in:	— (the joint critic `theory/verdicts/w21-joint-critic.md` W6-O2 fenced the statement's strength and directed that it not be promoted; it adjudicated the neighbouring positive rows, not this negative one)

16.5 What the negatives have in common

Read together, the ledger and the three obstructions show one pattern, and it is worth naming because it predicts where the next dead end will be. Every failed universality in this campaign was a statement trying to obtain a *number* from symmetry alone. The withdrawn orbit label wanted a complete invariant from a group quotient, and got one only after the diagonal action was taken seriously and the answer turned out to be a double coset. The withdrawn soft factor wanted a hard-independent coefficient from a kinematic difference, and lost it as soon as one asked what sits inside the current matrix element. The unrestricted universality claim wanted one coefficient for every local source, and a four-site operator moved that coefficient while leaving the hard process fixed. The literal memory conjecture wanted a displacement from a phase, when the displacement is a charge. The amputation conjecture wants a value from a normalisation, and normalisation supplies exactly one square root of it. The localisation hunt wanted outcome probabilities from a volume, and volumes consume multiplicities rather than producing them. The categorical hunt wanted a Taylor coefficient from fusion, and fusion is blind to any scalar dressing that is trivial at zero momentum.

The surviving architecture is the complement of that list, and it is exactly what the positive chapters prove. Symmetry — group-theoretic, categorical, or cohomological — delivers structure: which sectors exist, which transitions are allowed, which observables are quantised, and in what coset their spectrum lies. Dynamics delivers values: the two-body contact equation gives the soft slope $1/S$, the current and channel velocity give the residue $2iv_h$, and the scattering problem gives the transmission weights. Every proved statement in this labbook respects that division of labour, and every refuted one violated it. The practical rule extracted from the negatives is therefore simple: when a proposed theorem promises a number, one should be able to point at the dynamical input that carries it — and if the only inputs are a group, a category, a class definition or a volume, the number is not there.

17 Numerics: models, algorithms, tests, and results

17.1 What the numerics are for

The numerical work in this campaign plays two roles and only two.

The first is that of an *oracle*. Several of the theorems in Section 10 and Section 7 are built on structural facts about a specific model — a two-magnon scattering phase, a magnon dispersion, the position of a phase boundary — which the proofs consume as inputs. Where such an input can be computed exactly, it is computed, and the computation is kept next to the proof that uses it.

The second is that of a *falsifier*. A conjecture in this campaign becomes interesting only once someone has written down, in advance, a number that it forbids. The spin-s battery of Section 17.6 is the clearest instance: before any production run, the two competing hypotheses were recorded (soft phase slope $1/s$ and memory ratio $-1/s$ if the conjecture survives; 2 and -2 if it does not), together with a decision band of eight per cent. The runs were then executed once and the verdict read off. Nothing was retuned afterwards.

Two disciplines are enforced throughout.

Red–green development. Every checked quantity has a test that was written first and observed to fail first. Two of these failures are recorded in the source notes because they were instructive rather than clerical: the first ferromagnetic two-magnon estimator, a windowed centroid of the one-body density, was biased at the ten-per-cent level and returned the *wrong sign* for the hard-magnon displacement; and the first spin-s contact calculation symmetrised a wavefunction that is not symmetric, obtaining a spin-independent scattering length that would have falsified the conjecture for the wrong reason. In both cases the numerics caught the error before the algebra was redone, and in neither case was a tolerance moved to make a test pass. Tolerance constants are fixed in the test headers before the production runs and are quoted in this section wherever they carry weight.

Pre-registration and honest verdicts. A pass criterion that is chosen after seeing the data is not a criterion. Where the pre-registered band was missed, or where a diagnostic turned out to measure something other than what its name suggested, this is reported in Section 17.9 rather than smoothed over.

One boundary is absolute and is repeated here because it governs how every number below should be read: **no numerical result in this campaign promotes a claim.** A survived falsifier leaves a conjecture a conjecture. The one claim row that rests on numerics alone — the empirical XXZ memory scan — is held at [SKETCH] precisely because it is evidence and not a theorem, and the conjecture the spin-s battery failed to kill remains [CONJECTURE]. Section 18 states which corners of the triangle are carried by proof and which by evidence.

17.2 The codebase

The numerics live in a single Julia package, `TriangleMPS`, under `numerics/`. It splits into two implementation families that share conventions and estimators but nothing else.

The *exact-sector family* builds a Hamiltonian directly as a sparse matrix on an explicitly enumerated basis of a conserved sector, and evolves a state with a Krylov exponential. It uses no tensor-network library at all: sparse linear algebra, an eigen-solver and a matrix exponential are the whole toolkit. Every number it produces is exact up to a stated Krylov tolerance and a stated basis truncation.

The *uniform-MPS family* works in the thermodynamic limit and in windows cut out of it, and does use a tensor-network library (`TensorKit` for the graded tensors, `MPSKit` for the variational and time-evolution algorithms). It is the only part of the campaign where the accuracy is controlled by a bond dimension rather than by a basis size.

17.2.1 The three models

Three Hamiltonians carry all of the numerics.

(i) The easy-axis spin-$\frac{1}{2}$ XXZ ferromagnet with a frozen kink. On N sites,

$$H \;=\; -\sum_{x=1}^{N-1}\Big[\tfrac{J_\perp}{2}\big(S_x^+S_{x+1}^- + S_x^-S_{x+1}^+\big) + J_z\,S_x^zS_{x+1}^z\Big], \qquad \Delta := J_z/J_\perp > 1, \tag{351}$$

with $J_\perp = 1$ throughout, so that $J_z = \Delta$. Both terms are ferromagnetic, so the fully polarised states are exact ground states and z is the easy axis. Sites 1 and N carry *frozen* classical spins: they contribute their Ising bonds to their neighbours but cannot flip. With $\sigma_1 =\uparrow$, $\sigma_N =\downarrow$ this realises the one-kink boundary condition, and it makes the total S^z of the $L = N-2$ dynamical sites exactly conserved by construction rather than merely energetically. The magnon dispersion in this convention is

$$\omega(k) = J_z - J_\perp\cos k, \qquad v_g(k) = J_\perp\sin k, \tag{352}$$

gapped for $\Delta > 1$, and it is reproduced by the code against the exact open-chain eigenvalues $\omega_m = J_z - J_\perp\cos\big(\pi m/(L+1)\big)$ to 10^{-10}. The same construction is carried to general site spin s in the spin-s modules, with $S^z = +s$ frozen on the left end and $-s$ on the right.

(ii) The isotropic spin-S Heisenberg ferromagnet on a ring. With $n_x := S - S_x^z$ the on-site magnon number,

$$H_S \;=\; -J\sum_x\big(\mathbf{S}_x\cdot\mathbf{S}_{x+1} - S^2\big), \tag{353}$$

whose exact two-magnon form is a diagonal part $2JS\sum_x n_x - J\sum_{\text{bonds}} n_xn_{x+1}$ together with a hop amplitude $-\frac{J}{2}\sqrt{n_a(2S-n_a+1)}\sqrt{(n_b+1)(2S-n_b)}$ from site a to site b. The free hop is $-JS$; the hop that creates or destroys a doubly occupied site carries $-Jg$ with $g = \sqrt{S(2S-1)}$, which vanishes at $S = \frac{1}{2}$ (the hard-core case) and is the entire reason the higher spins behave differently. The one-magnon dispersion is $\omega(k) = 2JS(1-\cos k)$. For $S \ge 1$ the two-magnon basis has $N(N-1)/2 + N$ states, the extra N being the doubly occupied sites — a genuine second channel, not bookkeeping.

(iii) The anisotropic spin-1 λ–D chain, the showcase model. Note the sign: the overall coupling here is $+J$, antiferromagnetic, the opposite convention to Equation (351) and Equation (353).

$$H \;=\; J\sum_x\Big[S_x^xS_{x+1}^x + S_x^yS_{x+1}^y + \Delta\,S_x^zS_{x+1}^z + K\big(\mathbf{S}_x\cdot\mathbf{S}_{x+1}\big)^2\Big] + D\sum_x\big(S_x^z\big)^2, \tag{354}$$

with $J = 1$ and S^z eigenvalues $\{+1, 0, -1\}$. At $K = 0$ the (Δ, D) plane already contains the three phases the campaign needs inside a single Hamiltonian: a Néel phase with $\mathbb{Z}_2$ symmetry breaking and genuine kinks ($\Delta \gtrsim 1.19$ at $D = 0$), the Haldane symmetry-protected phase ($\Delta \approx 1$, $D \lesssim 0.97$), and a trivial large-D phase ($\Delta \approx 1$, $D \gtrsim 0.97$). The biquadratic coupling K is carried only so that the exactly solvable AKLT point $(\Delta, D, K) = (1, 0, \frac{1}{3})$ lies inside the same family and can serve as a closed-form calibration; every physics point of the campaign has $K = 0$. The two string unitaries are $U_\alpha = \exp(i\pi S^\alpha)$ and the two string order parameters are

$$\mathcal{O}^\alpha \;=\; -\lim_{r\to\infty}\Big\langle S_1^\alpha\Big(\textstyle\prod_{1<y<1+r} e^{i\pi S_y^\alpha}\Big)S_{1+r}^\alpha\Big\rangle, \qquad \alpha \in \{x, z\}, \tag{355}$$

normalised so that the AKLT value is $+4/9$. Note that U_x does not conserve S^z, so $\mathcal{O}^x$ is unavailable in the charge-graded code path; this is why the model is carried in two independent tensor backends (an ungraded one on $\mathbb{C}^3$, and a $U(1)$-graded one that additionally resolves the entanglement spectrum by total S^z), which are cross-checked against each other.

17.2.2 Sector construction and the controlled truncation

The exact-sector modules never diagonalise the full Hilbert space. For the kink problem the relevant statement, and it is a correction to the naive picture, is that a single-domain-wall configuration $\uparrow^a\downarrow^{L-a}$ has $n_{\text{down}} = L - a$, so *different kink positions live in different S^z sectors.* Each S^z sector therefore contains exactly one sharp kink, rigidly tied to the sector label, and the kink-plus-one-magnon space is the neighbouring S^z eigenspace restricted to configurations with one or three domain walls.

Domain-wall counting is a spin-$\frac{1}{2}$ notion, and the spin-s modules replace it by the invariant *up-variation*

$$D(n) \;:=\; \sum_x \max\bigl(0,\ n_x - n_{x+1}\bigr), \tag{356}$$

the total upward variation of the magnon number including the frozen ends. A monotone wall of any width has $D = 0$; a wall plus one magnon on either side has $D = 1$. At $s = \frac{1}{2}$ this reduces to $D = (\#\text{domain walls} - 1)/2$, so the cut $D \le 1$ reproduces the spin-$\frac{1}{2}$ three-wall basis configuration for configuration — something the test suite checks explicitly, together with the equality of the two operators' spectra.

The truncation to $D \le d_{\max}$ is not an exact sector, because the elementary exchange move changes the wall count by 0 or ± 2. It is, however, controlled in three separate senses. It is off-resonant: a configuration with k extra pairs of walls sits roughly kJ_z above the one-kink energy, one magnon gap away. It is Hermitian: the code builds PHP, so unitarity and energy conservation of the evolution remain exact and are measured (norm drift below 2×10^{-12} and energy drift below 10^{-9} across every production run). And it is measurable: the discarded matrix elements and the weight removed by the state-preparation projection are both reported. The decisive convergence check is a direct basis enlargement. At $s = 1$, $N = 46$, $\Delta = 6$, going from $d_{\max} = 1$ (dimension 7 128) to $d_{\max} = 2$ (dimension 1 021 680) moves the memory ratio by 1.2×10^{-4} — a factor 143 in basis size for a change in the fifth decimal.

One warning is worth restating, because the diagnostic invites misreading. The reported Frobenius norm of the discarded matrix elements is extensive and unnormalised; it grows with basis size and cannot be compared to unity or read as a relative error. In the accepted spin-$\frac{1}{2}$ runs its ratio to the retained hopping norm is already 0.99, while those runs agree with the exact reference to within a few per cent. What bounds the error is state-level: initial-state leakage, exact conservation, size convergence, the basis-enlargement comparison above, and the $s = \frac{1}{2}$ control on the same code path.

17.2.3 Time evolution

Sector ground states in the exact-sector family are obtained by dense symmetric diagonalisation below dimension 400 and by Lanczos above it, from a *deterministic* start vector (the sharp-wall basis vector plus a small reproducible perturbation), so that every run is bit-reproducible from the parameters recorded in its own output file.

Time evolution there is Krylov, not Trotter and not Runge–Kutta: one `KrylovKit` matrix-exponential call per step (`krylovdim` $= 40$, `tol` $= 10^{-12}$ or tighter), stepping the interacting run and, where a reference is needed, the free runs in lockstep. The step size is a fixed input; there is no adaptive step control, because the tolerance sits on the Krylov expansion and the resulting conservation is measured afterwards rather than assumed. The projected Hamiltonian is Hermitian, so the only error is that tolerance; accumulated over up to some 550 steps the measured norm drift stays below 1.8×10^{-12} and the energy drift below 6.1×10^{-10}.

The uniform-MPS family uses three different variational algorithms for three different geometries. Bulk ground states are found by a tangent-space uniform-MPS optimisation with the Galerkin gradient as the convergence measure; a stalled search is restarted from the stalled state, and if it still does not converge it is handed once to a Riemannian gradient descent with a different descent direction, the result being reported as measured either way. Convergence means the Galerkin gradient is below tolerance and nothing weaker. The kink band is obtained from a quasiparticle (tangent-space) excitation ansatz in the *topological* sector — left vacuum ψ_A, right vacuum ψ_B, the two broken Néel vacua — so that the excitation returned is a domain wall rather than a magnon. Finite open chains are prepared by two-site sweeping density-matrix renormalisation. Real-time evolution is two-site time-dependent variational principle at a fixed truncation rank, inside a finite window whose infinite environments are frozen.

The frozen-environment window is an honest limitation, not a formality: nothing leaving the window comes back, so a readout is only meaningful while the measured edge leakage is small. The transport records carry the leakage time series, and the analysis is supposed to restrict the readout to samples below a fixed leakage tolerance. Section 17.9 reports what happens when that restriction selects nothing.

17.2.4 Observables and estimators

Position estimators for a wall are all *windowed*, on a window W of half-width h centred on the initial wall. This is forced: an unwindowed first moment of the magnetisation gradient telescopes to the total magnetisation, which is exactly conserved, so a global centroid would return $\delta x \equiv 0$ identically. Three estimators are computed, all exact on a sharp wall, with $m(x) = \langle S^z_x \rangle$:

$$\hat{X}_1 = \frac{\sum_{x\in W}\big(x+\frac{1}{2}\big)\big[m(x)-m(x+1)\big]}{\sum_{x\in W}\big[m(x)-m(x+1)\big]} \qquad \text{(gradient centroid)}, \tag{357}$$

$$\hat{X}_2 = \frac{1}{2s}\sum_{x\in W} m(x) \;+\; \frac{x_a+x_b}{2} \qquad \text{(integrated magnetisation)}, \tag{358}$$

$$\hat{X}_3 = \text{the interpolated sign change of } m(x) \qquad \text{(zero crossing)}. \tag{359}$$

$\hat{X}_1$ and $\hat{X}_2$ are both linear in the state and both return the weight-averaged position on a mixture of sharp walls, but they weight the window differently, so their difference is a direct estimate of the systematic error and is reported with every result. $\hat{X}_3$ is deliberately inequivalent: on a two-branch mixture it is quantised and jumps as the branch weights cross one half, which makes it a useful diagnostic and a bad observable. Where a single number must be quoted the campaign quotes $\hat{X}_2$, which the truncation study finds some sixteen times less sensitive to the basis cut than $\hat{X}_1$.

Transmission and reflection are windowed magnon *numbers*, with a buffer b excluding the wall region,

$$R = \sum_{x\le X_{\text{ref}}-b}\big(s-m(x)\big), \qquad T = \sum_{x\ge X_{\text{ref}}+b}\big(s+m(x)\big), \qquad \text{trapped} = 1-T-R, \tag{360}$$

each of which vanishes identically on a clean wall of either branch. The memory itself is an asymptotic intercept difference: straight lines are fitted to $X(t)$ on a pre-collision and a post-collision window, both selected from the data by the condition that the trapped weight be below a fixed tolerance, and

$$\delta x \;=\; \big[a_{\mathrm{post}} + b_{\mathrm{post}} t_c\big] - \big[a_{\mathrm{pre}} + b_{\mathrm{pre}} t_c\big], \qquad t_c = \mathrm{standoff}/v_g. \tag{361}$$

For the two-magnon scattering runs the estimator is different and the difference mattered. The code measures the *chamber marginals* $(\langle x\rangle, \langle y\rangle)$ — the mean positions of the left and the right down spin — against a free *product* reference evolved in lockstep. No spatial window appears, so there is no window-boundary artefact, and the residual mis-ordering bias cancels between run and reference to first order. Semiclassically the measured shifts are momentum derivatives of the scattering phase,

$$\Delta_s = -\frac{\partial \delta}{\partial k_s}, \qquad \Delta_h = -\frac{\partial \delta}{\partial k_h}, \tag{362}$$

which is what makes a wavepacket clock into a measurement of the soft theorem's coefficients. The finite-packet bias of the centroid shift is linear in $1/\sigma_x^2$, so each kinematic point is run at $\sigma_x \in \{8, 11, 14\}$ and Richardson-extrapolated to zero width using measured data only, with the quoted error bar the spread over all width pairs.

The λ–D transport runs use a third set of estimators, matched to the windowed-charge definitions of Section 7. With the staggered density $n_x := (-1)^{x+1} S^z_x$ and a window $W = [a, b]$, the wall coordinate of the definition is

$$\mathfrak{X}_W \;=\; a - 1 + \frac{1}{2s} \sum_{x \in W} \big(n_x + s\big), \tag{363}$$

which saturates once the wall leaves W — hence the definition's demand that the core be padded from both edges. Alongside it the code computes an *s-free* centroid of the wall density $w_x = (n_x - n_{x+1})/(2s)$, whose normalisation divides out, and an s-free zero crossing. That the reported displacement is measured with an estimator not containing s is what makes the comparison against $2s$ a measurement rather than an identity.

The windowed charge law itself is obtained exactly rather than sampled. The windowed staggered charge $\sum_{x \in W} n_x$ is a sum of commuting on-site operators of integer spectrum, so its characteristic function evaluated on the integer grid contains its whole spectrum and a discrete Fourier inversion returns a genuine probability law. Two derived certificates are measured on the same state and the same window: the modulus of $\langle e^{2\pi i \hat{Q}_W} \rangle$, which must equal one if the windowed charge has spectrum in a single coset of $\mathbb{Z}$, and the modulus of $\langle e^{2\pi i \mathfrak{X}_W} \rangle$, which must be strictly below one whenever $2s \notin \mathbb{Z}$ because the wall coordinate has spectrum in $(2s)^{-1}\mathbb{Z}$. Same state, same window, opposite verdict: the charge is quantised and the wall position is not.

17.2.5 Ground-state diagnostics of the λ–D chain

For each parameter point the uniform-MPS driver records the energy density, the energy variance, the half-chain entanglement entropy and its Schmidt spectrum, the largest sub-unimodular transfer-matrix eigenvalue as a correlation length (nulled when the state fails to fill its bond space, since the transfer matrix then carries unphysical eigenvalues in the unsupported block), an independent exponential fit to the measured connected $\langle S^z S^z \rangle$, the staggered magnetisation, both string order parameters of Equation (355) together with their full profiles out to $r = 64$, and the first three relative splittings $(\lambda_i - \lambda_{i+1})/\lambda_i$ of the entanglement spectrum. Convergence means the Galerkin gradient is below tolerance and nothing weaker. Every point is recorded at $\chi \in \{16, 32, 48\}$ so that the drift is visible rather than hidden.

17.2.6 The test suite

The package's tests auto-discover every test file and run in one session. There are 418 assertion sites in 107 nested test sets across twelve files; because most of them sit inside loops over parameter grids, the suite executes 3842 checks, all passing, with nothing marked broken or skipped.

Two structural habits are worth naming. First, agreement is engineered to be a real check rather than a restatement: every sector construction is confronted with an *independently written* dense brute-force builder — a separate Kronecker-product routine per test file, not a shared helper — and the assertion is eigenvalue-by-eigenvalue spectrum equality, or in several cases set equality of the enumerated bases. Second, the falsifier tests are two-sided by construction. The decision is not "is the measurement near the prediction" but a genuine exclusive-or: a boolean for the conjecture surviving and a boolean for it being falsified are computed against the same band, and the test asserts that *exactly one* of them fires before asserting which. An inconclusive result, or one consistent with both hypotheses, therefore goes red rather than being written up as a pass. Elsewhere the same criterion is stated as a three-way verdict — survives, falsified, or inconclusive — and the inconclusive verdicts are recorded in the production output rather than dropped.

Coverage is by shard. `test_xxz_sector.jl` and `test_xxz_dynamics.jl` check the sector enumeration against brute-force configuration counting, the dispersion against the exact open-chain eigenvalues, and unitarity and energy conservation of the Krylov stepper. `test_memory_experiment.jl` checks the estimators of Equations (357) to (359) on analytically known states and the identity $T + R + \text{trapped} = 1$ to 10^{-10}. `test_fm_twomagnon.jl` carries the pre-registered displacement tolerances — 2×10^{-2} on the displacement, one per cent of the leading scattering length 2; 5×10^{-3} on velocities; 10^{-9} on conservation; 2×10^{-1} on the oracle truncation — none of which was changed after the fact. `test_spin1_twomagnon.jl` and `test_spins_twomagnon.jl` validate the spin-S sector Hamiltonian against a brute-force dense $(2S + 1)^N$ diagonalisation of the ring at $N = 6$, $S = 1$ and $N = 8$, $S = \frac{1}{2}$, validate the momentum blocks against the full sector spectrum, and carry the falsifier decision tests with their fixed criteria. `test_spin1_memory.jl` and `test_spins_memory.jl` check that the up-variation basis of Equation (356) agrees with the frozen spin-$\frac{1}{2}$ basis configuration for configuration, and hold the memory ratio decision tests. `test_lambdaD_groundstate.jl` pins the AKLT calibration to its closed forms, cross-checks the two tensor backends, and gates the phase diagnostics. `test_lambdaD_memory.jl` gates the kink dispersion, the memory coefficient and the edge contrast at production geometry.

The last of these is where the discipline earned its keep. A gate on the Haldane edge moment was failing at $|m_L| = 0.408$ against a threshold of 0.45. The threshold was *not* lowered. Diagnosis showed a finite-size edge–bulk overlap obeying

$$\tfrac{1}{2} - |m_L| \;\approx\; 0.70\, e^{-L/2\xi}, \qquad \xi \approx 6.03 \text{ sites}, \tag{364}$$

which reproduces every $L \geq 32$ to better than one per cent and reaches $|m_L| = 0.4965$ at $L = 64$; the test geometry was moved to the production size $L = 40$ and a new, cheap ground-state-only test set was added asserting the monotone approach $m_{24} < m_{40} < m_{64} < \frac{1}{2}$ and the fitted decay length. That test set would have identified the original failure immediately.

17.3 The λ–D chain: phases and the rigidity dichotomy

The first production programme establishes that one Hamiltonian family carries all three corners the campaign needs and that the diagnostics separate them.

Figure 1 fixes the four reference points and their calibration. Figures 2 and 3 follow the diagnostics across the two transitions. The headline is the one recorded in Section 14: across the

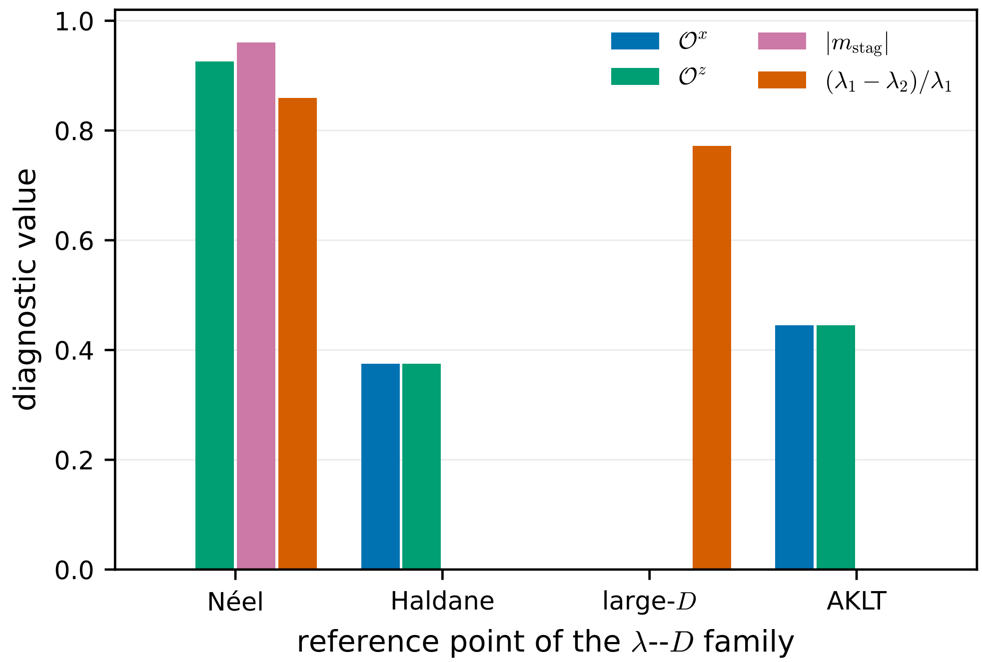

Figure 1: Ground-state diagnostics of the spin-1 λ–D chain Equation (354) at its four reference points, at the largest bond dimension computed ($\chi = 48$). The Néel point ($\Delta = 2.5$, $D = 0$) has $\mathcal{O}^x = 10^{-23}$ but $\mathcal{O}^z = 0.9255$ and staggered magnetisation 0.9603; the Haldane point ($\Delta = 1$, $D = 0$) has $\mathcal{O}^x = \mathcal{O}^z = 0.3743$ and an entanglement doublet that is exactly degenerate at the resolution of the calculation; the large-D point ($\Delta = 1$, $D = 2.5$) has both string orders zero and a split spectrum; the AKLT point ($\Delta = 1$, $D = 0$, $K = \frac{1}{3}$) reproduces its closed forms $\mathcal{O}^x = \mathcal{O}^z = 4/9$, $e = -2/3$ and $S = \log 2$ to the printed digits. The figure makes the operational point of the whole programme: $\mathcal{O}^z$ is *large* in the Néel phase and therefore does not distinguish it from Haldane, whereas $\mathcal{O}^x$ does.

Haldane boundary the entanglement doublet is exact until it is not, jumping by an $O(1)$ amount at a transition that a continuous bulk order parameter merely drifts through. The critical points bracketed by the jumps agree with the literature values $D_c \simeq 0.97$ and $\Delta_c \simeq 1.186$.

Three controls support the reading. The AKLT point is a zero-parameter test of nine closed-form numbers at once — energy density, correlation length, entropy, both string orders, exactly two Schmidt values each $1/\sqrt{2}$, a vanishing doublet splitting, vanishing staggered order, and the exact correlator $\langle S_1^z S_{1+r}^z \rangle = \frac{4}{3}(-\frac{1}{3})^r$ out to $r = 8$ — and it passes to the printed digits; in the charge-graded backend the two Schmidt states are additionally found to sit in S^z sectors exactly one unit apart, which is the edge doublet read straight off the charge labels. The Haldane doublet splitting reaches 9×10^{-14} at the largest bond dimension and stays below 5×10^{-12} across the entire Haldane side of the sweep, so the degeneracy is exact at the resolution of the calculation rather than merely small. And the Néel point is deliberately run once more on a *one-site* unit cell, which cannot represent a symmetry-broken state and is expected *not* to converge: that failure is itself the evidence that the breaking is physical, and the row is kept in the record with its diagnostics visibly nonsensical.

Two caveats belong with these figures. The bond dimensions available on the hardware used, $\chi \in \{16, 32, 48\}$, are not enough to resolve a critical point: near a transition the correlation length and the string order are χ-limited, which is why every point is recorded at all three bond dimensions

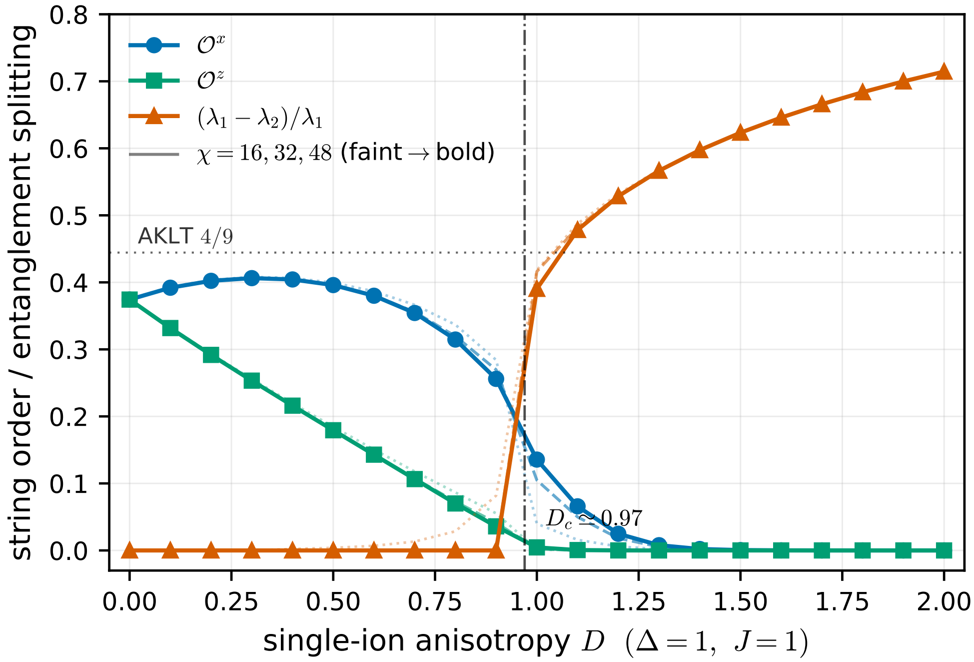

Figure 2: The Haldane-to-large-D sweep at $\Delta = 1$: string order parameters and the relative splitting of the leading entanglement doublet, each recorded at $\chi = 16, 32, 48$ (faint to bold) so that the bond-dimension drift is visible. This is the rigidity dichotomy in one panel. The bulk coefficient $\mathcal{O}^x$ *drifts continuously* from its Heisenberg value 0.374 down through 0.256 as D approaches the transition, while the entanglement doublet stays degenerate to below 5×10^{-12} over the whole Haldane side and then jumps by an $O(1)$ amount — from 4×10^{-12} at $D = 0.9$ to 0.39 at $D = 1.0$, eleven orders of magnitude in one step of the sweep — bracketing the literature value $D_c \simeq 0.97$. A quantised discrete datum survives exactly where a continuous one is already sliding.

rather than at the best one. And the reported correlation length is the maximum over *all* transfer-matrix channels, which need not be the longitudinal channel that a $\langle S^z S^z \rangle$ fit sees; in the large-D phase the two differ by more than a factor of two because the longest channel there is transverse.

campaign id:	— (calibration and phase data for Section 14)
proved in:	— (numerical record, no theorem asserted)
tested by:	`numerics/results/lambdaD-phase-points.json` (17 rows), `lambdaD-D-sweep.json` (63 rows), `lambdaD-delta-sweep.json` (75 rows); produced by `numerics/scripts/run_lambdaD_phases.jl`; gated by `numerics/test/test_lambdaD_groundstate.jl`
reviewed in:	bd lane `tns-f5r` wave 1

17.4 The λ–D kink: dispersion and memory coefficient

The second production programme puts a kink of the Néel phase in motion and measures what a magnon-like wavepacket does to it, in the non-integrable setting of Equation (354). It runs at $\Delta = 2.5$, $D = 0$, $J = 1$, deep in the Néel phase, where the measured staggered tail density is

$$s = m_{\text{stag}} = 0.960340, \tag{365}$$

an ordinary number rather than a half-integer. This matters for reading the prediction: the memory law of Section 7 predicts the coefficient $2s$, and $2s = 1.920680$ here.

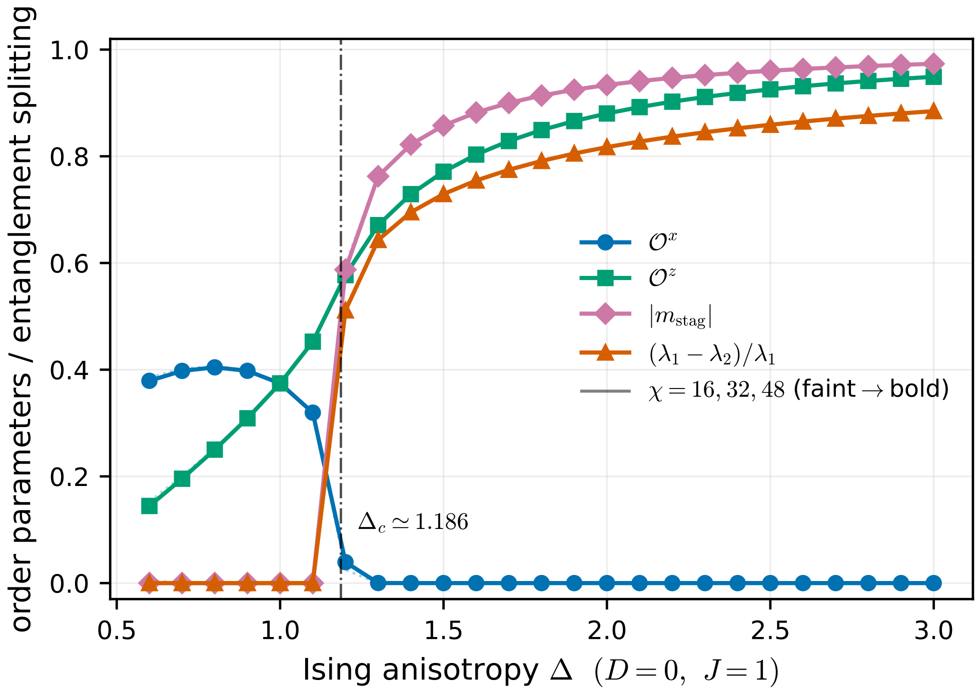

Figure 3: The Haldane-to-Néel sweep at $D = 0$, on a two-site unit cell so that the symmetry-broken state is representable. The staggered magnetisation and the entanglement splitting switch on together between $\Delta = 1.1$ (both below 10^{-13}) and $\Delta = 1.2$ (0.587 and 0.511 respectively), bracketing the literature value $\Delta_c \simeq 1.186$. $\mathcal{O}^x$ collapses across the same window ($0.319 \to 0.039 \to 1.5\times10^{-4}$ at $\Delta = 1.1, 1.2, 1.3$), whereas $\mathcal{O}^z$ *grows* straight through the transition ($0.452 \to 0.577 \to 0.671$) and is therefore useless as a phase discriminator here — the point already made statically in Figure 1.

Two points about Figure 4 matter for how the transport numbers should be read. The band is *computed*, not fitted: it comes from a variational quasiparticle solve at each momentum, and the group velocity is a central difference over three such solves. It is therefore fixed *before* any dynamics, and the measured wavepacket velocity is tested against it rather than fitted to it. The wavepacket itself is built from the variationally optimised excitation tensor at the working momentum; an undressed step from one vacuum to the other is kept only as a deliberately bad control, and it is bad by a wide margin — its energy exceeds the band by 9.8, against a gate of 1.0, where the dressed packet sits within 3.7×10^{-3} of $\omega(k_0)$.

Figure 5 then establishes that the memory coefficient extracted from the transport is converged in the one parameter that could plausibly have been controlling it. At the largest rank the run measures a mean escaped charge $\nu = -16.372936$ and, with the s-free centroid estimator, a wall displacement of $\delta x = 8.534665$ sites, giving

$$-\frac{\nu}{\delta x} = 1.918404 \qquad \text{against} \qquad 2s = 1.920680, \tag{366}$$

a deviation of 0.12 per cent. Because the estimator on the left does not contain s, this is a measurement of the coefficient and not an identity. The measured propagation velocity in the window, 2.1343, sits 2.2 per cent below the band value 2.1816 of Figure 4, which is the size one expects from a packet of finite width sampling the band curvature.

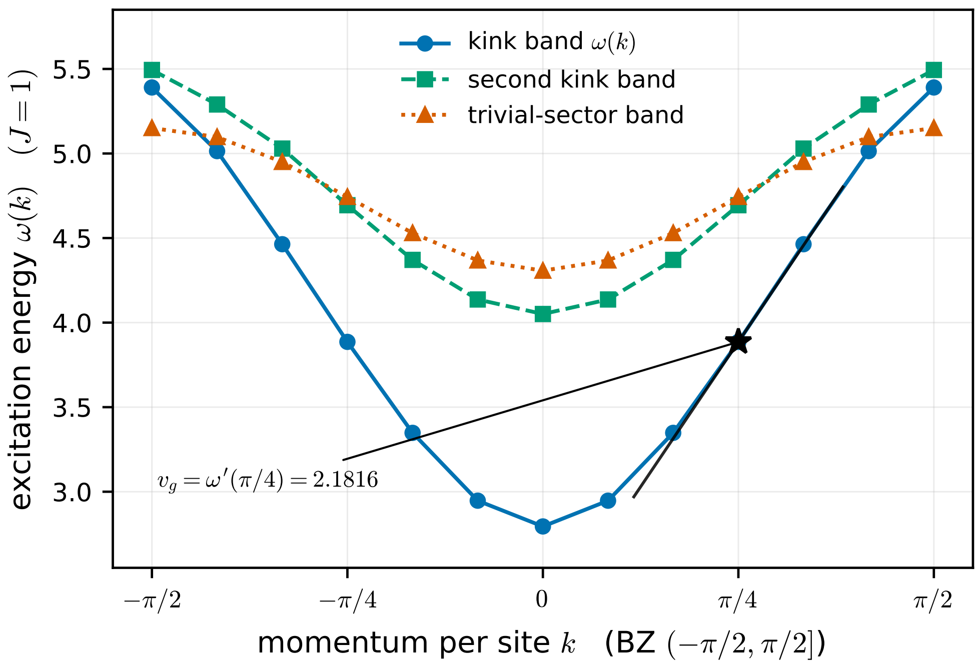

Figure 4: Kink dispersion in the topological sector of the λ–D chain at $\Delta = 2.5$, $D = 0$, obtained from a quasiparticle excitation ansatz with the two broken Néel vacua as left and right boundary conditions. Momentum is per site, so $\omega(k) = \omega(k+\pi)$ and the zone is $(-\pi/2, \pi/2]$. The kink band is gapped ($\Delta E = 2.7947$) with bandwidth 2.5957 and is cleanly separated from the second kink band and from the trivial-sector band over the whole zone — the separation that makes a single-band wavepacket meaningful. The straight line is the tangent at the working momentum $k_0 = \pi/4$, giving $\omega(k_0) = 3.886051$ and the group velocity $v_g = 2.181628$ against which the transport runs are calibrated. Three bond dimensions ($\chi = 16, 24, 32$) agree to nine significant figures, so only the largest is drawn.

The integer structure behind the law is certified separately, on the same state and the same window. The windowed charge has $|\langle e^{2\pi i \hat{Q}_W} \rangle| = 0.99995$ with a phase defect of 5×10^{-15} — its spectrum lies in one coset of $\mathbb{Z}$, so every escaped-charge outcome is an integer — while the wall coordinate has $|\langle e^{2\pi i \mathfrak{X}_W} \rangle| = 0.438$, comfortably below one, as it must be since $2s \notin \mathbb{Z}$. Displacing the window charge law's weight to $1/(2s)$ turns the inversion into an aliased signed measure with genuinely negative entries, and the test asserts that it does. The wall displacement is not quantised; the charge is.

This is evidence for the memory law of Section 7 in a model with no integrability to lean on, and it is not a proof of anything. Two limitations are stated in the source and should be stated here. First, the protocol measures the *bookkeeping* half of the memory statement: nothing scatters off the wall in this run, so the displacement is ballistic transport of a single packet, and a memory in the sense of a transient perturbation leaving a permanent offset would require a second excitation inside the window and is not attempted. Second, what is recorded is the exact single-time law of the windowed charge at each sample; that coincides with the ordered two-point measurement law of the theory only under the additional hypothesis named there, which is carried in the record as an assumption and not as a result.

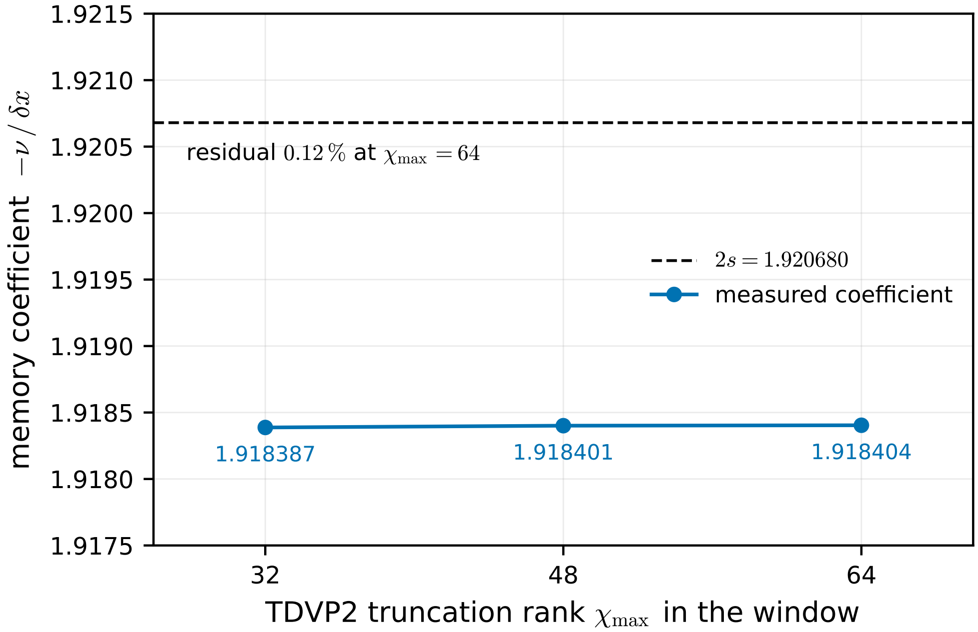

Figure 5: Truncation-rank convergence of the measured memory coefficient $-\delta x/\nu$ for the λ–D kink transport at $L = 48$, against the predicted $2s = 1.920680$ of Equation (365). Raising the two-site truncation rank inside the evolution window from $\chi_{\max} = 32$ to 64 moves the coefficient by 3.1×10^{-6}, from 1.9183874 to 1.9184040: the number is converged in the truncation. The residual gap to $2s$ is 0.12 per cent and is a property of the finite window and readout geometry, not of the truncation. The supporting quality gates are a norm drift of 1.4×10^{-6} and an energy drift of 2.6×10^{-4} over the readout interval.

campaign id:	— (evidence for the memory law of Section 7; no claim promoted)
proved in:	— (numerical record)
tested by:	`numerics/results/lambdaD-kink-dispersion.json` (3 rows), `lambdaD-kink-memory-convergence.json` (3 rows); produced by `numerics/scripts/run_lambdaD_memory.jl`; gated by `numerics/test/test_lambdaD_memory.jl`
reviewed in:	bd lane `tns-f5r` wave 2; `theory/lanes/blitz-2026-08-29/wave2-repair/SUMMARY.md`

17.5 The λ–D edge: the memory contrast between phases

The third λ–D experiment is the static counterpart of the transport run. A ground state is prepared with boundary fields $h_L = +\frac{1}{2}$ and $h_R = -\frac{1}{2}$ on a chain of $L = 40$ sites, the left field is switched off at $t = 0$, and the magnetisation of the left half is followed. The Haldane phase has a protected edge spin-$\frac{1}{2}$ and should hold its polarisation; the large-D phase has no edge mode and should not acquire one.

Figures 6 and 7 show the contrast. Read together with Equation (364), which extrapolates the edge moment to $\frac{1}{2}$ as $L \to \infty$ with a decay length matching the spin-1 Heisenberg correlation length, they are the numerical face of the rigidity statement of Section 14: a discrete, quantised boundary datum that a trivial phase simply does not have.

Four caveats travel with these runs and are recorded in the data file. The finite-chain Haldane retention is exact only up to the edge–edge splitting time, which grows like $e^{L/\xi}$. The right edge is pinned throughout. The choice $h_R = -h_L$ places both phases in the total-S^z zero sector, so that m_L is a genuine measurement rather than half of a conserved total — with $h_R = +h_L$ the

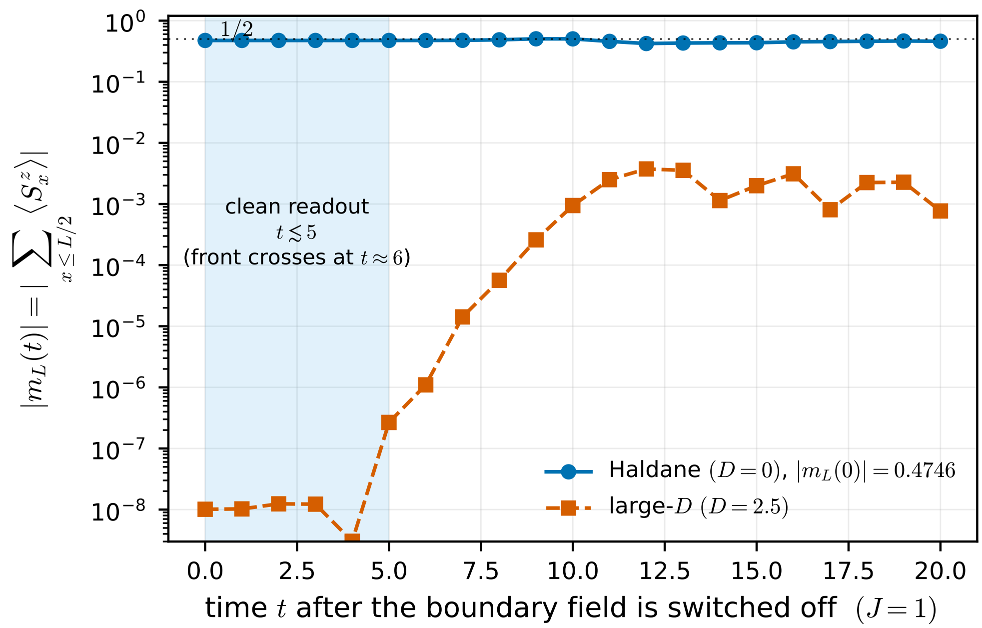

Figure 6: Edge memory after the left boundary field is switched off, on a logarithmic scale, for the Haldane point ($D = 0$) and the trivial large-D point ($D = 2.5$) of the same Hamiltonian family, at $L = 40$. The Haldane chain carries $|m_L(0)| = 0.4746$, close to the half unit expected of a protected edge spin-$\frac{1}{2}$, and holds it; the large-D chain never exceeds 3.7×10^{-3}, a contrast of a factor 127. The shaded band is the only honest readout window: a bulk magnon front of speed $v \approx 2.5$ reaches the window edge at $t \approx 6$, after which $|m_L|$ overshoots $\frac{1}{2}$ because bulk charge is crossing the window, not because the edge is decaying. Data outside the band, including the record's $t = 20$ "retention" field, is front-contaminated (see Section 17.9).

diagnostic would have been a symmetry artefact. And the *relative* retention of the large-D run is a ratio of numerical zeros and is meaningless; the absolute numbers are the ones to quote.

campaign id:	— (evidence for the boundary rigidity of Section 14)
proved in:	— (numerical record)
tested by:	`numerics/results/lambdaD-edge-memory.json` (2 rows); produced by `numerics/scripts/run_lambdaD_memory.jl`; gated by `numerics/test/test_lambdaD_memory.jl`
reviewed in:	bd lane `tns-f5r` wave 2; open defect `tns-nkf`

17.6 The spin-s falsifier battery

The conjecture at issue is the one discussed in Section 18: that the soft phase slope and the memory quantum are the same asymptotic charge datum $|q_{\text{hard}}|/s$. At $s = \frac{1}{2}$ both are 2, which is exactly the sort of coincidence that deserves to be attacked. The falsifier was stated in advance: if the spin-1 phase slope is not 1, the coincidence is numerology and the conjecture must be dropped.

The analytic input is a contact solution of the spin-S two-magnon problem, obtained without any integrability assumption by keeping the free extension Ψ *unsymmetrised* and matching on the physical chamber. Writing $K = k_1 + k_2$, $q = (k_1 - k_2)/2$, $c = \cos(K/2)$ and

$$n \;:=\; 2Sc\cos q - e^{iq}\big[(2S-1)c + \cos q\big], \tag{367}$$

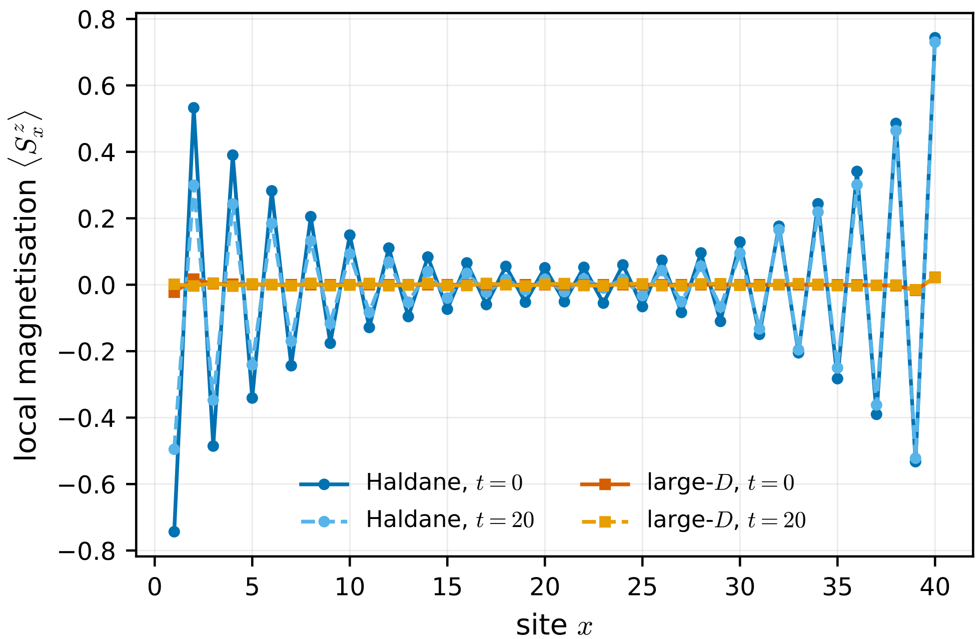

Figure 7: Local magnetisation profiles for the same two runs, at preparation and at the end of the evolution. The Haldane profile shows the staggered, exponentially decaying edge moment characteristic of a fractionalised boundary spin, essentially unchanged by the evolution; the large-D profile is flat at the resolution of the plot in both phases of the protocol. The right-hand rise in the Haldane curves is the *other* edge, which is pinned by a permanent field for the whole run.

the ratio is $S_{12} = n/(-\bar{n})$, manifestly of unit modulus, and the soft expansion gives

$$\frac{\partial \delta}{\partial k_s}\bigg|_{k_s=0} = \frac{1}{S}, \tag{368}$$

with all hard-momentum dependence cancelling. At $S = \frac{1}{2}$ the $(2S-1)c$ term vanishes and the expression collapses to the frozen $s = \frac{1}{2}$ oracle formula, checked to 10^{-12}.

Figures 8 and 9 show both halves. Three independent routes agree on the slope — an exact-eigenvector residual test at Bethe–Yang momenta, the ansatz-free ring extraction, and wavepacket dynamics — and across all 84 ring rows the extracted phase agrees with the analytic spin-S expression to 8×10^{-13}. The memory ratio is supported by size convergence (three chain lengths agreeing to five decimals), by sharp-versus-dressed initial walls agreeing to 0.5 per cent, and by the basis-enlargement check quoted in Section 17.2.

Two pieces of restraint belong with these numbers. The independent cross-check records four of its fourteen memory runs as *inconclusive* — all at the lower working momentum, where the packet is too broad in momentum for the asymptotic reading and the two linear estimators bracket the prediction with a spread of 0.17 — and those runs are kept in the published record with the diagnosis attached rather than dropped. And the largest spin was reached only at a reduced chain length: the corresponding sector needs of order twenty-four gigabytes at the production size, and an attempt there was killed by the operating system, so the $s = \frac{3}{2}$ point is a single run on a shorter chain and is the weakest of the battery.

The verdict is that the conjecture *survived* both falsifiers. It remains [CONJECTURE]. What the runs test sharply is the $1/s$ factor, since s is varied over $\{\frac{1}{2}, 1, \frac{3}{2}, 2\}$; what they did not test is the $|q_{\text{hard}}|$ factor, because every leg in every run carries unit charge. That gap was later attacked

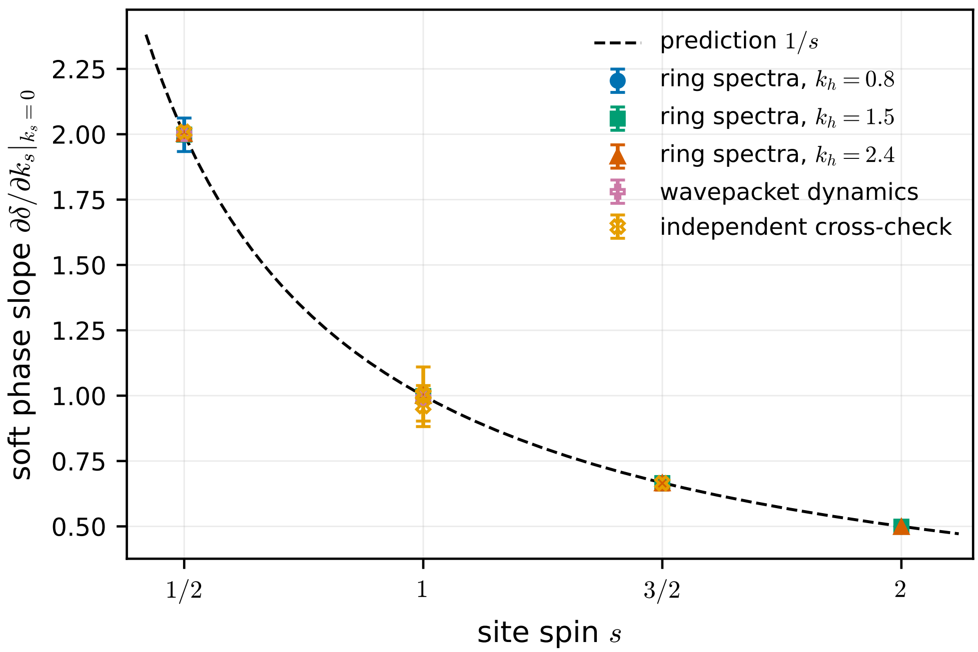

Figure 8: Measured soft phase slope $\partial\delta/\partial k_s|_0$ against site spin, versus the predicted $1/s$. The ring points are *ansatz-free*: exact total-momentum-block spectra of the two-magnon sector are inverted through $E = \omega(k_1) + \omega(k_2)$ and the Bethe–Yang quantisation $Nk_s = 2\pi n_s + \delta$, extrapolated over $N \in \{60, 90, 120, 180, 240, 360, 480\}$; no wavefunction ansatz and no S-matrix enters. Independent wavepacket dynamics and an independent cross-check by different methods are overlaid. Every one of the twelve ring determinations lies within 0.003 of $1/s$: 1.9979–2.0010 at $s = \frac{1}{2}$, 0.9972–0.9995 at $s = 1$, 0.6645–0.6654 at $s = \frac{3}{2}$, and 0.4978–0.4989 at $s = 2$. The prediction 2 for every spin — the outcome that would have killed the conjecture — is excluded by many standard deviations at $s \geq 1$.

directly: a three-magnon exact diagonalisation at $N = 100$ and $s = \frac{1}{2}$, in which the hard leg is a charge-two bound state, measured a slope of

$$4.070 \pm 0.090 \qquad \text{against} \qquad |q_{\text{hard}}|/s = 4, \tag{369}$$

inside a pre-registered absolute window of 0.35, with a dense-sector oracle running green and a deliberate mutation of the hopping running red. This too is finite-packet evidence at a single volume and a single packet width: there is no size or width extrapolation, and no charge-two kink-memory experiment exists. The conjecture stays where it was.

campaign id:	Bc
proved in:	— (conjecture; the slope half at unit charge is a separate proved row, see Section 10)
tested by:	`numerics/results/spin1-bc-falsifier.json` (84 ring, 16 dynamical, 22 memory rows); `spin1-bc-crosscheck.json`; `numerics/scripts/run_spin1_bc_falsifier.jl`, `run_spin1_bc_crosscheck.jl`; decision tests in `numerics/test/test_spins_twomagnon.jl`, `test_spins_memory.jl`, `test_spin1_twomagnon.jl`, `test_spin1_memory.jl`
reviewed in:	bd lane `tns-8e9`; charge-two lane `theory/lanes/blitz-2026-08-29/bc-charge2/`

17.7 The ferromagnetic two-magnon displacement scan

The soft theorem of Section 10 is derived algebraically. The displacement scan tests it dynamically, and the test is genuinely independent: the code never writes down a Bethe wavefunction.

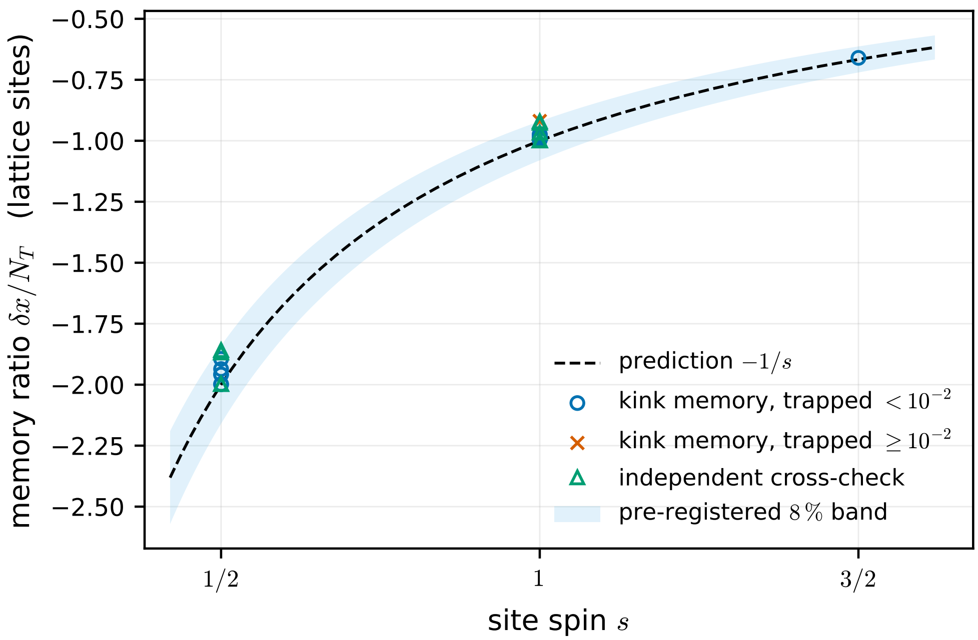

Figure 9: The memory half of the same battery: measured ratio $\delta x/N_T$ from spin-s kink-transport runs, against the predicted $-1/s$ and the pre-registered eight-per-cent decision band fixed before the first run. Circles are runs whose trapped weight is below 10^{-2}, so that the asymptotic condition behind the law is actually met; crosses are the runs where it is not. The nine spin-1 grid points average -0.97996, and the eight of them with small trapped weight average -0.98755, within 1.3 per cent of -1; four spin-$\frac{1}{2}$ controls on the same code path average -1.97817; the single spin-$\frac{3}{2}$ point gives -0.66015 against -0.66667 and is the sharpest of all because it has only partial transmission ($T = 0.383$), so the ratio is not near-saturated.

It enumerates down-spin pairs, assembles the two-magnon Hamiltonian from the model directly, launches two Gaussian packets in the incoming configuration — hard behind soft, so that the hard magnon overtakes — and measures how far each outgoing packet has been displaced relative to free propagation. By Equation (362) those displacements *are* the momentum derivatives of the scattering phase, so a sign error, a wrong branch or a swapped channel convention would show up immediately.

Figures 10 and 11 show the two displacements. Beyond the pointwise agreement, the coefficients of the soft expansion can be read off the dynamics with no S-matrix input at all, by combining runs at $+k_s$ and $-k_s$: the antisymmetric combination isolates the first hard invariant and the symmetric one the leading scattering length. Extrapolating the two available soft momenta at fixed hard momentum gives 2.147178 for the former, against 2.147197 from the exact lattice S-matrix and $2\cot(k_h/2) = 2.146852$ from the theorem, and the latter converges to 2 within 0.2 per cent. **The leading, hard-data independent scattering length 2 is a directly measured lattice displacement**, not an artefact of an ansatz.

The one place where the dynamical method is strictly weaker than the algebra is kinematic rather than analytic: as the hard momentum approaches π its group velocity vanishes, the incoming condition becomes almost unsatisfiable, and the collision takes $O(10^3)$ time units. The accuracy is controlled by a single dimensionless resolution — final separation over combined final packet width — and, because dispersive spreading grows with the propagation time, pushing the packets further

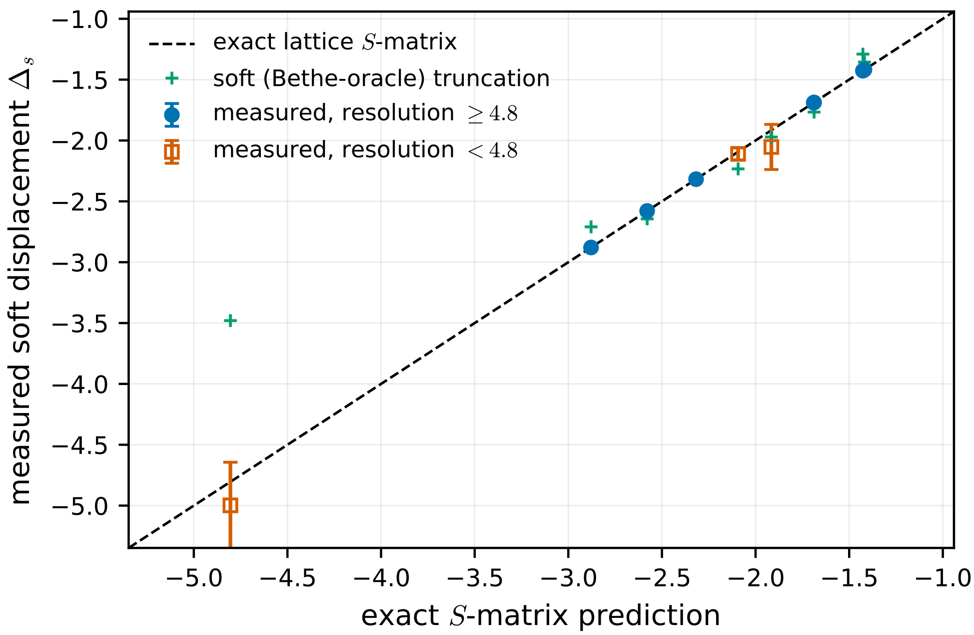

Figure 10: Measured soft-magnon displacement Δ_s against the exact lattice S-matrix prediction, for ten kinematic points of the spin-$\frac{1}{2}$ isotropic Heisenberg ferromagnet; the dashed line is equality. Each point is Richardson-extrapolated in $1/\sigma_x^2$ from packet widths $\sigma_x \in \{8, 11, 14\}$, with the error bar the spread over all width pairs. Every point agrees with the exact S-matrix inside its own error bar; where the wavepacket geometry is well resolved the deviation is 3×10^{-6} to 5×10^{-4}, i.e. four to six significant digits of a scattering-phase derivative recovered from real-time dynamics alone. The crosses are the soft truncation of the theorem, which deviates visibly and by design — most dramatically at the leftmost point, where the soft hierarchy is deliberately violated and the dynamics side with the exact S-matrix, confirming the theorem's stated validity domain rather than refuting it.

apart does not improve it; only fatter packets or a faster relative velocity do, at cost cubic in the packet width.

The related claim row `N1`, that excitation-ansatz magnon amplitudes reproduce the Bethe S-matrix in the soft limit, has *not* been run and remains [CONJECTURE] with no supporting data.

campaign id:	N1 (not run); supporting data for the two-magnon soft expansion of Section 10
proved in:	— (numerical record)
tested by:	`numerics/results/fm-displacement-scan.json` (10 summary rows over 30 evolutions); produced by `numerics/src/fm_twomagnon.jl`; gated by `numerics/test/test_fm_twomagnon.jl`
reviewed in:	`numerics/docs/fm-twomagnon-notes.md`

17.8 The XXZ kink-memory scan

The last family is the original spin-$\frac{1}{2}$ memory experiment: a magnon wavepacket sent at a kink in the easy-axis chain Equation (351), with the displacement of the kink measured before and after.

The bookkeeping argument is worth stating because it is what the data confirms. Before the collision the state is a kink at X plus one extra down spin in the up region. After full transmission the magnon is an *up* bubble in the down region, at the same total S^z. The wall must absorb the difference, and it can only do so by moving two sites. Hence a transmitted magnon displaces the

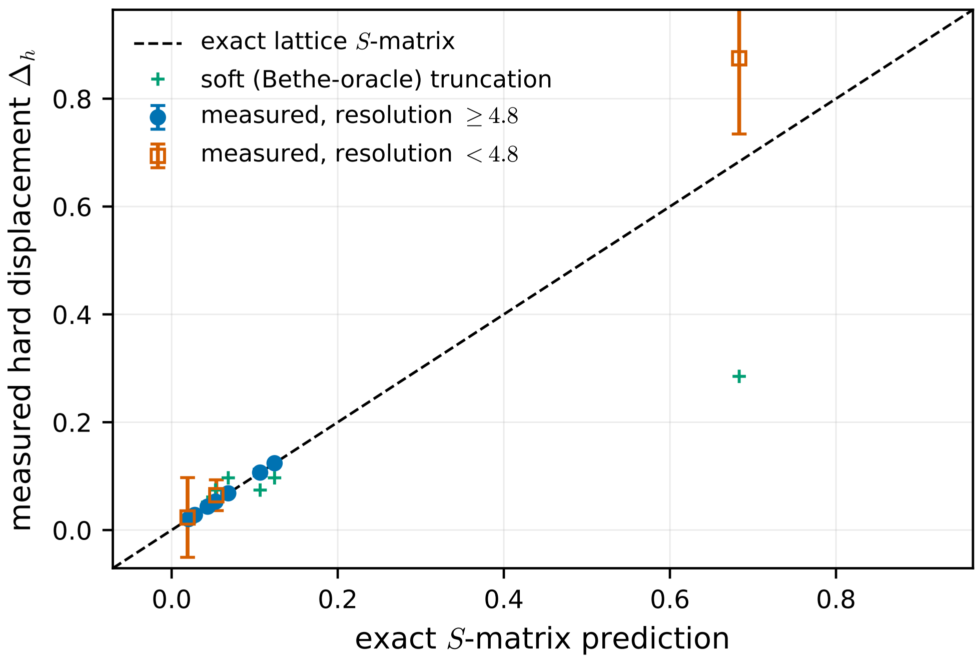

Figure 11: The same comparison for the hard-magnon displacement Δ_h. It is positive in all ten runs and one to two orders of magnitude smaller than Δ_s — the classical hard-sphere pattern, with the overtaken particle recoiling backwards by the contact length and the overtaking one jumping forwards. This is the structural content of the soft theorem made visible: the leading term of the phase carries no hard-momentum dependence and therefore cannot displace the hard packet at all, so Δ_h is quadratic in the soft momentum.

kink by exactly -2 sites, a reflected one by zero, and

$$\delta x \;=\; -2\,T, \tag{370}$$

so the memory is quantised and the whole of its magnitude sits in the transmission probability. On the lattice the collective coordinate can only shift in units of two sites and the continuous observable is the *weight* T, not the displacement.

Figure 12 is the quantitative statement and Figure 13 the physical one. A refuted prior expectation belongs here too, since it was recorded before the runs: it was predicted that at large anisotropy the kink would be an opaque barrier. It is not. Transmission tends to *one* as $\Delta \to \infty$. The mechanism is visible in the domain-wall algebra — the elementary exchange move takes a magnon adjacent to the wall on the up side directly into a bubble on the down side, displacing the wall by two, so transmission is a first-order, energy-conserving process inside the retained manifold, while the merge channel that produces reflection is off-resonant by one magnon gap. The easy-axis kink is nearly transparent to magnons, and the more so the deeper the easy axis. This is the numerical face of the decoupling statement of Section 6: the magnon couples to the wall's collective coordinate and to nothing else.

This is the evidence behind claim row `N2`, and it is held at [SKETCH]. The row says exactly what it is: an empirical scan matching the conditional memory expectation at the recorded precision, $\delta x \approx -2T$ to within 0.004 sites. It is numerical evidence, not a theorem, and it does not promote the conditional theorem it agrees with.

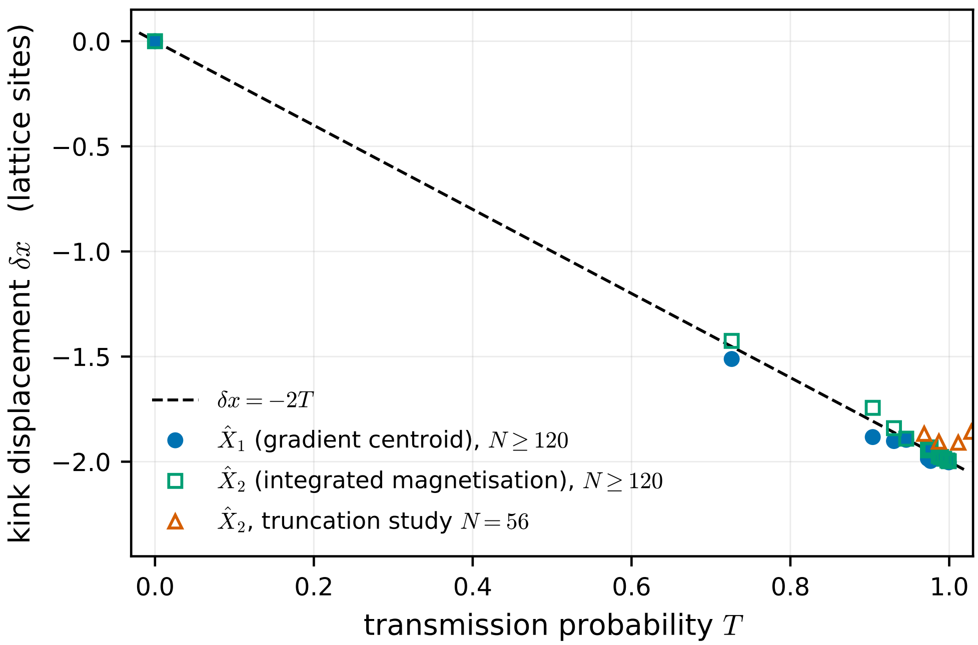

Figure 12: Kink displacement against transmission probability for the spin-$\frac{1}{2}$ XXZ memory scan, with the prediction $\delta x = -2T$ of Equation (370) as the dashed line. Production runs at $N \geq 120$ are shown for both linear estimators; the $N = 56$ points are the truncation study, whose geometry carries much larger systematics. Over the whole scan $|\delta x_1 + 2T| \leq 0.075$, and at the cleanest geometries — those with trapped weight below 10^{-6} — the residual is 0.004 sites. Three chain lengths $N = 120, 160, 200$ reproduce the same numbers to six significant figures, and an Ising control with the exchange switched off returns $T = 0$, $R = 1$ and $\delta x = 0$ exactly.

campaign id:	N2 [SKETCH]
proved in:	— (empirical scan only; the theorem it agrees with is conditional and lives in Section 7)
tested by:	`numerics/results/memory-scan-1.json` (21 runs); produced by `numerics/scripts/run_memory_scan.jl` over `numerics/src/xxz_sector.jl`, `xxz_dynamics.jl`, `memory_experiment.jl`; gated by `numerics/test/test_xxz_sector.jl`, `test_xxz_dynamics.jl`, `test_memory_experiment.jl`
reviewed in:	`numerics/docs/kink-sector-notes.md`

17.9 Honesty: defects, quarantine, and what none of this proves

A quarantined record

The λ–D transport production stage produced one record that is *not* a result and is not plotted anywhere in this section. The file `numerics/results/lambdaD-kink-memory.FAILED-NaN.json` is quarantined under a name that says so.

What happened is this. The stage ran two rows at $L = 72$, one for each initial dressing of the kink. The evolution itself was clean in both: energies, norms and position time series are finite throughout, out to $t = 14$. The first row, with a quasiparticle-dressed initial kink, completed normally and reported a coefficient of 1.9131 against $2s = 1.920680$. The second row, with a deliberately undressed sharp kink, has *every* analysis output non-finite: measured velocity, all three displacement estimators, the mean escaped charge and the coefficient.

The mechanism is visible in the record and is worth stating precisely, because it is more specific

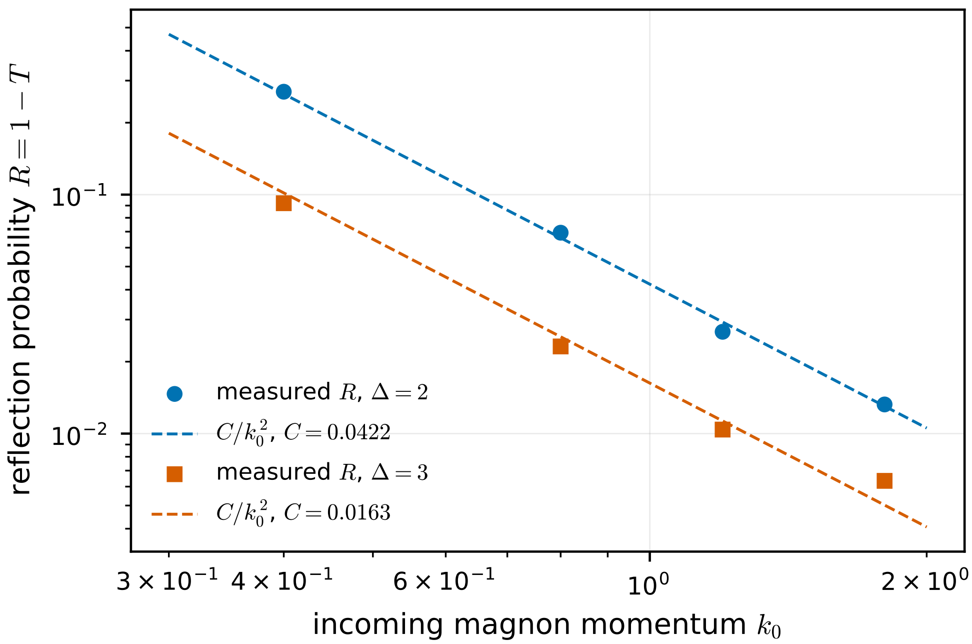

Figure 13: Reflection probability against incoming magnon momentum, on logarithmic axes, at two anisotropies, with fitted curves $C(\Delta)/k_0^2$. The measured R follows the one-dimensional low-energy threshold law with $C(2) \approx 0.042$ and $C(3) \approx 0.016$, so $T \to 0$ and, by Equation (370), $\delta x \to 0$ as $k_0 \to 0$: *a soft magnon leaves no memory.* This is the model's Adler-type zero, and it is the qualitative content of the soft theorem seen from the memory side. In the same regime $C(\Delta)$ falls like Δ^{-2}, so the kink becomes *more* transparent, and the memory closer to its quantum, the deeper the easy axis.

than "the fit broke". The readout interval is selected by requiring both that the wall be padded from the window edges and that the measured edge leakage stay below a fixed tolerance of 10^{-3}. For the sharp preparation the leakage starts at 5×10^{-3} — already five times the tolerance at $t = 0$ — and grows monotonically to 0.159. No sample ever qualifies, the recorded readout index range is empty, and the analysis correctly declines to compute a displacement outside the region where its definition applies, returning non-finite values instead. That branch is doing what its specification promises, and for an undressed junction, which is not a one-kink state at all, declining is arguably the right answer.

The defects are therefore not where one first looks. The first is that a production stage *wrote its output file and exited with status zero* while producing nothing but non-finite numbers; this is the same species of failure as a test gate that cannot fail, and it is why the record was committed silently. The second is that there is no finiteness gate anywhere in the driver. Both are tracked as an open priority-one defect (bd `tns-kng`), along with the requirement to rerun. A third question is open rather than defective: even the surviving row's readout interval was closed by the end of the integration rather than by the physics, so whether its numbers should be re-derived on a longer run is undecided.

Until this is resolved there is no `lambdaD-kink-memory.json`, and the converged coefficient of Figure 5 — which comes from a separate, regenerated convergence record at $L = 48$, independently recomputed from raw fields — is the only transport number this labbook quotes.

A mismeasured diagnostic

The edge-memory record carries a field named `retention`, evaluated at $t = 20$, reading 0.971 for the Haldane run. That number is front-contaminated and should not be quoted. As Figure 6 shows, the left-half magnetisation is frozen to within 7×10^{-6} for $t \leq 5$ and then a bulk magnon front crosses the window edge at $t \approx 6$, pushing $|m_L|$ past $\frac{1}{2}$ — bulk charge entering the window, not edge memory persisting. The honest readout window is $t \lesssim L/2v$, and the module docstring at present caveats only the far longer edge–edge splitting time. The quantities to quote are the initial, final and extremal values of m_L within the clean window, and the contrast between the phases, which is robust: a factor of 127 in the edge moment. This is an open documentation and record defect (bd `tns-nkf`).

Estimator hazards, recorded

Three estimator lessons are worth carrying, all of them learned by a test failing. A windowed centroid of the one-body density looks unimpeachable and is biased at the ten-per-cent level by dispersively spread tails, and it jumps discretely as an integer window edge moves — it produced the wrong *sign* for the hard-magnon displacement, which would have appeared to falsify a correct theorem. The zero-crossing wall estimator $\hat{X}_3$ is quantised on mixtures and jumps as branch weights cross one half; it is a diagnostic and must never be used interchangeably with the linear estimators at partial transmission. And the transmission and reflection integrals of Equation (360) are normalised only in the one-magnon sector: in an enlarged basis, virtual magnon pairs contribute their own up and down density and $T + R$ can exceed one, with the trapped weight going negative. Two runs in the scans exhibit exactly that, and are labelled accordingly.

A fourth lesson was promoted from a bug into an invariant. An early spin-s memory run returned a ratio of -1.104 where the raw before-and-after difference of the wall position was -1.02. The cause was geometric: the packet was not yet asymptotic at $t = 0$, so it already overlapped the reflected-weight region, the trapped weight never fell below tolerance, the data-driven pre-collision window degenerated to a three-point geometric fallback, and that window's spurious slope was extrapolated across the whole interval. The fix was not a tolerance but a hard precondition, now enforced by the code, that the standoff exceed the buffer by at least four packet widths — at which separation only 3×10^{-5} of the packet lies outside, an order of magnitude below the trapped tolerance. There is now a test asserting that the protocol *refuses* a non-asymptotic geometry, and the decision band was left unchanged.

Two further reporting rules are baked into the records, and a reader should know them. A quantity a given code path cannot represent is written as not-a-number, never as zero — so a zero in these files means a measurement, not an absence. And a bond whose numerical rank falls short of its nominal bond dimension is flagged, because the transfer spectrum then contains null-space artefacts; the affected correlation length is nulled while the raw value is retained for inspection, and there is a test asserting that the raw value is finite *and wrong.*

What is evidence and what is proof

To be explicit, because this is the point on which a reader could most easily be misled.

The numbers in Sections 17.3 to 17.8 are *evidence*. Some of it is very strong: a scattering-phase derivative reproduced to six significant figures from real-time dynamics, a memory coefficient converged in its truncation to six decimals, twelve independent determinations of a slope all within 0.003 of a prediction that a competing hypothesis puts a factor of two away. None of it is proof, and none of it changes a status.

Concretely: the conjecture attacked in Section 17.6 survived its falsifiers and remains [CONJECTURE]; the empirical scan of Section 17.8 is [SKETCH] and stays there; the row on excitation-ansatz amplitudes is [CONJECTURE] and has no data at all. The theorems these experiments accompany are proved, where they are proved, elsewhere in this document and by the arguments recorded there — and where they are conditional, as in Section 7, agreement with a numerical experiment does not discharge the condition.

17.10 Provenance of every record

For reference, each committed results file with the script that produced it and the tests that gate it. All paths are relative to the repository root.

`numerics/results/fm-displacement-scan.json` — produced by the displacement-scan driver in `numerics/src/fm_twomagnon.jl`; gated by `numerics/test/test_fm_twomagnon.jl`; documented in `numerics/docs/fm-twomagnon-notes.md`.

`numerics/results/memory-scan-1.json` — produced by `numerics/scripts/run_memory_scan.jl` over `numerics/src/xxz_sector.jl`, `xxz_dynamics.jl` and `memory_experiment.jl`; gated by `numerics/test/test_xxz_sector.jl`, `test_xxz_dynamics.jl` and `test_memory_experiment.jl`; documented in `numerics/docs/kink-sector-notes.md`.

`numerics/results/spin1-bc-falsifier.json` — produced by `numerics/scripts/run_spin1_bc_falsifier.jl` over `numerics/src/spin1_twomagnon.jl`, `spin1_collision.jl`, `spin1_memory.jl`, `spin1_memory_run.jl`, `spins_twomagnon.jl`, `spins_twomagnon_sector.jl`, `spins_twomagnon_collision.jl`, `spins_memory.jl`, `spins_memory_sector.jl` and `spins_memory_run.jl`; gated by `numerics/test/test_spin1_twomagnon.jl`, `test_spin1_memory.jl`, `test_spins_twomagnon.jl` and `test_spins_memory.jl`; documented in `numerics/docs/spin1-twomagnon-notes.md`.

`numerics/results/spin1-bc-crosscheck.json` — produced by `numerics/scripts/run_spin1_bc_crosscheck.jl`, by deliberately different methods from the file above; gated by the same test files.

`numerics/results/lambdaD-phase-points.json`, `lambdaD-D-sweep.json` and `lambdaD-delta-sweep.json` — produced by `numerics/scripts/run_lambdaD_phases.jl` over `numerics/src/lambdaD_model.jl` and `lambdaD_diagnostics.jl`; gated by `numerics/test/test_lambdaD_groundstate.jl`.

`numerics/results/lambdaD-kink-dispersion.json`, `lambdaD-kink-memory-convergence.json` and `lambdaD-edge-memory.json` — produced by `numerics/scripts/run_lambdaD_memory.jl` over `numerics/src/lambdaD_memory.jl`, `lambdaD_memory_run.jl` and `lambdaD_edge.jl`; gated by `numerics/test/test_lambdaD_memory.jl`.

`numerics/results/lambdaD-kink-memory.FAILED-NaN.json` — **quarantined**, produced by the same transport driver; not gated, not plotted, and not to be cited as a result. See Section 17.9 and bd `tns-kng`.

The figures of this section are regenerated from these records by `labbook/figures/make_figures.py`, which reads only the JSON files listed above and writes the PDFs referenced in the figure environments. Any rerun of the numerics must be followed by a rerun of that script, in the same commit, under the lockstep rule.

18 The state of the triangle

This closing section says, without hedging, where the programme stands. It reports statuses; it proves nothing. Each result is named as it is named in the section that owns it, and every status written here is the status carried in the campaign's claim ledger — including the ones that are uncomfortable.

18.1 The three corners

Corner A — asymptotic symmetry: complete. This corner is finished, and it is the only one of which that can be said. The truncated-symmetry identity, the unbroken endpoint theorem, the broken-case sector-jump theorem, and the Goldstone and lattice-Noether theorem are all [PROVED] (Section 5); their review loop converged with wording notes only, and their statuses are final as stated. The corner also delivered two corrections that the rest of the campaign has had to absorb. First, the coset space $(G_L \times G_R)/G_{\text{diag}}$ is *not* the classifying object: in the unbroken case the object that acts is the quotient of the group by the kernel of the virtual representation, and in the broken case the complete invariant of a vacuum pair is a double coset. Second, this corner does *not* by itself supply the soft edge: what it gives is an exact operator continuity equation, which entails no Adler zero and no universality.

Two boxes inside the corner remain [SKETCH] and are load-bearing wherever they are used: the split property, which is what a genuine edge Hilbert space realisation would need, and uniformity of the broken-case statement over a continuous vacuum manifold. They are flagged at every point of use rather than assumed silently.

Alongside the corner sits the tail-density result — for a circle-covariant injective MPS vacuum pair with opposite densities, $2\rho \in \mathbb{Z}$ — which is [PROVED] (Section 8), and the scope theorem (Section 8), also [PROVED], which says which symmetry directions can carry a calibrated memory at all: the charge lives on the centre of the unbroken subgroup, semisimple directions supply none, and a finite group supplies no current at all.

Corner B — memory: the strongest positive result, with one honest hole. The finite-window memory-index theorem is [PROVED] and unconditional (Section 8): with the circle-charge condition alone, the window charge's spectrum lies in one coset of $\mathbb{Z}$ and the two-projective-measurement increment is an integer. The window-charge identity is [PROVED] unconditionally as an identity (Section 9), and it is what makes the theorem more than bookkeeping: the window charge is not a new object but the windowed lift of the conserved first-moment wall coordinate, and its first corollary closes the obvious loophole by showing that no state carrying an escaping charged leg has time-uniform kink tails.

The infinite-volume law is [PROVED, CONDITIONAL] (Section 8): under the circle-charge condition and the relaxation hypothesis, every subsequential ordered protocol law is a probability on $\mathbb{Z}$ and the displacement is $\delta x = -(2s)^{-1} \sum_{\nu} \nu \, p_{\nu}$. Half of that hypothesis is free — its first clause is a theorem from separability and strong continuity alone, [PROVED] (Section 8) — so the real content sits in the remaining two clauses. In the easy-axis model, energy conservation bounds the *number* of domain walls uniformly in time, [PROVED] (Section 9); but wall number is not wall length, so this does not deliver the tightness clause. A sufficient no-runaway condition for tightness exists at [SKETCH], as does its composition with the mean-escape obstruction (Section 9).

The hole: clauses two and three of the relaxation hypothesis have not been established for any concrete dynamical model, the easy-axis kink model included. Until they are, this part of the corner prints in conditional form. The conditional theorem does not fall; what is missing is its first fully unconditional dynamical instance.

The older, channel-based route to the same conclusion is also on the books and is honest about its price. The general conditional memory law is [PROVED, CONDITIONAL] on the four-clause asymptotic hypothesis with displayed channel charges (Section 7); the Fano-graph reduction and the transmission amplitude with its soft zero are [PROVED] for the *projected* component of the kink model, and the full-chain statement is open. The brief's literal memory conjecture is [REFUTED], with the physical-current flux identity surviving as [PROVED]. The recorded easy-axis scan is [SKETCH], and remains so because it is numerical evidence rather than a theorem.

Corner C — the soft theorem: structure proved, value proved, general limit law one hypothesis short. The exact two-body content is done. Complete two-magnon resolution by direct diagonalisation, with no assumed Bethe completeness, is [PROVED]; the spin-$\frac{1}{2}$ soft expansion is [PROVED]; and the exact spin-S slope

$$\partial_{k_s}\delta_{\text{phys}}\big|_{k_s=0} = \frac{\operatorname{sgn}(v_h - v_s)}{S}$$

is [PROVED] at every site spin, without any integrability input (Section 10). The oracle cross-checks are [PROVED] and are used as checks, never as premises.

The kinematic apparatus is in place. The canonical two-magnon wave-operator decomposition is [PROVED], and the fixed-packet construction from exact band data is [PROVED, CONDITIONAL] — with the important difference, relative to its kink counterpart, that its hypotheses are instantiated on the ferromagnet, so the implication is not vacuous (Section 11). The ordered-limit packet theorem on the separated class is [PROVED].

The index apparatus is in place. The finite soft-index identity is [PROVED], first on a finite $SU(2)$ ring and then for an arbitrary compact Lie symmetry on any finite graph or periodic lattice, in both Ward registers; sector-label integrality is [PROVED] (Section 12). On the displayed separated-preparation subclass the matching theorem is [PROVED] with zero remainder, and with it the value statement and the exact scalar factorisation of the protocol datum (Section 12). The chain from symmetry to a number is therefore closed on that subclass, and closed in the right direction: matching transports an on-shell number inwards, and no value is stipulated anywhere.

What is not closed is the *general* limit law. The structural statement and the general value statement are both [SKETCH], and their shared missing input is a single named hypothesis — see the next subsection. The n-leg soft theorem remains [CONJECTURE] (Section 13). Two of its supporting rows are [CONJECTURE] (packet-smeared form-factor regularity; control of the order of the volume, width and soft limits) and the one-hard Ward reduction is [SKETCH], with its fixed-volume formulas established as an off-shell interpolation and the on-shell infinite-volume estimate open. The exact finite-sector Ward projection is [PROVED], carrying an erratum that must be read with it: its convenient scalar form is exact for one magnon and false for two, and the correct replacement is register-dependent.

The SPT dichotomy. The rebuilt rows are [PROVED] (Section 14): the multiplier cancellation for closed insertions, the edge-residue congruence, the twist and ordered-product relations, and the boundary-window memory theorem — the last unconditional at finite length and conditional in its limit clause. Bulk coefficient continuity along a common-gap path is likewise [PROVED], which is precisely what makes the dichotomy a dichotomy rather than a slogan. Two old rows are [REFUTED]: pointwise blindness of closed contractions to the projective class, and the all-orders claim that the class can appear in no coefficient at all. The dynamical instance — a nonzero edge-changing reflection amplitude for an explicit open boundary Hamiltonian — is [CONJECTURE], and the pieces it is missing are named: half-chain wave operators, the boundary asymptotic hypothesis, the on-shell reflection matrix, and nonvanishing.

Two spatial dimensions. Recorded, and deliberately kept separate from the one-dimensional programme (Section 15). The exact toric-code endpoint-label theorem and the finite planar window-integrality theorem are [PROVED]; the categorical selection theorem and isotopy flatness are [PROVED] as conditional implications on displayed hypotheses. The planar limit law is [SKETCH], capped there by a nonvacuity test: no nonzero model instance of its two substantive relaxation clauses has been exhibited.

18.2 The three edges

The edge statements below are the campaign's recorded ones. They are quoted faithfully; a remark at the end says how they should be read today.

Asymptotic symmetry ⇒ soft theorem: [conjecture]. This is the core proof obligation and it is not discharged. Corner A supplies only the exact lattice continuity equation — a statement about the true model that needs no tensor networks at all — and, in the ferromagnet, the Ward residue. It supplies no Adler zero and no universality. The live obligations are: the canonical wave-operator decomposition; packet-smeared infinite-volume form-factor regularity, including control at momenta of order $1/N$; the one-hard Ward reduction for two hard magnons into a three-magnon channel; the exhaustive normed reduction used by the no-contact factorisation; microscopic membership in the no-contact class; and the remainder and limit-order control. Separately, and settled negatively: process independence over unrestricted local sources is [REFUTED], so any statement along this edge must carry a source-class hypothesis.

Soft theorem ⇒ memory: [sketch]. The edge does not have the shape the brief conjectured. Its true shape is: soft behaviour of the transmission amplitude, hence a vanishing transmission probability at zero momentum — a memory Adler zero — hence the conditional displacement law $\delta x(k) = -T(k)/s$. The physical-current flux identity is [PROVED] and the charge bookkeeping is [PROVED] conditional on the asymptotic hypothesis; the middle step is conditional on the Fano-graph reduction, and the full-chain version on asymptotic completeness. Whether the zero and its coefficient are *universal* remains [CONJECTURE], and the universality counterexample of Section 13 is a standing warning here too.

Memory ⇒ asymptotic symmetry: [sketch], with a proved core. The core is [PROVED]: under stationary vacua and finite-range dynamics, the vacuum-pair label is rigid at finite time, and for a separated event the exact ledger

$$2s\,\delta x + (q_{\mathrm{out}} - q_{\mathrm{in}}) = 0$$

holds — reflection gives zero displacement, transmission gives $-1/s$. The older half-line formula is [REFUTED], wrong in sign and by a factor of two. The edge nevertheless stays at [SKETCH] for one stated reason: nobody has shown that a *measured* memory reconstructs an asymptotic-symmetry action or a classifying datum. The vacuum-pair label is read off, not shifted; the displacement records leg-charge transport inside it. The thermodynamic uniqueness and flatness statement that would sharpen this is [CONJECTURE], supported only by finite-volume evidence.

Remark 18.1 (How to read the edge block). The three edge statements above are quoted from the campaign's recorded edge block, which predates several later promotions — in particular the exact spin-S slope theorem, the finite soft-index identities in their general form, the separated-preparation matching theorem, and the two-magnon wave-operator construction. A refresh of that block is queued. The refresh will not upgrade an edge: none of those promotions supplies a universality statement or a reconstruction statement, which are what the first and third edges are missing. It will, however, shorten the first edge's list of live obligations, and it will make visible that the soft corner now reaches a number on a displayed subclass rather than only a shape.

18.3 The one remaining gap in the one-dimensional soft law

Strip away everything that is finished and one hypothesis is left standing between the campaign and a general one-dimensional soft limit law. It is the exhaustive decomposition hypothesis of Section 12: that, after the outer limit and uniformly in the hard variable, the limit datum splits as

$$\mathcal{A}_* \;=\; \mathcal{A}_*^{\mathrm{desc}} + \mathcal{A}_*^{\perp} + \mathcal{A}_*^{\mathrm{dir}} + \mathcal{A}_*^{\partial}, \qquad \mathcal{A}_*^{\mathrm{desc}}(\epsilon) = \int (e^{ik}-1)\, L(k,h)\, \big[2iv_h\ell_h\big]\, \mathrm{d}\mu_{*,\epsilon}(k,h),$$

with L continuously differentiable of the expected profile and the other three terms of second order in the soft scale. Grant that decomposition and the structural statement, the Adler zero, and — with the matching step and class membership — the value all follow. Withhold it and each of those stays at [SKETCH].

It is worth being precise about why this is the residue rather than one obstacle among many. The value half is no longer missing: the exact spin-S slope is proved, the matching hypothesis is proved with zero remainder on the separated-preparation subclass, and on that subclass the protocol datum has an exact scalar factorisation with a soft factor $(e^{ik}-1)$ and a quotient that extends continuously through zero. What the decomposition hypothesis would add is exactly the step from "the datum factorises on this displayed subclass" to "the datum decomposes, on the general protocol class, into a descendant piece of the expected shape and remainders that do not matter". Its components — orthogonal-current, direct-contact and window-boundary bounds — are named, and none of them is proved.

There is a second, independent gap, in the memory corner rather than the soft corner, and it should not be conflated with the first: the two substantive relaxation clauses have no unconditional dynamical instance (Section 9). The finite-window theorem is untouched by it; only the infinite-volume law prints conditionally.

18.4 Open problems

The following are the campaign's live questions, in rough order of how much each would change the picture.

1. **The exhaustive decomposition of the protocol datum.** Prove the hypothesis above, or find the class on which it fails. This is the single step between a subclass result and a general one-dimensional soft limit law.

2. **An unconditional dynamical instance of local relaxation.** Establish the convergence and tightness clauses for the easy-axis kink model. Every ingredient that has been proved so far — the free first clause, the domain-wall number bound, the no-runaway sufficiency lemma — approaches this without reaching it, and the wall-number-versus-wall-length distinction is exactly what stands in the way.

3. **Exhibit a member of the no-contact class.** The class is currently uninhabited as far as anyone has shown, at every density. Two rows — the conditional matched value and the amputation conjecture — predict what the class constant of any member must be, so both are constraints on future work rather than present evidence. Until a member exists, the universality statement is a theorem about a possibly empty class, and the labbook says so.

4. **The amputation constant.** The exact leg-conversion computation supplies only the square root of the order-parameter density factor. The amputation conjecture therefore requires a second factor from a mechanism that is *not* a leg normalisation; any proposed proof must be checked against that computation or it double-counts the leg.

5. **Full-chain asymptotic completeness for the kink model.** The Fano reduction and the soft zero are proved for the projected component only; the leakage off that component is measured but not controlled.

6. **Thermodynamic uniqueness and flatness of the kink family.** One zero-energy kink per regularised-charge sector, with no recoil. This is also what the memory-implies-symmetry edge needs.

7. **The higher-charge factor in the "two 2s" conjecture.** The unit-charge slope half is now a theorem at every site spin and the finite-volume falsifier for charge two was survived; the general charge factor, and the charge-two kink-memory test, are open.

8. **The dynamical SPT instance.** A nonzero edge-changing reflection amplitude on an open interval of momenta, which would turn the rigidity dichotomy into an observable event.

9. **Nonvacuity in two dimensions.** A nonzero model instance of the planar relaxation clauses, and a microscopic pure-endpoint instantiation of the categorical selection theorem beyond the toric code.

10. **The unrun numerical check.** Excitation-ansatz magnon amplitudes against the Bethe amplitude as the momentum goes to zero (Section 17).

18.5 The north star, and how far it stands

The campaign's stated target is a theorem, a verification against exact solvable-model data, and tensor-network numerics on a serious model illustrating the memory effect.

The theorem exists, but it is not the theorem the brief expected. The brief expected a soft theorem whose zero-frequency limit would quantize the memory. What was proved instead is that the memory is quantized *first*, by symmetry alone, with no scattering theory anywhere in the argument — and that the soft theorem's job is reduced to supplying the number that multiplies an already-integral ledger. That inversion of the burden of proof is the campaign's result, and it is a stronger statement than the one it replaces, because the quantization half now stands on an unconditional finite-window theorem instead of on an unproved factorisation.

The verification exists. The exact spin-S two-magnon slope is proved independently of integrability and agrees with the exact solvable-model data where those data exist; the oracle facts were recomputed rather than quoted; and the pre-registered falsifiers — including the charge-two slope test at finite volume — were survived at their declared criteria. Survival is recorded here as evidence, never as proof.

The numerics stand partly built (Section 17). Ground states of the non-integrable running model have been computed across its phases with a calibration point, giving the raw data for the rigidity dichotomy. The kink transport measurement that would show the quantized memory directly on that model was interrupted mid-flight and its partial results are explicitly not trusted; it is the first item to be redone.

So: two corners of the triangle carry proved theorems, the third carries proved theorems on a displayed subclass and one named hypothesis short of the general statement, and all three edges remain below the level of theorems — one conjecture and two sketches, each with its missing step named rather than gestured at. That is a smaller claim than “the infrared triangle is a theorem on the lattice”. It is also, unlike that claim, true.

19 Reduction to the standard continuum setting

19.1 The question

Every physicist who reads this work will ask the same thing first, and they will ask it before they ask anything else:

> Does this reduce, in the right limits, to what we already accept as a memory effect, a soft theorem, and an asymptotic symmetry?

The question is fair and it is answerable. This section answers it in the register in which it was asked: effective-field-theory matching, perturbation theory, stationary phase, collective coordinates. Four reductions are examined and each one ends in a verdict. Two of them come out well, one comes out badly, and one comes out well in a way that requires a correction to be issued rather than a confirmation to be announced. All four verdicts are stated plainly.

Remark 19.1 (Status of everything in this section). The reduction arguments below are *physical arguments, not part of the claims DAG.* They carry no status tag, they are not proved anywhere in the repository, and no result elsewhere in this labbook depends on them. Where a reduction consumes a lattice result, that result is a theorem owned by another section and is cited through its provenance block; where a reduction consumes a continuum equation, that equation is transcribed from the local source. The purpose of the section is to let a reader decide whether the lattice objects are the objects they already know.

Summary of the four verdicts.

soft theorem	Reduces, with caveats. The naive continuum limit is a free theory, but the soft *law* survives it intact and matches the accepted continuum magnon amplitude term by term. The soft-scaling exponent is $\sigma = 1$, not 2.
asymptotic symmetry	Does not reduce as stated. The naive one-dimensional specialisation of the continuum construction is exactly the object this campaign refuted. The charge-splitting half of the corner does reduce cleanly.
memory	The magnonics reduction is clean, with constants. The Fourier-residue reduction reduces in structure but not in content, for a reason the continuum literature predicts. The kink-model transmission coefficient does not reduce at all.
the edges	Reduce, with caveats. Every silent continuum assumption has a named lattice counterpart, and two of them the lattice removes outright rather than renames.

19.2 The soft theorem

What the lattice gives

The lattice side of this reduction is exact. For the fully polarised bilinear ferromagnet at any site spin S, the exact two-magnon coefficient ratio and its soft phase slope are theorems (Section 10): with $z_j = e^{ik_j}$, $a = 1 + z_1 z_2$, $b = z_1 + z_2$ and $\mu = (2S-1)a + b$,

$$S_{12}(k_1, k_2) = \frac{S\,a\,b - z_1\mu}{z_2\mu - S\,a\,b}, \qquad \left.\frac{\partial \delta_{\text{phys}}}{\partial k_s}\right|_{k_s=0} = \frac{\operatorname{sgn}(v_h - v_s)}{S}. \tag{371}$$

The soft magnon decouples in the plain sense that $S_{12} \to 1$; this is a limit, not an Adler-zero theorem.

campaign id:	S2-2body-S, S2-2body, OR2, D6–D8
proved in:	Section 10; `theory/spin-s-twomagnon.md`, `theory/oracle-bethe.md`
tested by:	—
reviewed in:	—

The naive continuum limit is free, and this is verified

Introduce a lattice constant a_ℓ and physical momenta $p = k/a_\ell$, and let $a_\ell \to 0$ at fixed p. At $S = \frac{1}{2}$ the exact phase is known in closed form through the rapidity $\lambda(k) = \frac{1}{2}\cot(k/2)$, and

$$\lambda(a_\ell p) = \frac{1}{a_\ell p} - \frac{a_\ell p}{12} + O(a_\ell^3), \qquad \lambda_1 - \lambda_2 = \frac{p_2 - p_1}{a_\ell\, p_1 p_2} + O(a_\ell), \tag{372}$$

so that

$$\delta_{12} = 2\arctan\frac{1}{\lambda_1 - \lambda_2} = 2\arctan\frac{a_\ell\, p_1 p_2}{p_2 - p_1} \xrightarrow[a_\ell \to 0]{} 0. \tag{373}$$

The rapidity difference diverges like $1/a_\ell$ and the scattering matrix tends to the identity. The naive continuum theory of lattice magnons is free. This is the one-dimensional face of Dyson's vanishing ferromagnetic magnon scattering length, and the continuum effective-theory literature reaches it independently: in the ferromagnet without external fields the scattering length vanishes, as a consequence of the derivative couplings demanded by Goldstone's theorem.

What survives: the doubly soft amplitude

The free limit is not the end of the story, because the soft theorem is not a statement about the size of the interaction. Expanding (371) to second order in *both* lattice momenta, the constant and linear parts of numerator and denominator cancel identically and what remains is

$$\boxed{S_{12}(k_1, k_2) = 1 - \frac{i}{S}\,\frac{k_1 k_2}{k_1 - k_2} + O(k^3).} \tag{374}$$

Two checks. Sending $k_1 = k_s \to 0$ at fixed $k_2 = k_h > 0$ returns the exact slope $1/S$ of (371). Setting $S = \frac{1}{2}$ returns the frozen spin-one-half coefficient 2 of Section 10.

In physical momenta, (374) reads

$$S_{12} = 1 - \frac{i}{M_0}\,\frac{p_1 p_2}{p_1 - p_2} + O(a_\ell^2), \qquad M_0 := \frac{S}{a_\ell}, \tag{375}$$

where M_0 is the spin, or magnetisation, density. Every microscopic parameter has dropped out. The exchange coupling J is absent, and the site spin appears only inside M_0.

Observation 19.2 (The soft slope is an inverse spin density)**.** The exact lattice slope $\mathrm{sgn}(v_h - v_s)/S$ is a slope per *lattice* momentum. Per physical momentum it is

$$\left.\frac{\partial \delta_{\mathrm{phys}}}{\partial p_s}\right|_{p_s = 0} = \frac{\mathrm{sgn}(v_h - v_s)}{M_0}, \tag{376}$$

and as a length — the Wigner displacement that the soft magnon imparts to the hard one — it is $a_\ell/S = 1/M_0$. The combination that stays finite in the continuum is the spin density, and the soft law is the statement that the displacement equals one inverse spin density.

This resolves the apparent paradox of (373). A literal $a_\ell \to 0$ at fixed S sends $M_0 \to \infty$, that is, it describes an infinitely stiff magnet, and of course the interaction vanishes there. In a real magnet M_0 is the saturation magnetisation, a finite measurable number, and the magnon effective theory is a low-energy expansion with a physical cutoff of order $1/a_\ell$ rather than a continuum limit. The soft slope is then a perfectly ordinary dimensionful Wilson coefficient.

Matching to the continuum magnon effective theory

Model the magnon as a Schrödinger field with $\omega(p) = p^2/2m$. From $\omega_S(k) = 2JS(1-\cos k) \simeq JSk^2$ one reads $m = 1/(2JSa_\ell^2)$. In one dimension the Born-level two-body scattering matrix for identical bosons is $S - 1 = imM/\big(2|p_1 - p_2|\big)$ with M the on-shell $2 \to 2$ amplitude, so (375) matches onto

$$M(p_1, p_2) = -\frac{2p_1p_2}{m\,M_0} = -4Ja_\ell^3\,p_1p_2. \tag{377}$$

Two consequences. First, writing the leading contact interactions as

$$\mathcal{L}_{\text{int}} = -\frac{g_0}{2}\big(\psi^\dagger\psi\big)^2 - \frac{g_2}{4}\Big[\big(\partial_x\psi^\dagger\big)^2\psi^2 + \text{h.c.}\Big] + \cdots\,, \tag{378}$$

the matching forces $g_0 = 0$ and $g_2 \propto 1/(mM_0) = 2Ja_\ell^3$: the scattering-length operator has vanishing Wilson coefficient and the leading magnon self-interaction carries two derivatives. That is Dyson's theorem, obtained here as a consequence of (374) rather than as an input. Second, the accepted continuum computation for the ferromagnet, with dispersion $\omega = (F^2/\Sigma)\boldsymbol{k}^2$, gives the tree amplitude

$$M\big[\pi(\boldsymbol{k}_1) + \pi(\boldsymbol{k}_2) \to \pi(\boldsymbol{k}_3) + \pi(\boldsymbol{k}_4)\big] = \frac{2F^2}{\Sigma^2}\,\big(\boldsymbol{k}_1\cdot\boldsymbol{k}_2\big), \tag{379}$$

stated in the source to agree with Dyson's microscopic analysis. The functional form of (377) and (379) is the same, with no constant piece in either; identifying Σ with the spin density M_0 and F^2 with the spin stiffness JS^2a_ℓ makes every parametric dependence agree, and leaves an overall factor of two that depends on normalisation conventions this section has not re-derived. That factor is recorded as an open point below.

Do the soft and continuum limits commute?

They do, and it is worth saying why the question has a slightly disappointing answer. The lattice constant enters the exact scattering matrix only through the change of variables $k = a_\ell p$, so there is nothing available to fail. On the exact spin-one-half function (373), both orders of the two limits give zero, and both orders of the two limits applied to the slope give zero as well.

The limit that genuinely fails to commute is a different pair, and it is the one that carries the physics.

Observation 19.3 (Soft against hard, not soft against continuum)**.** From (374),

$$\lim_{k_h\to 0}\left.\frac{\partial\delta_{\text{phys}}}{\partial k_s}\right|_{k_s=0} = \frac{1}{S}, \qquad \left.\frac{\partial}{\partial k_s}\Big[\lim_{k_h\to 0}\delta_{\text{phys}}\Big]\right|_{k_s=0} = 0. \tag{380}$$

The two orders differ by the entire content of the theorem.

This is the same non-uniformity that Section 10 records when it fixes the hard momentum away from the zone centre, and it is visible in the quadratic coefficient $|v_h|/\omega_h = \cot(k_h/2)$, which diverges as $k_h \to 0$. Its physical reading is that "soft" is a statement about the *ratio* $k_s/k_h = p_s/p_h$, which does not know about a_ℓ at all. That is exactly why the soft law survives a limit in which the scattering matrix becomes trivial: the theorem constrains a scale-invariant ratio, and only the overall size of the interaction carries the factor of a_ℓ.

Where the type-B bound sits, and a correction

The accepted continuum route to a soft theorem for a non-relativistic Goldstone runs through current conservation. With the broken current matrix element written as

$$\langle\Omega|J^\mu(x)|\theta(\boldsymbol{p})\rangle = e^{-ip\cdot x}\big[\,ip^\mu F_1(|\boldsymbol{p}|) + i\delta^{\mu 0}F_2(|\boldsymbol{p}|)\,\big], \qquad \omega^2 F_1 + \omega F_2 - \boldsymbol{p}^2 F_1 = 0, \tag{381}$$

the current matrix element between scattering states factorises on the Goldstone pole,

$$\langle\beta|J^\mu(0)|\alpha\rangle = \frac{i}{p^0-\omega(|\boldsymbol{p}|)}\langle\Omega|J^\mu(0)|\theta(\boldsymbol{p})\rangle\langle\beta+\theta(\boldsymbol{p})|\alpha\rangle + R^\mu(p), \tag{382}$$

and current conservation cancels that pole exactly, leaving

$$\langle\beta+\theta(\boldsymbol{p})|\alpha\rangle = \left.\frac{p^0R_0(p)+p^rR_r(p)}{\omega(|\boldsymbol{p}|)F_1+F_2}\right|_{p^0=\omega(|\boldsymbol{p}|)}. \tag{383}$$

The Adler zero then follows only under the additional assumption — flagged in the source as not following from standard polology — that $R^\mu(p)$ is non-singular as $\boldsymbol{p}\to 0$ on shell. Counting powers in (383) with a soft-scaling exponent σ defined by $\mathcal{A}\propto\varepsilon^\sigma$ under $\boldsymbol{p}\mapsto\varepsilon\boldsymbol{p}$ gives the two bounds

$$\sigma\geq\min(m,n+1)\ \ (\text{type } A_m,\ d\geq m+1), \qquad \sigma\geq\min(2m,n+1)\ \ (\text{type } B_{2m}), \tag{384}$$

where n is the degree of the theory's generalised *spatial* shift symmetry, that is, the degree of the polynomial $\alpha(x)=\epsilon_{\nu_1\ldots\nu_n}x^{\nu_1}\cdots x^{\nu_n}$ under which the Goldstone field shifts.

The ferromagnet magnon is type B_2, so $m=1$. Its broken generators produce only the ordinary constant shift of the Goldstone field, so $n=0$, and (384) gives $\sigma\geq\min(2,1)=1$. The exact lattice amplitude (374) vanishes *linearly* in either soft momentum, so $\sigma=1$ exactly. The mechanism is transparent in (383): with a quadratic dispersion the temporal piece $p^0R_0=\omega R_0$ is already of second order, but the spatial piece p^rR_r is of first order unless a spatial shift symmetry of degree at least one forces $R_r(0)=0$, and the Heisenberg ferromagnet has no such redundant symmetry.

Remark 19.4 (Correction to the campaign's reading of the type-B bound). An earlier internal digest recorded the expectation that the lattice soft theorem should realise the enhanced type-B value $\sigma=2$, and glossed the type-B bound as needing no additional shift symmetry. Both readings are wrong. Linear vanishing *is* $\sigma=1$. The two type-B_2 theories that do reach $\sigma=2$ in the source do so because they possess a linear spatial generalised shift symmetry, that is $n=1$, whereupon $\min(2m,n+1)=2$. What is genuinely special about type B is that $n=1$ *suffices* to reach $\sigma=2$, whereas type A_1 with $n=1$ still obtains only $\min(1,2)=1$. The lattice ferromagnet has $n=0$ and lands on $\sigma=1$, in agreement with the bound and with the exact two-magnon computation. No proved statement of this campaign changes; only the expectation was mistaken.

Observation 19.5 (Verdict for the soft theorem)**.** *Reduces, with caveats.* The exact lattice soft law reduces to the accepted continuum statement — a derivative-coupled type-B_2 Goldstone with vanishing scattering length, tree amplitude proportional to $\boldsymbol{k}_1\cdot\boldsymbol{k}_2$, and an ordinary Adler zero — with the slope $1/S$ per lattice momentum becoming one inverse spin density per physical momentum. The caveats are that the literal continuum limit at fixed site spin is free and the physical statement therefore requires a finite lattice constant; that the soft-scaling exponent is 1 and not 2; and that an overall factor of two in the amplitude match is a normalisation question left unchecked.

19.3 The asymptotic symmetry

The specialisation the continuum construction makes

A continuum asymptotic-symmetry construction assigns one charge per function on the celestial sphere. In one spatial dimension the celestial sphere degenerates to two points, left and right infinity, a function on it is a pair (g_L, g_R), and the diagonal subgroup acts trivially on the physical content of the pair. The specialisation is therefore

$$\mathcal{A} = (G_L \times G_R)/G_{\text{diag}}, \qquad \text{complete invariant} \quad [\, g_L g_R^{-1} \,]. \tag{385}$$

This is the correct guess, and it is the group this campaign started from.

It is the object this campaign refuted

The refutation is recorded in Section 5: the orbit of (385) is *not* the set of vacuum pairs, and $[\, g_L g_R^{-1} \,]$ is *not* the complete invariant. What the lattice supplies in its place are two different objects. In the unbroken case the orbit is $\mathcal{A}_{\text{eff}} = G/N_\alpha$, where N_α is the kernel of the virtual representation $\rho_\alpha \colon G \to \mathrm{PGL}(\chi)$ carried by the bond space; the stabiliser contains the diagonal and equals it exactly when N_α is trivial. In the broken case, and under transitivity, vacuum pairs form $(G/H_\alpha) \times (G/H_\alpha)$ with stabiliser $H_\alpha \times H_\beta$, and the complete diagonal invariant is a *double* coset in $H_\alpha \backslash G / H_\alpha$.

campaign id:	A2-orbit-r1 (refuted), A1(e), A2(e)
proved in:	Section 5; `theory/corner-a.md`, `theory/corner-a-kinks.md`
tested by:	`theory/checks/corner_a_check.py` C6–C8, C7, C11
reviewed in:	`theory/verdicts/corner-a-r3.md`

Observation 19.6 (Where the two pictures agree and where they part)**.** When the virtual representation is faithful and the symmetry is completely broken, N_α and H_α are trivial, both lattice classifiers collapse to G, and (385) is recovered. The continuum answer is therefore the lattice answer in the structureless case. The lattice adds three things with no continuum counterpart: the finite kernel N_α, produced by a projective representation on a finite-dimensional bond space that the continuum construction has no analogue of; the vacuum stabiliser H_α, which turns single cosets into double cosets; and the cohomology class $[\omega_\alpha] \in H^2(G, U(1))$, an invariant of the *representation* rather than of the group, which obstructs removing the multiplier and has no image in (385) at all. Conversely, the continuum has one piece of machinery with no lattice image, the logarithmic celestial Goldstone correlators, which rest on representation theory of the massless little group and should not be sought here.

The soft and hard charges do reduce

The other half of the corner reduces cleanly, and it is the half a referee will recognise immediately. The continuum construction defines a soft charge as the zero-frequency antisymmetric combination of Goldstone creation and annihilation operators, adds a hard charge built as a bilinear in the matter operators with kernel $q^\mu p^\nu \gamma_{\mu\nu}/(p\cdot q)$, and proves that the sum commutes with the scattering matrix between physical states. Its lattice mirror is built from proved machinery.

The modulated charge $Q[f;\xi] = \sum_x f(x) q_x(\xi)$ obeys the exact continuity identity

$$\big[H,\, Q[f;\xi]\big] = \sum_x (\Delta f)(x)\, j_{x|x+1}(\xi) \tag{386}$$

for every Lie-algebra element and every finite-range Hamiltonian, with the second difference of the profile in the role of the derivative of the celestial function; this is the lattice statement of current conservation and it involves no boundary at infinity. Acting with $Q[f;\xi]$ on the vacuum creates a magnon whose amplitude vanishes like the momentum, which is the soft charge; acting with the same charge on a prepared hard magnon deforms the hard leg, and the *failure* of the soft charge to be absorbed is exactly the soft factor,

$$Q_{k_s}|k_h\rangle - |B^{\mathrm{in}}\rangle = \big(1 - S_{12}\big)|P_{12}\rangle = -2ik_s|P_{12}\rangle + O(k_s^2) \qquad \text{at } S = \tfrac{1}{2}. \tag{387}$$

This is the lattice form of “soft charge plus hard charge commutes with the scattering matrix”, with the exact slope of Section 10 in the role of the continuum spin-rotation kernel. In both settings the soft factor is of order one and not of order $1/\omega$: the lattice sits in the soft-pion universality class, where the leading pole is absent, not in the photon or graviton class.

campaign id:	G0(e), G0(c), WI, AC-EX-2M-D29, S2-2body
proved in:	Section 4, Section 5, Section 10
tested by:	`theory/checks/corner_a_check.py` C3–C5, C9, C10
reviewed in:	—

Observation 19.7 (Verdict for the asymptotic symmetry)**.** *Does not reduce as stated.* The naive one-dimensional specialisation of the continuum asymptotic-symmetry group is literally the statement this campaign refuted. The correct lattice classifiers reduce to it in the structureless case, and the difference between them is precisely the structure the continuum construction cannot see. The charge-splitting half of the corner — continuity equation, Ward identity, soft and hard charges — reduces cleanly. For an external presentation this means Corner A should be offered as a correction of the naive specialisation, not as its confirmation.

19.4 The memory effect

There are two different accepted continuum statements here, and the lattice must be mapped onto both. They are not the same statement and they do not fare the same way.

The magnonics reduction, with constants

The accepted continuum result is that a spin wave passing a transverse domain wall in an easy-axis ferromagnet drives the wall. The linearised Landau–Lifshitz problem around the Walker profile becomes a Schrödinger equation with a reflectionless Pöschl–Teller potential,

$$q^2\varphi(\xi) = \Big[-\frac{d^2}{d\xi^2} - 2\,\mathrm{sech}^2\xi\Big]\varphi(\xi), \qquad \xi = \frac{z}{\Delta_W}, \qquad q^2 = \frac{\omega}{K} - 1, \tag{388}$$

whose scattering solution has unit transmission and a pure phase shift. The magnon's spin is $-\hbar$ on one side of the wall and $+\hbar$ on the other, so each transmitted magnon deposits $2\hbar$ of angular momentum on the wall, and the wall must move against the spin-wave direction at

$$V_{\mathrm{DW}} = -\frac{\rho^2}{2}\,V_g, \qquad V_g = \frac{\partial\omega}{\partial k}, \tag{389}$$

with ρ the dimensionless spin-wave amplitude.

The lattice side is the windowed memory theorem of Section 7. With the frozen windowed wall coordinate $\mathfrak{X}_W$ and asymptotic densities $\pm s$, charge conservation gives the displacement and the ledger

$$\delta x = -\frac{\langle N_T\rangle}{s}, \qquad 2s\,\delta x + \big(q_{\mathrm{out}} - q_{\mathrm{in}}\big) = 0, \qquad q_{\mathrm{in}} = -1, \quad q_{\mathrm{out}} = +1. \tag{390}$$

campaign id:	M-quant, M-quant-G, M-flux, D13(a), D14, D17, D18
proved in:	Section 7; `theory/memory-quantization.md`, `theory/memory-quantization-general.md`
tested by:	`theory/checks/mquant_check.py`, `theory/checks/mquant_general_check.py`
reviewed in:	corpus-r2, `theory/verdicts/mquant-g-r2.md`

These are the same equation. The lattice arithmetic $q_{\text{out}} - q_{\text{in}} = 2$ *is* the continuum statement that the magnon spin flips from $-\hbar$ to $+\hbar$. To see it, write $M_0 = s/a_\ell$ for the spin density — the same quantity that appeared in Observation 19.2 — and $\delta X = a_\ell\,\delta x$ for the physical displacement. Then $2s\,\delta x = 2M_0\,\delta X$ and (390) becomes

$$2M_0\,\delta X + \left(q_{\text{out}} - q_{\text{in}}\right) = 0 \quad\Longrightarrow\quad \delta X = -\frac{1}{M_0} \;\text{ per transmitted magnon,} \tag{391}$$

and, at a transmitted-magnon flux Φ,

$$V_{\text{DW}} = -\frac{\Phi}{M_0}. \tag{392}$$

It remains to evaluate the flux for the continuum spin wave. The magnetisation is a unit vector with transverse amplitude ρ in the asymptotic domains, so $m_z \simeq 1 - \rho^2/2$; each magnon removes one unit of spin, so the magnon number density is $n = M_0\rho^2/2$. The wall is reflectionless, so $\Phi = nV_g$, and (392) gives

$$V_{\text{DW}} = -\frac{\rho^2}{2}\,V_g, \tag{393}$$

which is (389) exactly, constants included, independently of the exchange coupling, the anisotropy, the wall profile and the magnon momentum.

Remark 19.8 (The two displacements are one inverse spin density). The physical displacement per transmitted magnon in (391) and the Wigner displacement of Observation 19.2 are numerically the same length, $1/M_0$. This gives the campaign's "two twos" cross-reference a continuum reading: both quantities are one inverse spin density. It is an observation about units and a physical argument only; the corresponding conjecture is unaffected and remains a conjecture.

The Fourier-residue chain

The canonical continuum argument writes the memory as the residue of a zero-frequency pole. Fourier transforming the radiative field by stationary phase and assuming that it approaches finite but different values at early and late retarded times, one obtains the memory as the coefficient of the pole in frequency, and identifies that coefficient with the soft factor. In slogan form, the Fourier transform of a pole in frequency space is a step function in time.

The residue *structure* reduces correctly. The exact finite-time flux identity of Section 7 reads

$$\delta x = \frac{1}{2s}\Big[\tilde{\jmath}_{a-1|a}(0) - \tilde{\jmath}_{b|b+1}(0)\Big], \tag{394}$$

which is precisely the zero-frequency weight of the physical boundary current, obtained by exact telescoping of the windowed coordinate; and the spectral dress of the memory observable writes the same number as a genuine direct-current residue, $\delta x = \frac{1}{2s}\sum_{x\in W}\big[m_x(+\infty) - m_x(-\infty)\big]$.

What does *not* reduce is the identification of that residue with the soft factor. The continuum chain works because the product of frequency and radiative field has a finite nonzero limit; there is a genuine pole, supplied by Weinberg's soft factor. The lattice soft factor has no pole: the scattering ratio tends to one and the soft-scaling exponent is 1. Multiplying a regular function by a vanishing frequency gives zero. The campaign's literal memory conjecture — displacement equals the zero-frequency limit of the soft factor — is refuted for exactly this reason, and what survives is the boundary-current weight (394) together with the conditional quantisation (390), into which soft data enter only through the transmission probability.

campaign id:	M (refuted), M-flux, D13(b)
proved in:	Section 7; `theory/memory-quantization.md` §1
tested by:	`theory/checks/mquant_check.py`
reviewed in:	corpus-r2

This is not a lattice pathology. It is what the continuum literature predicts for a derivative-coupled Goldstone. In the soft-pion triangle the leading soft theorem is absent and the memory consequently lives in the *subleading* falloff coefficient of the pion field, where it equals a linear functional of the escaped hard charge. The lattice reproduces that pattern exactly.

Observation 19.9 (Which lattice observable is which falloff coefficient)**.** The leading radiative coefficient of the continuum field corresponds to the outgoing one-magnon wave amplitudes on the two legs, that is to the reflection and transmission amplitudes; these carry the phase and no direct-current shift. The subleading coefficient, where the pion memory lives, corresponds to the *integrated* asymptotic tail content $\sum_{x>b}\big(\varrho(S^z_x)+s\big) = 2sN_T$. The continuum hard charge corresponds to $q_{\mathrm{out}} - q_{\mathrm{in}} = 2N_T$, and the continuum statement that the subleading coefficient equals a linear functional of the hard charge corresponds to $\delta x = -\langle N_T\rangle/s$. Both sides say the same thing: memory is a linear functional of the integrated hard charge that escaped, not the residue of a soft pole.

Observation 19.10 (What plays the role of the falloff assumption)**.** The continuum needs the radiative field to approach finite but different values at early and late retarded times, and it needs the frequency and radius limits taken in a definite order. The lattice counterparts are named hypotheses rather than footnotes: the summability class of Section 9, which requires the deviation of the magnetisation profile from its two asymptotic values to be summable on each side, is the lattice "finite but different values"; the wave-operator and local-decay hypothesis is the lattice "the radiation has left"; and the spectral dress requires the time derivative of each site magnetisation to be integrable in time. The lattice also states its own order-of-limits fence in the same place, and it is the exact counterpart of the continuum footnote: a plane-wave magnon is not summable, so every soft statement about memory must fix the wave packet first and take the momentum to zero afterwards, and the zero-frequency limit must be taken after the thermodynamic limit.

The kink model in its continuum scaling limit

The easy-axis kink model of Section 6 carries an unconditional projected transmission coefficient,

$$t(k) = \Big[1 + \frac{iJ^2}{4\,\omega(k)\,v(k)}\Big]^{-1}, \qquad T(k) = 16(\Delta-1)^2k^2 + O(k^4) \ \text{ as } k \to 0, \tag{395}$$

with crossover momentum $k_* = 1/\big(4(\Delta-1)\big)$. The natural continuum comparison is the calculation (388). The comparison has one clean half and one bad half.

The clean half is the dispersion. Writing the kink parameter $q = \Delta - \sqrt{\Delta^2-1}$ and $\Delta = 1+\epsilon$, the wall width in lattice units is $\xi_{\mathrm{DW}} = 1/\ln(1/q) \simeq 1/\sqrt{2\epsilon}$, matching the continuum $\Delta_W = \sqrt{A/K}$ up to a factor $\sqrt{2}$ under $A \sim Ja_\ell^2$ and $K \sim J(\Delta-1)$; and in the scaling limit $\Delta \to 1^+$, $k \to 0$ with $q_c := k\,\xi_{\mathrm{DW}}$ fixed,

$$\omega(k) = J(\Delta - \cos k) \simeq \omega_{\mathrm{gap}}\big(1+q_c^2\big), \tag{396}$$

which is exactly the continuum relation $q^2 = \omega/K - 1$. The lattice magnon sector reduces cleanly to the continuum Landau–Lifshitz magnon on a wide wall.

The bad half is the transmission. In the same scaling limit $v = J \sin k \simeq Jq_c\sqrt{2\epsilon}$ and $\omega \simeq J\epsilon(1+q_c^2)$, so

$$\frac{J^2}{4\omega v} \simeq \frac{1}{4\sqrt{2}\,\epsilon^{3/2} q_c(1+q_c^2)} \longrightarrow \infty, \qquad T \simeq \left[4\sqrt{2}\,\epsilon^{3/2} q_c(1+q_c^2)\right]^2 \longrightarrow 0. \tag{397}$$

The projected lattice wall becomes totally reflecting exactly where the continuum wall is exactly reflectionless.

Remark 19.11 (Why the transmission comparison fails, and what it costs)*.* The two computations are controlled in opposite regimes. The kink at $\Delta \to 1^+$ is a wide object: its exact zero-energy product family has $q \to 1$, and in the magnetisation product basis it is a broad superposition over configurations carrying many domain walls. The sector reduction that makes (395) exact keeps only the three-wall component, and the measured leakage into five-wall configurations grows like the inverse square of the anisotropy, from below one per cent at $\Delta = 8$ to about ten per cent at $\Delta = 2$; extrapolated to $\Delta \to 1^+$ it is of order one. Physically, the continuum reflectionlessness comes from the wall's translational zero mode sitting exactly at threshold, which is what makes the potential in (388) transparent; the truncated model does have a side-coupled kink level, but its detuning stays finite as the momentum vanishes, so the threshold cancellation is absent. At large anisotropy the two do agree: the lattice reflection probability falls off like the inverse square of the anisotropy and the wall is transparent for all momenta above the small crossover scale, which is the continuum reflectionless answer. The cost is a presentational one and it should be paid openly: (395) must not be offered as the lattice version of magnon-driven wall motion. The memory law (390) is that, it is a conservation law, and it is independent of the transmission probability; the transmission probability only sets the size of the effect, through the expected transmitted number, in the Ising regime.

Observation 19.12 (Verdict for memory)**.** The magnonics reduction *reduces cleanly*, with constants, and it is the strongest reduction in this section; its one caveat is that the lattice statement is conditional on an asymptotic-completeness hypothesis which the continuum derivation replaces by a closed-form solution, so that the lattice assumes what the continuum computes. The Fourier-residue reduction *reduces with caveats*: the direct-current structure and the falloff hypotheses have exact counterparts, but the identification of the memory with the residue of the soft factor does not reduce, and the campaign's refutation of that identification is what the continuum literature predicts for a derivative-coupled Goldstone. The transmission coefficient of the kink model *does not reduce.*

19.5 The edges

From Ward identity to soft theorem

The cleanest continuum precedent proves that the asymptotic-symmetry derivation of a soft theorem and the textbook broken-Ward–Takahashi derivation are the same derivation. Its recipe is short: take the Ward–Takahashi identity for the broken current, Fourier transform to an on-shell massless momentum, apply the reduction formula to the remaining fields, and take the soft limit; the contact terms on the right-hand side drop out, one term on the left gives the amplitude with one soft Goldstone, and the matter term gives the soft factor.

The lattice runs the same four steps, and the first two are theorems. The continuity identity (386) is the lattice conservation law and needs no boundary at infinity, matching the local continuum programme rather than a null-infinity construction. The truncated-symmetry identity of Section 5 telescopes the symmetry to exactly two boundary-bond insertions, which is the lattice contact term; and the lattice version is finite and computed, with both boundary terms retained, since discarding them is demonstrably false at any finite window. The third step replaces the plane-wave Fourier transform by momentum superpositions of window vectors and the reduction formula by the wave-operator construction of Section 11. The fourth step is a theorem at two legs: on the separated-preparation subclass the fixed-time readout has the exact first jet with phase slope $\mathrm{sgn}(v_h - v_s)/S$.

The caveat is the load-bearing one, and it is stated in Section 18: at n legs this is the general soft conjecture, and the hypothesis that supports it has no exhibited instance. The lattice therefore reproduces the continuum pattern as a template, and as a theorem at two legs, not as a theorem at n legs.

From soft theorem to memory

This edge does not reduce, and the reason has already been given: on the lattice the soft factor has no pole, so the Fourier residue of the soft factor is zero, and the corresponding conjecture is refuted. The surviving lattice edge runs through charge conservation instead, which is the soft-pion pattern rather than the graviton pattern. An external presentation may still use the slogan about poles and step functions expositorily, provided it says in the same breath that the lattice pole is absent and that the memory is therefore carried by integrated charge rather than by a residue.

The silent assumptions, matched

Every continuum derivation quoted in this section makes assumptions it does not prove. The following correspondence records, as fact, which named lattice hypothesis occupies each slot.

continuum assumption	**lattice counterpart, with status**
The remainder function is non-singular on shell as the momentum vanishes; the source flags this as not following from standard polology.	The remainder bounds of the uninstantiated reduction hypothesis, and the continuous-extension clause of the universality class. The telescoping identity is the mechanism intended to *prove* rather than assume this. The hypothesis has no exhibited instance; this is the campaign's advertised technical advantage and it is not yet cashed.
No singularities beyond the Goldstone pole; no bilinear current insertions on external legs.	The claim that the naive kinematic prefactor is a universal soft factor is [REFUTED]: hard data can enter at first order through the current. The lattice found the loophole rather than assuming it shut.
The radiative field approaches finite but different values at early and late retarded times.	The summability class of Section 9, the wave-operator and local-decay hypothesis, and integrability in time of each site magnetisation rate. The local-decay hypothesis is open on the full chain.
The product of frequency and radius is large when the limits are taken.	The order-of-limits fence: fix the wave packet, then take the momentum to zero; take the zero-frequency limit after the thermodynamic limit. Frozen in the definitions.
Stationary-phase mode expansion at large radius, with no zeroth-order term in the falloff expansion.	The wave-packet construction of Section 11, with disjoint velocity supports. Established as a construction; its interface with the fixed-time readout has named gaps.
No surface terms in the Ward identity.	The truncated-symmetry identity keeps both boundary-bond insertions *explicitly*, and setting them to zero is recorded as false at finite window. [PROVED], with the surface terms computed.
A hyperbolic foliation with prescribed falloff near timelike infinity, needed to make the hard charge a finite integral.	The windowed wall coordinate is a bounded local observable unconditionally, so the displacement exists on every state and at finite volume, with no asymptotic hypothesis and no order-of-limits clause. This slot the lattice removes rather than renames.
A physical-mode condition and a transversality condition, imposed by hand.	The admissible-profile conditions on the charge profile, with the type discipline that forbids quoting a fixed-momentum identity under a decaying-profile hypothesis. Frozen in the definitions.

Observation 19.13 (Verdict for the edges)**.** *Reduce, with caveats.* Every silent continuum assumption has a named lattice counterpart, and in two slots — windowed memory and surface terms — the lattice eliminates the assumption rather than renaming it. The caveats are that the symmetry-to-soft edge is a theorem only at two legs; that the soft-to-memory edge does not run

through a Fourier residue of the soft factor and must not be advertised as doing so; and that the regularity slot, the one this campaign claims as its chief contribution relative to the continuum, is still occupied by a hypothesis with no exhibited instance.

19.6 The definitional audit: what the discrete and continuous definitions say about each other

The four reductions above ask whether the campaign's *results* land on the accepted continuum statements. They do not answer a sharper and prior question: whether the campaign's *definitions* denote the objects whose names they borrow. That question was put separately, and it was put in the form that makes it hardest to dodge — compare each definition, clause by clause, with compact Hamiltonian lattice electrodynamics in three spatial dimensions, which is a regulator in which the Maxwell photon and the Weinberg pole demonstrably exist. Four independent audits were run, one of them hostile and run without sight of the other three. They agree on every overlapping item. This subsection records what they found.

Nothing here changes a status. No lane asked for one and none was needed: the statuses in the claims DAG were already scoped honestly. What the audit changes is the vocabulary the campaign is entitled to use.

Remark 19.14 (Register of this subsection)*.* As with the rest of this section, everything below is physical argument. Where a lattice-gauge fact is standard it is used as such; where a continuum equation appears it is transcribed from the local sources; where a limit is asserted it is asserted, not proved. The audit's value is negative and definitional: it tells the reader which sentences about this campaign would be false.

The Gauss-law nucleus, and the exact statement it supports

Fix a finite oriented cubic lattice of spacing a_ℓ. Each link carries the rotor Hilbert space $L^2(U(1))$, with holonomy $U_\ell = e^{i\theta_\ell}$ and integer electric flux $E_\ell = -i\,\partial/\partial\theta_\ell$, so that

$$[E_\ell, U_{\ell'}] = \delta_{\ell\ell'} U_\ell, \qquad \operatorname{spec} E_\ell = \mathbb{Z}. \tag{398}$$

Matter of integral charge sits on vertices with charge operator ρ_x. Write $s_{x\ell} = +1$ if x is the tail of ℓ, -1 if it is the head, and 0 otherwise. The Gauss operator and the physical subspace are

$$G_x = (\operatorname{div} E)_x - \rho_x, \qquad (\operatorname{div} E)_x = \sum_\ell s_{x\ell} E_\ell, \qquad G_x P_{\text{phys}} = 0. \tag{399}$$

The first thing the audit establishes is which lattice-QED operator the campaign's regional symmetry is allowed to be compared with, and it is not the obvious one. The full local gauge transformation $\Gamma[\varepsilon] = \exp\bigl(i \sum_x \varepsilon_x G_x\bigr)$ is a redundancy: $\Gamma[\varepsilon] P_{\text{phys}} = P_{\text{phys}}$, so its orbit is a point and it cannot be the nontrivial transplant of a truncated on-site symmetry. The correct counterpart is the matter-only regional rotation

$$W_R(\alpha) = \exp\Bigl(i\alpha \sum_{x\in R} \rho_x\Bigr), \tag{400}$$

the gauged descendant of a global on-site matter symmetry. This distinction carries the whole comparison.

With it, the match is exact and not merely suggestive. The finite-graph divergence theorem is an operator identity, every interior link occurring once with each sign,

$$\sum_{x\in R}(\operatorname{div} E)_x = \sum_{\ell\in\partial R}\sigma_{R\ell}E_\ell, \qquad \sigma_{R\ell}=\pm 1 \text{ as } \ell \text{ exits or enters } R, \tag{401}$$

and combining (399) with (401) on the physical subspace gives

$$\boxed{W_R(\alpha)P_{\text{phys}} = \prod_{\ell\in\partial R} e^{i\alpha\sigma_{R\ell}E_\ell}\,P_{\text{phys}}.} \tag{402}$$

This is the gauge-theory avatar of the truncated-symmetry identity of Section 5. In one spatial dimension the entering left link contributes $e^{-i\alpha E}$ and the exiting right link contributes $e^{+i\alpha E}$, which is exactly the orientation of the two virtual bond insertions — the inverse implementer on the left cut, the implementer on the right cut. The dictionary is one line,

$$V_\alpha\big(e^{i\alpha}\big)^{\sigma_{R\ell}} \;\longleftrightarrow\; e^{i\alpha\sigma_{R\ell}E_\ell}, \tag{403}$$

and it is quantitative, not pictorial. Two differences must be stated with it: the lattice-QED equality holds only after projection onto the Gauss-law physical subspace, whereas the campaign identity is a tensor identity before any local constraint is imposed; and the campaign identity starts from a genuine global on-site symmetry, with no local constraint anywhere in sight.

For an angle-dependent profile the identity acquires the term that the continuum calls soft. Summation by parts against a real profile ε_x, with $(d\varepsilon)_\ell=\varepsilon_y-\varepsilon_x$ on an internal link $\ell=(x\to y)$, gives the three operators

$$Q^\partial_R[\varepsilon]=\sum_{\ell\in\partial R}\sigma_{R\ell}\varepsilon_{x_{\text{in}}(\ell)}E_\ell, \qquad Q^H_R[\varepsilon]=\sum_{x\in R}\varepsilon_x\rho_x, \qquad Q^\nabla_R[\varepsilon]=\sum_{\ell\subset R}(d\varepsilon)_\ell E_\ell, \tag{404}$$

and Gauss's law then supplies the finite-spacing soft/hard dictionary

$$\boxed{Q^\partial_R[\varepsilon]\,P_{\text{phys}} = \big(Q^H_R[\varepsilon]+Q^\nabla_R[\varepsilon]\big)P_{\text{phys}}.} \tag{405}$$

For constant ε the gradient term vanishes and (405) collapses to (402). The converse statement is the honest one and it is worth displaying, because the naive slogan is false: the modulated matter rotation on its own obeys

$$e^{iQ^H_R[\varepsilon]}P_{\text{phys}} = e^{i\left(Q^\partial_R[\varepsilon]-Q^\nabla_R[\varepsilon]\right)}P_{\text{phys}}, \tag{406}$$

which is *not* boundary-only when $d\varepsilon\neq 0$. A claim that every modulated on-site operation telescopes purely to the cut would be wrong.

Observation 19.15 (The arrow to the accepted soft/hard split, and what it costs)**.** Under a limit sequence that keeps the ultraviolet, thermodynamic and null stages separate — send the spacing to zero at fixed physical sphere radius R, box size L and retarded time u, then $L\to\infty$, then $R\to\infty$ at fixed $u=t-R$, then $u\to-\infty$ — the three operators of (404) become

$$Q^\partial \longrightarrow Q^+_\varepsilon, \qquad Q^H \longrightarrow Q^+_H, \qquad Q^\nabla \longrightarrow Q^+_S, \tag{407}$$

with $Q^+_\varepsilon = Q^+_S + Q^+_H$ the single sphere-flux charge of the continuum construction, not a sphere flux plus a further hard part. Each arrow is a physical argument resting on Riemann-sum control of the flux, the prescribed falloffs, and the null-boundary assumptions; none is a theorem of this campaign. Two fences travel with them. At finite spacing Q^{∇} should be called a gradient term and not a soft-photon operator: softness needs the null and zero-frequency limits that come later, and the zero mode is itself defined as a frequency limit, $N_z = \lim_{\omega\to 0} \int du\, F^{(0)}_{uz} e^{i\omega u}$. And for massive charged matter the hard charge does not all cross null infinity, so the decomposition is null hard plus timelike hard plus soft, not the two null integrals of (407).

campaign id:	WI, A1, A2, D3, D4, D10
proved in:	Section 5, Section 4; audit in `theory/lanes/reduction/q1-gauss.md` §1–§3 and `theory/lanes/reduction/q4-adversarial-defs.md` §5
tested by:	—
reviewed in:	`theory/verdicts/reduction-defs-adjudication-r1.md` (N2)

The honest converse: the bond-implementer package does not transplant

Equation (402) is the one continuum-facing statement of the symmetry corner that survives intact. The rest of the package does not, and the failure is not a matter of technical difficulty: no limit sequence inside the bond-implementer definition repairs it, because four separate pieces of structure are wrong. Displaying them is the point.

what fails	why, at the level of the definition
the quotient	The campaign divides the symmetry group by $N_\alpha = \{g : \rho_\alpha(g) \text{ scalar}\}$, a state-level projective kernel: a phase invisible on one bond space. Lattice QED divides *allowed* boundary transformations by those acting *trivially on physical data*. A constant phase can be scalar on one fixed charge sector and still be physically measurable as a character across charged sectors, where it enters the hard soft factor. The two quotients discard different things.
locality	The campaign residue is a virtual insertion defined only through padded windows of a matrix-product state. The boundary flux $\sum_{\ell\in\partial R} \sigma_{R\ell} E_\ell$ is a physical boundary-link observable — or, after a subregion factorisation choice, an edge-mode generator. No canonical isomorphism identifies a χ-dimensional virtual space with $L^2(U(1))$ on a cut link.
boundary multiplicity	The campaign has one finite group at each of two ends of a one-dimensional cut. A three-dimensional region has $O(R^2/a_\ell^2)$ independent boundary samples and tends to a function group on the sphere; the continuum has one conserved charge for *every* boundary function. That enlargement is not contained in the definition as written, and the eventually-constant profile class, which has only two ends, cannot encode an angular function at all.
spectrum	Compact electric flux is unbounded and integer valued. A fixed finite-dimensional bond cannot realise the rotor algebra: $[E, U] = U$ makes the unitary U shift every flux eigenvalue up by one, which no finite spectrum admits. A genuine limit would need $\chi \to \infty$, a boundary-surface tensor product, and a proved convergence of the flux algebra. None of this is in the definition.
the extra datum	Abelian electric boundary charges commute; the finite electric boundary algebra carries no multiplier. The campaign's load-bearing extra datum is precisely a finite-dimensional projective multiplier, the cohomology class $[\omega_\alpha] \in H^2(G, U(1))$. That class is not the Gauss-charge algebra of this target, and the asymptotic-symmetry quotient does not manufacture it.

Two smaller points complete the picture. The padding that makes the window insertion map injective has no continuum falloff counterpart at all: at fixed padding depth its physical thickness even vanishes as the spacing goes to zero, and it implies neither the field falloffs nor smoothness of the boundary profile. And the finite-total-variation profile condition is a *one*-dimensional total-variation condition; a radially constant, angularly nonconstant profile in three dimensions generally fails its naive analogue, the appropriate sufficient replacement being angular C^2 regularity together with falloff-weighted bounds. What survives from the profile definition is structural discipline — a compactly supported profile gives a bona fide operator, a non-decaying profile must be defined by a limiting action, a distributional kernel is not itself an admissible operator — and not definitional equality.

Observation 19.16 (Verdict for the Gauss corner)**.** There is a real quantitative nucleus and it should be stated as exactly what it is: on the physical subspace, regional matter charge equals oriented cut-link electric flux, exactly at finite spacing, with the same orientation as the campaign's two bond insertions; and the weighted version splits the boundary charge into a hard matter term and a gradient term which, under the ordered limits, become the accepted hard and soft charges. This is a connection to accepted continuum electrodynamics, not an analogy. It is *not* an instantiation of the bond-implementer definition, which fails on quotient, locality, boundary multiplicity and spectrum, nor of the profile classes, nor of the modulated charge, which supplies the hard Noether part only. And it would still require a separate scattering and reduction argument to reach the Weinberg pole.

The memory verdict: correct rotor arithmetic, wrong observable

The accepted electromagnetic datum is a real transverse one-form. In the boundary gauge the local source gives the endpoint identity

$$e^2\partial_z N = \int_{-\infty}^{\infty} du\, F^{(0)}_{uz} = A^{(0)}_z\big|_{\mathcal{I}^+_+} - A^{(0)}_z\big|_{\mathcal{I}^+_-}, \tag{408}$$

and the operational readout attached to it is a probe kick or a relative phase: integrating the Lorentz force through the radiation epoch gives $m\,\Delta v_A = q_\mathrm{p}\int dt\, E_A = -q_\mathrm{p}\Delta A_A$. The lattice discretisation of (408) is the *link angle*: with $\theta_\ell = a_\ell e A_i + O(a_\ell^2 e)$ and a principal lift, $\Delta\widetilde{\theta}_\ell/(a_\ell e) \to \Delta A_i$ as $a_\ell e \to 0$, the compact period $2\pi/(a_\ell e)$ receding to infinity. That is the object a discretisation of memory has to reach.

The campaign's circle-charge integrality clause reaches something else, and it reaches it exactly. The compact rotor statement and the campaign clause are the same theorem:

$$e^{2\pi i n_\kappa} = e^{2\pi i\kappa} I \iff \operatorname{spec} n_\kappa \subset \kappa + \mathbb{Z} \qquad \text{versus} \qquad e^{2\pi i S^z_x} = cI \iff \operatorname{spec} S^z_x \subset \kappa + \mathbb{Z},\ c = e^{2\pi i\kappa}, \tag{409}$$

clause for clause, including the affine offset and its role as a background or twist label. A twisted rotor sector, a background electric flux, and the campaign's projective on-site phase all produce the same coset structure, and they cancel identically in a difference taken with the same normalisation. But in lattice electrodynamics the statement belongs to the *link electric field*, conjugate to a gauge angle; in the campaign it belongs to an *on-site matter charge*. Compactness alone identifies neither with the other, and the difference has a quantitative consequence: the link rotor has unbounded spectrum, so $a_\ell e\, n_\ell$ has vanishing level spacing and can tend to a real canonical momentum, whereas a fixed finite on-site spectrum gives $a_\ell e\, S^z \to 0$ and fills nothing. A real limit from on-site weights would need the available weight range to grow — a growing local representation, or a collective many-site sum with occupied weights of order $1/(a_\ell e)$ — and no such representation-scaling limit is stated.

What the protocol measures is therefore charge-transfer counting statistics. At fixed window and reference the windowed charge has spectrum in one affine coset, $\operatorname{spec}\hat{Q}_{W,c_0} \subset \kappa_{W,c_0} + \mathbb{Z}$ with $\kappa_{W,c_0} \equiv L\kappa + s(a+b-1-2c_0)$ modulo the integers; time evolution is an automorphism and preserves the spectrum; hence for two sequential projective readouts

$$\nu = q_- - q_+ \in \mathbb{Z}. \tag{410}$$

This is offset cancellation between two classical outcomes, not commutativity of two operators and not a dynamical memory theorem. It is a legitimate full-counting-statistics protocol; the lattice theorem that consumes it is correct; the naming is what fails.

Observation 19.17 (The free-Maxwell counterexample)**.** Separate the asymptotically free radiative sector of a scattering solution from the hard charged sector that fixes the constraint data. Free

Maxwell already carries infinitely many large-gauge symmetries, and one may choose data whose radiation-zone field has a smooth finite-energy pulse with a nonzero endpoint shift (408). This is accepted electromagnetic soft memory, measurable as a relative phase between adjacent charged probes. Now transplant the protocol. Applied to the separated radiative sector, the windowed *matter* charge has no charge to measure: every spectral projection is the zero-charge projection, every counting law is the point mass δ_0, and the protocol returns zero while the accepted memory is nonzero — the transplant is vacuous. Replacing the measured observable by the radiative memory coordinate does not repair it: that is a field quadrature with continuous spectrum, so the integrality clause fails outright and the support theorem disappears. Substituting photon number to recover an integer spectrum changes both the symmetry charge and the observable, and a sharp memory step is carried by an infrared coherent cloud whose photon number need not be finite even when its energy and its classical memory are, so the first-moment tightness clause can fail too. The same obstruction is plainer still in gravity, where the memory tensor varies continuously with the hard momenta and no integer on-site circle spectrum can be its discretisation.

campaign id:	D13, D26, D27, M-flux, M-INDEX-fin, M-INDEX-spec
proved in:	Section 7, Section 8; audit in `theory/lanes/reduction/q2-memory-defs.md` §2–§5 and `theory/lanes/reduction/q4-adversarial-defs.md` §2
tested by:	—
reviewed in:	`theory/verdicts/reduction-defs-adjudication-r1.md` (N1)

Two further mismatches are worth displaying because each is a named, closable gap rather than a vague worry. The first is the endpoint. A settled detector record implies the protocol's time-averaged endpoint,

$$\left|\frac{1}{T}\int_T^{2T} f(t)\,dt - f_+\right| \le \sup_{t\in[T,2T]} |f(t)-f_+| \longrightarrow 0, \tag{411}$$

but the converse is false: for $f(t)=\cos\omega t$ the same average is bounded by $2/(|\omega|T)$ and tends to zero although f never settles. The protocol can therefore assign an endpoint to a persistent local oscillation for which the accepted final reading does not exist. This is an infrared, long-time issue; no refinement of the spatial discretisation touches it, and closing it needs a Tauberian upgrade that is not proved. The second is conjugacy. If a projective measurement resolves the integer flux sharply, then for the angle unitary $U|n\rangle = |n+1\rangle$ and the associated dephasing map $\mathcal{D}$,

$$\mathrm{tr}\big[\mathcal{D}(\rho)\,U\big] = 0 \qquad \text{for every trace-class } \rho: \tag{412}$$

a sharp charge measurement destroys exactly the phase coherence one would need to read an angle. The campaign's protocol is one step further removed still, measuring on-site matter charge rather than link flux.

Observation 19.18 (The missing bridge, and the name that survives)**.** To pass from a hard charge history to a field memory one must impose Maxwell's constraint, specify early and late Coulombic data and boundary conditions, and invert the angular differential operator. Different angular current profiles can share one total escaped integer and yet have different $\Delta A_A(z,\bar{z})$; the map is not determined by the counting law. No such reconstruction is present in any of the audited definitions or claims, and no limit can manufacture it. This is the campaign's principal outstanding definitional debt, and it is the object the background framing always said memory should live in.

What does survive is a narrower and quite defensible name. The lattice displacement law $\delta x = -\langle N_T\rangle/s$ reduces cleanly, constants included, onto accepted magnon-driven domain-wall motion — that reduction is Section 19.4 above, and it is the strongest reduction in this section.

So "memory" is justified in the magnonics sense, as a permanent displacement of a real object produced by a passing wave. It is not justified as generic electromagnetic or gravitational radiative memory, and the package must not be advertised that way without the reconstruction theorem.

The soft-insertion verdict: which definition is the accepted one

Here the audit's finding is partly favourable and the favourable half should be said first. The campaign's asymptotic-leg definition — one additional delta-normalised asymptotic excitation of unit leg weight, with the same amputation applied to the amplitudes with and without it, and with an explicit statement that this is *not* a charge-created or current-created vector — is the accepted LSZ-type definition, transcribed into lattice-magnon language. Its normalisation fence is what makes it so, and it is exact: with $Q_k|\Omega\rangle = \sqrt{Z_\rho}\,|k\rangle$ one has

$$\frac{\|Q_q|h\rangle\|}{\sqrt{N}} = \sqrt{Z_\rho - 2/N} \longrightarrow \sqrt{Z_\rho}, \tag{413}$$

so a charge-created leg becomes the asymptotic leg only after this conversion in the reduction limit, and a common finite-state factor must never be used to set the leg constant.

The fixed-time protocol datum is a different object. It is deliberately pre-asymptotic: it applies the modulated charge to an already prepared hard packet at finite time and reads a normalised projection ratio, declining to define reduction exhaustiveness, wave operators, or on-shell matching. On the spin-one-half ferromagnet the difference from the incoming scattering wave is computed exactly,

$$Q_{k_s}|k_h\rangle - |B^{\text{in}}\rangle = \big(1 - S_{12}\big)|P_{12}\rangle = -2ik_s|P_{12}\rangle + O(k_s^2), \tag{414}$$

with relative size $\sqrt{2}\,|k_s|\big(1 + O(k_s)\big)$. Because the protocol ties its soft momentum to its soft scale ϵ, this is a first-order discrepancy — and the sought first jet is first order too, since $S_{\text{phys}} - 1 = 2i\,\text{sgn}(v_h - v_s)k_s + O(k_s^2)$. The two definitions can therefore agree on the intercept and disagree on the jet, which is the entire content of the theorem. This is why the unrestricted identification is left open rather than assumed.

The order in which the two can be reconciled is fixed, and the protocol gets it right. Writing the stages explicitly,

$$\underbrace{N \to \infty}_{\text{thermodynamic}} \prec \underbrace{T \to \pm\infty}_{\text{scattering}} \prec \underbrace{\big(W \uparrow \mathbb{Z},\ \sigma \downarrow 0\big)}_{\text{window and packet width}} \prec \underbrace{\epsilon \downarrow 0}_{\text{soft, last}}, \tag{415}$$

and the two definitions sit at two different points of that chain:

$$\text{fixed-time datum} \xrightarrow[\text{separated-preparation class only}]{\text{outer limit at fixed } \epsilon} \text{asymptotic-leg multiplier average} \xrightarrow{\epsilon\downarrow 0} \text{soft theorem.} \tag{416}$$

On the separated-preparation class — smooth compactly supported packets in one regular primitive two-magnon channel, with disjoint velocity supports at each fixed ϵ, growing initial separation, shrinking hard width, a settling time that grows, and the ring size chosen last — a norm bridge of the form

$$\big\|e^{-iH_S T} I_S F_R - I_S U_0(T) M_{S_{\text{phys}}} F_R\big\| \le K_{M,\epsilon}\, s_M(f_\epsilon)\, s_M(g_\sigma)\Big[(1+R)^{3-M} + (1+d_\epsilon T - R)^{3-M}\Big] \tag{417}$$

makes every readout error vanish at each fixed ϵ, and the protocol's ratio equals the accepted on-shell multiplier average exactly,

$$\mathcal{A}_*(\epsilon) = \int \big[S_{\text{phys}}(k,h) - 1\big]\, d\mu_{*,\epsilon}(k,h), \tag{418}$$

with *zero* remainder after the outer limit — so automatically $o(\epsilon)$ when the soft limit is finally taken. No bound uniform in ϵ is asserted or needed. Outside that class no such theorem exists and (414) is the control. The gap between the two definitions is therefore definitional, not merely technical.

campaign id:	D24, D25, D29, AC-EX-2M-D29, S-IDX-MATCH-HS-SEP, ML6-HS-SEP
proved in:	Section 10, Section 11, Section 12, Section 13; audit in `theory/lanes/reduction/q3-soft-defs.md` §2, §5
tested by:	—
reviewed in:	`theory/verdicts/reduction-defs-adjudication-r1.md` (N4)

Observation 19.19 (The Weinberg-pole control experiment)**.** The audit ran the obvious hostile test: put a real gauge photon through the campaign's soft-insertion definitions and see whether they notice. They do. In lattice electrodynamics with a hard charged leg of dispersion $E(\boldsymbol{p})$ and velocity $\boldsymbol{v}(\boldsymbol{p})$, the propagator adjacent to emission of a transverse photon of momentum $\boldsymbol{k}$ and energy ω_γ has denominator

$$\big(p^0+\omega_\gamma\big)^2 - E(\boldsymbol{p}+\boldsymbol{k})^2 = 2E(\boldsymbol{p})\big[\omega_\gamma - \boldsymbol{v}(\boldsymbol{p})\cdot\boldsymbol{k}\big] + O(|k|^2), \tag{419}$$

and the lattice Ward–Takahashi identity $\widehat{q}_\mu\Gamma^\mu(p+q,p) = eQ\big[G^{-1}(p+q) - G^{-1}(p)\big]$ ties the vertex numerator to the same derivative of the inverse propagator. Summing attachments over outgoing and incoming legs gives

$$\mathsf{S}_{\text{gauge}}(q) = \sum_i \eta_i\, eQ_i\,\frac{p_i\cdot\varepsilon}{p_i\cdot q} = \frac{\mathsf{S}_{-1}(\hat{k})}{|k|} + O(1), \qquad \eta_i = \pm 1 \text{ for outgoing/incoming,} \tag{420}$$

the leading term being a *Laurent* residue, not a Taylor derivative: $|k|\mathsf{S}_{\text{gauge}}$ has a finite directional limit while $\mathsf{S}_{\text{gauge}}$ has no value or derivative at zero momentum. This sharply violates the campaign's regularity hypotheses — the twice-differentiable soft map with zero intercept and finite contact first jet, and the conjectured linear vanishing of the soft multiplier. The definitions therefore reject the gauge amplitude, and they reject it for the right and observable reason, namely its negative first Laurent power. The complementary half of the control is equally sharp: a literal transplant of the campaign's modulated *matter* charge does not create the pole, and does not create the photon leg either, because on the physical subspace it is the hard part of a large-gauge charge by (405) and the soft radiative part is missing. The correct statement is thus narrow and true: the definitions distinguish the gauge and global infrared classes by scope and by failed regularity, not by one unified charge-insertion definition, and the absence of a $1/\omega$ pole in the one-dimensional spin chain is consistent with, and does not exclude, the standard pole in three spatial dimensions.

One further distinction from the same lane deserves recording, because the two vocabularies collide on the phrase "pure gauge". The campaign's Goldstone analysis calls a direction pure gauge when its tensor is a virtual coboundary $A_\alpha X - XA_\alpha$, which changes no local bulk quantity; the continuum calls the large-gauge mode locally pure gauge while *retaining* its asymptotic endpoint data as a physical charge and vacuum label. The exact statement common to both is only that a zero-momentum coboundary changes no local bulk field or state but can leave boundary data. The campaign's null direction is the analogue of a *proper*, small gauge redundancy; the soft photon is the boundary mode of a *broken large* one. Equating the two would reverse the broken and unbroken content of both definitions. An exact analogue would need an enlarged asymptotic or edge phase space in which the telescoping endpoint vectors are retained and charged rather than removed, and the current definitions do not construct one.

The hostile lane's ranked findings

The fourth audit was run adversarially and without sight of the other three. It was asked to find rats, it found some, and it ranked them by severity. The ranking is reproduced here in substance because the severities, not the prose, are the useful part.

severity	**object**	**finding**
fatal	the no-contact universality class	Source-dependent, and selected by conditions on the very amplitudes whose factorisation is at issue, so that the membership clauses substantially restate the desired conclusion. It has no proved microscopic member and it is not the asymptotic-state domain the continuum theorems use. The conditional implication built on it remains correct as an implication; it is not presently a nonvacuous continuum-style soft theorem, and must never be described as one.
fatal	the bond implementer as a QED boundary charge	Wrong quotient, wrong locality, wrong boundary multiplicity, wrong spectrum, as displayed above. No limit sequence inside the definition yields the unbounded integer flux algebra or the allowed-modulo-trivial gauge quotient.
fatal	the integrality-plus-counting package as generic memory	It measures integer compact-charge counting statistics. A free radiative sector can carry a nonzero soft endpoint shift while the matter-charge counting law is δ_0; substituting the field shift destroys integrality (Observation 19.17).
major	the wall coordinate, charged-dressing masquerade	The windowed observable tends to a soliton centre only for a fixed, centred core profile. Its physical change is $\Delta X + \Delta C_f + (\Delta\,\text{core charge})/(2M_0)$, so a change of charged core dressing at fixed geometric centre is indistinguishable from a translation. The stated packet-outside and padding conditions do not exclude it; a centring condition is owed.
major	the dynamical dress, undefined subtraction	The displayed dynamical coordinate is exactly the conserved regularised total magnetisation, so its displayed memory is identically zero — the trap the definition itself records. Invoking the asymptotic hypothesis cannot change the value of a formula: a leg-subtracted replacement operator must first be defined, and it is not.
major	the excitation ansatz called a "particle"	The ansatz supplies a tangent-space quotient and a boundary-remainder limit, not a spectral band, a resolvent pole, or a wave operator. Particle status is legitimate only where exactness is supplied independently, by the exact-band hypotheses or a separate spectral theorem. The audit checked and found no currently proved row that crosses this gap silently.
major	integrality read as an asymptotic QED charge	It captures the constant compact weight lattice. For a generic real angular profile the weighted hard sum is not integer valued, and the radiative term is not an on-site weight-lattice operator at all.
minor	"spectral dress" as a name	An exact Fourier rewrite of the finite-window response; the word can suggest a radiative-field or soft-amplitude identification that the identity does not supply.
minor	the projective phase in a gauge reading	Ordinary compact rotor flux has trivial phase; a nontrivial offset needs additional background or projective

Observation 19.20 (What survived the hostile hunt)**.** The summary sentence is the one worth remembering: *every definition is internally exact lattice mathematics; what fails are three continuum-facing identifications.* Specifically, the wall coordinate survives narrowly as a centred rigid-soliton collective coordinate; the spectral dress survives as exact endpoint-jump and Fourier-residue kinematics at fixed window with an integrable rate; the integrality clause survives as the correct compact circle arithmetic, matching a trivial-phase rotor flux; the counting protocol survives, renamed, as a coherent conditional two-time-measurement full-counting-statistics definition for lattice charge transfer; the locality, packet-norm, amputation and zero-intercept clauses survive as useful analytic definitions, the asymptotic-leg clause being the accepted reduction-type definition; the bond implementer survives internally as a valid finite endpoint construction on padded windows and states; and the truncated-symmetry identity survives as the exact constant-profile Gauss telescope of (402). What does not survive is the identification of the second group with the continuum objects whose names they carry.

Observation 19.21 (Naming discipline)**.** The audit's operative conclusion is a vocabulary rule, and it binds every external presentation of this campaign. The safe names are: a *collective-coordinate charge ledger* for the wall displacement and its conservation law; a *finite-window Fourier response* for the spectral dress; *compact-charge measurement statistics* for the integrality and counting package; and *conditional exact-band scattering* for the asymptotic-state construction. "Memory" is defensible in the magnonics sense only, as in Observation 19.18. Absent the missing bridges, the campaign may not describe the counting package as generic electromagnetic or gravitational radiative memory, may not use boundary-charge or asymptotic-gauge language for the bond implementer, and may not present the conditional factorisation implication as a nonvacuous continuum-style soft theorem. Three bridges would lift these restrictions, and only these three: a reconstruction theorem tying the charge ledger to the field-side memory; an exhibited microscopic member of the no-contact class; and a centring amendment to the wall coordinate.

campaign id:	D4, D5, D12, D13, D24, D26, D27, ML5-B, D24-VAL, S-general
proved in:	Section 5, Section 6, Section 8, Section 13; audit in `theory/lanes/reduction/q4-adversarial-defs.md` §6–§7
tested by:	—
reviewed in:	`theory/verdicts/reduction-defs-adjudication-r1.md` §1–§4

19.7 What is still owed

The reductions above rest in places on physical judgement rather than computation, and those places are listed here rather than left implicit.

1. The amplitude match of (377) against (379) agrees in form and in every parametric dependence and differs by an overall factor of two. That factor is believed to be a normalisation convention relating the amplitude to the scattering ratio for identical particles, together with the normalisation of the order-parameter density in the continuum source. It has not been checked.

2. The vanishing of the contact coupling is read off the leading doubly soft expansion. Higher orders in the lattice constant are not computed and could in principle generate a contact term at second order.

3. The assertion that the structureless case of the lattice classification is the correct continuum corner is a matching statement, not a theorem.

4. The magnon number density used to convert the continuum spin-wave amplitude into a magnon flux is the linear-spin-wave identification, exact to second order in the amplitude. The match of (389) is therefore a leading-order match, as is the continuum derivation itself.
5. The lattice memory law is conditional on an asymptotic-completeness hypothesis that is open on the full chain. A referee may reasonably say that the lattice has assumed the physics and derived the bookkeeping.
6. The statement that the sector-reduction leakage is of order one at $\Delta \to 1^+$ is an extrapolation from three measured anisotropies, not a bound.
7. The threshold explanation of the transmission mismatch in Remark 19.11 is a physical reading. What is established is only that the two computations disagree in the limit.
8. The assignment of degree zero to the ferromagnet's generalised spatial shift symmetry in Remark 19.4 is read off the structure of the broken generators; no Lie-algebraic classification has been run to exclude a redundant symmetry.
9. The three missing bridges of Observation 19.21 are owed and none is started. A reconstruction theorem tying the charge ledger to the field-side memory is the largest of them, and until it exists the counting package may be called radiative memory in no sense at all.
10. Every arrow of Observation 19.15 — the Riemann-sum convergence of the boundary flux, the spacelike-to-null deformation, and the hard/soft split at null infinity — is a matrix-element or classical statement. No operator-norm continuum limit of the unbounded flux is asserted, and the interacting scaling limit in which it would hold is not constructed.
11. The centring amendment owed to the wall coordinate is a change to a definition, not to a proof. Until it lands, every downstream use of that observable carries the charged-dressing ambiguity as an unstated hypothesis.

Finally, the single most exposed point. The memory corner is proved where the continuum computes, and computed where the continuum is proved: the memory law, which reduces cleanly, is conditional on an open asymptotic-completeness hypothesis, while the only unconditionally computed quantity in the kink model is computed in a regime disjoint from the one in which the continuum comparison lives. A reader who accepts the magnonics reduction will ask next what the lattice computes that is neither assumed nor restricted to a projection, and the honest answer today is the two-magnon soft slope of Section 10, and nothing else in the memory corner.